Meeting of the Board
15 – 18 July 2024
Songdo, Incheon, Republic of Korea
Provisional agenda item 10

GCF/B.39/02/Add.09/Rev.01

15 July 2024

Consideration of funding proposals – Addendum IX

Funding proposal package for FP234

Summary

This addendum contains the following seven parts:

a) A funding proposal titled "Tonga Coastal Resilience";
b) No-objection letter issued by the national designated authority(ies) or focal point(s);
c) Environmental and social report(s) disclosure;
d) Secretariat’s assessment;
e) Independent Technical Advisory Panel’s assessment;
f) Response from the accredited entity to the independent Technical Advisory Panel's assessment; and
g) Gender documentation.

It is noted that the environmental and social report(s) disclosure has been updated.

Table of Content

Funding Proposal

Project/Programme title:	Tonga Coastal Resilience
Country(ies):	Tonga
Accredited Entity:	United Nations Development Programme (UNDP)
Date of first submission:	[2017/06/05]
Date of current submission	[2024/05/02]
Version number	[V.011]

GREEN CLIMATE FUND

Contents

Note to Accredited Entities on the use of the funding proposal template
• Accredited Entities should provide summary information in the proposal with cross-reference to annexes such as feasibility studies, gender action plan, term sheet, etc. • Accredited Entities should ensure that annexes provided are consistent with the details provided in the funding proposal. Updates to the funding proposal and/or annexes must be reflected in all relevant documents. • The total number of pages for the funding proposal (excluding annexes) **should not exceed 60**. Proposals exceeding the prescribed length will not be assessed within the usual service standard time. • The recommended font is Arial, size 11. • Under the GCF Information Disclosure Policy, project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Accredited Entities are asked to fill out information on disclosure in section G.4.

Please submit the completed proposal to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

“FP-[Accredited Entity Short Name]-[Country/Region]-[YYYY/MM/DD]”

GREEN CLIMATE FUND

A

A. PROJECT/PROGRAMME SUMMARY

A.1. Project or programme	Project	**A.2. Public or private sector**	Public
A.3. Request for Proposals (RFP)	*If the funding proposal is being submitted in response to a specific GCF Request for Proposals, indicate which RFP it is targeted for. Please note that there is a separate template for the Simplified Approval Process and REDD+.* Not applicable		

A.4. Result area(s)

*Check the applicable GCF result area(s) that the overall proposed project/programme targets below. For each checked result area(s), indicate the estimated percentage of **GCF and Co-financers' contribution** devoted to it. The total of the percentages when summed should be 100% for GCF and Co-financers' contribution respectively.*

	GCF contribution	Co-financers' contribution[1]
Mitigation total	Enter number %	Enter number %
☐ Energy generation and access	Enter number %	Enter number %
☐ Low-emission transport	Enter number %	Enter number %
☐ Buildings, cities, industries and appliances	Enter number %	Enter number %
☐ Forestry and land use	Enter number %	Enter number %
Adaptation total	Enter number %	Enter number %
☒ Most vulnerable people and communities	60 %	85 %
☐ Health and well-being, and food and water security	Enter number %	Enter number %
☒ Infrastructure and built environment	40 %	15 %
☐ Ecosystems and ecosystem services	Enter number %	Enter number %

A.5. Expected mitigation outcome *(Core indicator 1: GHG emissions reduced, avoided or removed / sequestered)*	*Indicate greenhouse gas (GHG) emission reductions or removals in tCO2eq over total lifespan of the project/programme[2]*	**A.6. Expected adaptation outcome** *(Core indicator 2: direct and indirect beneficiaries reached)*	100,179	
			3,608 direct beneficiaries (of which 1,804 are women)	96,571 indirect beneficiaries expected (of which 48,289 are women)
			3.6% *Indicate % of direct beneficiaries vis-à-vis total population*	96% *Indicate % of indirect beneficiaries vis-à-vis total population*

A.7. Total financing (GCF + co-finance[3])	23,919,409 USD	**A.9. Project size**	Small (Upto USD 50 million)
A.8. Total GCF funding requested	22,656,409 USD *For multi-country proposals, please fill out annex 17.*		

[1] Co-financer's contribution means the financial resources required, whether Public Finance or Private Finance, in addition to the GCF contribution (i.e. GCF financial resources requested by the Accredited Entity) to implement the project or programme described in the funding proposal.
[2] The total lifespan of the project/programme is defined as the maximum number of years over which the outcomes of the investment are expected to be effective. This is different from the project/programme implementation period.
[3] Refer to the Policy of Co-financing of the GCF.

A.10. Financial instrument(s) requested for the GCF funding	*Mark all that apply and provide total amounts. The sum of all total amounts should be consistent with A.8.* ☒ Grant 22,656,409 ☐ Loan Enter number ☐ Guarantee Enter number ☐ Equity Enter number ☐ Results-based payment Enter number		
A.11. Implementation period	*Indicate the number of years and months the project/ programme is expected to be implemented.* 7 years	**A.12. Total lifespan**	*Indicate the maximum number of years over which the outcomes of the investment are expected to be effective, i.e. to lead to adaptation and/or mitigation results.* 40 years
A.13. Expected date of AE internal approval	*This is the date that the Accredited Entity obtained/will obtain its own approval to implement the project/ programme, if available.* Click or tap to enter a date.	**A.14. ESS category**	*Refer to the AE's safeguard policy and GCF ESS Standards to assess your FP category.* B
A.15. Has this FP been submitted as a CN before?	Yes ☒ No ☐	**A.16. Has Readiness or PPF support been used to prepare this FP?**	Yes ☐ No ☒
A.17. Is this FP included in the entity work programme?	Yes ☒ No ☐	**A.18. Is this FP included in the country programme?**	Yes ☒ No ☐
A.19. Complementarity and coherence	*Does the project/programme complement other climate finance funding (e.g. GEF, AF, CIF, etc.)? If yes, please elaborate in section B.1.* Yes ☒ No ☐		
A.20. Executing Entity information	*If not the Accredited Entity, please indicate the full legal name of the Executing Entity(ies) and provide its country of registration and ownership type. Note that there can be more than one Executing Entity. Also indicate if an Executing Entity is the National Designated Authority. Refer to the definition of Executing Entity in the Accreditation Master Agreement.* United Nations Development Programme (UNDP)		
A.21. Executive summary (max. 750 words, approximately 1.5 pages)			

Provide an executive summary of the project/programme including:
Climate change problem
Proposed interventions
Climate results/benefits

1. Climate change is expected to bring Tonga serious impacts that include more heavy rainfall events, severe tropical cyclones and extreme sea level events, which will require adaptation for people, infrastructure, and coastal ecosystems[4]. Tonga faces a potential long-term threat from permanent inundation and wave-driven flooding, and some studies have suggested that significant displacement of communities will take place (World Bank, 2021a). The largest island Tongatapu, faces significant frequent flooding risks, as well as permanent losses from sea level rise (SLR) in the absence of future adaptation measures. Tongatapu is Tonga's most populous island and has the highest population density holding around 74% of the population (74,320).[5] This figure is projected to grow over the next decade with Tongans commonly relocating from the outlying islands to Tongatapu. On Tongatapu there is clear evidence of the impacts of local sea level rise, and the permanent losses under a 0.5m SLR scenario are expected to be 6%, and under a 1.0m SLR scenario to be 25% [6]. Sea level rise hotspots present greater impacts in certain areas or regions over others, including the South Pacific Convergence Zone (SPCZ), with greater warming projected in the areas in and around Tonga (the Southwest SPCZ and the Northeast SPCZ)[7]. By the 2090s, sea level rise is estimated at around 59.5 cm increase using historical records only, or within the range of 40 cm–87 cm under RCP8.5 [8]. By 2150 SLR in Tonga is likely to have risen by 1.37 meters and up to 2m (under SSP3-7.0), and by up to 1m already by 2100.[9] With most of Tongatapu's most populated areas of the north and northeast coasts at only 1m above sea level, SLR is likely to see many settlements inundated in the absence of adequate adaptation strategies in place. Tongatapu faces significant frequent flooding risks, as well as permanent losses from SLR in the absence of future adaptation measures. At the same time, geographic isolation and economic vulnerabilities, including dependence on remittance and foreign aid, increase the challenges faced by communities and decision-makers and compound climate change risks.

2. The project therefore seeks to address coastal innundation in critical areas through immediate coasta protection measures in the northeastern parts of Tongatapu, while fostering long term adaptation measures through the provision of critical climate risk information and planning tools supported through key training, knowledge sharing and capacity. The project also propses a transformative adaptation approach through the establishment and support for multi-stakeholder decision making on critical adaptation needs and land use planning. Ongoing climate change threats, risk, and impacts to date have challenged Tonga and brought the issue of transformative adaptation to the forefront. Therefore the overall objective for the project is to support the transformative adaptation agenda setting through a participatory process; supporting development and spatial planning (community development plans, district plans and island strategic plans) to consider climate risks and promote appropriate and transformative adaptation options. Specifically, these are to be achieved through three key outputs:

3. **Output 1 Strengthened knowledge, capacity and engagement for incorporating climate risks into long-term adaptation planning supported through multi-sectoral, multi-stakeholder engagement and dialogue platform.** This component will (1.1.) establish a national multi-stakeholder engagement platform for dialogue on co-creating long-term climate change adaptation strategies and solutions including voluntary retreat; (1.2) develop village and district level participatory climate risk informed plans, and; (1.3) build the capacity of local government, village committees and NGOs to integrate climate risks and adaptation needs into community level planning, and inform future Community Development Plans (CDP).

   Output 2 Strengthened national and local capacities for effective monitoring and assessment of climate risks will (2.1) strengthen the mechanisms for collecting and analysing data and information for better-informed climate risk monitoring and coastal adaptation planning, and; (2.2) improve the knowledge base of multi-sectoral, multi-stakeholders on adaptation planning strategies for long-term resilient planning and transformative adaptation for Tonga based on climate risks and projections.

   Output 3 Reduced vulnerabilities of coastal communities in Hahake to climate hazards through coastal protection measures by undertaking (3.1) coastal protection measures constructed along 4.3km of coastline in Hahake, and; (3.2) sharing lessons learned and best practices in climate resilient coastal protection measures for scale-up at the national and regional level.

4. The project proposes to directly benefit 3608 people, or 3.6% of the population: <u>Output 1</u> supports building the capacity of local government, village committees and NGOs to integrate climate risks and adaptation needs into community-level planning, and inform future Community Development Plans (CDP) Under Output 1, 300 people (community members) are therefore expected to directly benefit from being trained in climate-risk and adaptation for updating/developing the CDPs/ ISPs. A further 25 government staff (MoI, MLNR, MoF & NP, MIEDECC, MIA) will directly benefit from training in incorporating climate change risks in planning and budgeting, supporting development planning; while under <u>Output 2</u> a further 25 government staff (including MLNR, MoI, MIEDECC) will directly benefit from technical training on climate risk monitoring and data. This figure of 3,608 direct beneficiaries also incorporates the populations directly adjacent to the coastal resilience works under <u>Output 3</u>, a total of 3,258 people:

Table 1 Output 3 beneficaries in Hahake

Census 2021			
Location	**Village population size**	**Female**	**Male**
Kolonga	1201	625	576
Makaunga	357	185	172
Talafo'ou	387	191	196
Navutoka	679	344	335
Manuka	302	150	152
Nukuleka	332	159	173
Total	**3258**	**1654**	**1604**

5. Indirect beneficiaries are the 96,571 remaining total population of Tonga (Tonga's population is 100,179 people as per 2021 census) considering those that will benefit indirectly from the adaptation planning and strategies of the island-level spatial plans developed, and the outcomes developed through the multistakeholder engagement platform. As the first to be updated, the climate-risk-informed planning piloted in Tongatapu will initiate and guide (under Activity 1.2), developing village and district level participatory climate-risk-informed plans, and; (under Activity 1.3) building the capacity of local government, village committees and NGOs to integrate climate risks and adaptation needs into community level planning, and informing future CDPs, (guiding the other 4 Island Divisions). In addition to the Strategic Development Plan for Tongatapu, the project supports other district plans to incorporate climate risk. The project supports the renewed versions of development and spatial planning (community development plans, district plans and island strategic plans) to consider climate risks and promote appropriate and transformative adaptation options. The project engages the participatory process workshops/consultations for developing village and district spatial plans for each of 5 islands and 5 Island-level graphical reports related to settlement patterns and climate risk for strategic long-term coastal adaptation planning developed. This will include consultations determining adaptation needs, gender needs, participatory vulnerability assessment and planning. The Island Plans incorporate the community development plans and district development plans. In addition to renewing the Island Strategic Development Plans – for Ha'apai (2017-2023), Niua (2015-2018), 'Eua (2015-2018), and Vava'u (2017-2023) the island spatial planning will be supported by the collection, interpretation and use of data on climate risks for coastal adaptation planning to produce scenario-based coastal risk maps and knowledge material for use. These will also be aided through the development of materials for developing village and district spatial plans (e.g. graphic reports developed from Output 2, visual aids to demonstrate climate change impacts) and a graphic report package, with visual aids to demonstrate climate change impacts.

6. There are overlaps between the project locations and the beneficiaries that they include under Output 3, and the beneficiaries covered by previously implemented projects. The Tonga Climate Resilience Sector Project (CRSP), Asian Development Bank, 2013-2019: contributed 2kms of revetment for coastal protection measures in the Hahake area. The current proposed Tonga Coastal Resilience project will complement this earlier work, adopting the lessons regarding revetment design and community consultation. The village of Navutoka (the largest in Hahake) is protected by the ABD revetment completed in 2019. However, Navutoka's inhabitants are also included as direct inhabitants to the proposed Tongal Coastal Resilience project, because unless the entire shoreline of Hahake is protected (as is the aim of Output 3's Activity 3.1), flooding can occur from marine water entering currently unprotected neighbouring shores and moving around to flood landward side of Navutoka. The proposed Tonga Coastal Resilience project closes off the gaps in seaward defenses to stop flooding of the entire, low lying Hahake area and thus all villages will have significant protection benefits. The ADB revetment protected from direct wave impacts, but did not protect from potential flooding that can occur if neighboring unprotected shores are significantly over topped. To provide comprehensive flood risk prevention the entire low laying Hahake foreshore requires improved protection. The UNDP Coastal Resilience Project is therefore designed to do this, as it will close gaps in the existing flood prevention infrastructure and intends to bring the entire foreshore to a more uniform flood protection design specification.

[4] ICEDS, *Infrastructure and Settlements Key Findings for the Pacific from the United Nations Intergovernmental Panel on Climate Change's (IPCC) Sixth Assessment Report (AR6) on Impacts, Adaptation and Vulnerability*, Pacific Factsheet: Infrastructure and Settlements (Australian National University (ANU), Canberra: Institute for Climate, Energy and Disaster Solutions (ICEDS), 2022) <https://www.pacificclimatechange.net/sites/default/files/documents/infrastructure-settlements-ipcc.pdf>.

[5] TSD, 'Population and Housing Census | Tonga Statistics Department', 2021 <https://tongastats.gov.to/census-2/> [accessed 18 September 2023]. Tonga Statistical Department 2021

[6] ADB, *Multi-Hazard Disaster Risk Assessment, Tongatapu*, Risk Assessment Summary Report (Asian Development Bank (ADB), 2021).

[7] IPCC, *Sixth Assessment Report Working Group I - the Physical Science Basis - IPCC* (Intergovernmental Panel on Climate Change (IPCC), 2022) <https://www.ipcc.ch/report/ar6/wg1/downloads/factsheets/IPCC_AR6_WGI_Regional_Fact_Sheet_Small_Islands.pdf>.

[8] World Bank, *Climate Risk Country Profile - Tonga* (Washington DC: World Bank Group, 2021).

[9] IPCC & NASA 2023. Projected Sea Level Rise Under Different SSP Scenarios: Nuku'alofa B, Sea Level Projection Tool, NASA Sea Level Change: IPCC AR6 Sea Level Projection Tool https://sealevel.nasa.gov/ipcc-ar6-sea-level-projection-tool?psmsl_id=1842&data_layer=scenario

B. PROJECT/PROGRAMME INFORMATION

B.1. Climate context (max. 1000 words, approximately 2 pages)

Climate change problem: Describe the climate change problem the proposal is expected to address. Describe the mitigation needs (GHG emissions profile) and/or adaptation needs (climate hazards and associates risks based on impacts, exposure, and vulnerabilities) that the proposed interventions are expected to address. Also describe the most likely scenario (prevailing conditions or other alternative) that would remain or continue in the absence of the proposed interventions. Include baseline information. The methodologies used to derive such information, including the mitigation and adaptation needs, should be included in the feasibility study.

Context: In describing the mitigation and/or adaptation needs, briefly describe the target region/area of the proposed interventions including information on the demographics, economy, topography, etc.

Related projects/interventions: Also describe any recent or ongoing projects/interventions that are related to the proposal from other domestic or international sources of funding, such as the Global Environment Facility, Adaptation Fund, Climate Investment Funds, etc., and how they will be complemented by this project/programme (e.g. scaling up, replication, etc.). Please identify current gaps and barriers regarding recent or ongoing projects and elaborate further how this project/programme complements or addresses these.

7. **Geographical setting**: The Kingdom of Tonga spreads across more than 170 small islands, many of which are uninhabited. Located in the western South Pacific - to the west of Niue, east of Fiji, and south of Samoa, Tonga covers around 747km^2 of land area that spreads across an estimated combined land and sea area of 720,000 km^2 [10]. Of the 176 islands, 36 are inhabited by a population totalling approximately 100,179 [11]. Tonga is situated at the subduction zone of the Indian-Australian and the Pacific tectonic plates and lies within the Ring of Fire where intense seismic activities occur. The islands are formed on the tops of two parallel sub-marine ridges stretching from Southwest to Northeast and enclosing a 50 km wide trough. The islands are divided into four main island groups: the Tongatapu group in the south (population 74,320); the Ha'apai group in the middle (population 5,665); the Vava'u group in the north (population 14,182); and the Niuas in the far north (population1,148).[12]

   Tongatapu is Tonga's most populous island with the highest population density. Of Tonga's total population of 100,179 people, around 74% (74,320) reside on Tongatapu Island, approximately half of which are concentrated within Tonga's capital city in the greater Nukualofa area [13]. The Island of Tongatapu has the largest land area of Tonga's islands (approximately 270km^2), and given the location of the main city of Nuku'alofa, Tongatapu also has the largest accumulation of built assets and infrastructure including the main airport and sea port. Despite being a raised atoll, Tongatapu Island is agriculturally highly productive land with an upper layer of volcanic soils derived from ancient volcanic ash falls.

8. **Migration and resettlement**: The population of Tongatapu is projected to grow over the next decade with Tongans commonly relocating from the outlying islands to Tongatapu. While all forms of internal migration occur in Tonga (rural to urban, rural to rural, urban to rural and urban to urban) the predominant trend in Tonga (and indeed globally) is rural to urban [14]. Comparisons from the 2016 Census data, for example, showed this trend of in-migration (11,153) to be far greater than out-migration (3,717). While in Vava'u, in-migration (1,713) was much lower than out-migration (6,039), and a similar trend was found in Ha'apai in-migration (1,551) and out-migration (5,003), 'Eua in-migration (1,739) and out-migration (1,535) and Ongo Niua in-migration (382) and out-migration (244) [15]. Paid work, education, health care and other opportunities associated with the capital city, drive relocation from other less well developed or productive parts of the archipelago. At the same time, it is expected that communities from other smaller, low laying islands and settlements will also become increasingly challenged by meteorological-ocean hazards associated with climate change, which in turn is expected to increasingly drive relocation.

   Displacement has also been an issue due to disaster impacts, such as the 2022 tsunami, and is likely to continue, with significant and long-term displacements clearly requiring multi-stakeholder and cross-ministerial response in the delivery of public services and the management of community interests and expectations. Prior to the 2022 tsunami, the need for government agencies to plan for the continued increase in the population of Nukualofa and Tongatapu and the implications of internal migration shifts for land use, spatial planning, housing and Government services was well-noted within Tonga's Migration and Sustainable

[10] World Bank, 'Tonga', *World Bank Climate Change Knowledge Portal*, 2021 <https://climateknowledgeportal.worldbank.org/> [accessed 20 September 2023].
[11] TSD, 'Population and Housing Census | Tonga Statistics Department', 2021 <https://tongastats.gov.to/census-2/> [accessed 18 September 2023].
[12] TSD, 'Population and Housing Census | Tonga Statistics Department', 2021 <https://tongastats.gov.to/census-2/> [accessed 18 September 2023].
[13] TSD, 'Population and Housing Census | Tonga Statistics Department', 2021 <https://tongastats.gov.to/census-2/> [accessed 18 September 2023].
[14] TWG MSDP, *Migration and Sustainable Development Policy* (Technical Working Group (TWG) of Tonga for the Migration and Sustainable Development Policy (MSDP), 2021) <https://crisisresponse.iom.int/sites/g/files/tmzbdl1481/files/appeal/documents/Tonga%20Migration%20and%20Sustainable%20Development%20Policy.pdf>.
[15] TWG MSDP, *Migration and Sustainable Development Policy* (Technical Working Group (TWG) of Tonga for the Migration and Sustainable Development Policy (MSDP), 2021) <https://crisisresponse.iom.int/sites/g/files/tmzbdl1481/files/appeal/documents/Tonga%20Migration%20and%20Sustainable%20Development%20Policy.pdf>.

Development Policy [16]. Aside from fast-hitting climate and non-climate related disaster events, climate-induced slow onset processes like coastal erosion, ocean acidification and other forms of environmental degradation in the medium term will drive both forced and voluntary internal movements in Tonga [17].

In extreme cases, destruction of habitat and agricultural viability reduces the prospects for return to the home environment and increases the impetus for long term internal and/or outward migration, or displacement [18]. Internal displacement and resettlement is further complicated by land ownership complexities in Tonga whereby under the Constitution of Tonga (the Constituition and the Land Act), all land in Tonga belongs to the King of Tonga. Land allotments are then inherited through families, but only to the eldest 'legitimate' son, and not to 'illegitimate' eldest sons or to women or younger male heirs. Notably, the influx of new settlers to Tongatapu is resulting in ever greater strain on land resources around the capital Nuku'alofa and increasing patterns of settlement into marginal, low-lying and highly flood exposed areas such as Nukuleka.

9. **Risk and vulnerability**: Tonga is one of the most vulnerable countries to climate change and disaster risks. The country ranks 6th most vulnerable and 5th most exposed (out of 185) globally [19]. The ND-GAIN Index score (out of 100) which summarizes a country's vulnerability to climate change and other global challenges in combination with its readiness to improve resilience scores Tonga on the ND-GAIN Index as 41.1, and its overall rank is 140 out of 185 [20]. Tonga's population already lives in a dynamic ecosystem, to which it has adapted, but climate change is likely to increase variability, pose new threats, and place stress on livelihoods. Potential threats to human well-being and natural ecosystems include increased prevalence of heat wave, intensified cyclones, saline intrusion, wave-driven flooding, and permanent inundation. As a result, biodiversity and the natural environment of Tonga are likely to face extreme pressure, as well as the losses of some species of fish, coral, birds, and terrestrial species. Communities are likely to need support to adapt and manage disaster risks facing their wellbeing, livelihoods, and infrastructure [21]. Geographic isolation and economic vulnerabilities, including dependence on remittance and foreign aid, will increase the challenges faced by communities and decision makers. [22]. The largest island Tongatapu, for example, where 74% of the population reside, faces significant frequent flooding risks, as well as permanent losses from sea level rise (SLR) in the absence of future adaptation measures. The permanent losses under a 0.5m SLR scenario are expected to be 6%, and under a 1.0m SLR scenario to be 25% [23]. Tonga has experienced a series of climate disasters in recent years; tropical cyclones Ian (2014), Gita (2018), Harold (2020), as well as non-climate related disasters with the Hunga-Tonga-Hunga-Ha'apai volcano eruption in 2022, which triggered tsunami waves throughout the Tonga archipelago. Combined, climate-induced and non-climate related disasters affected up to 80% of the population and wiped out between 12% to 40% of the GDP [24].

10. **Climate**: Tonga's climate is tropical with a wet season usually from November to April that has both moderate and variable rainfall, and dry season from May to October [25]. Mean annual precipitation averaged 1,666 millimetres (mm) over the period 1901–2019 and mean annual temperature in Tonga varies from 23°C to 26°C [26]. Tonga's weather is governed by a number of factors that include the trade winds and the movement of the South Pacific Convergence Zone (SPCZ), a zone of high-pressure rainfall that migrates across the Pacific south of the equator [27]. Year-to-year variability occurs under the influence of the El Niño Southern Oscillation (ENSO) in the south-east Pacific, which can bring prolonged drought conditions and contribute to a depletion of potable water, and tropical cyclones that occur during the wet season, causing extensive damage to local infrastructure, agriculture, and major food sources [28]. Overall, the Tonga archipelago has observed a historical warming of around 0.6°C between 1979 and 2018 but future trends are obscured by the inability of climate models to accurately simulate trends at sufficiently small spatial scales [29].

For western South-Pacific islands including Tonga, climate change is expected to bring more heavy rainfall events and severe tropical cyclones and extreme sea level events, which will require adaptation for people, infrastructure and coastal ecosystems

[16] TWG MSDP, *Migration and Sustainable Development Policy* (Technical Working Group (TWG) of Tonga for the Migration and Sustainable Development Policy (MSDP), 2021) <https://crisisresponse.iom.int/sites/g/files/tmzbdl1481/files/appeal/documents/Tonga%20Migration%20and%20Sustainable%20Development%20Policy.pdf>.
[17] TWG MSDP, *Migration and Sustainable Development Policy* (Technical Working Group (TWG) of Tonga for the Migration and Sustainable Development Policy (MSDP), 2021) <https://crisisresponse.iom.int/sites/g/files/tmzbdl1481/files/appeal/documents/Tonga%20Migration%20and%20Sustainable%20Development%20Policy.pdf>.
[18] TWG MSDP, *Migration and Sustainable Development Policy* (Technical Working Group (TWG) of Tonga for the Migration and Sustainable Development Policy (MSDP), 2021) <https://crisisresponse.iom.int/sites/g/files/tmzbdl1481/files/appeal/documents/Tonga%20Migration%20and%20Sustainable%20Development%20Policy.pdf>.
[19] UND, 'Rankings - Notre Dame Global Adaptation Initiative - University of Notre Dame', *Notre Dame Global Adaptation Initiative*, 2023 <https://gain.nd.edu/our-work/country-index/rankings/> [accessed 19 September 2023].
[20] UND, 'Rankings - Notre Dame Global Adaptation Initiative - University of Notre Dame', *Notre Dame Global Adaptation Initiative*, 2023 <https://gain.nd.edu/our-work/country-index/rankings/> [accessed 19 September 2023].
[21] World Bank, *Climate Risk Country Profile - Tonga* (Washington DC: World Bank Group, 2021).
[22] World Bank, *Climate Risk Country Profile - Tonga* (Washington DC: World Bank Group, 2021).
[23] ADB, *Multi-Hazard Disaster Risk Assessment, Tongatapu*, Risk Assessment Summary Report (Asian Development Bank (ADB), 2021).
[24] Asian Development ADB, 'Tonga at Risk of Sea Level Rise, Seismic Events, According to ADB Report', *Asian Development Bank*, 2023, Tonga <https://www.adb.org/news/tonga-risk-sea-level-rise-seismic-events-according-adb-report> [accessed 19 September 2023].
[25] World Bank, 'Tonga', *World Bank Climate Change Knowledge Portal*, 2021 <https://climateknowledgeportal.worldbank.org/> [accessed 20 September 2023].
[26] World Bank, 'Tonga', *World Bank Climate Change Knowledge Portal*, 2021 <https://climateknowledgeportal.worldbank.org/> [accessed 20 September 2023].
[27] World Bank, *Climate Risk Country Profile - Tonga* (Washington DC: World Bank Group, 2021).
[28] World Bank, *Climate Risk Country Profile - Tonga* (Washington DC: World Bank Group, 2021).
[29] World Bank, *Climate Risk Country Profile - Tonga* (Washington DC: World Bank Group, 2021).

[30]. For example, through the raising of dwellings and key infrastructure to reduce flood impacts; through planned inland relocation of communities and infrastructure, and; through green / grey protection approaches and infrastructure [31]. Currently, changes in rainfall patterns have resulted in almost-annual flood events that damage major assets and threaten livelihoods. Categories 4 and 5 tropical cyclones are increasing as a proportion of all cyclones and are severely impacting infrastructure in the Pacific. Rising sea levels will continue to drive an increase in the frequency of coastal inundation, damaging coastal infrastructure and settlements.

11. **Sea level rise (SLR):** According to Tonga's Third National Communication to the UNFCCC, Tonga already has witnessed general sea level rise of 6.4 mm per annum, using historical records from 1993–2007 [32]. The 2021 IPCC 6th Assessment Report indicates global mean sea level has risen approximately 20cm since 1901. Rates are accelerating and between 2006 and 2018 global average rates were 3.7mm / year. Although there is significant variability between locations, many countries in the Western Tropical Pacific including Tonga face pronounced accelerated sea level rise. Sea level rise hotspots, present greater impacts in certain areas or regions over others, including the South Pacific Convergence Zone (SPCZ), with greater warming projected in the areas in and around Tonga (the Southwest SPCZ and the Northeast SPCZ) [33]. By the 2090s, sea level rise is estimated at around 59.5 cm increase using historical records only, or within the range of 40 cm–87 cm under RCP8.5 [34]. Uncertainty in the projections of changes in the Antarctic ice sheet limit the certainty of knowledge of the range of this mean sea level change [35]. However, Tonga clearly faces a potential long-term threat from permanent inundation in coastal zones and across low-lying islands as well as wave-driven flooding, and some studies have suggested that significant displacement of communities could take place [36]. In combination with potential tectonic disturbances and cyclone activity, the risk to human communities of coastal submergence and wave inundation is among the highest in the world [37]. Under the recent AR6 IPCC projections [38] these show that even under relatively optimistic emissions reduction scenarios (mid-range SSP1-2.6) sea level will continue to rise for hundreds of years into the future and is likely to be 2m above current levels by 2300. Given current evidence, trajectories more consistent with SSP5-8.5 are more realistic; these indicate several meters (mid-range around 4.5m) above current levels by 2300.

12. On Tongatapu there is clear evidence of the impacts of local sea level rise. The Pacific Sea level and Geodetic monitoring Project's Tongatapu gauge shows a "raw" rate of 6.8mm / year for the period Jan 1993 to Aug 2021 [39]. This 28 year time series is still too short to account for long term seasonal fluctuations in sea level, but it is the relative rate over a meaningful monitoring period and is consistent with research [40] that has found consistently higher rates of sea level rise in the western tropical Pacific Islands.

[30] ICEDS, *Infrastructure and Settlements Key Findings for the Pacific from the United Nations Intergovernmental Panel on Climate Change's (IPCC) Sixth Assessment Report (AR6) on Impacts, Adaptation and Vulnerability*, Pacific Factsheet: Infrastructure and Settlements (Australian National University (ANU), Canberra: Institute for Climate, Energy and Disaster Solutions (ICEDS), 2022) <https://www.pacificclimatechange.net/sites/default/files/documents/infrastructure-settlements-ipcc.pdf>.

[31] ICEDS, *Infrastructure and Settlements Key Findings for the Pacific from the United Nations Intergovernmental Panel on Climate Change's (IPCC) Sixth Assessment Report (AR6) on Impacts, Adaptation and Vulnerability*, Pacific Factsheet: Infrastructure and Settlements (Australian National University (ANU), Canberra: Institute for Climate, Energy and Disaster Solutions (ICEDS), 2022) <https://www.pacificclimatechange.net/sites/default/files/documents/infrastructure-settlements-ipcc.pdf>.

[32] MEIDECC, *Third National Communication on Climate Change Report* (Minister for Meteorology, Energy, Information, Disaster Management, Climate Change and Communications, 2019).

[33] IPCC, *Sixth Assessment Report Working Group I - the Physical Science Basis - IPCC* (Intergovernmental Panel on Climate Change (IPCC), 2022) <https://www.ipcc.ch/report/ar6/wg1/downloads/factsheets/IPCC_AR6_WGI_Regional_Fact_Sheet_Small_Islands.pdf>.

[34] World Bank, *Climate Risk Country Profile - Tonga* (Washington DC: World Bank Group, 2021).

[35] World Bank, *Climate Risk Country Profile - Tonga* (Washington DC: World Bank Group, 2021).

[36] World Bank, *Climate Risk Country Profile - Tonga* (Washington DC: World Bank Group, 2021).

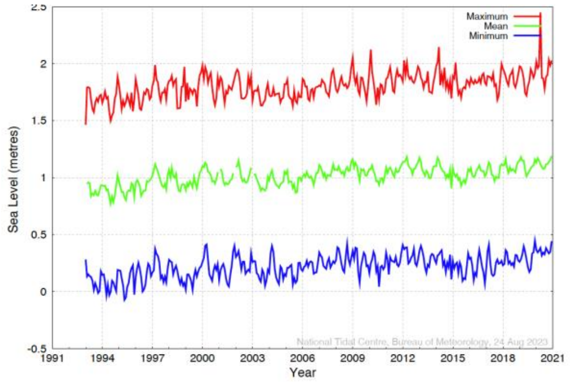
[37] Mélanie Becker, Mikhail Karpytchev, and Fabrice Papa, 'Chapter 7 - Hotspots of Relative Sea Level Rise in the Tropics', in *Tropical Extremes*, ed. by V. Venugopal and others (Elsevier, 2019), pp. 203–62 <https://doi.org/10.1016/B978-0-12-809248-4.00007-8>.

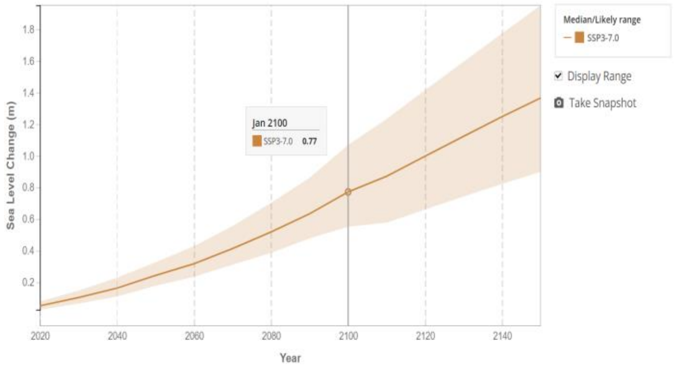
[38] IPCC, *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* (Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA,: [Masson-Delmotte, V., P. Zhai, A. Pirani, S.L. Connors, C. Péan, S. Berger, N. Caud, Y. Chen, L. Goldfarb, M.I. Gomis, M. Huang, K. Leitzell, E. Lonnoy, J.B.R. Matthews, T.K. Maycock, T. Waterfield, O. Yelekçi, R. Yu, and B. Zhou (eds.)], 2021), p. 2391 pp <doi:10.1017/9781009157896>.

[39] BoM, 'Pacific Sea Level Monitoring Project', 2023 <http://www.bom.gov.au/pacific/projects/pslm/#> [accessed 20 September 2023].

[40] Mélanie Becker, Mikhail Karpytchev, and Fabrice Papa, 'Chapter 7 - Hotspots of Relative Sea Level Rise in the Tropics', in *Tropical Extremes*, ed. by V. Venugopal and others (Elsevier, 2019), pp. 203–62 <https://doi.org/10.1016/B978-0-12-809248-4.00007-8>.

Figure 1 (Left) monthly sea level at Nuku'alofa, Tonga, from 1993-2021/ (Right) NASA's sea level projection tool output for Tongatapu SSP3-7.0





Monthly sea level at Nuku'alofa, Tonga from 1993-2021 shows an increase of 6.8mm / yr[41] and NASA's sea level projection tool[42] output for Tongatapu SSP3-7.0 shows a potential midrange rise of 0.77m by 2100. This extent of sea level rise will have catastrophic implications for low laying areas of Tongatapu Island such as village area of Nukuleka and west Kolonga on the Hahake Peninsula, much of which is already subject to damaging wave impacts and is extremely low-lying.

13. **Surface air temperature and sea-surface temperature** are projected to continue to increase over the course of the 21st century. Under all RCPs, the warming of up to 1.0°C is predicted before 2030; but after 2030 there is a growing difference in warming between each RCP. In Tonga by 2090, RCP8.5 results in a warming of 1.8–4.1°C while RCP2.6 gives a warming of 0.2–1.1°C. On the highest emissions pathway (RCP8.5) warming of around 2.6°C is projected by the end of the century for Tonga [43]. Projections for all emissions scenarios indicate that the annual average sea-surface temperature (*with medium confidence*) will increase in the future in Tonga. Of concern are even small increases in associated sea surface temperature and the potential increase in coral bleaching and mortality events. Given the dependency of local communities on reef resources and the defining role reef ecosystems play in protecting and maintaining soft shore environments throughout Tonga, climate change threats to reefs are extremely consequential. Nonetheless, research to define reef decline associated with climate change stress in Tonga specifically is virtually absent. The high resolution bathymetric LiDAR proposed here may act as an excellent baseline from which national monitoring may begin.
14. **Rainfall:** Due to the spatial distribution of the islands of Tonga, there is a disparity in the nature of rainfall in the country. Tonga's Third National Communication to the UNFCCC reports the annual mean rainfall at the five meteorological stations in Tonga between 1971–2007: Tongatapu reported an average of 1,721 mm, Vava'u an average of 2,150 mm, Ha'apai an average of 1,619 mm, Niua Fo'ou an average of 2,453 mm and Niua Toputapu an average of 2,374 mm [44]. Although there are differences between the islands, all of the islands in the Tongan archipelago receive very significant annual rainfall.
15. **Storms and Tropical cyclones:** Storms are by far the most common marine hazards affectng Tonga, as indicated below.

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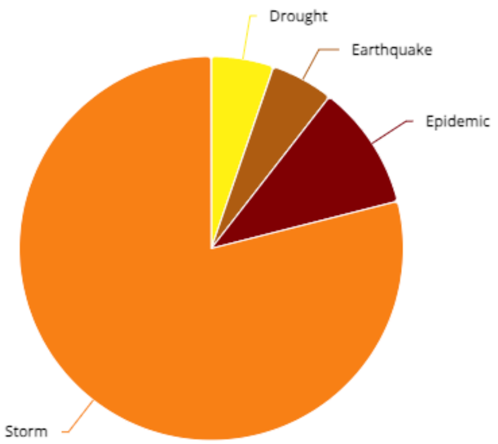
[41] BoM, 'Bureau of Meteorology Australia'. BoM, 'Pacific Sea Level Monitoring Project', 2023 <http://www.bom.gov.au/pacific/projects/pslm/#> [accessed 20 September 2023].
[42] IPCC & NASA 2023. Projected Sea Level Rise Under Different SSP Scenarios: Nuku'alofa B, Sea Level Projection Tool, NASA Sea Level Change: IPCC AR6 Sea Level Projection Tool https://sealevel.nasa.gov/ipcc-ar6-sea-level-projection-tool?psmsl_id=1842&data_layer=scenario
[43] World Bank, *Climate Risk Country Profile - Tonga* (Washington DC: World Bank Group, 2021).
[44] MEIDECC, *Third National Communication on Climate Change Report* (Minister for Meteorology, Energy, Information, Disaster Management, Climate Change and Communications, 2019).

Figure 2 Hazards by occurrence during 1980-2020

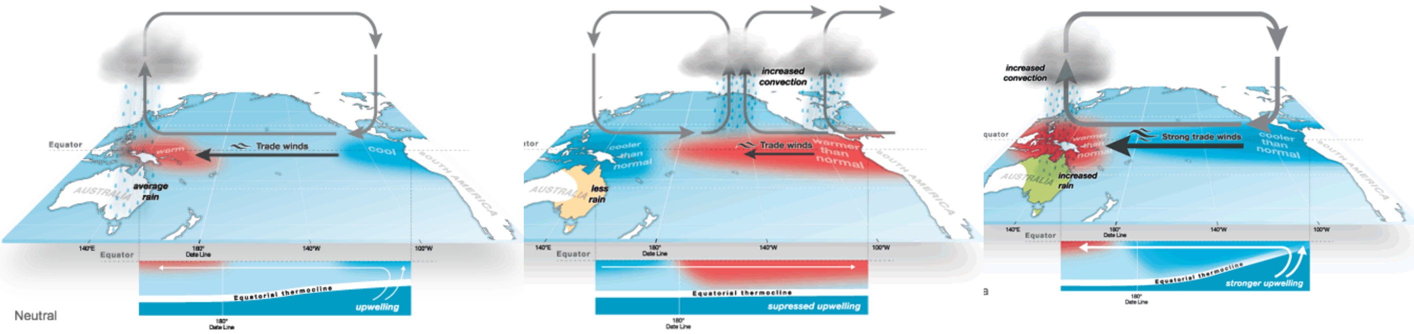


Source: World Bank 2021[48]

Tropical cyclones have huge impacts in Tonga, where a cyclone with a 100-year return period (or 50% chance of occuring within the current generaton) is likely to inflict damage equivailent to 6-% of the GDP [45]. The IPCC indicates that it is likely that the global number of major cyclones (Category 3 – 5) has increased over the last 4 decades [46]. Variability between ocean basins, and that the Southwest Pacific Ocean showed greater increases in Category 4 and 5 storms. These findings are consistent with long term global projections indicating that whilst the overall number of cyclones may decrease, the number of more intense category storms may increase. The recent record in Tonga is consistent with this; there were 85 cyclones developing within or crossing the Tongan archipelago between the years 1969 – 2011 with an approximate average of 20 cyclones per decade [47]. Of this figure of 85 cyclones during the 1969-2011 period, 55 cyclones occurred since 1981 and 19 of these were Category 3 or higher. There have been only three Category 5 tropical cyclones ever recorded in Tongan waters, TC Ron (1997),TC Ian (2014), and TC Harold (2020); it is notable that these have all occurred in the last twenty-five years, and two within the last decade.

16. **Climate variability**: compounds climate change impacts and vice versa. The presence of large-scale climate features and different sized land masses leads to profound regional variations in climate throughut the Pacific. Of these large-scale features, the El Niño-Southern Oscillation (ENSO) is the major cause of year-to-year climate variations [49]. Throughout the ion, ENSO affects the year-to-year risk of droughts, floods, tropical cyclones, extreme sea levels and coral bleaching. El Niño and La Niña events usually begin to develop between May and June and last until the following March to May. During an El Niño event, trade winds weaken or may even reverse, allowing the area of warmer than normal water to move into the central and eastern tropical Pacific Ocean. During a La Niña event, the Walker Circulation intensifies with greater convection over the western Pacific and stronger trade winds [50]. While such events tend to follow a typical pattern of development, the strength and timing of each event is different, as is the exact pattern of sea-surface temperature, wind and impacts. If neither phase is apparent conditions are termed ENSO neutral. In its neutral state (neither El Niño nor La Niña), trade winds blow east to west across the surface of the tropical Pacific Ocean, bringing warm moist air and warmer surface waters towards the western Pacific and keeping the central Pacific Ocean relatively cool [51].

Figure 3 El Niño-Southern Oscillation (ENSO) in 1. Neutral ENSO phase, 2. El Niño phase, and 3. La Niña phase



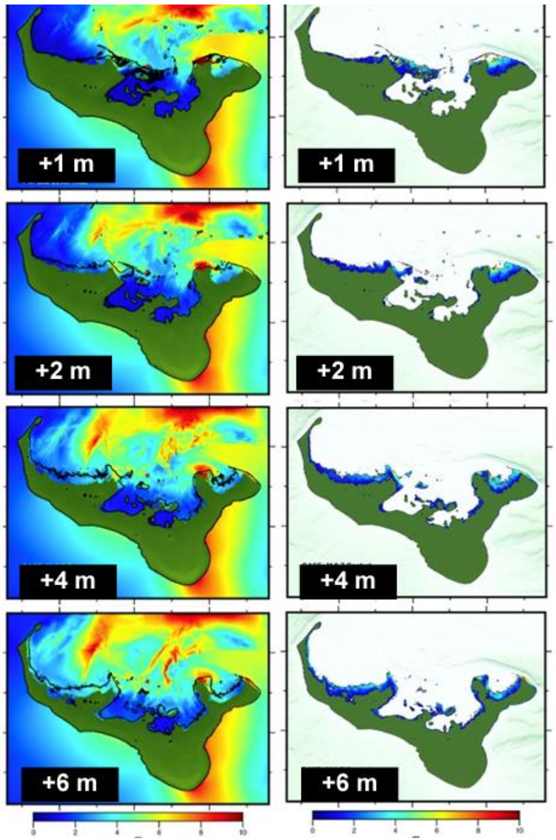
Source:BoM 2023[52].

[45] World Bank, 'Tonga', *World Bank Climate Change Knowledge Portal*, 2021 <https://climateknowledgeportal.worldbank.org/> [accessed 20 September 2023].
[46] *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*.
[47] World Bank, *Climate Risk Country Profile - Tonga* (Washington DC: World Bank Group, 2021).
[48] World Bank, *Climate Risk Country Profile - Tonga* (Washington DC: World Bank Group, 2021).
[49] WMO, 'El Niño/La Niña Southern Oscillation (ENSO)', *World Meteorological Organization*, 2018 <https://public.wmo.int/en/our-mandate/climate/el-ni%C3%B1ola-ni%C3%B1a-update> [accessed 19 September 2023].
[50] BoM, 'The Three Phases of ENSO', *Bureau of Meteorology (BoM) Australia* (Bureau of Meteorology, 2023) <http://www.bom.gov.au/climate/enso/history/ln-2010-12/three-phases-of-ENSO.shtml> [accessed 19 September 2023].
[51] BoM, 'The Three Phases of ENSO', *Bureau of Meteorology (BoM) Australia* (Bureau of Meteorology, 2023) <http://www.bom.gov.au/climate/enso/history/ln-2010-12/three-phases-of-ENSO.shtml> [accessed 19 September 2023].
[52] BoM,'The Three Phases of ENSO', *Bureau of Meteorology (BoM) Australia* (Bureau of Meteorology, 2023) <http://www.bom.gov.au/climate/enso/history/ln-2010-12/three-phases-of-ENSO.shtml> [accessed 19 September 2023].

El Niño and La Niña events have distinct impacts on the rainfall of most Pacific Island countries. Changes in sea-surface temperature and winds associated with El Niño and La Niña cause largescale shifts in rainfall patterns. ENSO also has a profound influence on sea level and the risk of tropical cyclones in the region and Tonga's climate pattern is very much affected by the El Niño phenomenon [53]. As the warm sea surface temperatures move east during El Niño, moisture and water vapour required for cloud formation also migrate eastward, influencing drought conditions in Tonga [54]. The last three major droughts that have occurred in Tonga in 1983, 1998 and 2006 have been directly linked to El Niño events around the same time [55]. Temperatures are also affected; year-to-year temperature variations along the equator can be larger than the average variations between the seasons. For Tonga, in the Southwestern Pacific Ocean, when there is the warm El Niño phase of ENSO (a deeper thermocline and decreased westward transport of water), the sea surface temperature increases to greater than normal in the Eastern Pacific. This shifts the prevailing rain pattern from the normal Western Pacific to the Central Pacific (as shown in the middle graphic).

17. **Combined climate change impacts and seismological hazards:** Notably also, the islands of Tonga are associated with the Tonga-Kermadec Ridge and Tonga Trench, which is a large submarine subduction zone and also one of the fastest moving and most tectonically active on Earth, forming part of the "Pacific Ring of Fire" of intense seismic activity. Tonga has several active sites of volcanism, and is subject to frequent tremors and earthquakes. Individual and island groups have been subject to tectonic uplift, subduction and tilting, however, limited monitoring infrastructure make evaluation of these dynamics very hard to determine. Tsunami inundation modelling, however, suggests that Tonga's islands are critically vulnerable to tsunamis, given their locations along the Tonga Trench. Tsunami modelling undertaken in 2011 [56] and then in 2021 [57] (see left figure) for Tongatapu shows that tsunami is also likely to impact the northern coastal areas of Nuku'alofa, and particularly Nukuleka, Talafo'ou, Navutoka, Manuka and Kolonga. When combined with predicted levels of sea level rise (SLR) under climate change impact projectios, tsunami modelling shows severe inundation throughout much of Nuku'alofa and the northern coast. These predicted impacts according to modelling for four sea level rise scenarios with tsunami heights generated of either +1m, +2m, +4m and +6m. The maximum tsunami amplitude is shown on the left side of the figure, and the overland flow depth on the right, for an 8.7 magnitude earthquake.

Figure 4 Modelling of tsunami events combined with Sea Level Rise (SLR) predictions for Tongatapu Island



Source: Borrero et al 2021[1]

18. Tonga's seismological activity recently resulted in the January 2022 violent eruption of the Hunga Tonga-Hunga Ha'apai submarine volcano 65km north of Tonga's capital Nuku'alofa, which produced a tsunami affecting Tongatapu and other islands of Tonga, as well as affecting Fiji, American Samoa, Vanuatu, New Zealand, Japan, the US, far-easter Russia, Chile and Peru. Hunga Tonga–Hunga Ha'apai's volcanic eruption plume reached a peak height of 58 kilometres into the atmosphere and sustained heights greater than 30 km. The eruption generated ocean-wide tsunamis, never before recorded in the Pacific instrumental record, and which were also undetected by tsunami warning, given that volcanically generated tsunamis still remain a 'blind spot' in our understanding of tsunami hazards [58]. Tsunami waves with run-up heights up to 45m struck the uninhabited island of Tofua in Tonga, and a 1.2m tsunami struck Nuku'alofa and Nukuleka and Manuka on Tongatapu [59].

[53] MEIDECC, *Third National Communication on Climate Change Report* (Minister for Meteorology, Energy, Information, Disaster Management, Climate Change and Communications, 2019).
[54] World Bank, *Climate Risk Country Profile - Tonga* (Washington DC: World Bank Group, 2021).
[55] World Bank, 'Tonga', *World Bank Climate Change Knowledge Portal*, 2021 <https://climateknowledgeportal.worldbank.org/> [accessed 20 September 2023].
[56] H Damlamian and others, *Tsunami Inundation Modelling Of Tongatapu, Kingdom of Tonga* (Secretariat of the Pacific Community (SPC) Geoscience Division (GSD), Geoscience Australia (GA), Kingdom of Tonga, National Disaster Management, 2011) <https://www.preventionweb.net/files/45270_219.pdf>.
[57] J.C Borrero, D Greer, and H Damlamian, 'Tsunami Hazard Assessment for Tongatapu, Tonga' (presented at the Australasian Coasts & Ports 2021 Conference, Christchurch, New Zealand, 2021).
[58] James P. Terry and others, 'Tonga Volcanic Eruption and Tsunami, January 2022: Globally the Most Significant Opportunity to Observe an Explosive and Tsunamigenic Submarine Eruption since AD 1883 Krakatau', *Geoscience Letters*, 9.1 (2022), 24 <https://doi.org/10.1186/s40562-022-00232-z>.
[59] BoM, 'National Tsunami Bulletin', *Bureau of Meteorology (Australia)*, 2022 <http://www.bom.gov.au/tsunami/national.shtml#nationalBulletin0> [accessed 18 September 2023].

19. The tsunami and ashfall following the eruption affected 85 per cent of the Tongan population directly, causing widespread damage to houses, schools, roads, and power and water supply networks [60] and setting a precedent of relocation for many settlements. In terms of loss and damage, the damage to the economy from the eruption and tsunami was estimated to be equivalent to US$182 million (approximately 421 million Tongan *Pa'anga*), or more than 36 percent of Tonga's GDP [61]. According to phone surveys in April-May [62] and July-August [63] of 2022, the tsunami disrupted livelihoods and access to health care; food insecurity; and caused loss of assets, where Tonga's poorest and most-vulnerable families were hit the hardest [64]. Overall, 2507 people (770 men and 828 women, 390 boys and 351 girls along with 168 babies under the age of 5) were directly affected by the tsunami generated as a result of the HTHH volcanic eruption [65]. Among the affected people, the Initial Damage Report (IDA) identified a total of 61 people, 18 female and 43 male having some form of disability. There were four deaths (3 females and 1 male) reported, three as a direct result of the tsunami, and one reported as due to the trauma of the event and 10 people were officially reported injured [66]. The tsunami also displaced coastal communities throughout Tongatapu, 'Eua and Ha'apai who evacuated their homes for safer areas, with around 1.9% of the total Tonga population displaced immediately following the event [67]. The tsunami waves largely affected the western coastal areas of Tongatapu and Ha'apai divisions, with Mango Island (Ha'apai) and 'Atataa Island displace, residents relocated to Tongatapu [68], while 'Eua, Vava'u and Niuas were less affected [69]. In total 293 houses were damaged and1,525 people were displaced by the impacts of the volcanic eruption and resulting tsunami [70]. Of the houses affected or damaged, the majority (174) were in Tongatapu, followed by 'Eua (75), Fonoifua and Pangai (16 each) and Mango (12). Limited flooding affected 3.5 percent of the total land area in the four priority divisions inundated in the days immediately following the tsunami [71]. These households were based in low lying coastal areas and low lying islands. Out of total damaged houses, 113 houses, around 35%, are women headed houses [72].

20. **Climate-sensitive economy:** World Bank data shows that Tonga has one of the world's smallest economies, as measured by GDP [73]. Of 31 thousand employed in the country, approximately 19.4 per cent work in agriculture, 30.9 per cent in industry and 49.8 per cent in services. In Tonga, the number of workers participating in the seasonal schemes in Australia and New Zealand is so considerable that it has led to a shortage of agricultural labour in some sending communities, especially during peak harvest times [74]. However, remittances from families overseas (most often in Australia and New Zealand) have been the largest contributor to Tonga's Gross National Product and a major source of its foreign exchange each-year [75], comprising around USD

[60] World Bank, 'Uncovering the Untold Impact of the 2022 Tonga Volcano and Tsunami: How Phone Surveys Reveal Crucial Insights', 2023 <https://blogs.worldbank.org/eastasiapacific/uncovering-untold-impact-2022-tonga-volcano-and-tsunami-how-phone-surveys-reveal> [accessed 18 September 2023].

[61] World Bank, 'Uncovering the Untold Impact of the 2022 Tonga Volcano and Tsunami: How Phone Surveys Reveal Crucial Insights', 2023 <https://blogs.worldbank.org/eastasiapacific/uncovering-untold-impact-2022-tonga-volcano-and-tsunami-how-phone-surveys-reveal> [accessed 18 September 2023].

[62] World Bank, 'Economic and Social Impacts of the Recent Crises in Tonga: Insights from the April-May 2022 Round of High Frequency Phone Surveys', 2022 <http://hdl.handle.net/10986/38273> [accessed 18 September 2023].

[63] World Bank, 'Economic and Social Impacts of the Recent Crises in Tonga : Insights from the July - August 2022 Round of High Frequency Phone Surveys - Round 2', *World Bank*, 2022 <https://documents.worldbank.org/en/publication/documents-reports/documentdetail/099900212142210482/P177189022159803d0834e0ae94a2732881> [accessed 18 September 2023].

[64] World Bank, 'Uncovering the Untold Impact of the 2022 Tonga Volcano and Tsunami: How Phone Surveys Reveal Crucial Insights', 2023 <https://blogs.worldbank.org/eastasiapacific/uncovering-untold-impact-2022-tonga-volcano-and-tsunami-how-phone-surveys-reveal> [accessed 18 September 2023].

[65] NEMO, *Initial Damage Assessment (IDA Report) Hunga Tonga-Hunga Ha'apai Volcanic Eruption and Tonga Tsunami (HTHH DIsaster)* (National Emergency Management Office (NEMO) Ministry of Meteorology, Energy, Information, Disaster Management, Environment, Climate Change, Communications, and CERT (MEIDECC), January 2022).

[66] NEMO, *Initial Damage Assessment (IDA Report) Hunga Tonga-Hunga Ha'apai Volcanic Eruption and Tonga Tsunami (HTHH DIsaster)* (National Emergency Management Office (NEMO) Ministry of Meteorology, Energy, Information, Disaster Management, Environment, Climate Change, Communications, and CERT (MEIDECC), January 2022).

[67] NEMO, *Initial Damage Assessment (IDA Report) Hunga Tonga-Hunga Ha'apai Volcanic Eruption and Tonga Tsunami (HTHH DIsaster)* (National Emergency Management Office (NEMO) Ministry of Meteorology, Energy, Information, Disaster Management, Environment, Climate Change, Communications, and CERT (MEIDECC), January 2022).

[68] NEMO, *Initial Damage Assessment (IDA Report) Hunga Tonga-Hunga Ha'apai Volcanic Eruption and Tonga Tsunami (HTHH DIsaster)* (National Emergency Management Office (NEMO) Ministry of Meteorology, Energy, Information, Disaster Management, Environment, Climate Change, Communications, and CERT (MEIDECC), January 2022).

[69] FAO, *Hunga Tonga-Hunga Ha'apai Volcano Eruption - Data in Emergencies Hazard Impact Assessment (DIEM-Impact) – Update No. 1, 17 February 2022 - Tonga | ReliefWeb* (United Nations Food and Agriculture Organisation (FAO), 17 February 2022) <https://reliefweb.int/report/tonga/hunga-tonga-hunga-ha-apai-volcano-eruption-data-emergencies-hazard-impact-assessment> [accessed 18 September 2023].

[70] OCHA, *Tonga: Volcanic Eruption Situation Report No.2 (As of 28 January 2022) - Tonga | ReliefWeb* (UN Office for the Coordination of Humanitarian Affairs, 28 January 2022) <https://reliefweb.int/report/tonga/tonga-volcanic-eruption-situation-report-no2-28-january-2022> [accessed 18 September 2023].

[71] FAO, *Hunga Tonga-Hunga Ha'apai Volcano Eruption - Data in Emergencies Hazard Impact Assessment (DIEM-Impact) – Update No. 1, 17 February 2022 - Tonga | ReliefWeb* (United Nations Food and Agriculture Organisation (FAO), 17 February 2022) <https://reliefweb.int/report/tonga/hunga-tonga-hunga-ha-apai-volcano-eruption-data-emergencies-hazard-impact-assessment> [accessed 18 September 2023].

[72] NEMO, *Initial Damage Assessment (IDA Report) Hunga Tonga-Hunga Ha'apai Volcanic Eruption and Tonga Tsunami (HTHH DIsaster)* (National Emergency Management Office (NEMO) Ministry of Meteorology, Energy, Information, Disaster Management, Environment, Climate Change, Communications, and CERT (MEIDECC), January 2022).

[73] World Bank 2023, 'World Bank Open Data', 2023 <https://data.worldbank.org> [accessed 19 September 2023].

[74] IOM & ILO, *Climate Change and Labour Mobility in Pacific Island Countries*, 14 September 2022 <http://www.ilo.org/suva/publications/WCMS_856083/lang--en/index.htm> [accessed 18 September 2023].

[75] Base, 'Remittances for Recovery in Tonga', *BASE*, 2022 <https://energy-base.org/news/remittances-for-recovery-dealing-with-the-aftermath-of-tongas-volcanic-eruption/> [accessed 18 September 2023].

194 million or 37.7% of GDP in 2020, the highest in the world [76], but which increased to 46.2% of GDP in 2021 [77]. Remittances have also been critical to strengthening resilience at the household, community and national levels and potentially constitute an alternative source of climate finance at local levels to build climate-resilient houses, improve the general quality of houses, purchase water tanks, improve property seawalls, as well as climate-proof community infrastructure [78]. Still, while international migration (out of Tonga) is significant with Tongan diaspora estimated to be 126,540 (more than the population residing within Tonga) [79], the population continues to grow with an annual average population growth of 0.3 per cent per annum over the last decade [80].

21. Overall, aside from the contribution from remittances which were 46.2% of GDP in 2021, [81] Tonga's economy is highly dependent on climate sensitive sectors such agriculture, fisheries, and tourism, and a limited resource base that is sensitive to external shocks [82]. In 2022, agriculture contributed to 19.93% of GDP, while industry contributed 19.44%, and services 60.63% [83]. Agriculture and fisheries: Tropical cyclones have significantly affected the agriculture and fisheries sectors. They have experienced immediate and long-term damages on crops, trees, fishing (boats, motors, and gear), livestock, tools and equipment for agriculture, as well as buildings and other infrastructure. Infrastructure: Most buildings and major infrastructure developments in urban and rural areas throughout Tonga occur on vulnerable, low lying coastal areas that are at risk to climate change. The services sector employ the majority of Tonga's population at 43%, however, the agricultural sector, which includes fisheries and forestry, is the leading productive sector in Tonga. The agricultural sector supports the majority of the population for subsistence and for cash income, employing a third of the labour force and accounting for at least 50% of the export earnings [84]. A large share of agricultural production is for subsistence and employs around 30% of the population, followed by industry which engages 27% [85]. Climatic conditions, concern for food security, availability of labour and technology, lack of technology and innovative practices, as well as market access, are all issues affecting agricultural production. The performance of the fisheries sector has fluctuated over time from boom-and-bust cycles due to the migration of the tuna species, market access, fish prices and policy. Tonga's road and other drainage systems remain underdeveloped and ill-suited to cope with intense and frequent rainfall, or storm surge. The lack of appropriate drainage on some roads in Vava'u, for example, also causes large amounts of sedimentation to flow into coastal waters during heavy rain. Water resources: a rise in sea level may provoke seawater intrusion, particularly in low lying coastal areas. Saltwater intrusion would be disastrous as it increases salinity in the groundwater therefore reducing the availability of sufficient freshwater for drinking purposes. A reduction in the area of fresh water due to increased drought, heat evaporaton and other factors such as erosion are a further risk. Biodiversity: biodiversity conservation is one of the most difficult environmental issues facing Tonga. The combined impacts of climate change and disasters on biodiversity only compound the scale of the challenge. For example, it takes 5 to 10 years for a coral reef which was destroyed by cyclone to recover.

22. **Long term transformative adaptation:** The project is designed to address underlying barriers to long term adaptation and specifically, undertake an approach leading to transformative adaptation. Ongoing climate change threats, risk, and impacts to date have challenged Tonga and brought the need for transformative adaptation to the forefront. Adaptive capacity gaps remain to support a deliberative transformative adaptation in Tonga due to the limited vertical and horizontal coordination between different levels of government that limits policy and implementation coherence, and the inadequate resource allocation to enhance overall resilience, specifically at the community level. There is also a a diversity of interests and beliefs that may not be in line with the long-term adaptation needs (including related to land-use). Complex land ownership arrangements and other social and institutional reasons are recognised as issues that need to be addressed to reach a sound, consensus-driven, climate-informed strategic approach to adaptation. Informing all stakeholders and obtaining consensus necessarily needs to be a bottom-up, dialogue-based process, and there is an urgent need to enable discussions on effective climate-informed land use planning that would provide a long-term solution for the vulnerable communities in Tonga. Dialogue for and on transformative adaptation has not yet happened within a multi-stakeholder forum context in Tonga. The project will therefore establish a national multi-stakeholder engagement platform for dialogue, which will provide an unprecedented programme of engagement on the impacts of climate change. The national multi-stakeholder engagement platform will draw on knowledge and experience from a range of sectors which can eventually inform policy options, and utilise knowledge, information and technical inputs on climate change impacts. This platform will work towards gaining interministerial and civil societal support for, as well as lasting agreement and commitment for climate change responses in Tonga. In the absence of this project, Tonga will continue to face these fundamental climate challenges without a coherent national approach to longer-term adaptation planning, including national land use planning.

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[76] IOM & ILO, *Climate Change and Labour Mobility in Pacific Island Countries*, 14 September 2022 <http://www.ilo.org/suva/publications/WCMS_856083/lang--en/index.htm> [accessed 18 September 2023].
[77] World Bank 2023, 'World Bank Open Data', 2023 <https://data.worldbank.org> [accessed 19 September 2023].
[78] IOM & ILO, *Climate Change and Labour Mobility in Pacific Island Countries*, 14 September 2022 <http://www.ilo.org/suva/publications/WCMS_856083/lang--en/index.htm> [accessed 18 September 2023].
[79] TWG MSDP, *Migration and Sustainable Development Policy* (Technical Working Group (TWG) of Tonga for the Migration and Sustainable Development Policy (MSDP), 2021) <https://crisisresponse.iom.int/sites/g/files/tmzbdl1481/files/appeal/documents/Tonga%20Migration%20and%20Sustainable%20Development%20Policy.pdf>.
[80] ILO, *Tonga: The Employment - Environment - Climate Nexus: Employment and Environmental Sustainability Factsheet* (International Labour Organisation (ILO), 30 November 2022) <http://www.ilo.org/asia/publications/issue-briefs/WCMS_862815/lang--en/index.htm> [accessed 18 September 2023].
[81] World Bank 2023, 'World Bank Open Data', 2023 <https://data.worldbank.org> [accessed 19 September 2023].
[82] World Bank, *Climate Risk Country Profile - Tonga* (Washington DC: World Bank Group, 2021).
[83] ADB, *Key Indicators for Asia and the Pacific 2023* (Asian Development Bank, 2023) <https://www.adb.org/publications/key-indicators-asia-and-pacific-2023> [accessed 19 September 2023].
[84] World Bank, *Climate Risk Country Profile - Tonga* (Washington DC: World Bank Group, 2021).
[85] World Bank 2023, 'World Bank Open Data', 2023 <https://data.worldbank.org> [accessed 19 September 2023].

23. **Supporting climate risk informed decision making:** Adaptive capacity gaps also remain inthe limited capacities within the public sector that may affect agenda setting, and the ability to anticipate, guide, and make decisions for long-term adaptation. There is a lack of localisation of knowledge and discourse on climate change, as well as an understanding and awareness of unique attributes of different islands, especially Tongatapu (such as changeability of reef-island, maintenance of sediment supply to island beaches). There is also currently a lack of climate informed land-use planning that takes into consideration the various risks such as future sea level rise and storm surge under different climate scenarios and return periods, which extends to the local levels.

24. The needs for strengthening knowledge, capacity and engagement for incorporating climate risks into long-term adaptation planning by developing capacity of local government, village committees, and NGOs will be assessed during the first part of the project. This will determine the approach for how local government, village committees, and NGOs can be supported to integrate climate change risks and adaptation needs into community level planning, in turn informing community development plans (CDPs), Island Development Plans (IDPs) and the development of climate risk informed spatial planning. At least 300 people are expected to be trained in climate-risk and adaptation for updating/developing the CDPs/ ISPs. Following which training materials and guidelines for incorporating climate-risk into the CDPs and IDP will be available to support those engaged in local level planning to integrate climate change risks in planning and budgeting.

25. Media and local NGOs and civil society organizations will be supported to conduct additional community outreach to socialise the issues informing transformational long-term adaptation approaches to addressing climate change. In collaboration with district and town officers awareness raising on risks, will also identify community priorities on land use and adaptation. The opinions and inputs from the public campaigns will be collated and debated within the multistakeholder engagement dialogue platform, to enable decision-making for long term, transformative adaptation. The project will build upon existing mechanisms for community engagement (e.g. exisiting civil society forums and community dialogue mechanisms) to raise public awareness on climate change and disaster risks that the country is facing. Mechanisms in which to engage dialogue for adaptation exists and traditional community structures also present an effective lens for discussing climate change adaptation. Tonga, for example, practices *Talanga* – an interactive and more purposeful way in which policy and planning related decision making can receive information from community levels[86] and engage communities. *Talanga* is a process of listening and speaking that enables people to 'story their issues' to provide information and lived experiences, that is often used in local debates to 'challenge' and critique assumptions and scrutinise powerful voices.[87] Other mechanisms include *Fono* which is a community meeting process (usually called by the estate owners/nobles) involving chiefs, and town officers.

26. **Supporting climate risk information and data:** In the absence of core technical data and analytical capability to assess and respond to the risks posed by climate change, the project also supports government agencies to be able to a) develop and interpret climate data and information, and b) incorporate this information into planning and decision making. While government agencies have a clear need for climate risk data, there is a lack of technical capacity (including data management, establishment of baseline data and, conducting monitoring and evaluation) for this work. Consequently, there are limitations to how comprehensively analysis of future risks and expected damages has been undertaken. A Capacity Needs Assessment with government agencies (MoI, MLNR, MoF & NP, MIEDECC, MIA) will inform the development of an overarching capacity building strategy and plan that is Government owned. The Assessment will feed into a National Capacity Building Strategy and Plan which will be needs-orientated to meet the existing and required future capacity gaps, and will avoid overlaps with other capacity development activites that the government is involved in. The Capacity Needs Asssessment will assess the key institutional capacity needs in the fields of technical capacity for closely monitoring the coastal dynamics of vulnerable coastlines, designing locally-appropriate protection measures and planning, and implementing the measures that are within technical and financial capabilities of the Government. The Capacity Needs Assesment will thereby determine the development of the long-term Capacity Building Strategy and Plan for government use on coastal and climate risk monitoring and analysis to inform coastal adaptation planning (e.g. including understanding and use of climate and risk infromation and data such as satellite imagery, aerial photography, drone imagery, LiDAR topography and bathymetry), and planning and budgeting for adaptation measures.

27. The Capacity Building Strategy and Plan will be an overarching Government-owned strategy to address the needs and gaps identified through the Capacity Needs Assessment of the various Government agencies and will incorporate the strategic approach for capacity building undertaken within all three components of the project. The Capacity Building Strategy and Plan will be an important tool for the differnet Government agencies to utilise in ensuring coordination among development partners and other Government initiatives that involve capacity building and trainings to ensure there is no overlap/relication and will ensure msustainability of the capacity building efforts of this and other projects. It is a tool that will be fully owned by the Government to take forward beyond the lifetime of this project in ensuring the capacities for resilience building in the country is an ongoing and sustained process. Government staff will also be trained in a) from technical training on climate risk monitoring and data (MLNR, MoI, MIEDECC) focused on improving LiDAR knowledge and capacities for interpreting key information, producing maps and knowledge material on scenario based coastal risk maps and b) incorporating these for decisions on long term settlement patterns and climate risk for strategic long-term coastal planning, supporting development planning (MoI, MLNR, MoF & NP, MIEDECC, MIA). Under the Capacity Building Strategy and Plan, technical officers and local government officers will be trained on coastal zone management, coastal processes, coastal planning and hazards management, and climate down-scaling with training and metoring undertaken and training materials developed and distributed to support government staff across ministries to be able to integrate climate risks into planning and budgeting within their work using these materials. Government staff will be provided with a package of training and capacity building relevant to the specific ministries so that

[86] Park, S., Owens, K., Summerfield-Ryan, O., Taylor, M., Ulloa, A. M, Viney, G, 'How Can Pacific Island Countries Meet Their Nationally Determined Contributions?', *NPJ Climate Action*, 2.1 (2023), 1–11 <https://doi.org/10.1038/s44168-023-00059-0>.

[87] Park, S., Owens, K., Summerfield-Ryan, O., Taylor, M., Ulloa, A. M, Viney, G, 'How Can Pacific Island Countries Meet Their Nationally Determined Contributions?', *NPJ Climate Action*, 2.1 (2023), 1–11 <https://doi.org/10.1038/s44168-023-00059-0>.

equipment and data can be fully utilized for use in generating climate and risk information and data such as satellite imagery, aerial photography, drone imagery, LiDAR topography and bathymetry. Training/mentoring and training materials will be available to support government staff engaged in planning to integrate climate change risks in planning and budgeting, supporting the development planning and decision-making. This will build Tonga's technical capacity in climate risk data collection and management, strengthening Tonga's ability to select long-term and medium-term adaptation interventions by prioritizing high-risk areas by strengthening national and local capacities for effective monitoring and assessment of climate risks. Without these critical and long-term-visioned project interventions, increasing threats will be continue to be faced by vulnerable coastal populations without any coherent mechanism for managed response, including voluntary retreat. This presents risks to people and property, and potentially social cohesion as climate hazards increase over time.

28. **Coastal vulnerabilities in Hahake**: The eastern area of Hahake is shown to the right and this is the proposed site of coastal protection works. The blue zone represents extremely low laying ground less than 1m above mean sea level. The shoreline adjacent to the blue zone across the Hahake peninsula is low lying and extremely exposed to wave overtopping and marine flooding. Two main areas of this shore remain unprotected at this time: Nukuleka in the west (2300m) and Kolonga in the northeast (2025m) where other parts of this shore have been subject to revetment work in 2018-19.

Figure 5 Map of Tonga showing Tongatapu Island where the capital Nuku'alofa is located.

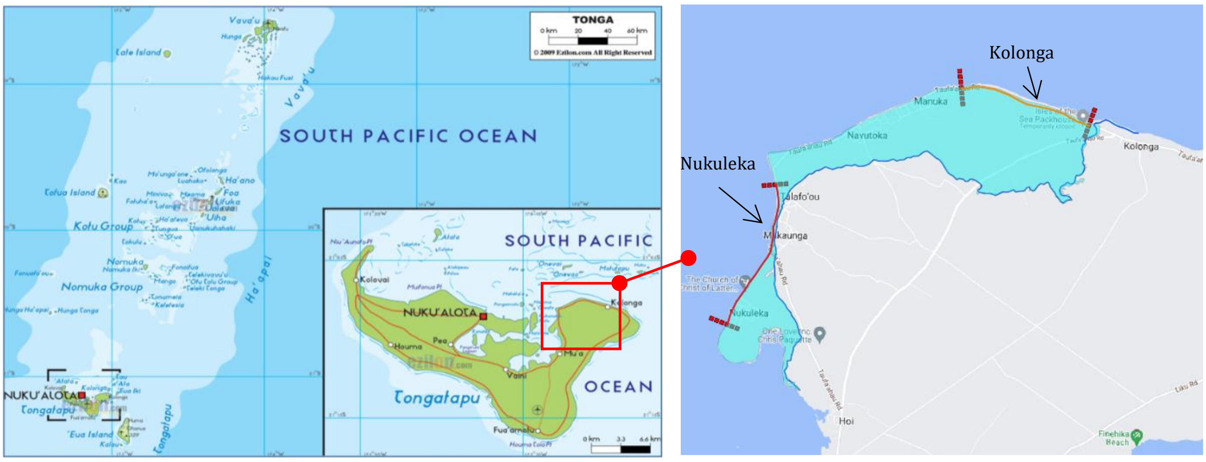
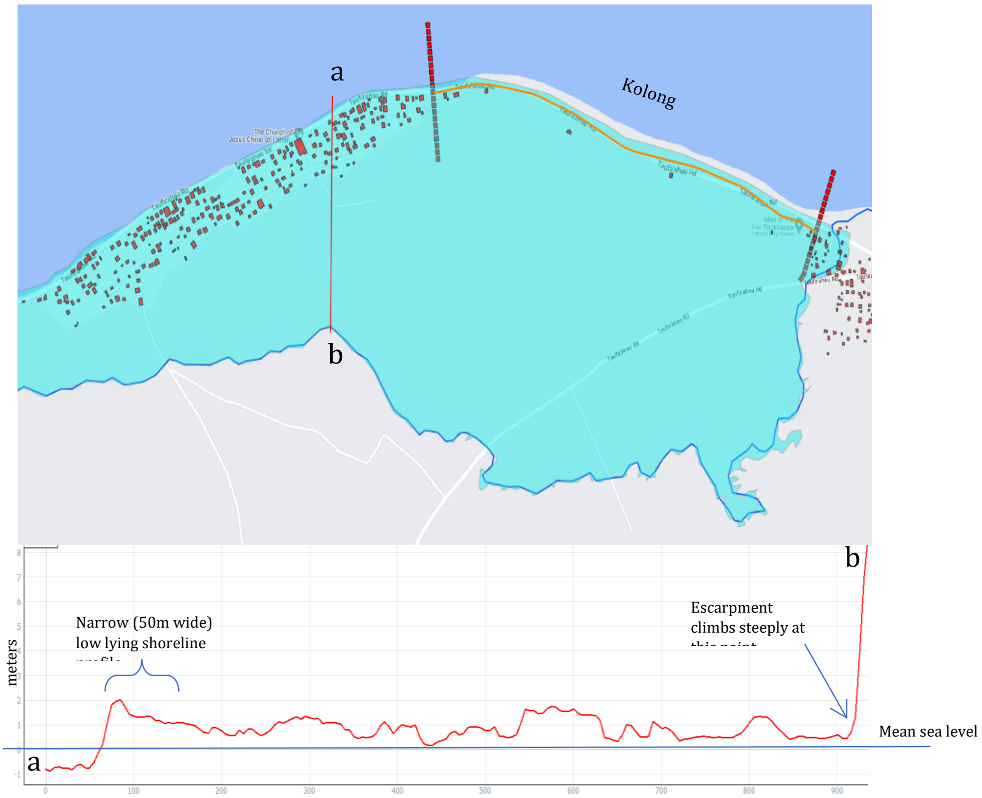


Figure 6 Detail of the low-lying Hahake Penisula on Tongatapu Island – land height against mean sea level



Detail of the extremely low topography of this blue zone adjacent to Manuka Village and the proposed site of coastal protective measures shows that much of the land is less than 1m above mean sea level. Any major breach of the unprotected shoreline in the Kolonga section will result in marine flooding of the villages throughout this area, likely destroying tree crops, food gardens and ground water resources.

29. The Nukuleka shoreline (left-side) is particularly vulnerable without adequate protection, having been subject to piecemeal engineering along its entire length, which has since mostly failed or collapsed. As such, this section of coastline is routinely over swept during storms, which causes marine flooding of homes and cuts the main access road which is also the designated evacuation route for the area. In Kolonga (right-side), the shoreline is functional but very low-lying, and it too, is regularly subject to over-wash damage that affects nearby assets and results in road closures. The profile below shows an elevation of less than 1 meter above median-sea level (MSL) with areas washed out and threatening the roadway. The Kolonga section of shoreline is particularly important to protect from increasing severity of storm waves and sea level rise, since if this shoreline fails and allows large volumes of marine water into the blue zone, it has the potential to flood the entire blue zone where the main villages of Manuka, Navutoka and Talafo'ou are located.

30. *Figure 7 Chainage along the Nukuleka proposed revetment site shown via elevation profile*

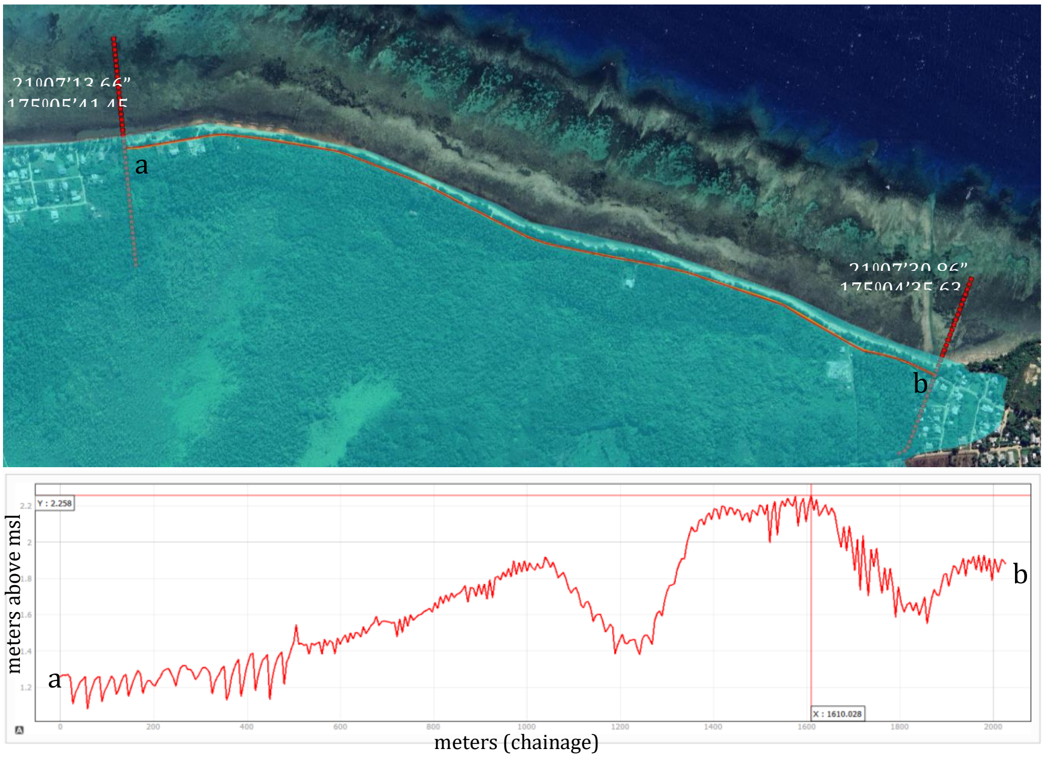
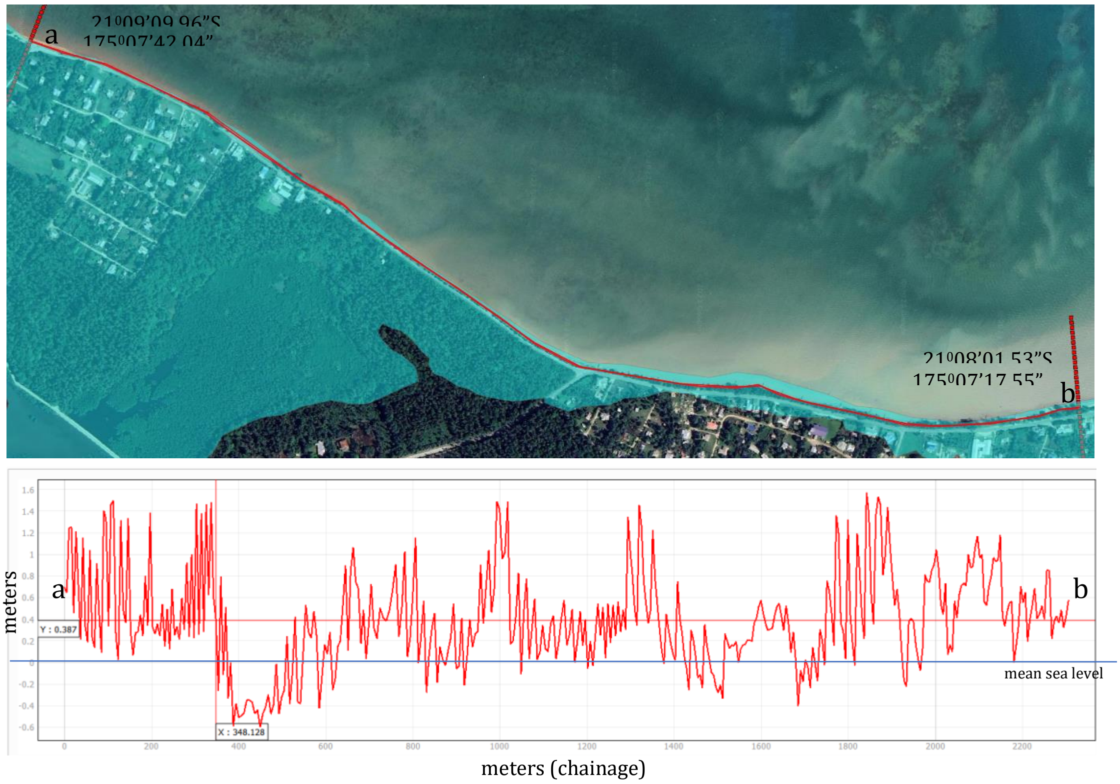


Figure 8 Chainage along the Kolonga proposed raised road site shown via elevation profile



This narrow (~50m) slightly raised berm feature doubles as a road and power line easement and is the main connecting road between Manuka (a) and Kolonga (b). Most importantly this narrow strip of land prevents marine water ingress into the extensive freshwater swamps landward. Whilst this nearshore feature is slightly more elevated than the Nukuleka shoreline the reef platform is comparatively narrow, and the site is far more exposed to larger deep water waves - making it very exposed to wave over wash and damage. In both cases these shorelines are extremely vulnerable to over wash and the very old Nukuleka seawall and ad hoc protective measures prevously built are in very poor condition. They were not subjected to appropriate engineering design and offer little protection during bad weather. Given this shoreline has been subject to hard engineering (albeit ad hoc/very poor engineering) for at least 60 years, thereby, there are few environmental risks to replacing the failing ad hoc protective measures with a well-designed revetment that is able to effectively reduce over wash and damage to property and assets such as the single access road.

31. In Kolonga this shoreline site is also very exposed to over wash events and road closure but this shoreline has only partially be subject to very old revetment works. These only occur in the western side over the first 200 – 250m after this the shoreline is degraded but still functional as a natural beach which is used by the local community for recreation and fishing that also has unusual geological formations, mangrove and seagrass grows in the nearby shallows. As such, it is not deemed advisable to replace this shoreline with a revetment and the best way to add overtopping protection and ensure a “last line of defense” would be to simply raise the road and construct a buried protective natural rock wall on the seaward edge of the raised road. In this way the natural shoreline is retained but much needed protection from over wash and storm waves would be achieved. It would also reduce damage to the main connecting road between Manuka and the other western villages and Kolonga. Kolonga is alternatively situated on higher ground with a rocky cliff shoreline and is used by the western low lying villages as an escape route for tsunami and cyclone alerts. Hence the serviceability of this connecting road is in itself an important consideration.

32. Without the proposed interventions, highly vulnerable coastlines on Tongatapu will remain without protective adaptation measures. In the low-lying areas of Tongatapu, sea-level rise and more intense cyclones are causing increased vulnerability for housing and infrastructure, and coastal land loss, with flow-on effects on livelihoods and food security. The Northwestern coastline of Tongatapu, the target location for physical coastal protection work under this project, is considered among the most at risk.

33. **Previous work addressing adaptation needs**: To date, the Government of Tonga has a number of critical policies and strategies upon which the proposed project builds. Notably:

- The Joint National Action Plan 2 on Climate Change and Disaster Risk Management (JNAP2) 2018-2028 is Tonga’s National Adaptation Plan (NAP) for climate change, sets out six policy objectives and targets an implementation strategy for the country to achieve its vision of a Resilient Tonga by 2035. JNAP2 is aligned with the Tonga Climate Change Policy and covers both climate change adaptation and disaster risk management. The six objectives are: 1) mainstreaming for a resilient Tonga, 2) implementation of a coordinated approach to research, monitoring and management of data and information, 3) Resilience-building response capacity, 4) Resilience-building actions, 5) Finance, and 6) Regional and international cooperation. These build on Tonga's first Joint National Action Plan on Climate Change Adaption and Disaster Risk Management (JNAP 1) in 2010 and existing capacities and enhancing institutions already in place at national, sectoral, community levels as well as the Outer Islands in planning processes for adaptation planning'.

- The Tonga Strategic Development Framework (TDSF) II, 2015–2025 guides the direction of infrastructure planning and development and sets out seven National Outcomes for a) a more inclusive, sustainable and dynamic knowledge-based economy; b) a more inclusive, sustainable and balanced urban and rural development across island groups; c) a more inclusive, sustainable and empowering human development with gender equality; d) a more inclusive, sustainable and responsive good-governance with law and order; e) an more inclusive, sustainable and successful provision and maintenance of infrastructure and technology; f) a more inclusive, sustainable and effective land administration, environment management, and resilience to climate and risk, and; g) a more inclusive, sustainable and consistent advancement of our external interests, security and sovereignty.
- 151 Community Development Plans (CDPs) and 6 District Development Plans, including Island Stategic Development Plans – for Ha'apai (2017-2023), Niua (2015-2018), 'Eua (2015-2018), and Vava'u (2017-2023), were developed by the Mainstreaming of Rural Development Innovation Trust (MORDI TT) and the Ministry for Internal Affairs, which seek to streamline strategic planning priorities under the Tonga Agriculture Sector Plan (TASP), the Tonga Fisheries Sector Plan (TFSP), the Tonga Tourism Road Map (TTRM), and the JNAP, as well as to streamline and ensure that International Development Partner efforts are not duplicated, or duplicating exisiting efforts.
- Migration and Sustainable Development Policy (2020) which provides the Government of Tonga with a policy planning framework to conceptualise and mainstream migration within Tonga's national development planning in a way that maximizes the potential gains of migration and minimizes the associated costs to address migration and incorporate it into the country's national development planning in a strategic and comprehensive manner. This includes regarding key aspects of migration driven by climate change, in terms of: Ensure access for all to adequate, safe, and affordable housing and basic services, including those who are internally displaced and those residing in informal settlements; enhancing inclusive and sustainable urbanization and the capacity for participatory, integrated, and sustainable human settlement planning and management, and; ensuring that all men and women, in particular the poor and the vulnerable, have equal rights to economic resources, ownership and control over land and other forms of property, inheritance, and natural resources in order to secure adequate, safe, and affordable housing and basic services.
- The Tonga National Infrastructure Investment Plan, 2021–2030 (NIIP3) was approved by the Cabinet in August 2021 and includes government priorities covering all infrastructure areas. The plan mainstreamed international infrastructure investment best practices into the Government of Tonga's integrated planning and project cycles, which include regular updates to ensure alignment with the second Tonga Strategic Development Framework (TSDF II), which covers 2015–2025, and with each administration's Government Priority Agenda (GPA). The Tonga National Infastructure Investment Plan 2021-2030 (NIIP 3) is aimed at strengthening the planning and budgeting system for infastructure investment and prioritises major infrastructure investments through its list of 13 priority projects. The NIIP processes—include tools and templates for project proposals and the project prioritization methodology. The Tonga National Infrastructure Investment Plan (2021-2030) includes an Hunga Tonga-Hunga Ha'apai Volcanic Eruption and Tsunami Update.
- National Women's Empowerment and Gender Equality Tonga Policy and Strategic Plan of Action 2019-2025 sets out to achieve gender equity by 2025, and ensure that "That all women, men, girls and boys as a whole achieve equal access to economic, social, political and religious opportunities and benefits." The policy seeks to achieve this, in part, through the provision of opportunities and support for women to develop their abilities to maximise their participation and benefits from society.
- The Kingdom of Tonga National Biodiversity Strategy and Action Plan (NBSAP) 2006, sets out to achieve the CBD Convention objectives, with key themes covering marine ecosystems, forest ecosystems, species conservation, and mainstreaming biodiversity conservation, amongst others - with specific emphasis on ecosystem approaches, agrobiodiversity and access and benefit sharing issues, local communities and civil society participation.
- The Kingdom of Tonga Ridge to Reef Strategic Action Framwork 2022-2030 under the GEF Pacific International Waters Ridge to Reef Regional Project, is aimed at achieving, by 2030, mainstreaming Ridge to Reef processes and objectives in national policy, frameworks, development activities and community led initiatives to ensure that water resources, environmental and social stability are the priority for a resilient Tonga.

34. In response to Tonga's urgent need for building coastal resilience to climate change and in accordance with JNAP 2 priorities, multiple donor-assisted and community-led projects have addressed coastal vulnerability issues. The most recent are:

- The Asian Development Bank (ADB) Multihazard Assessment (2021) of Tongatapu assessed multiple natural hazards in the capital city, Nuku'alofa, and the entire island where it is located, Tongatapu. Under the Tonga Integrated Urban Resilience Sector Project (TIURSP), ADB plan to support the Government to manage current flood risk and increase the resilience of the Nuku'alofa urban area. For this purpose, the 'Climate and Disaster Resilient Urban Development Strategy and Investment Plan' was developed, including a long-term adaptation pathway for Nuku'alofa which will enable continued development of the city in a manner resilient to natural hazards. To effectively prioritise components of the future strategy – e.g. investment in infrastructure for the most high-risk and/or ecologically sensitive locations, combined with other adaptation strategies such as changes in building policies, relocation or reclassification of urban areas, etc., the ADB and the GoT require a detailed assessment of hazard, exposure and risk for Tongatapu island. The project to complete these assessments is known as the Multi-Hazard Disaster Risk Assessment (MHDRA) provides the basis for decision making by estimating the cost of damage and number of people and assets at risk to inundation, tsunami, wind, and seismic hazards, including the impact of climate change, at a range of return periods.

35. Previous projects also include:

- Other projects such as the Australian Government's Governance for Resilient Development in the Pacific Program (Gov4Res) (Australian contribution $7.9 million, 2019-2023) supports national and local governments and communities, as well as regional organisations, to strengthen decision-making processes and governance systems towards risk informed and resilient development of which UNDP's proposed project will also support outcomes, given the enhanced access to technical capacity building and training for government staff, for climate-risk informed decision making under Output 2.

- Tonga Climate Resilience Sector Project (CRSP), Asian Development Bank, 2013-2019: This project was funded by the ADB (US$ 20 million) with co-financing from the Government of Tonga (GoT - US$ 3.88 million). The purpose of the project was to implement the Strategic Program for Climate Resilience (SPCR) prepared by GoT under phase II of the Pilot Program for Climate Resilience (PPCR). The CRSP Project contributed 2kms of revetment for coastal protection measures in the Hahake area, and the current project will complement the earlier work, and has adopted lessons regarding revetment design and community consultation.
- The IFAD Tonga Rural Innovation Project completed in 2017, supported the development of 151 Community Development Plans (CDPs) between 2015 and 2017, supporting 13,238 direct beneficiaries.

- EU funded (€12 million) Intra-ACP GCCA+ Pacific Adaptation to Climate Change and Resilience Building (PACRES), 2018 – 2023, working with the Ministry of Meteorology, Energy, Information, Disaster Management, Environment, Climate Change and Communication (MEIDECC), to build the capacity of Tongans to adapt to climate change andcontinue to build their resilience focused on the development of climate change and disaster resilience information andknowledge management products to enhance community awareness about the harmful impact of climatechange. This includes under JNAP 1, Tonga's climate change portal, hosted by MEIDECC: https://www.climatechange.gov.to/ Output 2 supports the climate change information and knowledge management products to enhance community awareness about the harmful impact of climate change developed under PACRES by providing the critical technical capacity for tools and for climate information and knowledge management.

- The Global Climate Change Alliance Plus Scaling Up Pacific Adaptation (GCCA+SUPA), 2019-2022: The overall objective of the project is for a holistic approach to coastal protection in northern Tongatapu to be adopted by government. The specific objective is to better equip communities to undertake small-scale coastal protection measures (hard and soft engineering). The three key result areas are (1) Conduct a coastal assessment, feasibility and conceptual design study for coastal protection along the entire north coast of Tongatapu (Niutoua to Ha'atafu); (2) Implement small-scale coastal protection and ecosystem-based measures in northwest Tongatapu (Sopu to Ha'atafu); and (3) Enhance awareness about the impact of climate change and natural disasters in Tonga.

- Trialling coastal protection measures in Western Tongatapu Project, European Union, 2016-2018: Funded by the European Union (EUR 550,000) and implemented by GIZ, this project has the overall objective of increasing resilience of six coastal communities in western Tongatapu to climate change impacts and sustaining livelihoods in those communities. There are four targeted outcomes: (i) Reduced climate change vulnerability and enhanced adaptive capacity, focusing on coastal protection and management; (ii) Enhanced education and awareness on coastal management in the context of climate change; (iii) Effective monitoring of the coastal protection measures, in collaboration with key stakeholders and other related projects and programs; (iv) Strengthened partnerships of project with communities. Building on the trials from the Eastern Tongatapu (see below), a suite of interventions are currently being designed which include: (i) construction of bamboo groynes; (ii) two new mangrove nurseries protected by the groynes; (iii) construction of geosynthetic sandbag defences to form secondary flood barriers within the backwaters of the wetland lagoon areas.

- Trialling Coastal Protection Measures in eastern Tongatapu, European Union, 2013-2015: This project was part of the Global Climate Change Alliance: Pacific Small Island States (GCCA: PSIS), funded by the European Union (EUR 500,000) and executed by the Pacific Community (SPC). The project focused on designing, building and monitoring 'hard' and 'soft' engineering measures working in combination along two coastal stretches in eastern Tongatapu. One measure consists of the construction of permeable groynes coupled with beach replenishment and coastal planting. The second measure involves constructing short offshore breakwaters combined with beach replenishment and coastal planting.

- The European Union funded GCCA: PSIS project in Tonga is 'Trialling coastal protection measures in eastern Tongatapu' 2013-2015 focusing on design, building and monitoring the success of 'hard' and 'soft' engineering measures working in combination along two coastal stretches. One measure consists of the construction of permeable groynes coupled with beach replenishment and coastal planting. The second measure involves constructing short offshore breakwaters combined with beachreplenishment and coastal planting.

- The Pacific Risk Resilience Programme (PRRP), Australian Govt, UNDP and Live and Learn Environmental Education, 2012-2018: The PRRP was a US $14 million program aimed at helping build the national and regional risk governance enabling environment to improve the resilience of Pacific communities to climate change and disasters. Lessons learned from the project included identifying the need to: build on national and sub-national linkages, foster a network of skilled and experienced personnel in Tonga, and ensure the project work is 'anchored' and has a sufficiently footprint within government structures to be sustained over time (PRRP mid-term evaluation 2017).

Other relevant prior projects include:

- Climate resilient marine spatial planning project (2022-2026) underway to assist in developing marine spatial plans (MSP) informed by the best scientific information available, including climate change scenarios, paving the way to a sustainable, inclusive, and resilient ocean-based economic development.

- Integrated Environmental Management of the Fanga'uta Lagoon Catchment as part of the GEF Ridge to Reef program and UNDP, 2015-2017: This program (US$ 1.76 million) is part of the Pacific Islands Ridge-to-Reef National Priorities program (R2R) of GEF. The R2R aims to maintain and enhance the ecosystem goods and services through integrated approaches to land, water, forest, biodiversity and coastal resource management that contribute to poverty reduction, sustainable livelihoods and climate resilience in Pacific Island countries. There is a second phase of the project funded by GEF (USD 16.25 million) 2023 - 2027 under MIEDECC and UNDP's Implementation which implements the Fanga'uta Lagoon Stewardship Plan and replicates lessons learned from the Tonga R2R Phase I to priority areas in Vava'u. Phase II conserves the wetland ecosystems that support the lives and livelihoods of local communities, the project aims to implement an integrated strategy to prevent and manage unsustainable and destructive activities that reduces the risks to Tonga's indigenous marine species and ecosystems from threats such as habitat loss, degradation, overexploitation, invasive species, pollution, and climate change. The objective of the project is "to effectively implement the Fanga'uta Stewardship Plan (FSP) for strengthened, integrated and inclusive management of the Fanga'uta Lagoon, and replicate successful strategies and lessons learned from the Tonga R2R Phase I to priority areas in Vava'u".

B.2 (a). Theory of change narrative and diagram (max. 1500 words, approximately 3 pages plus diagram)

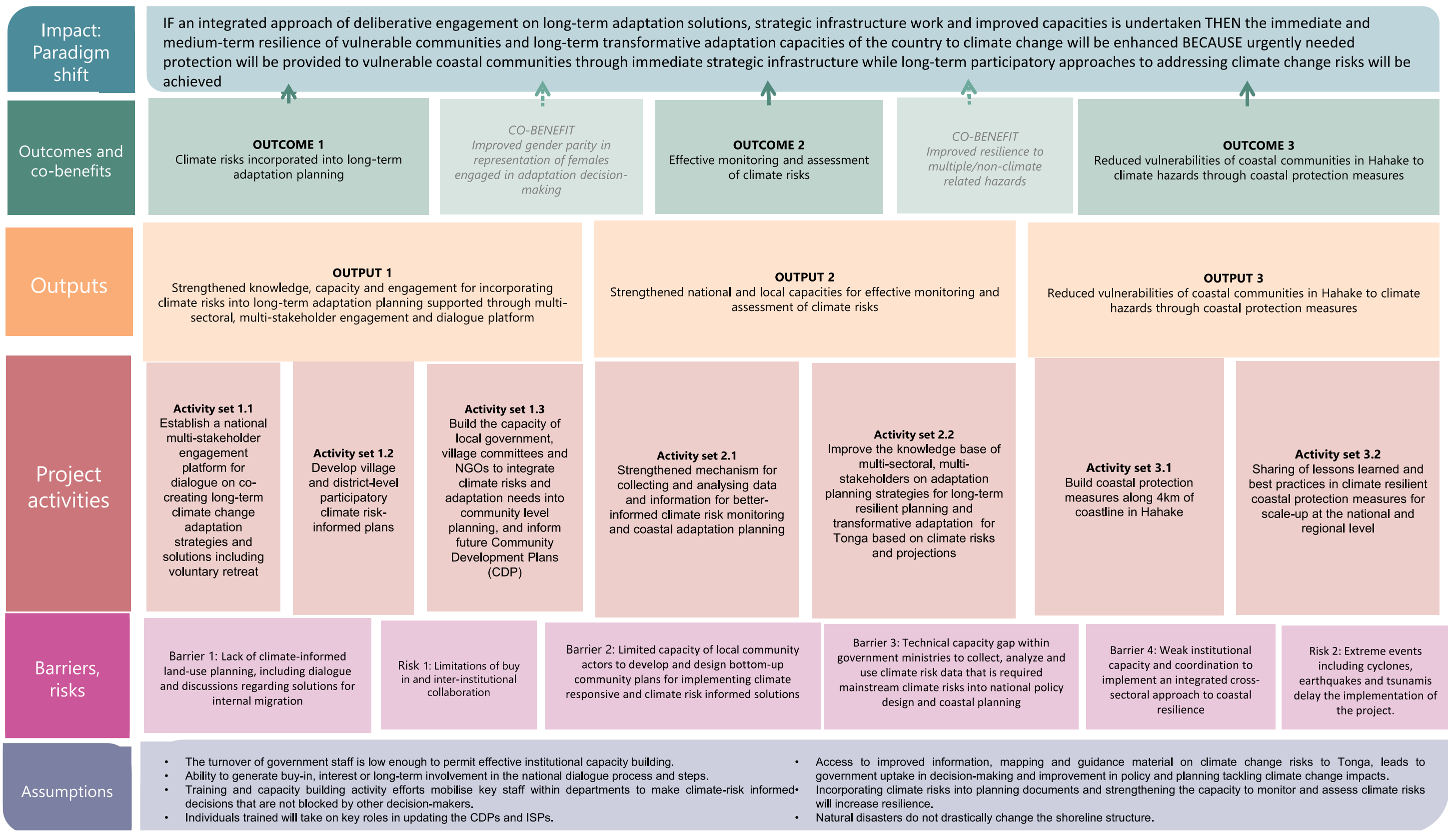
Present the theory of change (ToC) that contains a goal statement and describes how the proposed project/programme will contribute towards the goal statement by using results chain links from activities, outputs, to outcomes. By referring to the sample ToC diagram template available in the guidance note, present a ToC diagram (approximately 1 page) which visually represents the same logic in the narrative description. The ToC diagram and narrative may include a wide range of co-benefits[88] *as applicable in the context of the project/programme.*

Note all co-benefits will need to be further elaborated in section D.3 (sustainable development potential) and correspondent co-benefit indicators should be provided under section E.5 (project/programme specific indicators).

The theory of change should also include any relevant barriers (social, gender, fiscal, regulatory, technological, financial, ecological, institutional, etc.) that need to be addressed as well as risks and assumptions. Note that the assumptions can be elaborated further in sections E.3 (GCF outcome level: reduced reduced emissions and increased resilience) and E.5 (project/programme specific indicators) for each relevant indicator, as appropriate.

[88] GCF categorizes co-benefits into six areas which are: environmental, social, economic, gender, adaptation (relevant for pure mitigation projects) and mitigation (relevant for pure adaptation projects). Further guidance is available in the funding proposal (FP) guidance note.

Figure 9 Theory of Change



36. The *Theory of Change* for this project, shown schematically above, demonstrates how the implementation of project activities leads to the outputs of the project. The project will enable transformational change towards climate-responsive land use and long-term climate risk informed adaptation planning. It will do so through addressing key barriers to effective adaptation planning and putting in place means to avoid maladaptive measures in future.These outputs lead to the outcomes of building the long-term resilience of vulnerable coastal communities and transformative adaptation to the direct impacts of climate change in Tonga. Transformative adaptation aims to address root causes of vulnerabilities to climate change – including cultural, economic, environmental, and power relations by transforming them into more just, sustainable or resilient states.[89] The Theory of Change approach therefore involves an integrated set of outputs addressing key policy/regulatory, institutional and implementation/resourcing issues.

37. Ongoing climate change threats, risk, and impacts to date have challenged Tonga and brought the issue of transformative adaptation to the forefront. However, several adaptive capacity gaps remain to support a deliberative transformative adaptation in Tonga. These include:

38. limited capacities within the public sector that may affect agenda setting, and the ability to anticipate, guide, and make decisions for long-term adaptation;

39. limited vertical and horizontal coordination between different levels of government that limits policy and implementation coherence;

40. inadequate resource allocation to enhance overall resilience, specifically at the community level;

41. lack of localisation of knowledge and discourse on climate change, as well as an understanding and awareness of unique attributes of different islands, especially Tongatapu (such as changeability of reef-island, maintenance of sediment supply to island beaches), and;

42. a diversity of interests and beliefs that may not be in line with the long-term adaptation needs (including related to land-use).

43. The overall objective is to support the transformative adaptation agenda setting through a participatory process and build policy consensus, and promote planning coherence. Towards this end, the project will focus on a bottom-up development planning approach (community development plans, district plans and island strategic plans) to link development and spatial planning / land use management to promote transformative adaptation.

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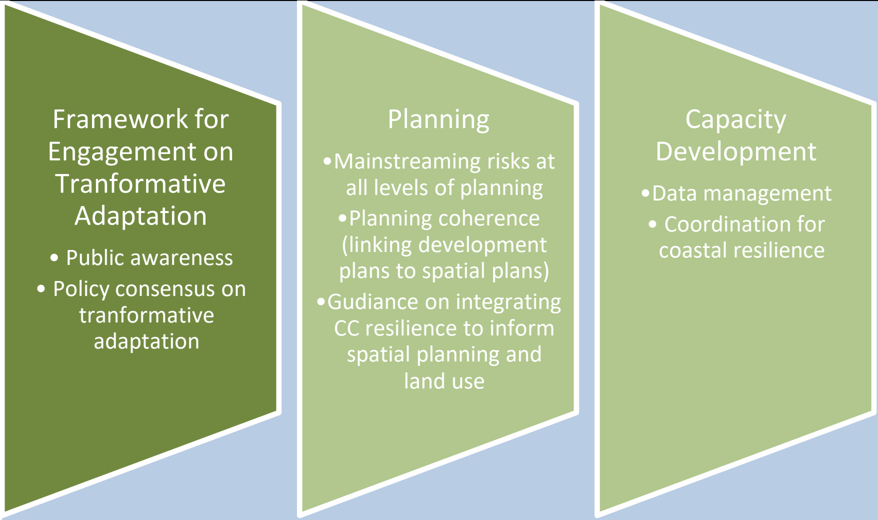
[89] Giacomo Fedele, Camila I. Donatti, Celia A. Harvey, Lee Hannah, David G. Hole, (2019), 'Transformative adaptation to climate change for sustainable social-ecological systems', Environmental Science & Policy, Volume 101, 2019, Pages 116-125, ISSN 1462-9011, https://www.sciencedirect.com/science/article/pii/S1462901119305337



44. Key barriers in the baseline addressed by the project: In the context of Tonga's existing programmes and commitments to reduce its vulnerability to climate change, there are several barriers that must be addressed in order to bring about transformational impact. Key barriers include:

    Barrier 1: Lack of climate-informed land-use planning, including dialogue and discussions regarding solutions for internal migration: Climate-informed land use plans ideally take into consideration various risks such as future sea-level rise and storm surge under different climate scenarios and return periods. In Tonga, some coastal communities located in highly exposed locations will likely have to move away before the sea-level reaches their doorsteps, or more realistically, damage from extreme wave events becomes too frequent to bear. Complex land ownership arrangements and other social and institutional reasons have prevented a sound, consensus-driven, climate-informed strategic land use plan from emerging. Informing all stakeholders and obtaining consensus necessarily needs to be a bottom-up, dialogue-based process, but a platform for enabling such dialogue does not currently exist. Therefore, there is an urgent need to address this barrier to enable island-wide discussions on effective climate-informed land use planning that would provide a long-term solution for the vulnerable communities in Tonga.

45. Barrier 2: Limited capacity of local community actors to develop and design bottom-up community plans for implementing climate responsive and climate risk informed solutions. At the same time, for building long-term resilience of the country to ever-increasing climate risks, it is critical that local communities' understanding, and capacities, grow in concurrence with technical capacity strengthening efforts for the central government agencies. Efforts to build capacity in Tongan communities and community/district agencies, as well as mobilizing necessary funding for community actions, have begun. However, currently, although Community Development Plans (CDP) have been developed for Tonga's communities they have not been able to incorporate climate risk so that communities can undertake adaptation planning.

    Output 1 will address these barriers by strengthening knowledge, capacity and engagement for incorporating climate risks into long-term adaptation planning supported through multi-sectoral, multi-stakeholder engagement and dialogue platform which will allow the co-creation of long-term climate change adaptation strategies including voluntary retreat. This output also develops critically needed capacity of local government, village committees, and NGOs, to integrate climate change risks and adaptation needs into community level planning, in turn informing community development plans (CDPs) and developing climate risk informed spatial planning.

46. Barrier 3: Technical capacity gap within government ministries to collect, analyze and use climate risk data that is required mainstream climate risks into national policy design and coastal planning: While government agencies have a clear need for climate risk data, there is a lack of technical capacity (including data management, establishment of baseline data and, conducting monitoring and evaluation) for this work. Consequently, there are limitations to how comprehensively analysis of future risks and expected damages has been undertaken. This barrier prevents an empirical evidence-based decision-making process from being carried out, meaning that adaptation decisions are often guided by budget availability rather than technical feasibility. This project seeks to build Tonga's technical capacity in climate risk data collection and management, strengthening Tonga's ability to select long-term and medium-term adaptation interventions by prioritizing high-risk areas. Increased technical capacity will also build a central repository of empirical experience in ecosystem-based coastal adaptation in Tonga, and the surrounding islands of the Pacific, through the project interventions.

    Output 2 addresses this barrier, by strengthening national and local capacities for effective monitoring and assessment of climate risks. This output enhances the mechanism for collecting and analysing data abd information for better informed climate risk monitoring and coastal adaptation planning, providing key capacity building, training and mentoring for coastal adaptation planning and monitoring and developing climate risk informed coastal plans. At the same time, Output 2 improves the knowledge base of multi-sectoral based, multistakeholders on adaptation planning strategies for long term resilient planning and transformative adaptation for Tonga, that are based on risks and projects. This includes improving LiDAR coverage for Tonga's Islands, and capacities for interpreting key information, producing maps and knowledge material on scenario based coastal risk maps, for decisions on long term settlement patterns and climate risk for strategic long-term coastal planning, while strengthening multistakeholder coordination and collaboration on coastal adaptation and planning integrating climate change.

47. Barrier 4: Weak institutional capacity and coordination to implement an integrated cross-sectoral approach to coastline resilience. Past approaches to coastal vulnerability reduction (i.e. isolated, engineering-oriented interventions), have had little coordination between ministries and between government and communities. Coastal assessments and construction have been typically outsourced to international technical agencies, leaving little need for agencies like Ministry of Land Survey and Natural Resources(MoLSNR), and Ministry of Meteorology, Energy, Information, Disaster Management, Environment, Climate Change and Communications (MEIDECC) (including their Departments – e.g. Tonga Meteorological Services and Department of Climate Change), to coordinate with each other. Yet, new coordination requirements emerge as Tonga embarks upon a new pathway to reduce long-term vulnerability and build resilience longterm. The Government of Tonga and communities must take on greater involvement and engagement to facilitate a proactive role in identifying and implementing solutions to the problems that Tonga faces nation-wide. For example, as the technical barrier for data collection on coastal parameters is addressed (see barrier #2), relevant agencies need to work closely together in sharing information about coastal processes, climate models and data on key coastal parameters. However, mechanisms to facilitate such multi-ministerial coordination, such as a protocol of data sharing and community engagement beyond the project lifespan, are currently weak.

Output 3 of the project, therefore, addresses the technical and financial barriers limiting investments in coastal protection that respond to Tonga's immediate adaptation needs for reducing vulnerabilities of coastal communities in Hahake to climate hazards through coastal protection measures. Coastal protection will be built along 4.3km of coastline. At the same time, this output develops long-term outlook and analysis of coastal protection measures through the sharing of the lessons learned on coastal protection measures and effectiveness, for potential scale-up at the national and regional levels. In addition, under previous projects, the process of support was such that the technical capacity needed for undertaking underlying assessments and designs was not sufficiently built within the ministries that are mandated to protect and manage coastal areas of the country. Coupled with the activities under Output 2, this Output will increase national capacities for understanding coastal processes, hydrodynamics, and oceanography. This, along with increased community awareness developed through Ouputs 1 and 2, will move the focus away from maladaptive coastal protection works, towards interventions based on sound information and technical design / construction.

B.2 (b). Outcome mapping to GCF results areas and co-benefit categorization

Fill in the GCF results area table below to map each project/programme outcome identified in section B.2(a) to the contributing GCF results area(s) by referring to the description of eight results areas provided in the guidance note.

Outcome number	**GCF Mitigation Results Area (MRA 1-4)**				**GCF Adaptation Results Area (ARA 1-4)**			
	MRA 1 Energy generation and access	**MRA 2** Low-emission transport	**MRA 3** Building, cities, industries, appliances	**MRA 4** Forestry and land use	**ARA 1** Most vulnerable people and communities	**ARA 2** Health, well-being, food and water security	**ARA 3** Infrastructure and built environment	**ARA 4** Ecosystems and ecosystem services
Outcome 1	☐	☐	☐	☐	☒	☐	☐	☐
Outcome 2	☐	☐	☐	☐	☒	☐	☐	☐
Outcome 3	☐	☐	☐	☐	☐	☐	☒	☐
Outcome 4	☐	☐	☐	☐	☐	☐	☐	☐
Outcome ...	☐	☐	☐	☐	☐	☐	☐	☐
Outcome ...	☐	☐	☐	☐	☐	☐	☐	☐

If any co-benefits have been identified in section B.2(a), fill in the Co-benefit table below to map each co-benefit to the corresponding category as defined in the FP guidance note.

Co-benefit number	**Co-benefit**					
	Environmental	**Social**	**Economic**	**Gender**	**Adaptation**	**Mitigation**
Co-benefit 1 Improved gender parity in representation of females engaged in	☐	☐	☐	☒	☐	☐

adaptation decision-making						
Co-benefit 2 Improved resilience to multiple/non-climate related hazard	☐	☒	☒	☐	☐	☐

B.3. Project/programme description (max. 2500 words, approximately 5 pages)

Define the project/programme. Describe the proposed set of components, outputs and activities that will be undertaken by the project/programme to attain the intended outcomes elaborated in Section B.2 (a). Please also elaborate how the project/programme activities will address the barriers described in Section B.2.

This section of the funding proposal should be consistent with sections C.2 (financing by component), E.5 (project/programme results level) and E.6 (project/programme activities and deliverables).

Referring to the feasibility study, describe why this set of interventions was selected instead of alternative solutions and how the project/programme can help unlock the needed support in a sustainable manner. Also identify trade-offs of the selected interventions, if applicable.

48. The **project objective** is to build the long-term resilience of vulnerable coastal communities to the direct impacts of climate change in Tonga, through an integrated approach that will enable transformational adaptation via climate-responsive and long-term climate risk informed land use and adaptation planning. The project therefore aims to build the long-term resilience of vulnerable coastal communities to the direct impacts of climate change in Tonga and build transformative adaptation capacities through strategic infrastructure work, long-term climate risk informed adaptation planning, and engagement on long-term adaptation solutions.

    This objective will be achieved through the three inter-related Outputs. The outputs and associated activities are detailed below:

49. **Output 1 Strengthened knowledge, capacity and engagement for incorporating climate risks into long-term adaptation planning supported through a multi-sectoral, multi-stakeholder engagement and dialogue platform**

    Almost 80% of the Tongan population live in limited areas along the coastline, many of which are highly exposed to coastal hazards. Available climate change projections suggest that, in future, the combined effects of sea-level rise and frequent wave overtopping events will render some of these areas unliveable or too unsafe or costly to live in. Yet, there is no long-term vision in the country with which to facilitate appropriate adaptation responses (including voluntary retreat if appropriate) because of the complex land ownership situations in Tonga. Transformative adaptation needs to be shaped through collective design, intention, and negotiation and requires an iterative process that will continuously take into consideration socio-political and economic aspirations of the people, and the increasing vulnerability to climate change and disaster risks. Establishing such a long-term vision is possible only with a long, dedicated process of raising awareness among all parties including the government, land owners and community members about accepting such a reality and building consensus. Without such a vision in place, in 50 years, many vulnerable Tongans will be forced to undertake unplanned or spontaneous relocation or migrate out from Tonga, and while preparedness planning for such a consequence is still limited, this situation is very likely to be disastrous. Hence, under this Output, the project will focus on raising awareness of people, officials, and decision makers by localizing climate change knowledge. This Output, is expected to make first steps required for the Tongan society to move towards making the necessary decisions and trade-offs to mitigate the effects of climate change on vulnerable coastal communities. Rather than a linear policy process, co-creation and implementation of practical solutions for better land use will be supported. The approach would ensure continuous engagement of all stakeholders rather than only providing inputs during the design stage of a policy process. As a first step to organizing a public campaign on transformative adaptation, clarification on the objective, scope, message/intention, and the target audience of the public campaign is needed during early implementation stages. Therefore, support will be provided to the government (including MEIDECC) to establish a multi-stakeholder platform to lead the engagement around transformative adaptation. The composition and the terms of reference of the multi-stakeholder platform, along with the stakeholder engagement plans for the public awareness campaigns will be developed in close collaboration with the government. The Terms of Reference will be revisited and updated during the latter period of the project in order to incorporate any lessons learnt, improvements based on how the establishment and convening of the national platform may have taken place.

50. <u>Activity 1.1. Establish a national multi-stakeholder engagement platform for dialogue on co-creating long-term climate change adaptation strategies and solutions including voluntary retreat.</u>

    The project will design a multi-sectoral and multi-actor stakeholder engagement plan and guidelines for a bottom-up national engagement process, by undertaking the following sub-activities:

    - Sub-Activity 1.1.1 Design of a multi-sectoral and multi-actor stakeholder engagement plan and guidelines for a bottom-up national engagement process.

- Sub-Activity 1.1.2 Facilitate a national awareness campaign on climate change and disaster risks and transformative adaptation strategies for multi-stakeholder dialogue co-creating long-term adaptation strategies and solutions (using material developed under Output 2)
- Sub-Activity 1.1.3 Develop a Terms of Reference for a National adaptation planning engagement platform specifying memberships, responsibilities and results
- Sub-Activity 1.1.4 Undertake meetings of the national engagement platform on long-term climate change adaptation as per the Terms of Reference
- Sub-Activity 1.1.5 Develop policy options to incorporate climate resilience and land use principles into national frameworks

The design of the multi-stakeholder engagement platform for dialogue on long-term climate change adaptation strategies and solutions, will be comprised of representatives selected (internally) from each GoT Ministry, with primarily members from MIEDECC, Ministry of Internal Affairs, PMO, Ministy of Infrastructure, Minstry of Works, Ministry of Finance and National Planning, Policy and Planning, Ministry of Fisheries, Ministry of Agriculture, Food and Forests, Ministry of Lands, Survey and Natural Resources, for example, as well as representatives from across Tonga's CSOs engaged with youth, inclusivity, gender, environment and other areas. Expertise on community dialogue and governance will be utilised to firstly design the multi-sectoral and multi-actor stakeholder engagement plan and guidelines specifically for this bottom up national engagement process, followed by the preparation of the terms of referemce for the dialogue platform itself. The national awareness campaign will be carried out based on the findings of the stakeholder engagement plan developed specifically for the national dialogue process, engaging widely with local, island and national level stakeholders on climate change risk to build the momentum for the_national multi-stakeholder engagement platform for dialogue. The various stages of stakeholder engagement and awareness campagin will utilise the data and information gathered and developed on the risks and vulnerabilities, future climate impacts, coastal dynamics etc from Output 2 as well as the local-level development plans as they come through from the other Activities under Output 1.

51. Since 2012, the Government, under the leadership of the Ministry of Internal Affairs (MIA) has promoted the concept of community-led development planning processes. This led to communities/villages across Tonga completing their Community Development Plan (CDP) for the first time in 2016-2017. CDPs are provided to all village committee, district and town officers. However, until now the CDPs have not incorporated the climate risk needed to address climate change adaptation priorities and propose solutions and opportunities to address the most urgent needs for building resilience. Activity 1.2 therefore suppports building the capacity of local government, village committees and NGOs to integrate climate risks and adaptation needs into community level planning, and inform future Community Development Plans (CDP). Data and information will be simplified and made accessible to the wider population (closely linked to activities under Outcome 2). The project specifically targets the CDPs and district plans for Tongatapu (excluding Nuku'alofa), which will pre-determine the selection of the village committees[90] in Tongatapu for the project (e.g. all in Tongatapu excluding Nuku'alofa). Beyond the targetting of CDPs and District Development Plans (DDPs) in Tongatapu, the project will support the development/updates of the Island Plans (for the five island cluster groups - Vava'u, Ha'apai, 'Eua, Tongatapu and Niuas) to incorporate climate risk considerations through guidelines, climate ifnormation data and graphical profiles:

52. <u>Activity 1.2 Develop village and district level participatory climate risk informed plans</u>

    - Sub-Activity 1.2.1 Support a participatory process for developing village and district spatial plans to incorporate climate risk planning and adaptation strategies and incorporating principles of land-use.

53. The project also supports the local governance capacity for adaptation planning and decision-making, mentoring and training to 300 people on integrating climate change risks into planning and budgeting (building the capacity of local government, village committees and NGOs to integrate climate risks and adaptation needs into community-level planning) over 4 years to inform future CDPs/ISPs. To assist government institutions and local government, village committees, and NGOs in the capacity to interpret climate risks into planning and to undertake budgetting, engage community-led decision-making, and to coordinate between government and other agencies to address climate risks<u> and inform future Community Development Plans (CDP), as well as SDPs, a climate responsive </u>Community Development Plan Guideline will assist the development of future CDPs (which are used to inform IDPs). This guideline includes participatory vulnerability assessment and planning, gender needs assessment, and adaptation needs assessment and planning. This will be supported through trainings and materials for key ministries to coordinate 'up and down' on climate change risk informed planning and budgeting through trainings and materials as well as island events (Vava'u, Ha'apai, 'Eua, Tongatapu and Niuas) for sharing lessons for producing climate responsive CDPs. These activities are also supported by the Strategy and Plan under Output 2.1, on climate risk monitoring and data and also informs the development of the CDP guideline.

54. <u>Activity 1.3 Build the capacity of local government, village committees and NGOs to integrate climate risks and adaptation needs into community level planning, and inform future Community Development Plans (CDP)</u>

    - Sub-Activity 1.3.1 Develop an updated climate-responsive Community Development Plan Guideline to inform the development of future CDPs that includes including participatory vulnerability assessment and planning, gender needs assessment, and adaptation needs assessment and planning.

[90] The village committees already exist and hold responsibilities for village level management areas such as local water supply, waste management, and town and district officers engage with these village committees.

- Sub-Activity 1.3.2 Design and implement a capacity building package for the Ministry of Internal Affairs on utilising the new climate-responsive CDP guideline
- Sub-Activity 1.3.3 Strengthen capacities of key ministries[91] for climate change risk informed planning and budgeting through trainings and materials
- Sub-Activity 1.3.4 Organize island events (Vava'u, Ha'apai, 'Eua, Tongatapu and Niuas) for sharing lessons for producing climate responsive CDPs

55. **Output 2 Strengthened national and local capacities for effective monitoring and assessment of climate risks**

56. Output 2 addresses the limited capacity and available data to collect, analyse and use climate risk to inform considered strategic spatial planning and inform adaptation in policy design and decision-making processes regarding land use and coastal management. Four government agencies have responsibilities in the broad areas of monitoring coastal processes and maintaining and managing coastal protection. These are: the Ministry of Lands, Survey and Natural Resources (MoLSNR); under MEIDECC the Department of Environment (for ecosystem monitoring), Department of Climate Change and the Meteorological Department (for climate modelling). These agencies, and the Government of Tonga in general, do not currently have access to the data and expertise to collect, manage and analyze in-depth data on coastal climate change risks. This activity therefore aims to build capacity in relation to climate risk data/interpretation in order to allow for better-informed coastal adaptation planning.

57. The Capacity Needs Asssessment will assess the key institutional capacity needs in the fields of technical capacity for closely monitoring the coastal dynamics of vulnerable coastlines, designing locally-appropriate protection measures and planning, and implementing the measures that are within technical and financial capabilities of the Government. The Capacity Needs Assesment will thereby determine the development of the over-arching long-term Capacity Building Strategy and Plan for government use on coastal and climate risk monitoring and analysis to inform coastal adaptation planning (e.g. including understanding and use of climate and risk infromation and data such as satellite imagery, aerial photography, drone imagery, LiDAR topography and bathymetry), and planning and budgeting for adaptation measures. Government staff will also be trained in a) from technical training on climate risk monitoring and data (MLNR, MoI, MIEDECC) focused on improving LiDAR knowledge and capacities for interpreting key information, producing maps and knowledge material on scenario based coastal risk maps and b) incorporating these for decisions on long term settlement patterns and climate risk for strategic long-term coastal planning, supporting development planning (MoI, MLNR, MoF & NP, MIEDECC, MIA).

58. Under the Capacity Building Strategy and Plan, technical officers and local government officers will be trained on coastal zone management, coastal processes, coastal planning and hazards management, and climate down-scaling with training and metoring undertaken and training materials developed and distributed to support government staff across ministries to be able to integrate climate risks into planning and budgeting within their work using these materials. Government staff will be provided with a package of training and capacity building relevant to the specific ministries so that equipment and data can be fully utilized for use in generating climate and risk information and data such as satellite imagery, aerial photography, drone imagery, LiDAR topography and bathymetry. Training/mentoring and training materials will be available to support government staff engaged in planning to integrate climate change risks in planning and budgeting, supporting the development planning and decision-making. The results of the Capacity Needs Assesment will thereby inform "Sub-Activity 2.1.1 Increase capacity of the Government to effectively collect, manage and analyse coastal monitoring data"; "Sub-Activity 2.1.2 Collecting, interpreting and using data on climate risks for coastal adaptation planning to produce scenario-based coastal risk maps and knowledge material for use under Output 1."; and "Sub-Activity 2.1.3 Improve multi-stakeholder coordination and collaboration on coastal adaptation and planning integrating climate change:

59. <u>Activity 2.1 Strengthened mechanism for collecting and analysing data and information for better-informed climate risk monitoring and coastal adaptation planning</u>

- Sub-Activity 2.1.1 Increase capacity of the Government to effectively collect, manage and analyse coastal monitoring data
- Sub-Activity 2.1.2 Collecting, interpreting and using data on climate risks for coastal adaptation planning to produce scenario-based coastal risk maps and knowledge material for use under Output 1.
- Sub-Activity 2.1.3 Improve multi-stakeholder coordination and collaboration on coastal adaptation and planning integrating climate change

60. Tonga has limited access to the fundamental baselines required to understand the impacts of sea level rise and marine hazards, namely high quality, accurate nearshore bathymetry (sea floor depth) and topography (land elevation). It is crucial that the relationship between sea level and land height is characterised accurately if Tonga is going to be equipped to make sound long term decisions regarding adaptation response and strategy. Likewise, the only way to accurately model wave impacts is if quality nearshore bathymetry and topography is available. In essence, it is not possible to design detailed adaptation to marine hazards at the engineering or strategic policy level unless these baselines are available as it is not possible to manage that which is not measured. Because of the need for high quality data for considering sea level rise or wave impacts, the Project intends to implement the necessary due diligence and collect this data via professional national survey design. This survey will complete gaps in current data coverage and will re-survey Tongatapu Island given existing data is older (2011) and comparatively course (up to 50cm vertical error) in comparison to contemporary techniques (better than 5cm vertical error). Additionally, Tongatapu

[91] Ministry of Internal Affairs, Ministry of Lands, Ministry of Finance and National Planning, Ministry of Meteorology, Energy, Information, Disaster Management, Environment, Climate Change and Communications, Ministry of Infrastructure

houses over 60% of the national population and many of the island's settlements are located in exposed nearshore locations with little topographic relief. It is imperative that accurate modelling is undertaken to explain the spatial and temporal dynamics of inundation in these locations and do so in the most detailed way possible. Sea level rise projections are becoming increasingly robust (and alarming) and in order for Tonga to make informed decisions the best possible explanation of such impacts must be made available at the Government, community and household scale.

61. UNDP learning from recent similar work in Tuvalu, a nation widely recognised globally in respect to the impacts of sea level rise, showed conclusively that local understanding of the extent of inundation due to sea level rise and the likely timeframes involved, was still poorly understood prior to the collection of LiDAR and associated inundation modelling in 2019.[92] Only once a clear empirical understanding became available did the Government embark on extensive and ambitious long term adaptation plans commensurate with what the science explained was the precise impacts they faced. This Project will leverage this new data in the same way and will use this to inform the national dialogue process to guide discussion of impacts and potential adaptation strategy and response. LiDAR survey work will capture gaps in outer island groups where low lying communities are predominent as well as the capital island Tongatapu where covergage is either missing or outdated. In addition to this survey, the project will also support the purchase of selected equipment (hardware and software) that will remain with the Government of Tonga to provide capability to carry on supplementary survey work as needed to track change and monitor over time. The project will cover a re-survey of Tongatapu and survey of all the low-lying islands that do not have current coverage. These include the following and details are included in the Budget Note #39 in Annex 4): Nomuka and lagoon, Mango Island and lagoon, Fomoifua Island and Tanoa Island and lagoon, Lalona Island and lagoon, Telekitonga Island and reef, Pangai Island complex and reef, Ofolanga and Mo'Unga'One, Niviva Island and lagoon, Lofanga Island and lagoon, Fotuha'a Island, Tungua, Ha'afeva, Ua Island and lagoon. Increasing capacity in analysis of the data will then bridge the current gap to enable these data to inform and guide climate risk driven modelling (coastal inundation, storm surge and hydrodynamic modelling), shoreline mapping, and vulnerability analysis of coastlines. As part of the activities under Output 2.1, indicative survey and equipment purchases to support improved coastal adaptation planning and monitoring capacity are included. This equipment will be accompanied by a full package of training and capacity building for departmental staff to ensure that equipment and data can be fully utilised. Furthermore, recognizing that the equipment and data, assessments and analyses must be shared across multiple ministries, GCF financing will also be used to assist in developing a long-term Capacity Building Strategy and Plan for government use on coastal and climate risk monitoring and analysis to inform coastal adaptation planning (e.g. including understanding and use of climate and risk infromation and data such as satellite imagery, aerial photography, drone imagery, LiDAR topography and bathymetry) for cross-ministerial cooperation in producing and utilizing data for the purpose of coastal protection. Where LiDAR equipment will support smaller monitoring and assessment work, this will be managed and maintained by the Ministry of Lands/Department of Geology who already have and maintain drones and other technical survey equipment.

62. The data collected through the project timeframe and beyond will also contribute to building the body of regional knowledge. The approach to facilitate data sharing amongst the key stakeholders in Tonga follows the lessons and approach model from UNDP's experience with the GCF-funded Tuvalu Coastal Adaptation Project (TCAP) to provide a mechanism to ensure data sustainability post-implementation. In the case of TCAP, after collecting high-quality baselines and undertaking wave hazard modeling, this data was used to generate a stand-alone, open online platform where all these data products can be viewed, interrogated, and accessed. The platform was developed, hosted, and maintained by SPC (a regional agency called the Pacific Community) and has been the key focus of more formalized capacity-building efforts in Tuvalu. SPC holds the regional mandate to provide, secure, and curate such products and data on behalf of the Pacific Island member States, of which Tonga is a full member. UNDP has a long-standing relationship with SPC and would seek to extend a further letter of agreement with SPC to contract their services to provide similar services to Tonga as we have done very successfully with Tuvalu. This builds sustainability because SPC is specifically tasked with data archiving and analysis services for the Pacific Islands. Once this Project's data and products are incorporated into SPC systems, they will provide access and support indefinitely. Tonga also has an established GIS office within its existing government infrastructure, and they, too, are specifically mandated locally to provide geospatial data archiving and curation services to the rest of the Government. UNDP has already established links with this office and will also ensure copies of all data and analysis products are uploaded to their systems. This dual regional and national approach ensures sustainability, and as we have seen with the TCAP data, SPC has independently, in partnership with Tuvalu, used these baselines for further tasks and products, namely the development of a national wave hazard early warning system.

63. Within the project, community-led decision-making will be informed by a solid baseline of technical information on a range of risk factors (coastal topography, wave modelling, rates of sea level rise etc.) and the capacity in government agencies and civil society to interpret this information in decision making. In addition to the Capacity Needs Assessment undertaken in Output 1 to establish the the capacity needs for strengthening the capacities of key Ministries (MoI, MLNR, MoF & NP, MEIDECC, MIA), a Capacity Needs Assessment is to be undertaken to inform the long-term Capacity Building Strategy and Plan specific to climate risk monitoring and data under Output 2. The Capacity Needs Asssessment under Output 2 (2.1.1) will assess the key institutional capacity needs in the fields of technical capacity for closely monitoring the coastal dynamics of vulnerable coastlines, design locally-appropriate protection measures and implement the measures that are within technical and financial capabilities of the Government. The capacity needs assessment will therefore also outline the gaps for the capacity building and training opportunities through regional placements in technical agencies in the Pacific established and supported (for example, SPC and others). The results of the Capacity Needs Assesment will thereby determine the development of the long-term Capacity Building Strategy and Plan for government use on coastal and climate risk monitoring and analysis to inform coastal adaptation planning (e.g. including understanding and use of climate and risk infromation and data such as satellite imagery, aerial photography, drone imagery, LiDAR topography and bathymetry). The Capacity Building Strategy and Plan, will

provide a package of training and capacity building relevant ministries' staff so that operation and coordination can be fully understood and implemented to ensure that equipment and data can be fully utilized for use in generating climate and risk information and data such as satellite imagery, aerial photography, drone imagery, LiDAR topography and bathymetry. Under the Capacity Building Strategy and Plan, technical officers and local government officers will be trained on coastal zone management, coastal processes, coastal planning and hazards management, and climate down-scaling with training and metoring undertaken and training materials developed and distributed to support government staff across ministries to be able to integrate climate risks into planning and budgeting within their work using these materials. The results of the Capacity Needs Assesment will thereby inform "Sub-Activity 2.1.1 Increase capacity of the Government to effectively collect, manage and analyse coastal monitoring data"; "Sub-Activity 2.1.2 Collecting, interpreting and using data on climate risks for coastal adaptation planning to produce scenario-based coastal risk maps and knowledge material for use under Output 1."; and "Sub-Activity 2.1.3 Improve multi-stakeholder coordination and collaboration on coastal adaptation and planning integrating climate change.

64. <u>Activity 2.1 Strengthened mechanism for collecting and analysing data and information for better-informed climate risk monitoring and coastal adaptation planning</u>

- Sub-Activity 2.1.1 Increase capacity of the Government to effectively collect, manage and analyse coastal monitoring data
- Sub-Activity 2.1.2 Collecting, interpreting and using data on climate risks for coastal adaptation planning to produce scenario-based coastal risk maps and knowledge material for use under Output 1.
- Sub-Activity 2.1.3 Improve multi-stakeholder coordination and collaboration on coastal adaptation and planning integrating climate change

65. <u>Activity 2.2 Improve the knowledge base of multi-sectoral, multi-stakeholders on adaptation planning strategies for long-term resilient planning and transformative adaptation for Tonga based on climate risks and projections</u>

- Sub-Activity 2.2.1 Support the use and interpretation of data collected in Output 2.1 by providing providing easily understood materials for multiple stakeholder types that consider strategic long-term coastal adaptation and inform the national multi-stakeholder adaptation planning engagement platform

66. **Output 3 Reduced vulnerabilities of coastal communities in Hahake to climate hazards through coastal protection measures**

The north-eastern coastline of Tongatapu is considered among the most at risk of wave and erosion-driven land loss and damages to key coastal infrastructure in Tonga. The target area is experiencing wave-induced erosion and overtopping, putting houses and coastal infrastructure at risk. Not only do such events undermine the structural integrity of the road itself, the absence of a drainage system in the road design causes sea waterlogging in the landward areas beyond the road. Waterlogging in the landward side damages coastal assets and causes erosional scarp undercutting of the road surface. With the projected sea level rise and episodes of intensive rainfall events in the future, coastal assets such as the road and houses are at risk from both the seaward and landward sides.

67. Supplementing the work in Outputs 1 and 2, Output 3 will implement coastal protection measures in the Hahake region of north-eastern Tongatapu Island. This forms an important part of the project's integrated approach responding to the urgent need highlighted by the Government of Tonga to immediately mitigate the likelihood of coastal innundation throughout the Hahake penisula, as well as prevent the consequences of climate-induced erosion and wave-overtopping events that are threatening the lives and assets of the most vulnerable coastal communities residing in areas subject to innundation. Coastal protection under Output 3 builds upon and completes exisiting invetments of the 2km revetment (completed by ADB), providing together complete continuation of coastal protection measures throughout the low lying sections of the Hahake coastline.

68. Output 3 will enhance coastal protection in Hahake, where climate change induced coastal erosion and wave overtopping is already causing land loss and damage to coastal infrastructure. At the end of the project, it is envisaged that an additional 4.3 km of vulnerable coastline will have been equipped with a range of coastal adaptation measures (including rock revetments) preventing erosion, wave overtopping and/or subsequent coastal flooding that are expected to worsen under projected climate change scenarios.

69. <u>Activity 3.1. Coastal protection measures built along 4.3km of coastline in Hahake</u>

- Sub-Activity 3.1.1 Detailed design and site-specific assessments of coastal protection measures finalized for Hahake
- Sub-Activity 3.1.2 Construct coastal protection measures in Hahake

70. The selection of site specific solutions has been based on the assessments of multiple factors, including the existence of pre-feasibility studies on a range of technical parameters relevant to the project (See the reports from Webb 2016 and Lewis 2017). Noting that Nukuleka has been previously subject to engineering (now failing), the footprint of which the project proposed to build over limits environmental impacts and stabilizes the shoreline, providing safer community access and reducing suspended sediment release during erosive events. The primary studies for the Nukuleka revetment are those compiled between 2015 and 2017 during the implementation of the ADB-funded CRSP Project. CRSP was limited by budget to undertaking only approximately 2km of rock revetment (the chosen option) on the Navutoka Village shore but undertook studies for the geomorphological assessment, options justification, ESIA, and detailed designs for all of Nukuleka. The Coastal Resilience project will build on these with the project's ESIA and development consent process on engineering designs, modeling, and

BoQs. CRSP's geomorphological assessment also considered the Kolonga (raised road) section of protective infrastructure, but designs were not developed. Thus, an independent options assessment, ESIA, development consent process, engineering design, modeling, and BoQs will be required. The raised road approach will occur within the existing road easement, which is already an engineered setting. This approach also avoids direct disturbance of the foreshore or marine environment; thus, impacts are minimal. The areas with the revetment constructed under the CRSP project avoided inundation during the recent tsunami event, whereas those areas still unprotected suffered marine incursion, flooding and physical damage to infrastructure. It is expected that the coastal infrastructure measures under Output 3 will potentially provide additional benefits in protecting land, soil, and surface and groundwater quality in the adjacent inland area to the site through the prevention of inundation as per potential inundation. This area serves residential households and provides some small crops.

71. Following UNDP's experience in the region and its policies and guidelines, the project is designed to engage the services of a consulting company/firm to provide design and supervision services for the construction component. While it has been envisaged as one company/firm providing the services, a market and technical analysis will be done at the start of implementation to ascertain whether two separate firms will be required for the two separate areas of construction or not. The contract/s will be for the duration from the start of design until the end of construction with the Terms of Reference detailing the expected services during design and construction. The firm/s will be expected to have a lead engineer who will provide consistent guidance and support throughout the implementation of the construction component of the project supporting the PMU. In addition, UNDP will be providing technical oversight to the design, safeguards and construction itself with the use of its in-house expertise and externally engaged experts as required. Tonga has workforce capacity, machinery, and materials to implement hard foreshore protective works as has been already demonstrated to date, by various engineering efforts undertaken around Tongatapu, including the recent successful construction of the CRSP revetment. The main obstacle of some of this work has been (as evidenced in other locations such as Nukuleka) poor attention to design, construction methodology and materials selection. Thus, in respect to coastal protection capacity building the key gap is international standards of design, construction methodology and materials quality. This Project therefore intends to address this via a learning whilst doing approach; by seeking to use local suppliers and manage their work under competent Coastal Engineering supervision and undertaking a best practice work strategy with close adherence to international standards. A local company is expected to be selected with international experts to provide supervision, particularly on design, construction methodology and materials quality. Note this is also reflected in the construction phase's conservative implementation timelines. This approach to engage local suppliers and construction is also more cost effective rather than securing a large multinational, import workforce, using imported materials whereby budgets would rise substantially (although likely securing fast, efficient infrastructure outcomes, zero capacity would remain following the project. UNDP is seeking to achieve meaningful capacity building and achieve the infrastructure work to appropriate standards as an integrated program of work with the intent of leaving behind hands on experience. Key components of this will be to demonstrate where Tonga has adequate capacity (e.g. the ability to implement large scale capital works) and where it does not (e.g. detailed hydrodynamic modeling and construction design). With such understanding the local authorities are better able to strategically seek international assistance in those areas where sustained local capacity is hard to maintain and alternatively use local providers and capacity where it is appropriate.

72. The site selection shares synergies with previous project work, notably CRSP, alongside engagement within the local community, particularly for the section in Nukuleka to Talafo'ou, and the EU GCCA measures adjacent to the coastal protection measures being applied between Manuu to Kolonga, accompanied by a strong commitment from the Government of Tonga. For example, some areas of the wider Hahake coastal area have benefitted from existing adaptation measures, including coastal groynes and sea walls of differing design and effectiveness. The most recent of these have been rock revetments supported through the ADB-funded CRSP project. The areas covered by this earlier work are shown by the green line, while the orange line on the right shows the coastal protection measures and raised road area, and the red line shows the site for the construction of the revetment. The pink line in the middle section refers to the (ad hoc) revetment installed over years by the Government of Tonga that attempts to secure some of this coast line, but of which is in disrepair. The current project is designed to supplement this earlier work through using compatible design specifications based on a 40-year design-life for the proposed rock revetments. Further technical detail on these coastal works is presented in the Feasibility Study (Annex 2). The coastal protection measures for Mankua to Kolonga under sub-activity 3.1.2 also compliment the small-scale coastal protection and ecosystem-based measures in northwest Tongatapu (Sopu to Ha'atafu) established under the GCCA+ in that both sets of measures will enhance the level of resilience provided by the other.

Figure 10 Location of site for construction of 2.3kms of reventment in Nukuleka to Talafo'ou (left side, red line) and location site for 2kms of coastal protection measures (raised road) for the Manuka to Kolonga shoreline (right side, orange line)

73. At the southernmost end of the Hahake Peninsula, 2.3 km of rock revetment will be installed along the coast between the villages of Talafo'ou and Nukuleka to protect the low-lying and highly vulnerable shoreline. On the north-eastern side, coastal protection measures for the Manuka to Kolonga shoreline will consist of over topping protection but not a foreshore revetment. A foreshore revetment is not considered appropriate at this location (Kolonga) because unlike Talafo'ou to Nukuleka the northeastern section has not been subject to previous extensive foreshore revetment works. This section of coast remains as an important recreational shoreline that has some important strand vegetation and geological features. This means that while this shore is stressed and in places erosive, it has a greater measure of ecosystem-integrity based resilience than the Nukuleka foreshore which has seen various and often ad hoc, poorly designed (and failing) structures constructed over decades and has no residual soft shoreline left due to its erosive nature.

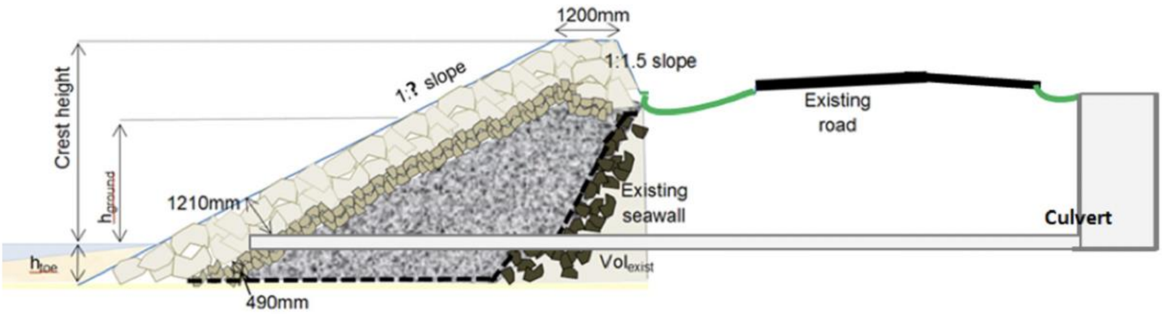
Figure 11 Shoreline between Nukuleka and Talafo'ou

74. The Kolonga section will consist of raising the roadway easement (approximately 1m but to be determined via modelling) and resealing the surface to reinstate the elevated roadway. A two-dimensional wave modelling to determine wave height and runup dynamics will be undertaken to inform infrastructure design criteria as part of the activity on detailed design. The shoreward flank of the raised road would also be armored (with local limestone) as a mostly buried last line of defense given it is possible this shore will become progressively more erosive. This work will be implemented on the existing road easement and is specifically conceived to provide much needed overtopping protection but not disturb the existing foreshore system or associated stand vegetation. The main objective of this approach is to protect the low-lying agriculturally important areas on the landward side of the road, nearby assets and the roadway itself from marine over wash as it is currently very exposed, while also ensuring that the existing ecological integrity of this area of coast is undisturbed and can continue to provide critical services of ecosystem-based resilience. The landward freshwater swamp areas are extensive, used for swamp taro and tree crop cultivation as well as fresh groundwater supply. If a major breach occurred on this shore and allowed large volumes of marine waters to enter this could have far reaching damage and implications to many of the villages in the Hahake peninsula. The raised roadway would also prevent routine cutting of this road during wave events as sand, rocks and other debris are frequently thrown over the road. The approach would retain this important accessway and emergency evacuation route (e.g. tsunami warnings) and also protect the mains electricity lines which run parallel to the road.

Figure 12 Shoreline for Manuka to Kolonga

The revetment will be based on an existing proven design illustrated below (as already constructed in the adjacent coastal area under ADB), and remaining intact and robust post-tsunami:

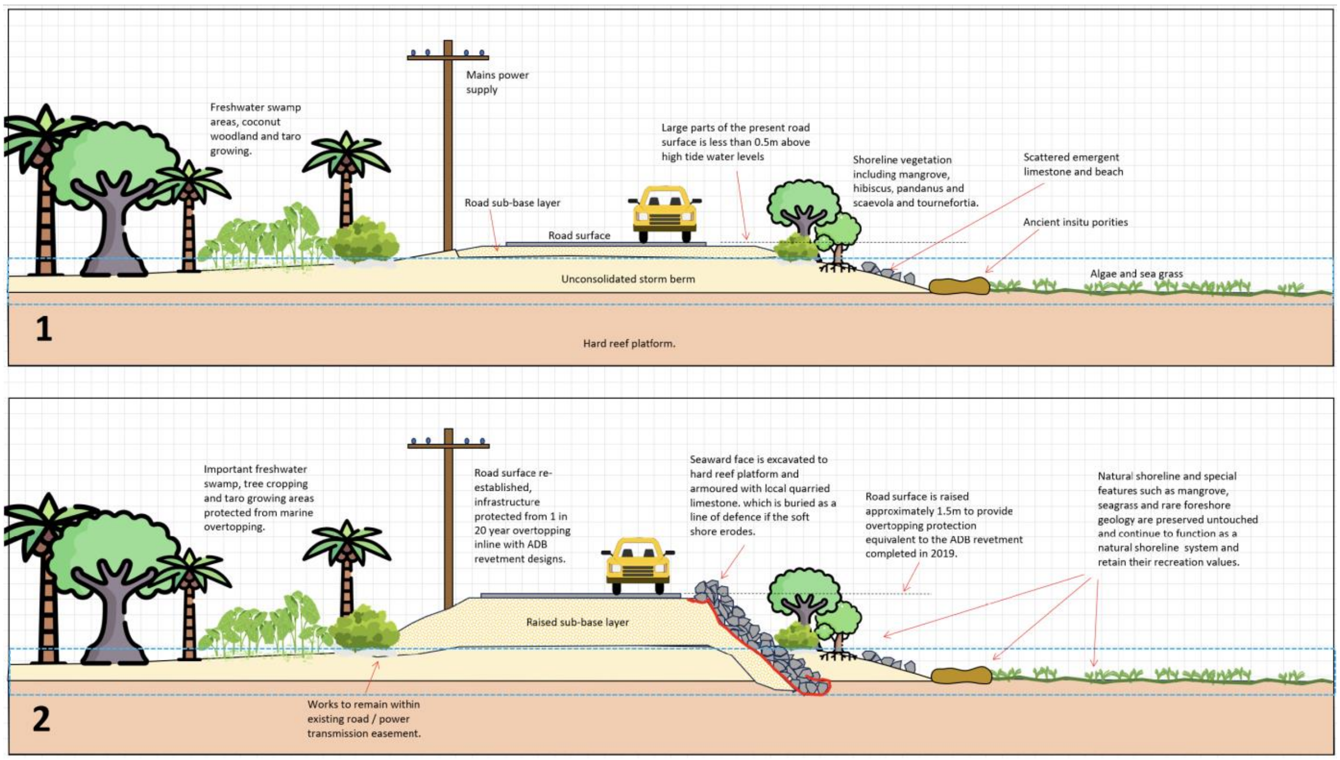
Figure 13 Proposed rock revetment for Sub-activity 3.1.2 Construction of 2.3kms of revetment Nukuleka - Talafo'ou



Legend	Rock Class	Type	D_{15} (mm)	D_{50} (mm)	D_{85} (mm)
	Class I	Armour Stone	528	605	677
	Class II	Secondary Armour	104	180	250
	Class III	Filter Layer	20	24	28
	-	Backfill	-	-	-
	-	Geotextile Layer	-	-	-

75. At the north eastern end of the Hahake Peninsula (shown above in Figure 11, 2kms of coastal protection measures developed for the Manuka to Kolonga shoreline on right side, orange line), adaptation measures will put in place approximately 2kms of coastal protection measures.

Figure 14 Schematic of the current situation (Panel 1) and shoreline protection construction (Panel 2) of 2kms of coastal protection measures developed for Manuka to Kolonga shoreline



For the shoreline between Manuka and Kolanga, the road surface would be raised approximately 1.5m to provide overtopping protection equivalent to the ADB revetment completed in 2019. The seaward face would be excavated to a hard reef platform and armoured with local quarried limestone, which is buried as a line of defence if the soft shore erodes. The road surface would be re-established, so that infrastructure is protected from a 1 in 20 year overtopping in line with the ADB revetment designs. Finally, the important freshwater swamp, tree cropping, and taro growing areas, would be protected from marine overtopping. This shoreline will not be disturbed via these proposed protective works that are designed to remain within the road easement and therefore not interfere with the natural shoreline processes.

76. To date, no licenses and permits have been applied for (as outlined in Annex 9). The area of development is Government held and confirmation from MLSLR for a Development Consent permit is still required as per the Act. The application will also require EIA and stakeholder consultation reports. Licenses and permits, such as the EIA approval required for coastal works will be applied for in the early implementation phase. These permits will be applied for at the implementation stage as the Detailed Project Proposal including technical specifications and design drawings must accompany the Application as required under sections 32 and 36 of the National Spatial Planning and Management Act 2012. The Development Consent for Coastal Protection Development issued by the MLSLR requires the documentation: a) detailed project proposal including technical specification and design drawings, b) Environmental Impact Assessment (EIA), c) Land ownership information/certificate, d) evidence of consultation with relevant authorities and stakeholders. Government has endorsed this Project and is aware that all Project capital works are located over the footprint of existing government infrastructure easements. Nonetheless a formal process of Development Consent via formal application in Tonga (i.e. formal permission to construct new infrastructure) will need to be undertaken. The DA process (Development Application) will occur only once more detailed analysis of the two works sites has been completed by Project technical staff and provisional designs adequate for the DA process can be submitted. It is expected this will also trigger the local ESIA process which will also be implemented by the Project. Project Technical Staff are expected to be secured in Q3 – 4 of year 1 and provisional designs are expected to be underway by Q1 - 2 year 2, this would coincide with submission of the relevant documents to the Government of Tonga to trigger the development application, associated ESIA process and ultimately resulting development consent (licensing).
77. An Operations and Maintenance Plan specific to the coastal protection measures will be developed as part of the design of the infrastructure by the Design and Supervision Engineers. This will be updated with a more robust Operations and Maintenance Strategy and Plan which will be developed for the post-construction O&M by the Design and Supervision Engineers following the completion of construction once the as-built designs are completed. This will form a part of the scope of work for the Design and Supervision Engineering Firm. It will be a part of the handover of the assets to the responsible Government agency, which is the Ministry of Infrastructure. The O&M Strategy and Plan will include the details of construction standards and will serve as a manual and knowledge tool for the Government together with what will be produced under Activity 3.1 to sustain the technical knowledge and lessons learnt for coastal protection measures.

78. Activity 3.2 Sharing of lessons learned and best practices in climate resilient coastal protection measures for scale-up at the national and regional level

In Tonga, and in the Pacific region in general, effective coastal protection measures are limited in number and diversity. The limitation is even more severe when one considers design standards that incorporate future climate risks. The coastal protection measures proposed in this project incorporates future climate risks reflected in the analysis of significant wave heights. But more importantly, this project is considered transformational in its approach of combining short-term and long-term measures (in developing a climate resilient land use plan) to adapt to climate risks, and this is truly innovative in the context of the Pacific region. Overall lessons of the technical effectiveness of the coastal protection measures implemented in Activity 3.1, in conjunction with the process of and results from Outputs 1 and 2, will be disseminated regionally and internationally through national/regional/international conference proceedings and other media, as well as providing guidelines and standardsfor coastal protection measures. As many countries in the Pacific will be facing a difficult decision to consider relocating some of the most exposed coastal populations in the future, and most countries today are not prepared institutionally, socially, and legally to do so, the experience from Tonga will make significant contributions to the regional body of knowledge and insights in the Pacific, for neighbouring Pacific SIDS governments and policy decision-makers.The technical evaluations undertaken of the design for coastal measures in Sub-activity 3.2.1 will inform the guidelines and standards developed as part of Aub-activity 3.2.3.

The following are covered under Activity 3.2 in the following activities:

- Sub-Activity 3.2.1 Technical evaluations of the design and effectiveness of the coastal protection measures conducted
- Sub-Activity 3.2.2. Production of and publication of a report on lessons learned
- Sub-Activity 3.2.3 Organize an international /regional conference for sharing lessons learned in climate resilient coastal protection and management

The Conference event will contribute to building of adaptation capacity throughout Tonga at varied levels, government, academic institutions and NGOs community representatives, through their active participation in the Conference event as a knowledge sharing activity sharing on the lessons learned from designing and constructing and evaluating effectiveness, of coastal resilience infrastructure (from Output 3), but also provides an opportunity to widely share LiDAR data and mapping developed and put to use in planning (from Output 2). While ultimately, the event serves to capacity-build beneficiaries (e.g. local government, village committee members) from across Tonga's 5 island groups, it also provides for knowledge sharing with neighbouring Pacific SIDS, offering the potential for scale-up regionally of the lessons learned within the project in the long-term, and supporting the paradigm shift for scale, replicability and sustainability at E2, as well also as the transformative elements described at D2.

B.4. Implementation arrangements (max. 1500 words, approximately 3 pages plus diagrams)

Provide a description of the project/programme implementation structure, outlining legal, contractual, institutional and financial arrangements from and between the GCF, the Accredited Entity (AE) and/or the Executing Entity(ies) (EE) or any third parties (if applicable) and beneficiaries.

Provide information on governance arrangements (supervisory boards, consultative groups among others) set to oversee and guide project implementation. Provide a composition of the decision-making body and oversight function, particularly for Enhanced Direct Access (EDA) proposals.

Provide information on the financial flows and implementation arrangements (legal and contractual) between the AE and the EE, between the EE or any third party and beneficiaries. For EEs that will administer GCF funds, indicate if a Capacity Assessment has been carried out. Where applicable, summarize the results of the assessment.

Describe the experience and track record of the AE and EEs with respect to the activities (sector and country/region) that they are expected to undertake in the proposed project/programme.

Provide a diagram(s) or organogram(s) that maps such arrangements including the governance structure, legal arrangements, and the flow and reflow of funds between entities.

79. The project will be implemented following UNDP's Direct Implementation Modality (DIM), which is the modality whereby UNDP takes on the role of Implementing Partner. This role is reflected in the Standard Basic Assistance Agreement (SBAA) signed between UNDP and the Government of Tonga signed on 2013, the Country Programme Action Plan (CPAP), and policies and procedures as outlined in the UNDP POPP (see https://info.undp.org/global/popp/ppm/Pages/Defining-a-Project.aspx). In DIM modality, UNDP applies its technical and administrative capacity and assumes the responsibility for mobilizing and applying effectively the required inputs in order to reach the expected outputs. UNDP assumes overall management responsibility and accountability for project implementation. Accordingly, UNDP must follow all policies and procedures established for its own operations. The responsibility for the execution of the DIM projects rests with UNDP. However, Government ownership of the project is ensured in the DIM modality, and the project implementation is directly co-located with the Government of Tonga.

DIM modality is justified given the limited capacity of the GoT in procurement and financial management, as found in the results of the UNDP Micro-HACT Assessment undertaken in 2023 and the overall risk assessmen being moderate.The 2023 Micro-

HACT assessement results stipulated that the project should be implemented under DIM. DIM will accordingly be applied in a way that maximizes cost-effectiveness while tailoring the capacity development of counterpart government institutions.

80. Furthermore, UNDP provides a three-tier oversight and quality assurance role involving UNDP staff in Country Offices and at regional and headquarters levels. This includes management of funds, programme quality assurance, fiduciary risk management, timely delivery of financial and programme reports to GCF and other requirements as per the AMA.The quality assurance role supports the Project Board by carrying out objective and independent project oversight and monitoring functions. This role ensures appropriate project management milestones are managed and completed. Project Assurance must be independent of the Project Management function; the Project Board cannot delegate any of its quality assurance responsibilities to the Project Manager. The project assurance role is covered by the accredited entity fee provided by the GCF. As an Accredited Entity to the GCF, UNDP is required to deliver GCF-specific oversight and quality assurance services including: (i) Day-to-day oversight supervision, (ii) Oversight of project completion, (iii) Oversight of project reporting. UNDP's responsibilities are outlined in the AMA that has been entered into between GCF and UNDP and will also be outlined in the FAA for this project. The FAA and AMA will govern UNDP's responsibilities for GCF. The 'Development Partner' role of UNDP is to represent the interests of the parties, which provide funding and/or technical expertise to the project (designing, developing, facilitating, procuring, implementing) and is covered by the accredited entity fee provided by the GCF. The Development Partner's primary function within the Board is to provide guidance regarding the technical feasibility of the project.

81. **General roles and responsibilities in the projects' governance mechanism**

<u>Implementing Partner</u>: The Implementing Partner for this project is the UNDP Pacific Multi-Country Office (MCO). The Implementing Partner is the entity to which the UNDP Administrator has entrusted the implementation of UNDP assistance specified in the signed project document along with the assumption of full responsibility and accountability for the effective use of UNDP resources and the delivery of outputs, as set forth in this document.

The Implementing Partner is responsible for executing this project. Specific tasks include:

- Project planning, coordination, management, monitoring, evaluation and reporting. This includes providing all required information and data necessary for timely, comprehensive and evidence-based project reporting, including results and financial data, as necessary. The Implementing Partner will strive to ensure project-level M&E is undertaken by national institutes and is aligned with national systems so that the data used and generated by the project supports national systems.
- Overseeing the management of project risks as included in this project proposal and new risks that may emerge during project implementation;
- Procurement of goods and services, including human resources;
- Financial management, including overseeing financial expenditures against project budgets;
- Approving and signing the multiyear workplan;
- Approving and signing the combined delivery report at the end of the year; and,
- Signing the financial report or the funding authorization and certificate of expenditures.

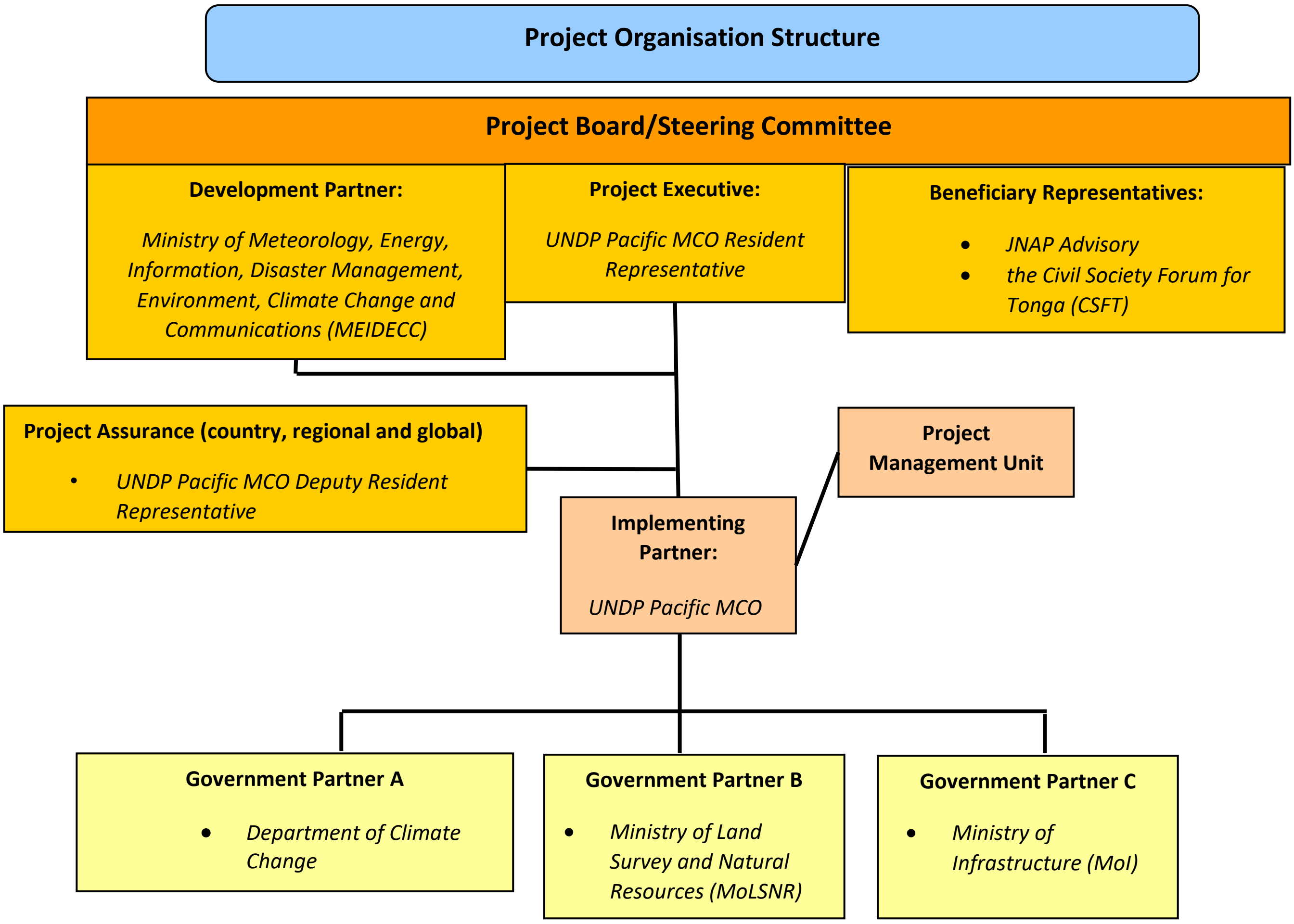
82. The implementation of the project will be carried out in collaboration and consultation with the following Government agencies: Department of Climate Change under the Ministry of Meteorology, Energy, Information, Disaster Management, Environment, Climate Change and Communications (MEIDECC) and with the Ministry of Lands and Natural Resources (MLNR) and Ministry of Infrastructure (MoI).

83. <u>Project stakeholders and target groups</u>: The project engages the Joint National Adaptation Plan (JNAP) Advisory for multisectoral multistakeholder involvement across government as well as the community level directly and the Tonga Civil Society Forum for Tonga (CFST), the Prime Minister's Office (National Planning and Local Government), and the Ministry of Internal Affairs (MIA).

84. <u>UNDP:</u> UNDP is accountable to the GCF for the implementation of this project. This includes oversight of project execution to ensure that the project is being carried out in accordance with agreed standards and provisions. UNDP is responsible for delivering GCF project cycle management services comprising project approval and start-up, project supervision and oversight, and project completion and evaluation. UNDP is responsible for the Project Assurance role of the Project Board. As UNDP is the implementing partner for this project, *a* strict firewall will be maintained between the delivery of project oversight and quality assurance performed by UNDP and project execution undertaken by UNDP. The segregation of functions and firewall provisions within UNDP in this case is described in the next section.

85. Project governance structure

Figure 15 Project Organisation Structure



86. The UNDP Resident Representative assumes full responsibility and accountability for oversight and quality assurance of this Project and ensures its timely implementation in compliance with the GCF-specific requirements and UNDP's Programme and Operations Policies and Procedures (POPP), its Financial Regulations and Rules and Internal Control Framework. A representative of the UNDP Country Office will assume the assurance role and will present assurance findings to the Project Board, and therefore attends Project Board meetings as a non-voting member.

87. Segregation of duties and firewalls vis-à-vis UNDP representation on the project board

UNDP's implementation oversight role in the project – as represented in the project board and via the project assurance function – is performed by UNDP Resident Representative and UNDP Deputy Resident Representative. UNDP's execution role in the project is performed by UNDP Project Manager, who will report to UNDP Resilience and Climate Change Team Lead of the UNDP Pacific MCO.

88. Roles and Responsiblities of the Project Organization Strucutre

Project Board: All UNDP projects must be governed by a multi-stakeholder board or committee established to review performance based on monitoring and evaluation, and implementation issues to ensure quality delivery of results. The Project Board is the most senior, dedicated oversight body for a project. The Project Board is responsible for making, by consensus, management decisions when guidance is required by the Project Manager. In order to ensure UNDP's ultimate accountability, Project Board decisions should be made in accordance with standards that shall ensure management for development results, best value money, fairness, integrity, transparency and effective international competition.

The three prominent (mandatory) roles of the project board are:

- High-level oversight of the execution of the project by the Implementing Partner, UNDP, (as explained in the "Provide Oversight" section of the POPP). This is the primary function of the project board and includes annual (and as-needed) assessments of any major risks to the project, and decisions/agreements on any management actions or remedial measures to address them effectively. The Project Board reviews evidence of project performance based on monitoring, evaluation and reporting, including progress reports, evaluations, risk register and the combined delivery report. The Project Board is the main body responsible for taking corrective action as needed to ensure the project achieves the desired results.
- Approval of strategic project execution decisions of the Implementing Partner, UNDP, with a view to assess and manage risks, monitor and ensure the overall achievement of projected results and impacts and ensure long term sustainability of project execution decisions of the Implementing Partner (as explained in the "Manage Change" section of the UNDP POPP).
- The Project Board is responsible for approval of multi-year and annual workplans and budgets. The multi-year and annual workplans and budgets will become effective when presented and approved by the project board. Any changes and reallocation of resources will be first reviewed and approved by the project board and communicated in writing to GCF.

89. The Project Board is responsible for providing overall guidance and direction to the project, ensuring it remains within any specified constraints, and provides overall oversight of the project implementation. The Project Board is also responsible for reviewing project performance based on monitoring, evaluation and reporting, including progress reports, risk register and the combined delivery report; and for making management decisions by consensus. In order to ensure UNDP's ultimate accountability, Project Board decisions are made in accordance with standards that shall ensure management for development results, best value money, fairness, integrity, transparency and effective international competition. In case consensus cannot be reached within the Board, the UNDP representative on the board will mediate to find consensus and, if this cannot be found, will take the final decision to ensure project implementation is not unduly delayed.

90. The Project Board undertakes Risk Management, by providing guidance on evolving or materialized project risks and agree on possible mitigation and management actions to address specific risks. The Project Board also reviews and updates the project risk register and associated management plans based on the information prepared by the Implementing Partner. This includes risks related that can be directly managed by this project, as well as contextual risks that may affect project delivery or continued UNDP compliance and reputation but are outside of the control of the project. For example, social and environmental risks associated with co-financed activities or activities taking place in the project's area of influence that have implications for the project.

91. Project Assurance: Project assurance is the responsibility of each project board member; however, UNDP has a distinct assurance role for all UNDP projects in carrying out objective and independent project oversight and monitoring functions. UNDP performs the quality assurance role and supports the Project Board and Project Management Unit by carrying out objective and independent project oversight and monitoring functions. This role ensures appropriate project management milestones are managed and completed. The Project Board cannot delegate any of its quality assurance responsibilities to the Project Manager. UNDP provides a three – tier oversight services involving the UNDP Country Offices and UNDP at regional and headquarters levels. Project assurance is totally independent of the Project Management function. A designated representative of UNDP playing the project assurance role is expected to attend all board meetings and support board processes as a non-voting representative. It should be noted that while in certain cases UNDP's project assurance role across the project may encompass activities happening at several levels (e.g. global, regional), at least one UNDP representative playing that function must, as part of their duties, specifically attend board meeting and provide board members with the required documentation required to perform their duties.

92. Frequency and Composition of the Project Board: The project board will meet at least once annually towards the end of the implementation year (November) to review progress and approve the next year's workplan. The composition of the Project Board must include individuals assigned to the following three roles. The project's Inception Report will provide details of the establishment of the Project Board, as well as updates and any changes to the Project Board composition:
    a. Project Executive.
    b. Beneiciary Representative(s) are: Individuals or groups representing the interests of those groups of stakeholders who will ultimately benefit from the project. Their primary function within the board is to ensure the realization of project results from the perspective of project beneficiaries. Often representatives from civil society, industry associations, or other government entities benefiting from the project can fulfil this role. There can be multiple beneficiary representatives in a Project Board. The Beneficiary representative (s) is/are: the JNAP Advisory and the Civil Society Forum for Tonga (CSFT).
    c. Development Partner(s): Individuals or groups representing the interests of the parties concerned that provide funding, strategic guidance and/or technical expertise to the project. The Development Partner(s) is/are: the *Ministry of Meteorology, Energy, Information, Disaster Management, Environment, Climate Change and Communications (MEIDECC).*

93. The Project Management Unit (PMU): All the PMU positions will be located in Nuku'alofa, Tonga, and none are based in UNDP Pacific Multi-Country Office (MCO). The majority of roles based full time within an office located in Nuku'alofa. However, some technical roles are to be part-time and remote (homebased) with travel to Tonga. The project's Inception Report will provide details of the establishment of the PMU, any changes and updates to the PMU structure. The Project Management Unit will comprise of the following national and international positions:
    - Full-time - International: Project Manager, based in-country.
    - Full-time – National, based in-country: Deputy Project Manager; Gender and Safeguards Officer; Communications Officer; Administrative Assistant; Finance Officer; Monitoring and Evaluation Officer; Driver.
    - Part-time – International, intermittently based in-country: a Chief Technical Advisor, Procurement Technical Advisor
    - Furthermore, a Project Implementation Unit (PIU) will be set-up within the Ministry of Infrastructure dedicated to the construction component of the project with the following expertise: Design and supervision engineering firm personnel, Safeguards specialist (construction), Safeguards Officer (construction). Procurement Technical Advisor

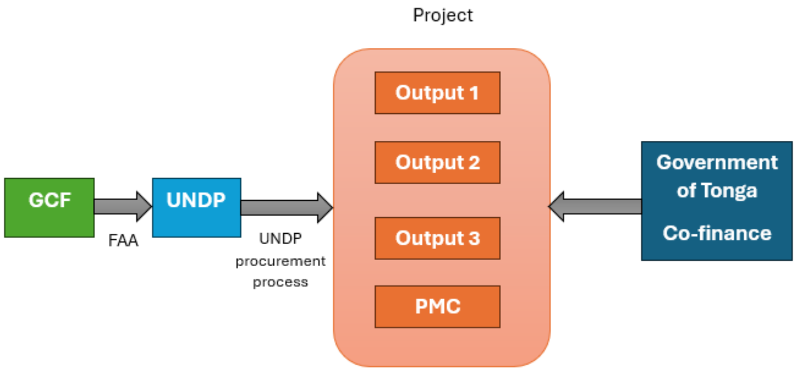
94. The Project Manager: The Project Manager (PM) (also called project coordinator) is the senior most representative of the Project Management Unit (PMU) and is responsible for the overall day-to-day management of the project on behalf of UNDP, including the mobilization of all project inputs, supervision over project staff, consultants and sub-contractors. The Project Manager will attend all board meetings and support board processes as a non-voting representative. The Project Manager will run the project on a day-to-day basis within the constraints laid down by the Project Board, and prepare key deliverables and documents to the board for their review and approval, including progress reports, annual work plans, adjustments to tolerance levels and risk registers. The Project Manager function will end when the final project terminal evaluation report and other project closure activities and documentation required by the GCF and UNDP, has been completed and submitted to UNDP. The Project Manager's prime responsibility is to ensure that the project produces the results specified in the project document, to the required standard of quality and within the specified constraints of time and cost, as well as managing the overall conduct of the project; Implementing activities by mobilizing goods and services; Checking on progress and plan deviations; Ensuring that changes are controlled and problems addressed; Monitoring progress and risks; and Reporting on progress including measures to address challenges and opportunities.

95. Fund flows

Upon approval of the project by the GCF Board, GCF funding will be received by the administrative agent (UNDP) as the AE on a standard Funded Activity Agreement (FAA) signed between the GCF and UNDP (see the figure below). Considering the latest Mico-HACT findings for MEIDECC and taking into account the procurement burdens entailed by what this project seeks to do, the project will be implemented as per UNDP's Direct implementation Modality (DIM). Tonga's Government co-finance will be independently invested to project outcomes, in parallel to GCF grant during implementation. UNDP has received co-financing commitment through an official letter from the Government of Tonga Ministry of Finance confirming commitment to co-finance the GCF Coastal Resilience Project. The Government co-finance is captured in the Project Document which is the legal document signed between the Government and UNDP for the project. The Board will ensure as part of its functions and responsibilities tracking of the co-financing amounts and its utilization as per the approved Work Plan of the project. There is no fund flow through the AE on co-finance.

96. The fund flow for the project is shown below.

Figure 16 Project Fund Flow



B.5. Justification for GCF funding request (max. 1000 words, approximately 2 pages)

Explain why the project/programme requires GCF funding to address mitigation or adaptation measures, i.e. Why is the project/programme not currently being financed by public and/or private sector? Which market failure is being addressed with GCF funding? Are there any other domestic or international sources of financing?

Explain why the proposed financial instruments were selected in light of the proposed activities and the overall financing package. i.e. What is the coherence between activities financed by grants and those financed by reimbursable funds? How were co-financing amounts and prices determined? How does the concessionality of the GCF financing compare to that of the co-financing? If applicable, provide a short market read on the prevailing of the pricing and/or financial markets for similar projects/programmes.

Justify why the level of concessionality of the GCF financial instrument(s) is the minimum required to make the investment viable. Additionally, how does the financial structure and the proposed pricing fit with the concept of minimum concessionality? Who benefits from concessionality?

In your answer, please consider the risk sharing structure between the public and private sectors, the barriers to investment and the indebtedness of the recipient. Please reference relevant annexes, such as the feasibility study, economic analysis or financial analysis when appropriate.

97. Grant financing is requested from the GCF to: create transformation in the current practices of Tonga's national land use planning towards long-term adaptation-focused and climate risk-sensitive planning; and to develop an effective climate risk monitoring system that allows for bottom-up community-level planning and community resilience building; as well as to reduce the impacts of climate change (sea level rise, erosion and wave overtopping) and protect the lives and assets of the vulnerable communities in the most exposed section of the north-eastern Tongatapu coastline.

    Although Tonga is not classified as a least developed country (LDC), its environmental and economic vulnerability has previously been higher than the LDC average as well as compared to other SIDS, largely due to its proneness to disasters [93].

98. Tonga is a Small Developing Island State (SIDS) with a total population of 100,179 [94], which is considered to be highly vulnerable to disasters and climate change impacts that will require higher capital spending on climate-resilient infrastructure projects, which could increase resilience to disasters at a faster pace [95]. Tonga's population already lives in a dynamic ecosystem, to which it has adapted, but climate change is likely to increase variability, pose new threats, and place stress on livelihoods. Communities are likely to need support to adapt and manage disaster risks facing their wellbeing, livelihoods, and infrastructure [96]. The country ranks 6th most vulnerable and 5th most exposure (out of 185) globally [97]. The ND-GAIN Index score (out of 100) which summarizes a country's vulnerability to climate change and other global challenges in combination with its readiness to improve resilience scores Tonga on the ND-GAIN Index as 41.1, and its overall rank is 140 out of 185 [98]. Geographic isolation and economic vulnerabilities, including dependence on remittance and foreign aid, will increase the challenges faced by communities and decision makers [99].

99. Tonga's economic instability has been further compounded by its exposure to disasters. The Global Climate Risk Index 2020 cited Tonga as one of the countries experiencing the highest levels of climate related loss as a percentage of GDP over the period 1999 - 2018 [100]. The largest island Tongatapu, for example, where 74% of the population reside, faces significant frequent flooding risks, as well as permanent losses from sea level rise (SLR) in the absence of future adaptation measures. The permanent losses under a 0.5m SLR scenario are expected to be 6%, and under a 1.0m SLR scenario to be 25% [101]. Tonga has also experienced a series of climate disasters in recent years; tropical cyclones Ian (2014), Gita (2018), Harold (2020), as well as non-climate related disasters with the Hunga-Tonga-Hunga-Ha'apai volcano eruption in 2022, which triggered tsunami waves throughout the Tonga archipelago. Combined, climate-induced and non-climate related disasters affected up to 80% of the population and wiped out between 12% to 40% of the GDP [102].

    Notably, during recent years Tonga's economy has been affected by both climate and non-climate related disaster impacts, with a downtrend in GDP following the Hunga Tonga-Hunga Ha'apai volcanic eruption and tsunami in 2022. Similarly, Tonga's economy previously suffered from climate-related impacts in the form of severe devastation from Tropical Cyclone (TC) Gita in 2018, and TC Harold in 2020. Tonga's tourism sector, for example, was badly affected by the Hunga Tonga-Hunga Ha'apai volcanic eruption and tsunami, as well as COVID-19 with visitor numbers 'falling to zero' in 2020, while the fisheries sector was also affected by lack of transport access to markets [103].

100. Given the combined factors of high upfront investments required for coastal adaptation, and the high debt levels of Tonga, the GoT and local communities have been constrained in their ability to implement critical interventions that are necessary for long-term coastal resilience, protection and management. Efforts to-date have been undertaken at a slow pace and are fragmented.

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[93] A Benfield, *2014 Annual Global Climate and Catastrophe Report*, 2014.
[94] TSD, 'Population and Housing Census | Tonga Statistics Department', 2021 <https://tongastats.gov.to/census-2/> [accessed 18 September 2023]. Tonga Statistical Department 2021
[95] World Bank-IMF, *Joint World Bank-IMF Debt Sustainability Analysis* (International Development Association, World Bank and International Monetary Fund (IMF), 2021) <https://documents1.worldbank.org/curated/en/657451613142701980/pdf/Tonga-Joint-World-Bank-IMF-Debt-Sustainability-Analysis.pdf>.
[96] World Bank, *Climate Risk Country Profile - Tonga* (Washington DC: World Bank Group, 2021).
[97] UND, 'Rankings - Notre Dame Global Adaptation Initiative - University of Notre Dame', *Notre Dame Global Adaptation Initiative*, 2023 <https://gain.nd.edu/our-work/country-index/rankings/> [accessed 19 September 2023].
[98] UND, 'Rankings - Notre Dame Global Adaptation Initiative - University of Notre Dame', *Notre Dame Global Adaptation Initiative*, 2023 <https://gain.nd.edu/our-work/country-index/rankings/> [accessed 19 September 2023].
[99] World Bank, *Climate Risk Country Profile - Tonga* (Washington DC: World Bank Group, 2021).
[100] D Eckstein and others, 'Global Climate Risk Index 2020 - World - Germanwatch', 2019 <https://reliefweb.int/report/world/global-climate-risk-index-2020> [accessed 19 September 2023].
[101] ADB, *Multi-Hazard Disaster Risk Assessment, Tongatapu*, Risk Assessment Summary Report (Asian Development Bank (ADB), 2021).
[102] ADB, *Multi-Hazard Disaster Risk Assessment, Tongatapu*, Risk Assessment Summary Report (Asian Development Bank (ADB), 2021).
[103] GoT, *Government of Tonga Budget Statement for the Year Ending 30th June, 2020* (Ministry of Finance, Government of Tonga, 2020).

Past initiatives have been either small scale or piecemeal interventions rather than comprehensive solutions integrated across island groups. Moreover, the capacity to develop, monitor and maintain climate risk data that help generate best practices for coastal protection is limited. Limitations in budgets have driven what has been done rather than what needed to be done in order for transformation to occur. Inevitably, without GCF support, such sub-optimal practices are likely to continue in the foreseeable future on multiple islands. It is against this background that a GCF grant (100% concessionality) is requested for the proposed project so that, in conjunction with government co-financing, Tonga can take the necessary comprehensive and systemic steps to build long-term and medium-term coastal resilience in the most vulnerable and high-value areas. The project places emphasis on inclusive consultative processes and deep engagement with key stakeholders in order to engender commitment and 'buy-in' to support the transformative change described in sections D.2 and E.2. In particular, this focus on coordination and cross-sectoral approaches, steers the GoT towards adoption of fundamental policy changes at national level to address climate risks. These initiatives, taken together, support the paradigm shift towards national commitment towards climate change risks and coastal adaptation.

B.6. Exit strategy (max. 500 words, approximately 1 page)

Explain how the project/programme will successfully exit once implementation is completed, including how results and benefits will continue beyond the project/programme period and how the contribution to paradigm shift will be maintained.

Include information pertaining to the longer-term ownership, project/programme exit strategy, operations and maintenance of investments (e.g. key infrastructure, assets, contractual arrangements). In case of private sector, please describe the GCF's financial exit strategy through Initial Public Offerings, trade sales, etc.

Provide information on additional actions to be undertaken by public and private sector or civil society as part of the project/programme to ensure sustainability of the results attained.

101. The project is designed to establish systems that are embedded in local policies, plans and government procedures in an enduring way that lasts beyond the term of the project. This is to be achieved through prolonged and meaningful engagement with the relevant stakeholders, in order to ensure that there is extensive support ('buy-in) for the agreed decisions and policies, as well as maintenance commitments for coastal infrastructure.

102. The overall design of this project, which has attempted to move from a piecemeal approach to a more comprehensive framework should lead to replacing the short-term reactive approach with a longer-term, planned, proactive, and sustainable approach. Transformative adaptation aims to address root causes of vulnerabilities to climate change – including cultural, economic, environmental, and power relations by transforming them into more just, sustainable or resilient states.[104] In the SIDS context, transformational long-term approaches to adaptation address vulnerability to climate change and disaster risks should be managed deliberately through "collective design, intentionality and political negotiation."[105] Given the SIDS context, transformational long-term approaches to adaptation to address vulnerability to climate change and disaster risks should be engaged by facilitating discussion for long-term transformational adaptation planning, with the access and capacity to ground decisions based on climate risk information. The overall transformational adaptation aspect that frames this project, is therefore via facilitation for long-term transformational adaptation planning (undertaking the discussion required for collective design and political negotiation), with the access and capacity to ground decisions based on climate risk information. The discussion undertaken within the multi-stakeholder platform is designed to establish a soundly based national approach to land-use/adaptation decision making that will be employed over the medium term – and well beyond the conclusion of the project. Its aim is to establish a national level approach to land use and adaptation throughout Tonga's small islands, and to empower communities to become directly involved in decision in their own localities. By doing so, the project is designed to establish a soundly based national approach to land-use/adaptation decision-making that will be employed over the medium term to long term, well beyond the project's conclusion. It aims to establish a national-level approach to land use and adaptation throughout Tonga's small islands and empower communities to become directly involved in local decisions. At the same time, in the final year of the project, a technical assessment will be carried out by an expert to review the effectiveness of the coastal protection measures put in place in the project, which will inform the capturing of best practices for coastal measures under Output 3. Therefore, these coastal protection measures will also create a body of knowledge developed through these interventions that will contribute towards more sustained coastal resilient solutions across the Pacific where other islands are confronted with similar climate challenges.

103. Community-led decision-making will be informed by a solid baseline of technical information on a range of risk factors (coastal topography, wave modelling, rates of sea level rise etc.) and the capacity in government agencies and civil society to interpret this information in decision making. This is expected to remain valid throughout the project, though ongoing training / capacity building is likely to be required in the medium term. Similarly, there will be an ongoing need for data collection /monitoring after project closure. Outputs 1 and 2 incorporate significant investment in capacity development across a range of fields including stakeholder engagement, community governance, and technical knowledge relating to data collection and analysis. While the specific training under this project will cease at project end, the work is designed to mesh with the work of capacity development agencies operating in the Pacific Islands region (notably, SPC and the University of the South Pacific) which provide ongoing training services. GCF investments in the technical capacity building, particularly for coastal monitoring, assessments and

[104] Giacomo Fedele, Camila I. Donatti, Celia A. Harvey, Lee Hannah, David G. Hole, (2019), 'Transformative adaptation to climate change for sustainable social-ecological systems', Environmental Science & Policy, Volume 101, 2019, Pages 116-125, ISSN 1462-9011, https://www.sciencedirect.com/science/article/pii/S1462901119305337

[105] Idowu Ajibade & Ellis Adjei Adams (2019), 'Planning principles and assessment of transformational adaptation: towards a refined ethical approach, Climate and Development, 11:10, 850-862, https://www.tandfonline.com/doi/full/10.1080/17565529.2019.1580557

analyses, will enable the Government to detect a long-term trend in coastal processes or a sudden change after a cyclone event and design and execute counter measures accordingly. While it is still likely that the Government would have to rely on external support for large-scale engineered solutions, enhanced technical capacity will enable them to tackle the issue more proactively and strategically prioritise what works should be undertaken with limited resources.

104. For Output 3, the project design is based on the premise that the responsibility for monitoring, maintenance and upkeep of hard engineering structures should be with MEIDECC (as expressed in the letter of co-financing in Annex 13), with technical assistance from the Ministry of Infrastructure (MoI). Any interventions relating to the protection and management of the coastal zone also needs to be fit for purpose and the interventions need to be congruent with community expectations. This community involvement and participation in the project concept development, design and implementation (including maintenance and monitoring of revetments and additional mixed approaches such as revegetation) will ensure that climate change adaptation interventions in the coastal zone are operated and maintained in a long term sustainable manner. The overall approach to design of coastal protection measures under the project is to take into account at an entire geomorphological section of coastline and understanding the processes and coastal dynamics provides a good sustainable approach for the MEIDECC and the MoI to use in the future. This should lead the way to replace the short term reactive approach with a longer term, planned and proactive and sustainable approach.

Moreover, in the final year of the project, a technical assessment will be carried out by an expert to review the effectiveness of the coastal protection measures put in place in the project, which will inform the capturing of best practice for coastal measures under Output 3. These coastal protection measures will therefore also create a body of knowledge developed through these interventions that will contribute towards more sustained coastal resilient solutions across the Pacific where other islands are confronted with similar climate challenges.

C. FINANCING INFORMATION

C.1. Total financing

(a) Requested GCF funding (i + ii + iii + iv + v + vi + vii)	Total amount	Currency
	22.66	million USD ($)

GCF financial instrument	Amount	Tenor	Grace period	Pricing
(i) Senior loans	Enter amount	Enter years	Enter years	Enter %
(ii) Subordinated loans	Enter amount	Enter years	Enter years	Enter %
(iii) Equity	Enter amount			Enter % equity return
(iv) Guarantees	Enter amount	Enter years		
(v) Reimbursable grants	Enter amount			
(vi) Grants	22.66			
(vii) Results-based payments	Enter amount			

(b) Co-financing information	Total amount	Currency
	1.263	million USD ($)

Name of institution	Financial instrument	Amount	Currency	Tenor & grace	Pricing	Seniority
Ministry of Finance	Grant	1.2	million USD ($)	Enter years Enter years	Enter%	Options
UNDP	Grant	0.063	million USD ($)	Enter years Enter years	Enter%	Options
Click here to enter text.	Options	Enter amount	Options	Enter years Enter years	Enter%	Options
Click here to	Options	Enter amount	Options	Enter years Enter years	Enter%	Options

enter text.						

	Amount	Currency
1. Total financing (c) = (a)+(b)	23.92	million USD ($)

(d) Other financing arrangements and contributions (max. 250 words, approximately 0.5 page)	*Please explain if any of the financing parties including the AE would benefit from any type of guarantee (e.g. sovereign guarantee, MIGA guarantee).* *Please also explain other contributions such as in-kind contributions including tax exemptions and contributions of assets.* *Please also include parallel financing associated with this project or programme (refer to the co-financing policy).* 105. UNDP projects in Tonga are exempted from taxation in line with conditions stipulated in the Standard Basic Assistance Agreement (SBAA), for activities related to procurement of good and services through UNDP. Section 7 of the Convention on the Privileges and Immunities of the United Nations provides, inter alia, that the United Nations, including its subsidiary organs, is exempt from all direct taxes, except charges for utilities services, and is exempt from customs duties and charges of a similar nature in respect of articles imported or exported for its official use. 106. According to the UN Convention on Privileges and Immunities, UN or UNDP are not, as a general rule, exempted from indirect taxes. However, Section 8 of the Convention states the following: "… when the United Nations is making important purchases for official use of property on which such duties and taxes have been charged or are chargeable, Members will, whenever possible, make appropriate administrative arrangements for the remission or return of the amount of duty or tax." As such, the Government of Tonga has put in place the national regulation on Consumption Tax (2020) where UNDP has been granted tax exemption by the Government of Tonga, and they are currently in the process of issuing a letter confirming that the regulation would cover this project.

C.2. Financing by component

Please provide an estimate of the total cost per component and output as outlined in section B.3. above and disaggregate by source of financing. More than one co-financing institution can fund a single component or output. Provide the summarised cost estimates in the table below and the detailed budget plan as annex 4.

Output	Activity	Indicative cost million USD ($)	GCF financing		Co-financing		
			Amount million USD ($)	Financial Instrument	Amount million USD ($)	Financial Instrument	Name of Institutions
1. Strengthened knowledge, capacity and engagement for incorporating climate risks into long-term adaptation planning supported through a multi-sectoral, multi-stakeholder engagement and dialogue platform	1.1 Establish a national multi-stakeholder engagement platform for dialogue on co-creating long-term climate change adaptation strategies and solutions including voluntary retreat	1.74	1.60	Grant	0.14	Grants	Ministry of Finance
	1.2 Develop village and district level participatory climate risk	0.83	0.75	Grants	0.08	Grants	Ministry of Finance

	informed plans						
	1.3 Build the capacity of local government, village committees and NGOs to integrate climate risks and adaptation needs into community level planning, and inform future Community Development Plans (CDP)	1.64	1.13	Grants	0.51	Grants	Ministry of Finance
2. Strengthened national and local capacities for effective monitoring and assessment of climate risks	2.1 Strengthened mechanism for collecting and analysing data and information for better-informed climate risk monitoring and coastal adaptation planning	6.70	6.47	Grants	0.23	Grants	Ministry of Finance
	2.2 Improve the knowledge base of multi-sectoral, multi-stakeholders on adaptation planning strategies for long-term resilient planning and transformative	1.02	0.99	Grants	0.03	Grants	Ministry of Finance
3. Reduced vulnerabilities of coastal communities in Hahake to climate hazards through coastal	3.1 Build coastal protection measures along 4.3 km of coastline in Hahake	9.92	9.85	Grants	0.07	Grants	Ministry of Finance
	3.2 Sharing of lessons learned and best	1.06	0.86	Grants	0.14	Grants	Ministry of Finance

protection measures	practices in climate resilient coastal protection measures for scale-up at the national and regional level						
				Grants	0.06	Grants	UNDP
4. Project Management	Project Management Costs	1.01	1.01	Grants			
Indicative total cost (USD)		23.92	22.66		1.26		

This table should match the one presented in the term sheet and be consistent with information presented in other annexes including the detailed budget plan and implementation timetable.

In case of a multi-country/region programme, specify indicative requested GCF funding amount for each country in annex 17, if available.

C.3 Capacity building and technology development/transfer (max. 250 words, approximately 0.5 page)

C.3.1 Does GCF funding finance capacity building activities?	Yes ☒ No ☐
C.3.2. Does GCF funding finance technology development/transfer?	Yes ☒ No ☐

If the project/programme is expected to support capacity building and technology development/transfer, please provide a brief description of these activities and qualify the total requested GCF funding amount for these activities to the extent possible.

107. The following sub-activities of the project under Outputs 1 and 2, support capacity and technology development/transfer, specifically:

Output 1 Strengthened knowledge, capacity and engagement for incorporating climate risks into long-term adaptation planning supported through multi-sectoral, multi-stakeholder engagement and dialogue platform. This output will build the capacity of local government, village committees and NGOs to integrate climate risks and adaptation needs into community level planning, and inform future Community Development Plans (CDP). USD 920,563

108. **Output 2 Strengthened national and local capacities for effective monitoring and assessment of climate risks** will (2.1) strengthen the mechanisms for collecting and analysing data and information for better-informed climate risk monitoring and coastal adaptation planning, and; (2.2) improve the knowledge base of multi-sectoral, multi-stakeholders on adaptation planning strategies for long-term resilient planning and transformative adaptation for Tonga based on climate risks and projections. This output enhances the mechanism for collecting and analysing data and information for better-informed climate risk monitoring and coastal adaptation planning, providing key capacity building, training and mentoring for coastal adaptation planning and monitoring and developing climate risk-informed coastal plans. At the same time, improving the knowledge base of multi-sectoral based, multi-stakeholders on adaptation planning strategies for long-term resilient planning and transformative adaptation for Tonga, which are based on risks and projects. This includes improving LiDAR coverage for Tonga's Islands, and capacities for interpreting key information, producing maps and knowledge material on scenario-based coastal risk maps, for decisions on long-term settlement patterns and climate risk for strategic long-term coastal planning while strengthening multistakeholder coordination and collaboration on coastal adaptation and planning integrating climate change. MLSNR and MIEDECC do not currently have access to the data and expertise to collect, manage and analyze in-depth data on coastal climate change risks. This activity therefore aims to build capacity in relation to climate risk data/interpretation in order to allow for better-informed coastal adaptation planning. Baseline data on coastal topography bathymetry and tide/wave behaviour will be procured from external providers. LiDAR survey work will capture outer island groups and

selected areas of Tongatapu where covergage is either missing or outdated. In addition to this survey, the project will also support the purchase of selected equipment (hardware and software) that will remain with the Government of Tonga to provide capability to carry on supplementary survey work as needed throughout the project, and subsequently. The budget allocated for procurement and deployment of LiDAR includes the costs of conducting the LiDAR survey as well as capacity building of the Government departments, especially the GIS Office. The project will collect this data in partnership with both the local Tongan authority (the GIS Office) and the SPC. Recognizing the limited geospatial capacities within Tonga, the SPC is a major link in terms of sustainability and how this data is archived and protected for continued use, access provision and ongoing availability. Housing data within the SPC builds sustainability given that the SPC is specifically tasked to carry out data archiving and analysis services for the Pacific Islands, and once this Project's data and products are incorporated into SPC systems they will provide access and support indefinitely. The SPC carries the regional mandate to supply, hold and undertake analysis of such data, a role that it already performs on behalf of Tonga. Therefore, data and analysis products produced via the project will be part of this ongoing (perpetual) agreement that predates and supersedes the short-term project timeframes. This model was successfully used for the LiDAR data collected in Tuvalu. Increasing capacity in analysis of the data will then bridge the current gap to enable these data to inform and guide climate risk driven modelling (coastal inundation, storm surge and hydrodynamic modelling), shoreline mapping, and vulnerability analysis of coastlines. Recognizing that the equipment and data, assessments and analyses must be shared across multiple ministries, GCF financing will also be used to assist in developing a long-term Capacity Building Strategy and Plan for government use on coastal and climate risk monitoring and analysis to inform coastal adaptation planning (e.g. including understanding and use of climate and risk information and data such as satellite imagery, aerial photography, drone imagery, LiDAR topography and bathymetry) for cross-ministerial cooperation in producing and utilizing data for the purpose of coastal protection. USD 8,356,893.

109. Total for capacity building and technology transfer is USD 9,277,456

D

CTED PERFORMANCE AGAINST INVESTMENT CRITERIA

This section refers to the performance of the project/programme against the investment criteria as set out in the GCF's Initial Investment Framework.

D.1. Impact potential (max. 500 words, approximately 1 page)

Describe the potential of the project/programme to contribute to the achievement of the Fund's objectives and result areas. As applicable, describe the envisaged project/programme benefits for mitigation and/or adaptation. Provide the intended outcomes for mitigation by elaborating on how the project/programme contributes to low-emission sustainable development pathways. Provide the intended outcomes for adaptation by elaborating on how the project/programme contributes to increased climate-resilient sustainable development. Calculations should be provided as an annex. This should be consistent with section E.3 reporting GCF's core indicators.

110. The overall objective of the project is to support the transformative adaptation agenda setting through a participatory process that builds policy consensus and promotes planning coherence. Towards this end, the project will focus on the bottom-up development planning process (community development plans, district plans and island strategic plans) and linking development planning with spatial planning and land use management to promote transformative adaptation through the incorporation of climate risk planning incorporated into the Tongatapu Island strategic development plan, given the incorporation of climate-risk-informed district spatial plan piloted in Tongatupu, which will serve as a pilot under the project to guide other Island-level spatial development plans.

111. Indirectly the project benefits the broader population given the community development planning and plans supported through guidance, packaging, outreach, training, and equipment for incorporating climate risk. At the same time, establishing the national multi-stakeholder engagement platform for dialogue on co-creating long-term climate change adaptation strategies and solutions will also deliver indirect benefits so that the project impacts Tonga's total population.[106] The project is expected to support multiple district spatial plans by providing data information and interpretation for incorporating climate risk; however, the initial piloting will be the Island-level plan for Tongatapu. Along with the Strategic Development Plan for Tongatapu, the Island Strategic Development Plans – for Ha'apai (2017-2023), Niua (2015-2018), 'Eua (2015-2018), and Vava'u (2017-2023), will be updated to consider climate risks and promote appropriate and transformative adaptation options. In addition to renewing the Island Strategic Development Plans – for Ha'apai (2017-2023), Niua (2015-2018), 'Eua (2015-2018), and Vava'u (2017-2023), the island spatial planning will be supported by the collection, interpretation and use of data on climate risks for coastal adaptation planning to produce scenario-based coastal risk maps and knowledge material for use.

112. Directly, the project will benefit 3,608 people, or 3.6% of the population. Under Output 1, a further 300 people (community members) are therefore expected to directly benefit from being trained in climate-risk and adaptation for updating/developing the CDPs/ ISPs. A further 25 government staff (MoI, MLNR, MoF & NP, MIEDECC, MIA) will directly benefit from the project through training in incorporating climate change risks in planning and budgeting, supporting development planning; while a further 25 government staff (MLNR, MoI, MIEDECC) will directly benefit from technical training on climate risk monitoring and data. 3,258 direct beneficiaries will receive protection through coastal protection measures (including rock revetments) in north-eastern Tongatapu to reduce their vulnerability to climate-induced erosion and wave overtopping events (Output 3).

96. The project activities will increase the generation and use of information on coastal processes, oceans and climate in decision-making by strengthening institutional capacity, human resources, awareness and knowledge for resilient coastal management. For example, an increased volume of coastal data can improve the investment decision on coastal protection and its design to reduce the risk of locking finite Government financial resources into poorly planned infrastructure projects that did not factor climate vulnerabilities into the project design. By increasing data collection on various coastal dynamics and improving modeling capacity, this project assists GoT in avoiding the risk of maladaptation. However, the project also ensures physical assets are made more resilient to climate variability and change, considering human benefits. Analysis was undertaken based on the number of people expected to benefit directly from the coastal protection measures under Output 3. The economic analysis shows the project to be economically efficient and robust to cost increases and benefits with very high confidence that the project is economically efficient.

113. Finally, the proposed interventions will contribute to a body of knowledge about coastline resilience. The best practices that will be identified and replicated across Tonga will strengthen long-term investment in adaptive capacity and reduce exposure to ongoing climate risks. The mutual reinforcement of both outputs will ensure Tonga has extensive experiences in hard and soft engineering practices aligned with institutional and community-based knowledge.

[106] TSD, 'Population and Housing Census | Tonga Statistics Department', 2021 <https://tongastats.gov.to/census-2/> [accessed 18 September 2023]. Tonga Statistical Department 2021

D.2. Paradigm shift potential (max. 500 words, approximately 1 page)

Paradigm shift potential is defined as 'degree to which the proposed activity can catalyse impact beyond a one-off project or programme investment'. In this section, elaborate on the contribution to paradigm shift and how the proposed project/programme aims to contribute towards it based on the theory of change described in section B2(a). Also describe how and to what extent the project/programme will be able to promote or contribute to paradigm shift through the below.

- *Potential for scaling up and replication*
- *Potential for knowledge sharing and learning*
- *Contribution to the creation of an enabling environment*
- *Contribution to the regulatory framework and policies*
- *Overall contribution to climate-resilient development pathways consistent with relevant national climate change adaptation strategies and plans*

114. The overall vision for the project is to address a suite of existing conditions in Tonga to create a fundamental shift enabling Tonga to address climate change risks, as they relate to land-use and coastal risks, through spatial adaptation plans and policy with broad public and political acceptance address key climate risks, supported by processes to gather and interpret critical data over time, and provide for communities to guide their own future through Community Development Plans, as well as constructing practical coastal protection measures along the Hahake coastline.

115. The paradigm shift potential from these interventions is illustrated below. The different elements fit together to generate high level national policy, in-depth public engagement and dialogue, the capacity to collect and interpret relevant high quality data, community-led adaptation planning, and models for coastal protection The package of activities also represent the highest priorities of the Government of Tonga, consistent with the National Adaptation Plan (JNAP II) and companion documents.

Initial Paradigm (Baseline)	**Future Paradigm (post-project)**
No mechanism for decision-making on land-use planning in the face of future climate risks	**Spatial adaptation plans and policy with broad public and political acceptance address key climate risks**
Lack of base data and capacity to interpret climate risks to coastal property, infrastructure, and livelihoods	**Strong cohort of officials and community members with capacity to interpret data and climate projections**
No mechanism for addressing coastal climate risks at community level	**Community development plans incorporate relevant climate risks and define future resilience pathways**
Limited understanding of coastal adaptation infrastructure	**Models available for coastal protection measures**

116. The national multi-stakeholder engagement platform for dialogue on co-creating long-term climate change adaptation strategies and solutions including voluntary retreat, comprehensive consultation process, and transparent consideration of future climate projections is highly applicable to other Pacific SIDS, with potential for application in 14 other Pacific Island countries. The dialogue will be as much evidence-based as possible equipped with the information about projected climate risks, under different climate scenarios, such as sea-level rise and extreme events such as tropical cyclones. Such information currently does not exist in the country and will be collected through activities under Component 2.

117. On a technical level, the in-depth process for knowledge transfer and building capacity to manage and interpret climate data is readily scalable for other SIDS in the Pacific islands region. Technical capacity building will assist, *inter alia*, the MIEDECC, and MoLSNR to monitor coastal, wave and current dynamics on a continual basis contributing to the body of knowledge about the coastlines of the country. Uninterrupted data collection will improve the accuracy of wave modelling and other analysis to provide decision makers, technical officers and community members with information about future coastal risks that was previously unavailable in the country. It is particularly important to highlight that a strong focus on technical capacity building for data collection, synthesis and analysis will assist government departments to collect information about, not only the performance of the infrastructure built with GCF financial assistance, but also the interactions between coastal ecosystems and the engineered infrastructure. It is expected that this evidence will feed into community awareness raising activities that will take place throughout the course of the project implementation, underpinning the importance of continued community engagement in ecosystem restoration, management and monitoring.

118. The M&E plan (Section E5) will include provisions for generation of lessons learned and best practices (reports, publications, and other communication and knowledge products for various media) to not only support adaptive project

management but also to inform learning across national/sub-national/community levels within the country and region. The three outputs combined will identify best practices for coastal protection that can facilitate replication of project results beyond the immediate project target communities. Drawing from the site-specific adaptation interventions applied in specific scenarios, the Government of Tonga will improve their capacity to plan and implement interventions on other vulnerable coastlines not addressed in this project. Other inhabited islands in many Pacific SIDS face similar challenges to their coastlines. Tonga's experience in comprehensively removing infrastructural, technical, and community barriers through GCF support will be shared in regional forums and replicated by other SIDS, thereby establishing Tonga as a regional leader in coastal protection methods.

119. At least one international/regional conference will be organized for the specific purpose of sharing lessons with the international communities, but particularly other SIDS. Prior to the conference, a technical assessment will be carried out to assess the technical performance of GCF investments in terms of reduced erosion or damages from storm surges. These, and lessons from the national dialogue engagement processes, will provide models that offer learning to other SIDS.

The following areas of the project are particularly important in offering knowledge and learning opportunities:

- Knowledge about future climate risks on coastal areas under different climate scenarios and the need for voluntary retreat from highly exposed coastal areas for some communities; knowledge about the impact of future internal migration on food security of the country (as people would need to move into areas where there are existing economic activities such as agriculture).
- Technical knowledge of the construction and monitoring of coastal protection works.
- Accumulation of data, information and local capacity to interpret climate risks, and coastal processes.
- Learning opportunity about community-based vulnerability assessment, adaptation priority setting and resource mobilization.

120. The Project will serve as an enabling environment for communities to govern, manage and implement their own climate change adaptation projects. The government alone cannot change the way people act; initiatives need to be developed that spur people to take action, building their capacity to intervene, to be innovative and to cope in the face of coastal erosion and climate change. the project will support the revision of the national land use strategy that incorporates long-term climate risks and impacts to identify and select long-term adaptation strategies. This will be achieved by the establishment of a national multi-stakeholder engagement platform for dialogue on co-creating long-term climate change adaptation strategies and solutions that will enable island-wide discussions and agreements on issues and concerns regarding land use that need to be urgently addressed in order to develop long-term transformative adaptation strategies, including that of voluntary retreat. In this way, the project will remove key barriers that have hindered transformation in Tonga, foster national dialogue, and create a paradigm shift in national land use planning through the revised national land use plan that focuses on long-term adaptation solutions. This activity will, under the leadership MEIDECC and MIA, help create a platform that brings together community members, estate owners and other stakeholders such as NGOs and facilitate evidence-based dialogue on long-term adaptation options, with respect to land use of the country. The national dialogue process is unprecedented in the Pacific Islands region, in that it brings a whole country approach to enable all citizens to participate in development of key national policy to address climate risks.

D.3. Sustainable development (max. 500 words, approximately 1 page)

Describe the wider benefits and priorities of the project/programme in relation to the Sustainable Development Goals and provide the potential in terms of:

- *Environmental co-benefits*
- *Social co-benefits including health co-benefit*
- *Economic co-benefits*
- *Gender-sensitive development benefit*

121. The proposed GCF project will contribute towards the following Sustainable Development Goals (SDGs) in Tonga. Of particular relevance is the contribution the project makes to SDG Goal 13 on taking urgent action to combat climate change and its impacts. The project directly contributes towards target 13.1 on strengthening resilience and adaptive capacity to climate related hazards and natural disasters in all countries and targets 13b, promoting mechanisms for raising capacity for effective climate change-related planning and management in least developed countries and small island developing states, including focusing on women, youth and local and marginalized communities. The project is also a direct contribution to Tonga's aspirations with respect to Goal 11: making cities and human settlements inclusive, safe, resilient and sustainable (especially target 11.5) by 2030, significantly reduce the number of deaths and the number of people affected and substantially decrease the direct economic losses relative to global gross domestic product caused by disasters, including water-related disasters, with a focus on protecting the poor and people in vulnerable situations.

122. **Economic co-benefits:** In addition to the primary benefits of coastal protection, construction under Output 3 of the project will create short- and medium-term job opportunities for local labor force, especially youth and women. During the construction of revetments in north-eastern Tongatapu, it is expected that approximately 50 short-term jobs will be created, with the expected male/female distribution to be 35/15. Of these, approximately 10 men and 5 women are expected to undertake skilled labor work including carpentry, heavy machinery operations, survey assistance, etc. This will provide formal employment and cash-income opportunities for both skilled and low skilled/vulnerable people and have a substantial

impact on the unemployment rate of Tonga that was around 2.1% in 2021 [107]. While the male-female distribution presented above is based on past experience in Tonga, women and youth will be given priority in participation of these job opportunities to ensure their representation.

123. The project has significant **environmental co-benefits** arising from improved coastal protection, increased biodiversity and reduced salinity levels in groundwater. Wave overtopping events and subsequent waterlogging can be a cause of saline water intrusion into the fragile groundwater resources in Tongatapu, where groundwater is still used for drinking purposes, or aggravate bacterial contamination by accelerating seepage from septic tanks and livestock pens. Many makeshift seawalls that have been privately built have collapsed due to the constant wave energy along the coasts of Tonga. However, due to the collapsed seawalls, the foreshore is rough, unsafe and poses very limited access to the sea throughout most sections, and upon which many people rely on for their livelihoods. Coastal protection measures that are appropriately designed and constructed are expected to ameliorate these environmental problems.

124. The enhanced capacity to monitor coastal systems supported under Output 2 will have wider benefits in terms of a) establishing a topographic/bathymetric baseline for coastal systems b) building a body of trained personnel able to use and interpret climate risk and potentially apply these skills across other fields. In this way the project supports the use of data over the medium term for uses beyond the immediate needs of the project.

125. The project will generate a range of **social co-benefits** that will positively affect the overall well-being of the Tongan people. The project interventions will contribute towards minimal disruption to the lives and livelihoods of the people. Reduced disruptions of economic activities, education, public and medical services, and cultural activities are directly linked to the level of resilience in society. Incidents of coastal inundation and waterlogging have direct impact on public health through the increased incidents of water borne diseases and overstretching the highly limited public health facilities in the country. GCF resources will contribute to the realization of these social benefits by stopping, or significantly slowing down, the level of erosion and wave overtopping events. These benefits will accrue over different time periods. The coastal works constructed under Output 3 will have immediate benefits to the local population. Conversely the fundamental policies emerging through the national multi-stakeholder engagement platform for dialogue (in Output 1) will benefit the whole population over the medium term. Similarly the the capacity development work and community adaptation planning under Output 2 will be realised in the medium term.

126. Coastal protection in north-eastern Tongatapu will also protect the road used by local communities as the only evacuation route to higher grounds in the south of the island. Without GCF support, it is possible that the next large-scale cyclone will undermine the infrastructure, causing a significant risk to the local population. The protection measures will also offer similar benefits in the case of tsunami. In addition, the implications of protecting low-lying coastal areas from encroaching impact of sea-level rise and erosion go well beyond what is captured in the range of social benefits such as education, health, or even safety. The project strives to protect the country's land and its cultural heritage by preventing, or significantly delaying, the scenario of parts of Tongan islands becoming uninhabitable. Through this project, it assures intangible benefits to the people of Tonga including the sense of dignity people derive from living in a country or culture to which they belong and the sense of self-fulfilment people derive from the ability to make decisions about their own future.

127. **Gender** issues will be mainstreamed at all levels during the project implementation in order to ensure promotion of gender equality through women and youth empowerment. The project has taken into consideration the different roles of men, women and children in society and how gender equality influences vulnerability to climate change and extreme events. Women and youth groups face more limited opportunities for engaging in economic activities and this has been identified as one of the key sources of vulnerability for these groups.In this regard, particular focus will be brought to ensuring strong participation of women and youth in the national dialogue, and related consltation / mangagement processes under Output 1.The project makes considerable efforts to promote women's participation and engagement in climate resilience - including developing long-term adaptation solutions through design thinking workshops, enhancing the capacity of women to shape local development and spaital planning.

128. This project has been designed with attention paid to linking increased opportunities by enhancing skillsets of women and youth in areas of land management activities, climate risk data monitoring and community adaptation planning. Women's groups are expected to receive specific support in building their capacity for participating in the implementation of community-level planning under Output 1, which will underpin the notion that women are an important stakeholder and agent of change in the society.

129. Throughout the course of the project, several impact assessments are planned. These assessments offer important opportunities to understand, not only the impact of the project activities in coastal vulnerability reduction in general, but also the ways in which women and other interest groups contributed to the achievement of the project impacts.

[107] World Bank 2023, 'World Bank Open Data', 2023 <https://data.worldbank.org> [accessed 19 September 2023].

D.4. Needs of recipient (max. 500 words, approximately 1 page)

Describe the scale and intensity of vulnerability of the country and beneficiary groups and elaborate how the project/programme addresses the issue (e.g. the level of exposure to climate risks for beneficiary country and groups, overall income level, etc.). Describe how the project/programme addresses the following needs:

- *Vulnerability of the country and/or specific vulnerable groups, including gender aspects (for adaptation only)*
- *Economic and social development level of the country and the affected population*
- *Absence of alternative sources of financing (e.g. fiscal or balance of payments gap that prevents government from addressing the needs of the country; and lack of depth and history in the local capital market)*
- *Need for strengthening institutions and implementation capacity*

130. Settlements in Tonga, are concentrated in coastal areas with significant percentages of the population living and earning their livelihoods within one kilometre of the coast. As a result, critical infrastructure, such as hospitals, overnment buildings, schools, and places of employment are located in the coastal zone. These infrastructure and livelihoods are at risk to coastal erosion caused by inundation during storms. Moreover, freshwater resources are negatively impacted by sea-level rise due to increa ed saltwater intrusion in freshwater lens and water cisterns. This reduces the supply of fresh drinking water, particularly in low-lying areas.

131. The largest island in Tonga is Tongatapu, where 74% of the population reside, faces significant frequent flooding risks, as well as permanent losses from sea level rise (SLR) in the absence of future adaptation measures. The permanent losses under a 0.5m SLR scenario, for example, are expected to be 6%, and under a 1.0m SLR scenario to be 25% [108]. Tonga has experienced a series of climate disasters in recent years; tropical cyclones Ian (2014), Gita (2018), Harold (2020), as well as non-climate related disasters with the Hunga-Tonga-Hunga-Ha'apai volcano eruption in 2022, which triggered tsunami waves throughout the Tonga archipelago. Combined, climate-induced and non-climate related disasters affected up to 80% of the population and wiped out between 12% to 40% of the GDP [109]. The geophysical features of the islands, characterized by their small land area with low elevation, make most Tongan islands particularly vulnerable to sea-level rise and coastal erosion. For example, an increase of 0.3 m and 1 m in mean sea level (MSL) would cause land loss of 3.1 and 10.3 km^2, respectively, or 1.1 and 3.9% of the total area of Tongatapu Island. Under the two scenarios, about 2,700 and 9,000 people would be affected, or 4.3 and 14.2% of the total population of Tongatapu, respectively. In the case of an extreme event, about 20,000 people currently living in the low-lying areas of Tongatapu that can be flooded by a storm surge of 2.8 m, which was recorded during Cyclone Isaac in 1982. If a storm surge of the same degree occurs in conjunction with a 0.3 m sea level rise, 27.9 km^2 (11% of Tongatapu Island) and 23,470 people (37% of the Tongatapu's population) would be at risk. These estimates would increase to 37.3 km^2 (14%) and 29,560 people (46%) under the 1m sea level rise scenario. The risk of inundation and flooding is further intensified by social factors, with women being disproportionality affected by climate hazards.

132. Much of the impact of climate change is felt by individual households and small businesses. The vast majority of households do not have the financial capital to implement household-level interventions for climate change adaptation. The limited disposable income of most Tongan households means that tendencies for short-term gain take precedence over investment into longer-term measures for climate resilience. Households are not able to save for contingencies, nor are they able to proactively implement interventions that will reduce their vulnerability to the effects of climate change. With the Government of Tonga being unable to implement the large-scale erosion/flood protection infrastructure that would be required to protect communities given their limited financial resources, both communities and infrastructure along the vulnerable coastlines will consequently remain exposed to erosion and flood risks during extreme rainfall events. For these reasons, the project is specifically designed to engage civil society, including communities and households, in the national dialogue, stakeholder engagement and community adaptation planning processes.

133. Although the incidence of absolute poverty is low in Tonga, many experience economic hardships, where they do not have sufficient opportunities to access financial resources to cover basic needs such as medical and education expenditures. This is due to the high economic cost of the smallness of the country. The national revenue base is heavily influenced by the fluctuations of the global economy, relying heavily on remittance payments and foreign donor aid. World Bank data shows that Tonga has one of the world's smallest economies, as measured by GDP [110]. Remittances have been the largest contributor to Tonga's Gross National Product and a major source of its foreign exchange year-over-year [111], comprising around USD 194 million or 37.7% of GDP in 2020, the highest in the world [112], but which increased to 46.2% of GDP in 2021 [113]. However, Tonga's GDP decreased to 469 million USD in 2021, down from 485 million USD in 2020 [114]. GDP growth was 0.7% in 2019 and was subsequently adversely affected by COVID-19, with GDP falling to -1.2% in 2020 [115].

[108] ADB, *Multi-Hazard Disaster Risk Assessment, Tongatapu*, Risk Assessment Summary Report (Asian Development Bank (ADB), 2021).
[109] ADB, *Multi-Hazard Disaster Risk Assessment, Tongatapu*, Risk Assessment Summary Report (Asian Development Bank (ADB), 2021).
[110] World Bank 2023, 'World Bank Open Data', 2023 <https://data.worldbank.org> [accessed 19 September 2023].
[111] Base, 'Remittances for Recovery in Tonga', *BASE*, 2022 <https://energy-base.org/news/remittances-for-recovery-dealing-with-the-aftermath-of-tongas-volcanic-eruption/> [accessed 18 September 2023].
[112] IOM & ILO, *Climate Change and Labour Mobility in Pacific Island Countries*, 14 September 2022 <http://www.ilo.org/suva/publications/WCMS_856083/lang--en/index.htm> [accessed 18 September 2023].
[113] World Bank 2023, 'World Bank Open Data', 2023 <https://data.worldbank.org> [accessed 19 September 2023].
[114] World Bank 2023, 'World Bank Open Data', 2023 <https://data.worldbank.org> [accessed 19 September 2023].
[115] GoT, *Government of Tonga Budget Statement for Year Ending 30th June 2022* (Nuku'alofa, Tonga: Ministry of Finance, Government of Tonga, 2022) <http://www.finance.gov.to/sites/default/files/2021-11/Budget%20Statement%202021-2022.pdf#>.

The Hunga Tonga-Hunga Ha'apai volcanic eruption and tsunami reduced GDP growth to -2.5% [116]. Although forecasts suggest that real GDP will grow with Tonga's GDP growth expected at 2.5% in 2023, and 3.2% in 2024 according to the Asian Development Bank [117], although Government of Tonga figures put this at a projected annual average growth of 3.4% for 2023 through to 2025 (FY) [118].

134. Despite the minimal contributions to global greenhouse gas emissions, Tonga is disproportionately burdened with the significant impacts from climate change risks. The root cause of this adverse condition is its high exposure and vulnerability to climate hazards, combined with limited adaptive capacity. A number of environmental, economic, and socio-political factors contribute to its vulnerabilities, and leads to increased risks of climate change impacts in Tonga.

135. Absence of alternative sources of financing: Current budgetary capability of GoT can only focus on short- and medium-term coastal protection efforts; with GCF financing, long-term institutional capacities will be expanded so that the GoT can play a more proactive role in identifying solutions to climate change induced problems. In the absence of significant cost-recovery potential, the additional investment required to build resilience to climate change in coastal region of Tonga is prohibitive for a small island country. The project is economically viable (refer Annex 3: economic analysis) and advances financial viability of the investments beyond project duration.

136. The need for strengthening institutions and implementation capacity**:** GoT has identified key institutional capacity needs in the fields of: Technical capacity to closely monitor the coastal dynamics of vulnerable coastlines, design locally-appropriate protection measures and implement the measures that are within technical and financial capabilities of the Government; Capacity to interpret climate change risks and engage in community-led decision making, and; Lack of coordination between government and other agencies in addressing climate risks.

D.5. Country ownership (max. 500 words, approximately 1 page)

Please describe how the beneficiary country takes ownership of and implements the funded project/programme. Describe the following:

- *Existing national climate strategy*
- *Existing GCF country programme*
- *Relevance to and alignment with existing policies such as Nationally Determined Contributions (NDCs), Nationally Appropriate Mitigation Actions (NAMAs), and National Adaptation Plans (NAPs)*
- *NDCs, NAMAs, and NAPs*
- *Capacity of Accredited Entities or Executing Entities to deliver*
- *Role of National Designated Authority*
- *Engagement with civil society organizations and other relevant stakeholders, including indigenous peoples, women and other vulnerable groups*

137. Efforts to address climate change vulnerabilities and enhance resilience in Tonga are outlined in the following key national documents:

- National Climate Change Policy of 2006 and 2016
- Joint National Action Plan on Climate Change Adaptation and Disaster Risk Management 2010–2015 (JNAP1); 2018-2028 (JNAP2)
- Third National Communication Report on Climate Change
- Enhanced Nationally Determined Contributions (NDC) 2020
- Tonga Strategic Development Framework I (2011-2014)
- Tongan Strategic Development Framework II 2015-2025: A more progressive Tonga: Enhancing Our Inheritance
- The Tonga Climate Change Policy (2016): A Resilient Tonga by 2035

138. Tonga Strategic Development Framework, 2015 – 2025: The Tonga Strategic Development Framework (TSDF II) 2015-2025 prioritizes climate change as one of the seven National Outcomes. It states a 'more inclusive, sustainable and effective land and environmental management and resilience to climate and risk'. It is strongly supported by three of five pillars of the framework. These include Pillar 2 – Social Institution, 4 – Infrastructure & Technology, and 5 – Natural Resources and Environment. In Pillar 2, GoT has successfully outlined needs and priorities identified by village community groups and populations through Community Development Plans (CDPs). To date, CDPs from all 173 communities have been completed or in the process of finalization. Local town councils and committees have been established with high level representatives chairing these committees. In Pillar 4, GoT underscores the importance of building island infrastructure capacity to adapt to the impacts of climate change. This includes constructing new public works and buildings to support community resilience to extreme weather events. Finally, Pillar 5 emphasizes improved land use planning, administration and management and improved resilience to the impact of climate change.

139. National Climate Change Policy: In 2016, the GoT completed a comprehensive update to its initial National Climate Change Policy to produce the "National Climate Change Policy - A Resilient Tonga by 2035" (NCCP 2016). It is a five-year policy

[116] GoT, *Government of Tonga Budget Statement for the Year Ending 30th June, 2023* (Ministry of Finance, Government of Tonga, 2023).
[117] ADB, 'Tonga: Economy', *Asian Development Bank: Asian Development Outlook 2023*, 2023, Tonga <https://www.adb.org/countries/tonga/economy> [accessed 19 September 2023].
[118] GoT, *Government of Tonga Budget Statement for the Year Ending 30th June, 2023*.

that sets out to achieve the vision of a resilient Tonga by 2035 through specific environmental, social and economic targets, ultimately aimed at increasing Tonga's resilience to the impacts of climate change. The GoT considers the policy goals to be very ambitious but achievable through an integrated approach to adaptation, mitigation, and disaster risk reduction. The NCCP specifically sets out reducing coastal vulnerability as primary goals. These goals include: (ii) redesigned, resilient, roads, coastal areas, buildings, and other infrastructure.

140. Third National Communication Report to the United Framework Convention on Climate Change Under the obligations assigned to parties under the UNFCCC, Tonga submitted its first national communication report in May 2005 and the second national communication (SNC) report in 2012. Tonga's Third National Communication was then submitted in 2020. The Vulnerability and Adaptation Assessment (V&A) chapter outlines Tonga's vulnerabilities to the adverse impacts of climate change on vulnerable sectors including agriculture, fisheries, coastal areas, water resources, lands, infrastructure, disaster risk management, biodiversity and human health. The key findings reported the vulnerabilities of the low-lying coastal areas in Tonga and the adaptation measures in place to counter the adverse impacts of sea level rise and storm surges such as coastal erosion and inundation. While key priorities highlighted the need to a) Formulate integrated coastal management plan; b) Climate proof planning, design, decision making on every development on the coast; c) Integrate climate change issues and disaster risks into the Environment Impact Assessment Process Conduct LIDAR survey to reduce coastal erosion in Tongatapu & Ha'apai; d) Conduct coastal feasibility studies and design of most appropriate measures to vulnerable communities on the coast, and; e) Promote coastal reforestation and afforestation.

141. The Joint National Action Plan 2 on Climate Change and Disaster Risk Management (JNAP 2) 2018-2028 sets out six policy objectives and targets and an implementation strategy for the country to achieve its vision of a Resilient Tonga by 2035. JNAP2 is aligned with the Tonga Climate Change Policy and covers both climate change adaptation and disaster risk management. The six objectives are: 1) Mainstreaming for a resilient Tonga, 2) Implement a coordinated approach to research, monitoring and management of data and information, 3) Resilience-building response capacity, 4) Resilience-building actions, 5) Finance, and 6) Regional and international cooperation. These build on Tonga's first Joint National Action Plan on Climate Change Adaption and Disaster Risk Management (JNAP 1) in 2010.

142. Tonga's National Designated Authority (NDA), ensures that this proposal is consistent with Tonga's national climate change strategies and plans, including the screening by the JNAP Secretariat for any proposals to the GCF. In initial concept stages the JNAP Task Force provides recommendations to the NDA based on their screening of the concept.

143. Furthermore, the project will provide a mechanism for collecting and analysing data and information for better-informed climate risk monitoring and coastal adaptation planning, providing key capacity building, training and mentoring for coastal adaptation planning and monitoring and developing climate risk-informed coastal plans that will support the JNAP 2's need for data collection and monitoring to progress Objective 2 of the JNAP 2.

144. The comparative advantage of the UNDP Pacific Multi-Country Office in Fiji (hereafter "UNDP MCO" or "Country Office") in the implementation of this project lies in its experience with effective facilitation of partnerships with other UN Agencies and regional organizations. In particular, UNDP has experience in collaborative partnerships with agencies that are party to the Council of Regional Organizations of the Pacific, such as SPREP, SPC/SOPAC, SPTO and several NGOs.

UNDP also has the required operational, financial and technical capacities to effectively manage and guide this project in Tonga under the umbrella of the United Nations Development Assistance Framework. The UNDP national level support to Tonga is detailed in the UNDP Sub-Regional Programme Document for Pacific Island Countries with one focus being 'Environmental management, climate change and disaster risk management'. In particular, "improved resilience of PICTs, with particular focus on communities, through integrated implementation of sustainable environment management, climate change adaptation/mitigation and disaster risk management" is Sub-Regional Outcome 4.

145. Technical aspects of project implementation is supported by a dedicated technical advisor based in Bangkok and a global senior technical advisor. The global network of the regional based advisors enables sharing and dissemination of knowledge beyond the country and region, drawing on best practices from projects with similar elements. UNDP's comparative advantage for GCF focal areas lies in its global network of country offices –such as the UNDP Pacific Office in Fiji – and its experience in supporting integrated policy development, human resources development and institutional strengthening as well as promotion of NGO and community participation. This experience means that UNDP is well-placed to assist Tonga in designing and implementing this project in a manner that is consistent with the LDCF mandate as well as national development planning.

146. UNDP's added value is also evident as described:

- Accountability: a track record of quality management of development finance as well as M&E and reporting on project implementation;
- Technical Expertise: a large number of experienced and qualified staff with expertise in a number of relevant fields (e.g. climate change adaptation, development planning) in country offices and regional headquarters, as well as a world-wide knowledge network of specialists;
- Regional and global cooperation: experience with developing synergies and cooperation at the regional and global levels, including through initiatives for North-South and South-South collaboration; and
- Coordination with other UN agencies: a mandate to support coordination and collaboration between other UN agencies as leader of the UN Development Group.
- UNDP has a proven ability to: i) formulate project proposals; ii) collaborate with development partners and donors; iii) mobilize resources for development implementation; iv) monitor, evaluate and report on results; v) support and further develop national/local capacities for implementation; and vi) contribute to ongoing learning

and improvement of processes. UNDP's track record of effective coordination of development planning and implementation; both with GoT and other development partners, makes it ideally placed to support the implementation of this project.

147. Several rounds of discussions with stakeholders on adaptation priorities and options in Tongatapu, Ha'apai and Vava'u inform the design proposed project. The project builds on past community consultations and ongoing partnership of the Government of Tonga with, inter alia, UNDP and SPREP and other multilateral agencies such as ADB and EU.

148. The first rounds of consultations were undertaken during 2017, and then again in 2023:

In north-eastern Tongatapu, Government has undertaken a series of consultations but most intensively in February and March 2017 with assistance from ADB and EU. In addition to identifying the priority of foreshore protection, the consultations placed a particular emphasis on seeking commitments from community members in continuous engagement in O&M. a series of multi-stakeholder consultations led by MEIDECC, with the participation of CSOs, NGOs, the publics, Town Officers and District officers were held from 8-21 February 2017 in all of the project target areas. In the consultations community members reiterated the importance of selected interventions of this project as well as the need for a coastal protection measure. Representation of women and youth were ensured during the stakeholder consultations and inputs from the residents on the coastal protection measures were collected. A final appraisal meeting was held in the capital on 9 March, which resulted in the endorsement of the project for submission.

149. These earlier consultations were then updated further during 2023. In August 2023, ongoing consultations were held to discuss the proposal development and design with key stakeholders in the MIEDECC, MoF, MoI, MIA, MoNSLR, as well as with Tonga's key CSOs (details of these consultation are included in Annex 7). Consultations were also held with community residents in the Hahake areas for site 1 and 2, for adjustments allowing for both the post-tsunami context, and then clarifications to proposed designs (September-October 2023).

150. These consultations have contributed, particularly to the following outcomes that are considered essential for successful project design and implementation:

- Identification/verification of climate change priority actions as a result of participatory, gender-responsive consultations;
- Increased awareness among local communities, NGOs and local/national governments about specific interventions and their limitations;
- Preparing the local leaders and communities and seeking their commitment for their active participation in project implementation, maintenance and monitoring;
- Facilitating understanding about women's specific needs and the specific roles they play in climate change adaptation in general and in project implementation in particular;
- Acknowledging the leading role of the youth in monitoring activities and data collection;
- Obtaining agreement about using the existing CDP platforms for facilitating long-term adaptation actions;
- Agreeing on the specific roles of government ministries, local governments, NGOs, and local communities in project implementation.

D.6. Efficiency and effectiveness (max`. 500 words, approximately 1 page)

Describe how the financial structure is adequate and reasonable in order to achieve the proposal's objectives, including addressing existing bottlenecks and/or barriers, and providing the minimum concessionality to ensure the project is viable without crowding out private and other public investments. Refer to section B.5 on the justification of GCF funding requested as necessary.

Please describe the efficiency and effectiveness of the proposed project/programme, taking into account the total financing and mitigation/ adaptation outcome the project/programme aims to achieve, and explain how this compares to an appropriate benchmark.

Please specify the expected economic rate of return based on a comparison of the scenarios with and without the project/programme.

Please specify the expected financial rate of return with and without the Fund's support to illustrate the need for GCF funding to illustrate overall cost effectiveness.

Please explain how best available technologies and practices have been considered and applied. If applicable, specify the innovations/modifications/adjustments that are made based on industry best practices.

151. The project will reduce the vulnerability of critical coastline areas to erosion and inundation in Hahake (north-eastern Tongatapu) and increase long-term resilience of communities in Tongatapu, thereby contributing to enhanced national and community-level capacity for managing the impact of extreme weather events in the long-run. This objective will be achieved through the following considerations that have been reflected in the design of the project:

- Selection of tested, locally appropriate options, based on detailed site-specific assessments and public consultations;

- Integrating capacity building of technical ministries and local communities for ensuring their long-term engagement not only for monitoring of the GCF investments but also for replicating the project results beyond the project geographical boundaries and timeframe;
- Mobilizing community support for implementing local land management and ecosystem restoration measures within the context of the existing local development planning and execution mechanism; and
- A comprehensive approach to removal of multiple barriers within a single project framework.

The effectiveness of the engineered coastal protection solutions has been tested in Tonga and elsewhere. A successful example of (coral) rock revetment is found in the capital city of Nuku'alofa in Tonga. It was constructed in 1983 following the destruction of the old sea wall by 1982 Cyclone Isaac. The old wall was also made of coral blocks but was thin and nearly vertical. When the storm surge from Cyclone Isaac resulted in over-topping the offshore reef, parts of the sea wall were destroyed immediately, while other parts collapsed due to the washing out of the fine fill material behind the wall. The 1983 revetment was built by the Government of Tonga with funding from Germany. Today, after more than 30 years of construction, the revetment structure remains in place[119]. Similarly, as presented in Section F.2. Technical Evaluations, revetments built in the 1960's in the vicinity of Nukuleka and Navutoka in north-eastern Tongatapu have been identified as extremely successful, and the shoreline position has been unexpectedly stable across this shore over the last 50 years, largely due to the revetment. These also cover the adjacent ADB-constructed revetment that although only completed in recent years, has already endured tsunami impact and remained undamaged and provided effective protection in comparison to areas where there is no such revetment.

152. Alternative solutions have been considered in the design of this proposed project. Two alternatives to the project that were considered were the general options of "retreat" and "accommodate". The former entails protection of possessions and infrastructure or reduction in injury and fatalities by removing them from hazard and establishing a setback zone to restrict future development in high hazard areas. The latter, in this particular case, relies on families either already planning to rebuild their houses or asking them to consider elevating houses so that possessions and infrastructure are protected from inundation-related damages.

153. A cost-benefit analysis conducted in PASAP, using data for Lifuka, Ha'apai, reveals that the retreat option consistently offered the lowest net costs (highest net benefit) for all damage scenarios, with the benefits increasing as the damage scenario worsens. While the PASAP analysis recognizes that these options are not mutually exclusive, the benefit-cost ratios of the other options, i.e., protection and elevation of houses, were sensitive to various assumptions made in the analysis such as damage scenarios, discount rates used, level of home elevation, etc. However, the options of retreat and elevation of houses were considered less suitable in reducing the imminent risks of climate change for practical reasons. Voluntary retreat, while demonstrating relative economic advantages, was considered by most community members as impractical given the complex land ownership situations in Tonga. On the other hand, the "accommodate" option by elevating houses would be achieved only slowly as such a solution can be practically promoted as the existing houses become required of renovation. Moreover, considering that some assets are located very close to the shoreline, the solution of elevating buildings alone was considered ineffective.

154. ADB has also investigated multiple alternative options for protecting assets in the north-eastern coast of Tongatapu. The option of moving key infrastructure (i.e. road) more inland was considered but rejected on the same grounds of land tenure issues. As the issue associated with land ownership is uniform across the country, the option of moving important assets out of harm's way, although it may be considered economically a preferred option, is likely to face similar practical difficulties in the short-run. However, it is important to emphasize that voluntary retreat will be one of the inevitable options that the Tonga community is likely have to adopt as sea level continues to rise. To increase the long-term preparedness, the platform established in Output 1, will facilitate the discussion on these long-term adaptation options. This is a reflection of the acute realization that the measures proposed in this project would be of limited value in isolation; a mix of short- and long-term solutions is critical for meeting the project objective.

155. For building the national and local capacity for effective monitoring, maintenance and community adaptation actions, it is proposed in this project that extensive training will target the administrators of coastal resources. Data will be obtained through engaging third party firms to collect key LiDAR data sets. This will be supplemented by the purchase of equipment (hardware and software) that will enable GoT to carry out ongoing data collection on a smaller scale. A package of training / capacity building will accompany these purchases to ensure relevant capability at national level.

156. Setting the conditions for continuous community engagement in local adaptation actions, primarily in the form of O&M of the hard-engineered solutions and climate risk based land management practices, is an integral part of the project design which contributes to the overall efficiency and effectiveness of the project. As detailed in Stakeholder Engagement Plan in Annex XIII, community members will play a key role in periodic cleaning of culverts/drainage in north-eastern Tongatapu for its long-term functioning and visual monitoring of the revetment structure especially after cyclone events. Without these voluntary services provided by the beneficiary communities, the long-term costs of these structures would be higher. Moreover, complementary capacity building targeting community members to mobilize additional resources, within the framework of the Community Development Plan, will further contribute to the sustainability of project results and thus to improving the cost-effectiveness of the project.

[119] Helu, S. (1991). Seawall construction and coastal development at Nuku'alofa, Tonga (abstract). In: Workshop on Coastal Processes in the South Pacific Island Nations, Lae, Papua New Guinea, 1–8 October 1987, *SOPAC Technical Bulletin* 7: 193.194. Published by the SOPAC Technical Secretariat in 1991, Suva, Fiji. Available online at: http://ict.sopac.org/VirLib/TB0007.pdf .

157. Reflecting the commitment of the Government of Tonga for sustained O&M, co-financing of US$225,000 has been agreed upon during the seven-year project implementation period. In addition, the Government has made a commitment to financing an amount of US$150,000/year for post-project O&M for the coastal protection measures for their expected lifecycles, *ca* 40 years. This will cover inspections, repairs, materials, and plant hire.

158. The economic analysis of the proposed project was carried out in accordance with the GCF guidance on the preparation of economic and financial analysis. The economic efficiency of the investment was determined by computing the net present value (NPV) with an assumed 6% discount rate, and the economic internal rate of return (EIRR). The proposed project is of the nature of a public good. The project itself will not generate any cash flow – for example in the form of a fee or charge for the flood protection service provided by the flood control infrastructure. In the absence of the generation of cash flow, the conduct of a financial analysis is not appropriate (as is the case for public good investment) and has indeed not been conducted in the context of this project.

159. Economic values (costs and benefits) are all measured in real terms of 2023. Economic costs of the project are net of taxes, duties, and price contingencies. Furthermore, the analysis assumes a shadow wage rate conversion factor of 1.00 for unskilled and semi-skilled labor in Tonga. This would appear appropriate given an unemployment rate of approximately 3% in Tonga in 2022. Similarly, a shadow exchange rate factor of 1.0 for traded goods is used in the analysis. These assumptions allow the use of financial cost as a measure of the economic cost of the project.

160. The economic analysis presents the EIRR and NPV for the entire project – and all of its components – as a whole. It should also be noted that while the costs associated with the "soft" measures of the proposed project are included in the analysis (such as the strengthening of national and local capacities and improving the knowledge base), the benefits of these measures taken as stand alone are not explicitly captured in the analysis. In other words, the cost-benefit analysis focuses on estimating the expected benefits to be delivered by Output 3.1 (engineered flood protection measures) for which benefits can be clearly identified and quantified, while accounting for the whole cost (the cost of all activities) of the project – thus implicitly assuming that all activities are conducive to the delivery of the long-term sustainable benefits of Output 3.1. If any other activities or soft measures were to deliver benefits on their own (as stand alone activity) in addition to and indepently from Output 3.1, these benefits are not accounted for in the analysis. To this extent, the quantified expected benefits included in this analysis are in all likelihood an under-estimate of the true economic benefits of the project.

161. Project expected benefits were estimated by comparing expected (future) damages in a scenario without project versus a scenario with project. .

162. The without-project scenario represents a continuation of the existing situation while the with-project scenario represents the project investment scenario. Both scenarios are dynamic in nature in that both projected population growth and economic (GDP) growth are included in the analysis. Hence, in the absence of the project, economic damages from extreme events are assumed to increase both as a result of the projected growth in population and as a result of the projected economic growth. All other things being equal, population growth increases the number of assets exposed to potential damages while economic growth increases the economic value of those assets.

163. The proposed project delivers a NPV of $9.91 million and EIRR of 8.52%. Hence, for all values of the discount rate below 8.52% the NPV remains positive, and for all values of the discount rate greater than 8.52%, the NPV is negative. The NPV equals 0 upon using a discount rate of 8.52%.

164. A sensitivity analysis was conducted to assess the sensitivity of NPV and IRR to cost increases and benefits decreases. The sensitivity analysis includes cost increases ranging from +10% up to +30%; benefits decrease of -10% down to -30%; and finally simultaneous increases of costs and decreases of benefits.

165. The economic analysis shows the project to be economically efficient and robust to cost increases and benefits decrease reaching approximately 20%. The analysis suggests that particular attention should be paid to controlling costs to the levels presented in the existing budget and avoid significant increases. On the other hand, it is also important to remind the reader that assumptions were made so as to under-estimate the true economic benefits of the project. There is thus very high confidence that the project is economically efficient.

166. The design of the proposed GCF project incorporates lessons and best practices from several other projects to bring about transformative impact that is technically proven, efficient and sustainable.

167. **Selection of specific type of coastal protection solutions in Tonga's Tongatapu Island**: The design of the proposed GCF project is based on the best knowledge available on coastal protection in the region and in Tonga informed by previous assessments and current relevant projects. The three broad strategies for coastal vulnerability – retreat, accommodate and protect – have been reviewed and analyzed for the target site. The options selected are a result of weighing multiple options against these three strategies and in considerations of community views. Potential coastal protection options were also informed by assessment studies which evaluated the technical feasibility, social acceptance, and economic implications of each option. Based on a review of the reports as well as an assessment of local environmental, bathymetric and oceanic conditions, rock revetment was proposed for north-eastern Tongatapu;

168. The review and selection of the above-mentioned options were done with a full recognition that, presently, international best practices lean towards the implementation of coastal protection strategies that aim to work with the natural processes rather than challenging the forces of the ocean head on.

169. **Monitoring of the coastlines with state-of-art technologies**: This project will also assist the Government of Tonga build technical capacity to monitor coastal changes over time using a combination of outsourced survey work and access to

developing technologies. Light Detection and Ranging (LiDAR) surveys of outer islands will be provided through private providers to ensure that data can be produced and analysed in time to be available to support other parts of the project (in particular the national dialogue under Output 1).

170. At the same time, the project will take advantage of recent technological advancements, through supporting the purchase of selected LiDAR equipment suitable for small scale surveys as a follow-up to the major survey. The Government of Tonga will gain access to Light Detection and Ranging (LiDAR) camera mounted on a small unmanned aerial vehicle ("drone"). UNDP has already started a private sector partnership in Maldives where the similar technology is used for mapping the coastal areas of their islands. Having access to this technology will enable the Government of Tonga to continue to accumulate data over time on key topographic and bathymetric parameters. A package of training will be provided in relation to the hardware and software associated with this equipment. This is a shift in the way coastal vulnerability is addressed by small island developing states: they now can be proactively part of the solutions to the problem that is the highest national priority for most of them. Previously, their ability to address the problem was dependent on donor resource availability and outside technical support for such survey work and data analysis. This approach will profoundly enhance the government's' ability to understand coastal changes and processes over time and design solutions that are effective and sustainable.

171. **Use of existing community-level financing options for sustainability and engagement**: The importance of local governance and the effective use of available domestic finance has long been recognized in the region. In Tonga, the advent of a bottom-up development planning approach, characterized by the Community Development Plans is likely to offer new opportunities for community-based adaptation actions in the future. By building on baseline initiatives such as UNDP PRRP's integration of community-based disaster risks management approach into CDPs, GCF support will strengthen the capacity of communities to access domestic/external funding sources such as GEF SGP for advancing community actions for climate resilience. By building the community capacity for accessing additional climate resources, it is expected that community-level adaptation actions such as ecosystem restoration activities, gained through the participation in the project implementation, will be sustained and replicated beyond the project timeframe and geographical boundary.

172. The proposed project will integrate best practices and lessons learned from SPREP and UNDP, two leading development organizations in the South Pacific Region. Consequently, the investments in this project will not only be replicated in other SIDS in the region, but will also catalyze further investments that will help scale up this nationwide approach. Lessons learned from the GCF project will provide the basis for detailed technical approaches to coastal resilience. These lessons will be shared nationally, through awareness campaigns as well as internationally to contribute to current knowledge on building climate resilience.

E

LOGICAL FRAMEWORK

*This section refers to the project/programme's logical framework in accordance with the **GCF's Integrated Results Management Framework** to which the project/programme contributes as a whole, including in respect of any co-financing.*

E.1. Project/Programme Focus

Please indicate whether this proposal is for a mitigation or adaptation project/programme. For cross-cutting proposals, select both.

☐ Reduced emissions (mitigation)

☒ Increased resilience (adaptation)

E.2. GCF Impact level: Paradigm shift potential (max 600 words, approximately 1-2 pages)

This section of the logical framework is meant to help a project/programme monitor and assess how it contributes to the paradigm shift described in section D.2 above by applying three assessment dimensions - scale, replicability, and sustainability.

Accordingly, for each assessment dimension (see the definition per assessment in the accompanying guidance note), describe the current state (baseline) and the potential scenario (target) and rate the current state (baseline) by using the three-point-scale rating (low, medium, and high) provided in the guidance note. Also describe how the project/programme will contribute to that shift/ transformation under respective assessment dimensions (scale, replicability and sustainability). In doing so, please refer to section B.2(a) (theory of change).

Assessment Dimension	Current state (baseline)		Potential target scenario (Description)	How the project/programme will contribute (Description)
	Description	Rating		
Scale	In Tonga, adaptation to climate change has occurred through a series of one-off project interventions that focus on single localities without reference to broader consideration of the long-term impacts of climate change on coastal areas, land use and sustainable livelihoods.	<u>Low</u>	The 7-year project will reach communities under Output 1, to drive accountable engagement on decision-making for long-term adaptation planning. The paradigm shift involves having the dialogue to underpin future decisions on spatial land planning in the face of sea level rise. Output 2 equips the government with the ability to map climate change impacts, risks and vulnerabilities and provides the critical data to plan for long-term adaptation solutions that aim to avoid maladaptive decisions. Allied to this will be a baseline of fundamental information on coastal topography and bathymetry nationwide. A cohort of officials and community representatives will have the	*Describe key applicable outputs and or resulting outcomes relevant to increasing (scaling up) quantifiable results within and beyond the scope of the intervention.* The project will increase the ability for Tonga's government to undertake climate-responsive planning through access to, and ability to utilise information for climate risk informed decision making. The project will support a series of interventions that address different levels: <u>National dialogue</u> A comprehensive National Dialogue will be convened to provide a platform to consider the implications of climate change – especially sea level rise – at a national scale. This will be based on inclusive cross-

			capacity to use this data to inform planning and policy making adaptation responses. The Government of Tonga will be in a position to make long term decisions on approaches to coastal adaptation, including the potential for voluntary retreat and its implications on land use and land ownership (e.g. trade-offs between different uses and beneficiaries). Documented examples of good practice will be available to guide coastal adaptation responses, including infrastructure.	sector consultations and engagement and be directed towards agreement to long-term policy solutions. <u>Technical inputs and capacity</u> The National Dialogue will receive technical inputs on climate change impacts (e.g. wave/hazard mapping in face of sea level rise). To this end the project will commission one large scale LiDAR survey and support local capacity in data collection and interpretation. <u>Community led adaptation</u> The project will facilitate review of Community Development Plans and District Plans to incorporate gender-responsive community led adaptation responses. <u>Coastal infrastructure models</u> The project will support coastal infrastructure that will both protect local communities and provide models for different approaches.
Replicability	Presently there is (limited technical capacity in terms of expertise and access to data to effectively model vulnerable land areas for planning purposes), limited coordinated engagement to facilitate dialogue from local to national level). No comprehensive nation-wide cross sectoral engagement on impacts of climate change has yet taken place in Tonga (or in other Pacific SIDS), so there has been no opportunity to replicate the approach. Infrastructure for coastal protection has been applied in Tonga (and elsewhere in the Pacific Islands region) but with	<u>Medium</u>	Output 2 undertakes national capacity building in multiple areas, based on capacity needs assessment. At the same time, the Output hands over tools and datasets, while strategically mentoring and training key staff. Documentation of good practices in coastal infrastructure will also provide models for wider distribution. Equally, the process for achieving agreement – through a dedicated National Dialogue platform on long term adaptation – will also be of wider interest in the face of sea-level rise would be of considerable interest to other SIDS in the Pacific Islands region and potentially elsewhere (the process – at national and community level - may be seen to be as important as the outcome).	*Describe key applicable outputs and resulting outcomes that will be replicated to other sectors, markets, geographical regions, or countries.* The project will provide important models for potential replication in several areas. Firstly – the National Dialogue Platform is unique in the Pacific region in terms of a) the extent of its national coverage, b) commitment to broad engagement, and c) being informed by up-to-date high quality information on hazards. Secondly – the outcome of the process will be a national policy (plus supporting documentation potentially including legal instruments) that will guide long-term decisions on adaptation. Both the above are novel and innovative approaches, for Tonga and the region, so the lessons learned will be captured through a structured process during the

	limited assessment of their design, effectiveness and longevity.			project. These will be disseminated as the project concludes through a variety of media, as well as dedicated knowledge sharing events. Models for coastal infrastructure can be used to inform future coastal protection in Tonga (including outer islands) and potentially more widely in the Pacific Islands region.
Sustainability	The Government of Tonga has made multiple commitments to addressing climate change through national statements and strategies - but these largely reflect national aspirations rather than specific actions and lack indication of allocation of resources while coordination between institutions/Departments is also limited.	Medium	The project aims to mainstream climate change hazards into key policies and plans from national level though to community level. This is a means to ensure that meaningful changes are adopted with respect to adaptation planning. In addition, the project will provide enhanced capacity at government and community level to collect, interpret and use information / data on climate change hazards.	*Describe key applicable outputs and resulting outcomes that will be sustained beyond the project/programme period.* The project will support and facilitate the National Dialogue Platform and parallel community engagement in adaptation planning. The project will provide critical data and information for climate risk planning and decision making (e.g. such as supporting coastal LiDAR surveys and the purchase of equipment to maintain current knowledge on key coastal parameters), while also building capacity in these areas as well as data interpretation. It is intended that this knowledge will be sustained through the project term and beyond.

E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)

Select appropriate IRMF core and supplementary indicators to monitor project/programme progress. More than one IRMF (core and or supplementary) indicators may be selected as applicable for each GCF results area and project/programme outcome (as defined in the table in section B.2(b)). If IRMF indicators are unable to measure any given project/programme outcomes, project/programme-specific indicators should be developed under section E.5 (project/programme specific indicators).

GCF Result Area	IRMF Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final[120]	

[120] The final target means the target at the end of project/programme implementation period. However, for core indicator 1 (GHG emission reduction), please also provide the target value at the end of the total lifespan period which is defined as the maximum number of years over which the impacts of the investment are expected to be effective.

		Sources of information and methods used to collect and report data /information to measure progress against targets	*The starting point or current value of the indicators before the implementation of the project*	*The estimated value of the indicator at the mid-point of the implementation*	*The estimated value of the indicator at the completion of the implementation*	*Externalities and factors outside project management's control that may impact the outcomes* *Data sources and methodologies applied for estimating baseline and targets*
ARA1 Most vulnerable people and communities	Core 2: Direct and indirect beneficiaries reached	Minutes of trainings and consultations with communities Beneficiaries feedback surveys Lists of attendees (disaggregated by gender) Training/mentoring materials available for government staff to integrate climate change risks in planning and budgeting Availability of district adaptation plans developed. Available Island Strategic Development Plans – for Ha'apai (2017-2023), Niua	*0* (Current population is not covered by climate risk informed district adaptation plans)	350 direct beneficiaries[121] 40,071 indirect beneficiaries[122]	3,608 direct beneficiaries (of which 1,804 are women)[123] 96,871 indirect beneficiaries expected (of which 48436 are women)[124]	Potential changes in settlement populations (with the inward and outward migration of people to Tongatapu) Based on 2021 Census data

[121] By the mid-term, 300 people (community members) will have benefited from some training activities in climate-risk and adaptation for updating/developing the CDPs/ ISPs, and 25 government staff will have benefited from some training in incorporating climate change risks in planning and budgeting, supporting development planning; while a further 25 government staff will have benefited from some technical training on climate risk monitoring and data.

[122] The 40,071 population is expected to benefit from by the project mid-term from the piloting of a climate-risk-informed Tongatapu Island Development Plan, which brings together the Island's population, omitting those already benefiting from climate-risk-informed planning under urban resilience planning for the greater Nuku'alofa area. The figure of 40,071 is the total population of Tongatapu 74,320 (2021 Census), excluding the population within Nuku'alofa (greater region/urban agglomeration) 34,249 (2021 Census), or around 6,134 households (ADB, 2023); where disaster and climate risk planning, including disaster risk management plans, are already addressed within urban master planning (Climate and Disaster Resilient Urban Development Strategy and Investment Plan for Greater Nuku'alofa)[1].

[123] After construction work under Output 3 is completed, 3,258 people will benefit from increased resilience to coastal climate change impacts from coastal protection measures, and 350 people will have benefited from training**.** This includes 300 people trained in climate-risk and adaptation for updating/developing the CDPs/ ISPs (from project indicator under Output 2) and 50 Government agency staff trained or receiving technical capacity building support under Outputs 1 and 2.

[124] Indirect beneficiaries are the population of Tonga's island groups of Tongatapu, Ha'apai, Niua, 'Eua, and Vava'u given that they will benefit indirectly from the adaptation planning and strategies of the island-level spatial plans developed following piloting in Tongatapu, as well as indirectly benefiting from the outcomes developed through the multistakeholder engagement platform. In addition to the Strategic Development Plan for Tongatapu, the project supports other district plans to incorporate climate risk. In addition to renewing the Island Strategic Development Plans – for Ha'apai (2017-2023), Niua (2015-2018), 'Eua (2015-2018), and Vava'u (2017-2023) the island spatial planning will be supported by the collection, interpretation, and use of data on climate risks for coastal adaptation planning to produce scenario-based coastal risk maps and knowledge material for use. These will also be supported through the development of materials for developing village and district spatial plans (e.g. graphic reports developed from Output 2, visual aids to demonstrate climate change impacts) and a graphic report package", with visual aids to demonstrate climate change impacts, as well as the participatory process workshops/consultations for developing village and district spatial plans for each of 6 islands and 6 Island-level graphical reports related to settlement patterns and climate risk for strategic long-term coastal adaptation planning developed. This will include consultations determining adaptation needs, gender needs, participatory vulnerability assessment and planning. The Island Plans incorporate the community development plans and district development plans.

		(2015-2018), 'Eua (2015-2018), and Vava'u (2017-2023) and Tongatapu that integrate climate change and adaptation concerns. MTE and TE Census data (update changes)				
ARA3 Intrastructure and built environment	Core 3: Value of physical assets made more resilient to the effects of climate change and/or more able to reduce GHG emissions	Independent evaluations (Interim Evaluation and Terminal Evaluation) Project progress reports (including Annual Progress Report) Site visits	The value for this indicator will be determined by mid-term. -	The value for this indicator will be determined by mid-term.	The value for this indicator will be determined by mid-term.	

E.4. GCF Outcome level: Enabling environment (IRMF core indicators 5-8 as applicable)

Select at least two relevant IRMF core (enabling environment) indicators to monitor and elaborate the baseline context and project/programme's targeted outcome against the respective indicators. Rate the current state (baseline) vis-à-vis the target scenario and select the geographical scope of the outcome to be assessed. Describe how the project/programme will contribute towards the target scenario. Refer to a case example in the accompanying guidance to complete this section.

Core Indicator	Baseline context (description)	Rating for current state (baseline)	Target scenario (description)	How the project will contribute	Coverage
Core Indicator 5: Degree to which GCF investments contribute to strengthening	Lack of coherent national approach to climate risks and the implications for current/future land use in Tonga	low	Long term engaged discussion informing decision making is established through the ongoing meetings of the national multi-stakeholder engagement platform for	Address recognized barriers to addressing climate risk and adaptation: • Provide resources to support the national multi-stakeholder engagement	National level (one country)

institutional and regulatory frameworks for low emission climate-resilient development pathways in a country-driven manner			dialogue on co-creating long-term climate change adaptation strategies and solutions including voluntary retreat; Inclusive and gender responsive climate change adaptation responses developed at national, island and local level through the national engagement platform for raising awareness and co-creating adaption solutions.	platform for dialogue on co-creating long-term climate change adaptation strategies and solutions process. • Implement technical and social capacity building at community and government level	
Core indicator 8: Degree to which GCF investments contribute to effective knowledge generation and learning processes, and use of good practices, methodologies and standards	Currently no knowledge management and sharing by Government of Tonga on good practices in coastal protection	low	The project will present best practice coastal protection options as part of training undertaken with government officers	The project will showcase best practice contributions through the regional sharing conference with PIC SIDS, and produce a shared publication of good coastal practices that that can be used throughout the Pacific SIDS	Multi-countries
Core Indicator 6: Degree to which GCF investments contribute to technology deployment, dissemination, development or transfer and innovation	Some current access to coverage of LDAR, multihazard risk assessments and other types of information for coastal adaptation planning, but limited translation and up to date information	medium	Initial climate impact and risk monitoring and informed decision-making undertaken with new updated LiDAR coverage	Interpreted and up to date, Tonga-wide LiDAR coverage completed and translated to provide the basis for spatial planning and strategic planning decision-making, including adaptation options for resettlement and avoiding maladaptation.	National level (one country)

E.5. Project/programme specific indicators (project outcomes and outputs)						
*This section should list out project/programme-specific performance indicators (outcomes and outputs) that are not covered in sections above (E.1-E.4). List down tailored indicators to monitor /track progress against relevant project/programme results (outcomes/outputs). AEs have the freedom to decide against which outcomes they would like to set project/programme specific indicators. If any co-benefits are identified in sections B.2(a)(b), and D.3, AEs are encouraged to add and monitor co-benefit indicators under the "**Project/programme co-benefit indicators**" section in table below. Add rows as needed.* *Please number each outcome and output as shown below to indicate association of outputs to the contributing outcome. The numbering for outputs under this section should correspond to the output numbering in annex 4 (detailed budget plan).*						
Project/programme results (outcomes/ outputs)	**Project/programme specific Indicator**	**Means of Verification (MoV)**	**Baseline**	**Target**		**Assumptions / Note**
				Mid-term	**Final**	
		Sources of information and methods used to collect and report data/information to measure progress against targets	*The starting point or current value of the indicators before the implementation of the project*	*The estimated value of the indicator at the mid-point of the implementation*	*The estimated value of the indicator at the completion of the implementation*	*Externalities and factors outside project management's control that may impact on the Component* *Data sources and methodologies applied for estimating baseline and targets*
Output 1 Strengthened knowledge, capacity and engagement for incorporating climate risks into long-term adaptation planning supported through multi-sectoral, multi-stakeholder engagement and dialogue platform	*Level of community participation in the multi-sector national dialogue platform on long-term climate change adaptation strategies and solutions*	*Minutes of meetings* *Attendees list* *Workshop reports* *Progress/Evaluation Reports (Annual Progress Report, Interim Evaluation, Terminal Evaluation)*	*0% of villages in Tonga have yet to participate in the national awareness campaign or sub-national level consultations on the multi-sector national dialogue platform*	*At least 40% of villages in Tonga participate in the national awareness campaign or sub-national level consultations on the multi-sector national dialogue platform*	*At least 80% of villages in Tonga participate in the national awareness campaign or sub-national level consultations on the multi-sector national dialogue platform*	*Inability to generate buy-in, interest or long-term involvement in the national dialogue process and steps.*
	Number of national multi-sectoral, multi-stakeholder engagement	• *Minutes of meetings*	*Currently no national mechanism for, or*	*No national mechanism for, or dialogue on*	*At least 2 regular meetings held for the*	*Commitment from all critical parties, including landowners*

	and dialogue platform meetings held (e.g. at least 1 meeting per year for last 2 years of the project lifetime (or as indicated in the final TOR for the national dialogue platform)	• *Attendee lists*	*dialogue on co-creating long-term transformative climate change adaptation*	*co-creating long-term transformative climate change adaptation*	*multistakeholder platform for dialogue (according to the agreed schedule/at least 1 meeting per year for last 2 years of the project lifetime)*	
Output 2 Strengthened national and local capacities for effective monitoring and assessment of climate risks	*Availability of a guideline for local planning that incorporates climate risks and adaptation*	*Number of and availability of plans and policies*	*Currently no guidelines available or climate-risk informed plans*	*A draft guideline available and piloted for local planning*	*At least 1 guideline adopted for local planning*	
	Number of Island development plans that integrate and address climate change, gender, and adaptation concerns	*Availability of gender-sensitive spatial planning guideline considering climate risk and adaptation strategies developed*	*Currently no guidelines available or climate-risk informed plans*	*1 island development plans integrate and addresses climate change, gender and adaptation concerns*	*5 island development plans integrate and address climate change, gender and adaptation concerns*	*The project team will sensitize and raise awareness on the importance of women's participation in defining the spatial plans for long-term adaptation*
	Number of individuals who are trained (disaggregated by gender) in climate risks and adaptation for updating/developing the CDPs /ISPs	*Consultation minutes and attendee lists for training undertaken* *Feedback analysis of trained individuals*	*0 community members are trained on updating CDPs and accessing finance for CCA priority actions*	*100 community members are trained on updating CDPs and accessing finance for CCA priority actions*	*At least 300 people trained in climate-risk and adaptation for updating/develop ing the CDPs/ ISPs*	*Individuals trained will take on key roles in updating the CDPs and ISPs.*

	Number of climate risk informed Island plans	*Island graphical reports on settlement patterns and climate risk for long-term coastal adaptation planning available* *Minutes of coordinating body/committee established for multi-stakeholder coordination and collaboration on coastal adaptation and planning integrating climate change*	*0 Island-level graphical profiles related to settlement patterns and climate risk for strategic long-term coastal adaptation planning developed.*	*0 Island-level graphical profiles related to settlement patterns and climate risk for strategic long-term coastal adaptation planning developed.*	*5 Island-level graphical profiles related to settlement patterns and climate risk for strategic long-term coastal adaptation planning developed*	*LiDAR coverage can be undertaken of all site/unaffected by weather/clouds*
Output 3 Reduced vulnerabilities of coastal communities in Hahake to climate hazards through coastal protection measures	*Length of coastal protection measures developed for Nukuleka to Talafo'ou and Manuka to Kolonga shoreline*	*Progress and Evaluation Reports:* *i) Technical design evaluation report and* *ii) Implementation report by assessment/ construction vendor* *iii) Annual progress report*	*2km of revetment constructed along NE Tongatapu (ADB funded)*	*2km of revetment constructed along NE Tongatapu (ADB funded)* *Technical design evaluation report completed*	*6.3km of revetment constructed along NE Tongatapu in total comprising of:* *2.3kms of rock revetment built between Nukuleka and Talafo'ou* *2kms of coastal protection measures constructed Manuka to Kolonga*	*Natural hazards do not drastically change the shoreline structure*
	Number of publications shared	*Availability of published report on lessons learned in*	*Limited published information available locally documenting*	*Limited published information available locally*	*1 published report on lessons learned (on coastal*	

		Government of Tonga's climate change website	*lessons learned on coastal protection measures and effectiveness*	*documenting lessons learned on coastal protection measures and effectiveness*	*protection measures and effectiveness) available*	
	Number of conferences organized for sharing lessons and best practices in climate resilient coastal protection measures	*Conference attendees list and conference report on lessons sharing conference*	*Limited sharing of lessons learned on coastal protection measures and effectiveness relevant to Pacific SIDS*	*Limited sharing of lessons learned on coastal protection measures and effectiveness relevant to Pacific SIDS*	*1 conference organized (inclusive of 1 conference report) on best practices in climate resilient coastal protection measures*	*Willingness for attendees to participate in lessons learned sharing*
Project/programme co-benefit indicators						
Co-benefit 1 *Improved gender parity in representation of females engaged in adaptation decision-making*	*Increase in adaptation solutions that respond to women's priorities in local planning tools or through design thinking workshops.*	*Meeting notes* *Planning documents (CDPs, ISPs etc)*	*Adaptation priorities in local planning tools are not differentiated by gender*	*A minimum of 10 percent of adaptation actions as per the local planning tools responds to priorities expressed by women*	*30 to 40 percent of adaptation actions as per the local planning tools responds to priorities expressed by women*	*The project makes considerable efforts to promote women's participation and engagement in climate resilience - including developing long-term adaptation solutions through design thinking workshops, enhancing the capacity of women to shape local development and spatial planning*
Co-benefit 2 Improved resilience to multi-hazards	*Number of plans considering coastal inundation events developed*	*Island graphical reports on settlement patterns and climate risk for long-term coastal adaptation planning available*	*0 Island-level graphical profiles related to settlement patterns and coastal inundation risk for strategic long-term*	*0 Island-level graphical profiles related to settlement patterns and coastal inundation risk*	*5 Island-level graphical profiles related to settlement patterns and climate risk*	*Planning for coastal inundation generated as an impact of climate change also generates resilience benefits toward tsunami inundation*

	Kilometers of coastal protection to multi-hazards (e.g. tsunami impacts) in Hahake	*Site survey*	*coastal adaptation planning developed.* *2km of coastal revetment in Nukuleka*	*for strategic long-term coastal adaptation planning developed.* *2km of coastal revetment in Nukuleka*	*incorporating coastal inundation for strategic long-term coastal adaptation planning developed* *6.3km of revetment constructed along NE Tongatapu in total comprising of:* *2.3kms of rock revetment built between Nukuleka and Talafo'ou* *2kms of coastal protection measures constructed Manuka to Kolonga*	*Climate change coastal protection measures offer some protection for other non-climate change related hazards such as tsunamis*

E.6. Project/programme activities and deliverables

All project activities should be listed here with a description and sub-activities. Significant deliverables should be reflected in annex 5 implementation timetable. Add rows as needed.

Please number the activities as shown below to indicate association of activities to the related outputs provided above in section E.5. Similarly, please number sub-activities as shown below to associate to the related activity.

Activities	Description	Sub-activities	Deliverables
Activity 1.1	Establish a national multi-stakeholder engagement platform for dialogue on co-	Sub-activity 1.1.1 Design of a multi-sectoral and multi-actor stakeholder	• Design and establishment of a multi-sectoral and multi-actor stakeholder

	creating long-term climate change adaptation strategies and solutions including voluntary retreat.	engagement plan and guidelines for a bottom-up national engagement process.	engagement plan and guidelines for a bottom-up national engagement process • Specific guidance on engaging women, girls and other vulnerable groups within engagement plan and guidelines for a bottom-up national multi-sectoral and multi-actor adaptation planning engagement process developed
		Sub-activity 1.1.2 Facilitate a national awareness campaign on climate change and disaster risks and transformative adaptation strategies for multi-stakeholder dialogue co-creating long-term adaptation strategies and solutions (using material developed under Output 2)	• National awareness campaign undertaken on climate risk and long-term transformative adaptation strategies. • Engagement and socialisation on climate change risk and long term transformative adaptation strategies and the multi-sectoral and multi-actor stakeholder bottom-up national engagement process undertaken at local, island and national levels. • National awareness campaign materials (that also incorporate women-led talk shows, or women's forum or media advertisements targeting women and their role in shaping national adaption strategies) developed • 3 Campaign activities targeting women on climate change risk and co-creating long term transformative adaptation strategies. • Design thinking workshops held for women/women leaders/women organisations) on climate change risk and co-creating long term transformative adaptation strategies.

			• Awareness consultative/workshop engagement activities with local, island and national level stakeholders on climate change risk and co-creating long term transformative adaptation strategies held
		Sub-activity 1.1.3 Develop a Terms of Reference for a National adaptation planning engagement platform specifying memberships, responsibilities and results	• Consultation undertaken on the memberships, responsibilities and results that should be incorporated within the ToR for the national adaptation planning engagement platform • A (gender sensitive) ToR for the national adaptation planning engagement platform that defines resposibilities, results and specifies memberships (including Women's Affairs Division/Ministry of Internal Affairs, Women's organizations) developed.
		Sub-activity 1.1.4 Undertake meetings of the national engagement platform on long-term climate change adaptation as per the Terms of Reference	• Regular meetings held (on the schedule defined at 1.1.3.2) for national engagement on long term transformative adaptation strategies (at least 1 meeting per year for last 2 years of the project lifetime).
		Sub-activity 1.1.5 Develop policy options to incorporate climate resilience and land use principles into national frameworks	• Guidance for land use considering climate risk and incorporating climate resilience developed • Land Use Policy update consulted and developed, prepared ready for submission to Cabinet • Gender-responsive policy options to incorporate climate resilience and

			land use principles into national frameworks developed.
Activity 1.2	Develop village and district level participatory climate risk informed plans	Sub-activity 1.2.1 Support a participatory process for developing village and district spatial plans to incorporate climate risk planning and adaptation strategies and incorporating principles of land-use.	• Guidelines for relevant offices and departments to include climate change considerations in bottom-up planning developed • Gender-sensitive guidelines for participatory village and district spatial plans, targetting 50% of female participation developed • 1 village and district level participatory process for developing a climate-risk informed district spatial plan piloted in Hahake. • Awareness consultative/workshop engagement activities with local, island and national level stakeholders on climate change risk and co-creating long term transformative adaptation strategies undertaken
Activity 1.3	Build the capacity of local government, village committees and NGOs to integrate climate risks and adaptation needs into community level planning, and inform future Community Development Plans (CDP)	Sub-activity 1.3.1 Develop an updated climate-responsive Community Development Plan Guideline to inform the development of future CDPs that includes including participatory vulnerability assessment and planning, gender needs assessment, and adaptation needs assessment and planning.	• Facilitate community outreach and engagement on adaptation • Collection and collation of local priorities to inform policy action through the gender-sensitive climate responsive CDP guideline • Updated gender-sensitive climate-responsive CDP guidelines developed.
		Sub-activity 1.3.2 Design and implement a capacity building package for the	• Develop gender-sensitive capacity building package on the new gender

		Ministry of Internal Affairs on utilising the new climate-responsive CDP guideline	and climate-responsive CDP guideline. • Develop training materials and hold training sessions/mentoring for government staff (men and women) on integrating climate change in planning and budgeting.
		Sub-Activity 1.3.3 Strengthen capacities of key ministries for climate change risk informed planning and budgeting through trainings and materials	• Guidelines and other official documents to support government staff to integrate climate change risks in planning and budgeting that incorporate the inclusion of equitable participation of decision makers and officials (men and women) in the capacity development trainings developed
		Sub-Activity 1.3.4 Organize island events (Vava'u, Ha'apai, 'Eua, Tongatapu and Niuas) for sharing lessons for producing climate responsive CDPs	• Organise and hold 5 island events for knowledge sharing on climate-responsive CDP with voices of and women's specific issues highlighted.
Activity 2.1	Strengthened mechanism for collecting and analysing data and information for better-informed climate risk monitoring and coastal adaptation planning	Sub-Activity 2.1.1 Increase capacity of the Government to effectively collect, manage and analyse coastal monitoring data	• Develop a long-term Capacity Building Strategy and Plan for government use on coastal and climate risk monitoring and analysis to inform coastal adaptation planning (e.g. including understanding and use of climate and risk infromation and data such as satellite imagery, aerial photography, drone imagery, LiDAR topography and bathymetry) • Capacity needs assessment to inform the long-term Capacity Building Strategy and Plan undertaken • Long-term Capacity Building Strategy and Plan for government use on coastal and climate risk monitoring

			and analysis to inform coastal adaptation planning developed • Capacity building and training opportunities through regional placements in technical agencies in the Pacific established and supported (for example, SPC and others) • Training/mentoring undertaken and training materials available to support government staff to integrate climate change risks in planning and budgeting • Technical officers and local government officers trained on coastal zone management, coastal processes, coastal planning and hazards management, and climate down-scaling
		Sub-Activity 2.1.2 Collecting, interpreting and using data on climate risks for coastal adaptation planning to produce scenario-based coastal risk maps and knowledge material for use under Output 1.	• Critical equipment for improved coastal adaptation planning and monitoring capacity purchase for use • Baselines available through LiDAR surveys that cover country-wide critical areas • Scenario-based coastal risk maps and knowledge material for use under Output 1 produced
		Sub-Activity 2.1.3 Improve multi-stakeholder coordination and collaboration on coastal adaptation and planning integrating climate change	• Capacity building to integrate data developed in activity 2.1.2 through mentoring and training for intergovernmental multistakeholder coordination undertaken • Coordinating body/committee for multi-stakeholder coordination and collaboration on coastal adaptation

			and planning integrating climate change established.
Activity 2.2	Improve the knowledge base of multi-sectoral, multi-stakeholders on adaptation planning strategies for long-term resilient planning and transformative adaptation for Tonga based on climate risks and projections	Sub-Activity 2.2.1 Support the use and interpretation of data collected in 2.1 by providing easily understood materials for multiple stakeholder types that consider strategic long-term coastal adaptation and inform the national multi-stakeholder adaptation planning engagement platform	• Consultative assessment on the trade-offs between adaptation solutions and current land use considering food security, water security and livelihoods as well as incorporating cultural sensitivities and traditional knowledge • 5 Island-level graphical reports related to settlement patterns and climate risk for strategic long-term coastal adaptation planning produced.
Activity 3.1	Build coastal protection measures along 4.3 km of coastline in Hahake	Sub-Activity 3.1.1 Detailed design and site-specific assessments of coastal protection measures finalized for Hahake	• Site-specific assessments of coastal protection measures developed and finalized for Nukuleka - Talafo'ou, Hahake (hard reventment) and coastal protection measures for the Manuka to Kolonga shoreline • Detailed design of coastal protection measures developed for Nukuleka - Talafo'ou, Hahake (hard reventment) • Design of coastal protection measures developed for Manuka to Kolonga shoreline
		Sub-Activity 3.1.2 Construct coastal protection measures in Hahake	• Construction of 2.3kms of reventment (Nukuleka - Talafo'ou) • Construction of 2kms of coastal protection measures developed for Manuka to Kolonga shoreline
Activity 3.2	Sharing of lessons learned and best practices in climate resilient coastal protection measures for scale up at the national and regional level	Sub-Activity 3.2.1 Technical evaluations of the design and effectiveness of the coastal protection measures conducted	• Conduct technical evaluation of the design and effectiveness of the coastal protection measures • Production of a report on lessons learned

		Sub-Activity 3.2.2 Production of and publication of a report on lessons learned	• A lessons learned (on coastal protection measures and effectiveness) report published with guidelines and standards for design and construction of coastal protection measures
		Sub-Activity 3.1.3 Organize an international /regional conference for sharing lessons learned in climate resilient coastal protection and management	• Conference on best practices and lessons learned in climate resilient coastal protection measures relevant to Pacific SIDS held with national and regional participants

E.7. Monitoring, reporting and evaluation arrangements (max. 500 words, approximately 1 page)

Besides the arrangements (e.g. annual performance reports) laid out in Accreditation Master Agreement (AMA), please give a summary of the project/programme specific arrangements for monitoring, reporting and evaluation including a description of the monitoring and reporting system that will be used to assess the climate results of the proposed activity. Please also summarize the types of interim and final evaluations. Describe Accredited Entity (AE) project reporting relationships, including to the National Designated Authority (NDA)/Focal Point and between AE and Executing Entity (EE) as relevant, identifying reporting obligations from the EE to the AE. This should relate to the frequency of reporting on project indicators, implementation challenges and financial status. Please note that interim and final evaluations are expected to embed an assessment of project/programme's contributions to a paradigm shift and enabling environment using a three-point scale rating. Refer to the guidance note for the summary requirements and factor in additional evaluation /assessment activities under this section accordingly.

173. Project-level monitoring and evaluation will be undertaken in compliance with the UNDP POPP and the UNDP Evaluation Policy, and also in line with GCF Evaluation Policy and IRMF.

    The primary responsibility for day-to-day project monitoring and implementation rests with the Project Manager. The Project Manager will develop annual work plans to ensure the efficient implementation of the project. The Project Manager will inform the Project Board and the UNDP Pacific Multi-Country Office of any delays or difficulties during implementation, including the implementation of the M&E plan, so that the appropriate support and corrective measures can be adopted. The Project Manager will also ensure that all project staff maintain a high level of transparency, responsibility and accountability in monitoring and reporting project results.

174. An independent mid-term re-evaluation will be undertaken, and the findings and responses outlined in the management response will be incorporated as recommendations for enhanced implementation during the final half of the project's duration. The terms of reference, the review process and the final evaluation report will follow the standard templates and guidance available on the UNDP Evaluation Resource Center. The final evaluation report will be cleared by the UNDP Country Office and the UNDP Regional Technical Adviser and will be approved by the Project Board. The final evaluation report will be available in English.

175. An independent terminal evaluation (TE) will take place when project activities have concluded The terms of reference, the evaluation process and the final TE report will follow the standard templates and guidance available on the UNDP Evaluation Resource Center. The final TE report will be cleared by the UNDP Pacific Multi-Country Office and the UNDP Regional Technical Adviser and will be approved by the Project Board. The TE report will be available in English.

176. The UNDP Pacific Multi-Country Office will include the planned project evaluations in the UNDP Pacific Multi-Country Office evaluation plan, and will upload the evaluation report in English and the management response to the public UNDP Evaluation Resource Centre (ERC) (www.erc.undp.org).

177. The UNDP Pacific Multi-Country Office will retain all M&E records for this project for up to seven years after project financial closure in order to support ex-post evaluations.

178. A detailed M&E budget (1-2% of the total GCF grant), monitoring plan and evaluation plan will be included in the UNDP project document.

179. Monitoring, reporting and evaluation arrangements will comply with the relevant GCF policies and Accreditation Master Agreement signed between GCF and UNDP.

180. An ex-post technical assessment will be carried out with respect to the coastal protection measures under output 3. Lessons learned across all Outputs will be disseminated through national / regional/ international conference proceedings and other media.

181. The UNDP Pacific Multi-Country Office will support the Project Manager as needed, including through annual supervision missions. The UNDP Pacific Multi-Country Office Office is responsible for complying with UNDP project-level M&E requirements as outlined in the UNDP POPP. Additional M&E and implementation quality assurance and troubleshooting support will be provided by the UNDP Regional Technical Advisor as needed. The project target groups and stakeholders including the NDA Focal Point will be involved as much as possible in project-level M&E.

182. A project inception workshop will be held after the UNDP project document has been signed by all relevant parties to: a) re-orient project stakeholders to the project strategy and discuss any changes in the overall context that influence project implementation; b) discuss the roles and responsibilities of the project team, including reporting and communication lines and conflict resolution mechanisms; c) review the results framework and discuss reporting, monitoring and evaluation roles and responsibilities and finalize the M&E plan; d) review financial reporting procedures and mandatory requirements, and agree on the arrangements for the annual audit; e) plan and schedule Project Board meetings and finalize the first year annual work plan. The final inception report will be cleared by the UNDP Pacific Multi-Country Office and the UNDP Regional Technical Advisor and approved by the Project Board and submitted to the GCF within 6 months after the Funded Activity Agreements effectiveness date.

183. A project implementation report will be prepared for each year of project implementation. The Project Manager, the UNDP Pacific Multi-Country Office , and the UNDP Regional Technical Advisor will provide objective input to the annual GCF APR. The Project Manager will ensure that the indicators included in the project results framework are monitored annually well in advance of the GCF APR submission deadline. Progress against the gender action plan, stakeholder engagement, social and environmental safeguards updates, challenges and delays must also be monitored by the Project Manager and reported in the GCF APR. The GCF APR will be shared with the Project Board and other stakeholders.

ASSESSMENT AND MANAGEMENT

F.1. Risk factors and mitigations measures (max. 3 pages)

Please describe financial, technical, operational, macroeconomic/political, money laundering/terrorist financing (ML/TF), sanctions, prohibited practices, and other risks that might prevent the project/programme objectives from being achieved. Also describe the proposed risk mitigation measures. Insert additional rows if necessary.

For probability: High has significant probability, Medium has moderate probability, Low has negligible probability

For impact: High has significant impact, Medium has moderate impact, Low has negligible impact

Prohibited practices include abuse, conflict of interest, corruption, retaliation against whistleblowers or witnesses, as well as fraudulent, coercive, collusive, and obstructive practices

Selected Risk Factor 1

Category	Probability	Impact
Technical and operational	Medium	Medium

Description

Limited human resources in government ministries, agencies and contractors delay project activities.

Mitigation Measure(s)

Please describe how the identified risk will be mitigated or managed. Do the mitigation measures lower the probability of risk occurring? If so, to what level?

Detailed consultations have taken place between MEIDECC and other government agencies, and UNDP to identify operational capacity limitation in implementing the project of this scale. Through the adoption of a DIM delivery, UNDP can ensure that the project team is sufficient to provide the additional resources, engagement and mentoring required to prevent delays to the project. Accordingly, the decision to use the DIM modality was selected to overcome any operational capacity limitations for the scale of the project. As a DIM project, being under the direct management of UNDP, project delivery will be closely monitored and necessary resources deployed. The PMU will monitor project progress as well as emerging risks and if necessary adopt adaptive management strategies to address any issues. The Project Management Unit which will support day-to-day execution of project activities will also be established with adequate resources and human capacity to fulfill operational and technical capacities needs. Furthermore, the Project Board will also oversee monitoring of the risk and ensure that risk mitigation measures are adhered to continuously throughout the project. continuously throughout the project. continuously throughout the project. continuously throughout the project.

Selected Risk Factor 2

Category	Probability	Impact
Technical and operational	Low	Medium

Description

Community commitment for engaging in project activities and post-project monitoring and actions is not sustained benefits of adaptation measures and project interventions are not immediately evident.

Mitigation Measure(s)

Please describe how the identified risk will be mitigated or managed. Do the mitigation measures lower the probability of risk occurring? If so, to what level?

The greatest benefits of the project will become apparent if and when there is a significant climate event, such as cyclone or tsunami, until the benefits of the project may not be as apparent to the community and the need for post-project monitoring not as urgently felt. Coastal protection needs are identified as a priority action in consultations and in Community Development Plans. Furthermore, community commitments are reiterated in the rounds of consultations that took place for the preparation of this proposal. Therefore, the probability of this risk materializing is considered low. However, the engagement of NGOs is thought to be critical as they play an important role in mobilizing and supporting community members, and the project has dedicated elements that are specifically building NGO capacity so that they will continue to play a role in community engagement.

Participatory impact assessment is scheduled towards the end of the project to present evidence-based impact of the project to maintain communities' continued interests and commitment in project participation.

Selected Risk Factor 3

Category	Probability	Impact
Technical and operational	Low	Low

Description

Project interventions are not implemented in a gender- and culturally-sensitive manner, undermining the overall effectiveness of the project.

Mitigation Measure(s)

Please describe how the identified risk will be mitigated or managed. Do the mitigation measures lower the probability of risk occurring? If so, to what level?

For long-term sustainable land planning to be accepted, gender and cultural needs must be recognized and reflected in the project implementation and its outcomes. Poor gender outcomes, particularly in terms of SEAH, could result in the project's sustainability and potential upscaling to be reduced. The Government has therefore taken a leadership in making sure that the proposed project has been designed in a gender-responsive manner. Community consultations have been carried out in such a way to ensure that different needs of men and women are captured and reflected in the design. A Gender Assessment and Action Plan (Annex XIII) is one of the outputs of the gender-responsive project design. Members of the Project Management Unit will be sensitized, at the beginning of the project, about the Action Plan and general gender-responsive approach to project implementation.

Selected Risk Factor 4

Category	Probability	Impact
Technical and operational	Low	Low

Description

Competing mandates and weak institutional coordination to implement an integrated cross-sectoral approach to coastline resilience

Mitigation Measure(s)

Please describe how the identified risk will be mitigated or managed. Do the mitigation measures lower the probability of risk occurring? If so, to what level?

The project has been designed specifically to remove this barrier. Because multi-sectoral responses to coastline management is a new area for the Government of Tonga, this existing risk is being targeted specifically in the project design. As such, the mitigation measures to this risk are the full implementation of Activity 2.1 and 2.2. While the CCD in MEIDECC has been tasked with multi-sectoral coordination in the implementation of JNAP, and they have received a positive review for their performance in this regard, multi-ministerial information sharing for the purpose of coastal resilience building is a new area. To facilitate inter-ministerial coordination, an SOP will be established, under Activity 2.1, for data and information sharing about coastal processes. Engagement of community, NGOs and local and national government in coastal resilience building will be promoted through, inter alia, Activity 2.2 where a capacity building package specifically for improving community-level adaptation planning process will be implemented. The project will also utilize the JNAP Taskforce as the primary mechanism for multi-ministerial coordination and information sharing for the project, and it has been working well in the last 8 years of JNAP implementation. Project-financed staff members, especially the Project Manager, will be specifically responsible for ensuring continued information sharing in a transparent manner as per the Term of Reference of the position.

Selected Risk Factor 5

Category	Probability	Impact
Technical and operational	High	Medium

Description

The national dialogue on climate-informed land use does not yield any agreements, especially from landowners, for the revision of the national land use plan.

Mitigation Measure(s)

Please describe how the identified risk will be mitigated or managed. Do the mitigation measures lower the probability of risk occurring? If so, to what level?

This risk is recognised due to the lack of tangible change arising from past efforts in changing the land ownership architecture. The national dialogue will increase awareness of climate change and the various options available to Tonga, but there is no guarantee that the actions, particularly those that involve voluntary relocation ie leaving currently owned land, will be wholly accepted – particularly during the reasonably short time frame of the project. Land tenure is complex in Tonga and past efforts in changing the land ownership architecture, though very limited in number, did not result in a tangible change. None the less, even if agreements are not reached within the project timeframe, the national dialogue will set the groundwork and create the framework for the inevitably required actions. For a country like Tonga where there is a significant concentration of populations in highly exposed areas, many of which will face a significant risk of inundation and damages from coastal hazards in the future, facilitating a carefully planned, evidence-based dialogue to gradually open the possibility of voluntary retreat is the only long-term adaptation option. To maximize the probability of success, the project will put in place a range of measures including the use of various visual materials (videos, 3D images, flyers), social mobilization tools (radio and TV programs), engagement of NGOs/CSOs in the activities, and a nation-wide campaign.

Selected Risk Factor 6

Category	Probability	Impact
Technical and operational	Low	Low

Description

Financial risks may impact the implementation of project activities (for example, foreign exchange risk through currency fluctuation).

Mitigation Measure(s)

Please describe how the identified risk will be mitigated or managed. Do the mitigation measures lower the probability of risk occurring? If so, to what level?

All standard procurement and financial risks have been considered in the design of the project and UNDP delivers oversight and quality assurance services for procurement and financial management of the project guided by UNDP financial rules and regulations. Project financing and budget has been designed in consideration of inflation rates (forecasting for 2024+), current commodity rates, exchange rate fluctuations with contingencies in place to mitigate the financial risks that may impact the implementation of project activities.

Selected Risk Factor 7

Category	Probability	Impact
Technical and operational	Low	High

Description

Extreme events, including cyclones, earthquakes and tsunamis may impact costal protection efforts under activity 3.1 that could delay the implementation of the project through delays or damage during construction.

Mitigation Measure(s)

Please describe how the identified risk will be mitigated or managed. Do the mitigation measures lower the probability of risk occurring? If so, to what level?

Given the regular seismic events affecting Tonga (earthquakes, volcanos, and tsunamis) and cyclones, there is a potential risk that these may occur during the project's implementation, including during the construction periods. Tonga's early warning systems allow for measures to be taken to evacuate workers in the incidence of flood inundation, cyclones, and tsunamis. In the incidence of disaster and climate change hazards, the rock materials (for the rock revetment and road raising) are considered durable and robust, and potential damage during the construction phase is minimal.

The Contracted Party undertaking construction is obliged to undertake construction in a way that leaves the works least vulnerable to such events or disturbances during construction. The Contracted Party is also obliged to respond to weather warnings (e.g. cyclone) by securing the worksite, assets and materials accordingly.

Selected Risk Factor 8

Category	Probability	Impact
Technical and operational	Low	High

Description

Limited workforce/labour, equipment and construction material (rocks for revetment) available for construction in Hahake coast

Mitigation Measure(s)

Please describe how the identified risk will be mitigated or managed. Do the mitigation measures lower the probability of risk occurring? If so, to what level?

There is a limited amount of workforce/labour, equipment and construction material (rocks for revetment) available in Tonga for the type of construction work required in Hahake coast, however, the project is expected to rely on local capacities/companies to be involved in the construction works. This is based on past similar work undertaken in the country by local companies under ADB's CRSP project and the lessons learnt there. Preliminary assessments have been undertaken in terms of availability and quality of construction material (rocks) in the main quarry in Tonga and close monitoring of the availability will continue during project approval and implementation phases. The construction timeline allows for staggering of construction between the two areas in Hahake coast to allow for the shortage of workforce/labour and equipment due to competing demands in the country. The Government ownership and coordination will be enhanced through the Project Board in aligning the timelines for construction and other large scale construction projects that may be undertaken in the country.

Selected Risk Factor 9		
Category	Probability	Impact
Sanctions	Low	Medium
Description		
Project engage with entities or individuals listed on UN sanctions		
Mitigation Measure(s)		
This project is implemented by UNDP under the Direct Implementation Modality (DIM) and will be guided by UNDP's policy framework. UNDP, by virtue of being an organ of the United Nations and therefore being part of the same entity, is also beholden to the principles on countering the financing of terrorism. UNDP systematically screens all entities being considered to be contracted by or partner with UNDP against the Consolidated UN Security Council Sanctions List (https://www.un.org/securitycouncil/content/un-sc-consolidated-list). As part of its due-diligence, UNDP Pacific Multi Country Office in Suva will ensure to screen all entities (private sector, NGOs, CSOs) or individuals against the Consolidated UN Security Council Sanctions List before engaging with them for the project. Bidding documents will also require individuals and legal entities for self-declaration for not being in the UN Security Council Sanctions List. While selecting partners and private sector entities, Capacity Assessments tools are also used to identify if these entities appear in the Consolidated UN Security Council Sanctions List.		
Selected Risk Factor 10		
Category	Probability	Impact
ML/FT	Low	Medium
Description		
Money laundering and terrorist financing risks		
Mitigation Measure(s)		
This project is implemented by UNDP under the Direct Implementation Modality (DIM) and will be guided by UNDP's policy framework. UNDP, by virtue of being an organ of the United Nations and therefore being part of the same entity, is also beholden to the principles on countering the financing of terrorism. UNDP systematically screens all entities being considered to be contracted by or partner with UNDP against the Consolidated UN Security Council Sanctions List. Please refer to the UN Security Council Consolidated Sanctions List available at https://www.un.org/securitycouncil/content/un-sc-consolidated-list for the latest information on sanctions. UNDP, through LexisNexis system, screens potential partners/vendors (including consultants) against anti-money laundering and countering of Terror Finance. In cases, where there is Positive Match, UNDP escalate such instances to Regional Bureau or HQ for decision making. UNDP also monitors and track such risks as part of CO integrated Programme Risk register as well as Project Risk Register where status of risk treatment actions is regularly monitored, updated and reported.		
Selected Risk Factor 11		
Category	Probability	Impact
Prohibited practices	Low	Medium
Description		
Prohibited Practices Risks		
Mitigation Measure(s)		
UNDP applies a zero tolerance policy in relation to fraud and corruption. The guiding principles of UNDP's commitment to prevent, identify and address all acts of fraud and corruption have been laid out in the UNDP Policy against Fraud and Corrupt Practices (the "FCP Policy"), which applies to all activities and operations of UNDP, including projects and programmes. The fundamental principles of UNDP's FCP Policy serve as the basis for and are integrated in UNDP's policy frameworks in relation to procurement, financial management, internal control and accountability and staff rules and regulations.		

Under the terms of the FCP Policy and related policies, UNDP has a commitment, when developing a new programme or project, to ensure that risks related to fraud and corrupt practices are fully identified and considered in the programme/project design and processes and that adequate and effective measures to mitigate such risks are put in place. In this light, the processes and requirements for UNDP's partnership capacity assessment (PCAT) require UNDP to carry out a thorough due diligence of the risks in relation to the programme/project it intends to engage in, as well as in relation to the potential partners who will be involved in the implementation of such programmes/projects. Such capacity assessments include, among other things, a screening of all entities being considered to be contracted by or partner with UNDP against the Consolidated UN Security Council Sanctions List (to which UNDP, by virtue of being an organ of the United Nations and therefore being part of the same entity, is also beholden), an assessment of any history of fraud, corruption or other fraudulent practices and/or potential conflicts of interest. Based on this capacity assessment, UNDP then builds an internal control component into the design of the project and ensures that the financial management of the programme/project contains adequate safeguards to prevent, monitor and address any risks and acts of fraud, corruption or AML/CFT that may be identified.

UNDP’s internal Partner Capacity Assessment Tool (PCAT) was applied to assess the capacities of MEIDECC, MLNR and MoI, the Government agencies that are planned to to be engaged in the project. The assessment includes a question regarding a history of fraud, corruption, money laundering, financing terrorism and the like. For all three partners, the assessment did not find evidence for such past practices. Further, a Harmonized Approach to Cash Transfers (HACT) assessment was conducted for MEIDECC, the lead government agency for this project, and they were assessed for the existence of policies against fraud and corruption, policies for reporting wrongdoing including misuse of funds. After assessing the PCAT and HACT results, as well as the size of the proposed project, the Government and UNDP agreed to execute this project under UNDP’s DIM modality, and hence, UNDP’s Enterprise Risk Management (ERM) Policy framework, which includes the AML/CFT policies and the Policy Against Fraud and Corrupt Practices, will be applied during the implementation.

As part of its incident management procedures, UNDP also establishes project-level grievance redress mechanism; where a stakeholder response mechanism allows grievances from direct/in-direct beneficiaries are duly reported and efforts are made to resolve the concerns. Communities / beneficiaries are also informed about OAI helpline to report incidences of fraud, corruption, prohibited practices and any wrong-doing for direct reporting.

F

GCF POLICIES AND STANDARDS

G.1. Environmental and social risk assessment (max. 750 words, approximately 1.5 pages)

Provide the environmental and social risk category assigned to the proposal as a result of screening and the rationale for assigning such category. Present also the environmental and social assessment and management instruments developed for the proposal (for example, Environment and Social Impact Assessment (ESIA), Environment and Social Management Framework (ESMF), Environmental and Social Management Plan (ESMP), Environmental and Social Management System (ESMS), environmental and social audits, etc.). Provide a summary of the main outcomes of these instruments. Present the key environmental and social risks and impacts and the measures on how the project/programme will avoid, minimize and mitigate negative impacts at each stage (e.g. preparation, implementation and operation), in accordance with GCF's Environmental and Social Safeguard (ESS) standards. If the proposed project or programme involves investments through financial intermediations, describe the due diligence and management plans by the Executing Entities (EEs) and the oversight and supervision arrangements. Describe the capacity of the EEs to implement the ESMP and ESMF and arrangements for compliance monitoring, supervision and reporting. Include a description of the project/programme-level grievance redress mechanism, a summary of the extent of multi-stakeholder consultations undertaken for the project/programme, the plan of the Accredited Entity (AE) and EEs to continue to engage the stakeholders throughout project implementation, and the manner and timing of disclosure of the applicable safeguards reports following the requirements of the GCF Information Disclosure Policy and Environmental and Social Policy.

Describe any potential impacts on indigenous peoples and the measures to address these impacts including the development of an Indigenous Peoples Plan and the process for meaningful consultation leading to free, prior and informed consent, pursuant to the GCF Indigenous Peoples Policy.

Attach the appropriate assessment and management instruments or other applicable studies, depending on the environmental and social risk category as annex 6.

184. Potential environmental effects arise in relation to the coastal works proposed under output 3, while social considerations are relevant across the project as a whole.

185. Water Quality Management Measures - although sediment movement may have transient affects on marine and wetland water, the project is expected to improve water quality in Tonga overall through improved engineering and drainage management in the project site. By following the management measures set out in the ESMF, the construction of coastal works, designing sediment retention infrastructure and drainage will not have a significant impact on water quality across the broader area. A standardized water quality monitoring program has been developed for the project.

186. Erosion, Drainage and Sediment Control - coastal excavations and soil disturbance during all activities (if undertaken during wet/storm event periods) and drainage works can result in the loss of soil stability, soil erosion and soil productivity. However, all sediment removed from the coastal zone, wetland and drainage work will be assessed, and where practicable, will be reused behind coastal structures or other beneficial reuse options. Effective and efficient mitigation measures will not only reduce impacts but could improve the conditions above the baseline.

187. Noise and Vibration Management Measures - all construction and operation activities have the potential to cause noise nuisance. No blasting is required in this project. Vibration disturbance to nearby residents and sensitive habitats is likely to be caused by the use of vibrating and excavation equipment. A standardized noise-monitoring program developed for this project will ensure equipment and machinery is maintained regularly, operated appropriately, and that potentially noisy construction activity will be conducted out during daylight hours for minimal disruption.

188. Air Quality Management Measures - a standardized air monitoring program will ensure minimal impact on air quality from the release of dust/particles during construction in sensitive areas and from construction vehicle emissions as well as quick corrective actions to respond to complaints.

189. Flora and Fauna - the project is unlikely to have direct impacts on flora and fauna species found in Tonga or on the national parks and reserves (both marine and terrestrial). As part of the project site is situated in the outer area of the Fanga'uta [lagoon] Stewardship Plan (FSP) The project will be implemented in consultation with the FSP Steering Committee. The project will also work alongside the relevant Coastal Community Management Committee(s) in relation to managing any transient effects on community fisheries interests. Protection of the flora and fauna will be ensured through maintaining clear boundaries for any vegetation clearance when required, avoiding introduction of new invasive/weed species or increase in existing weed species proliferation during construction activities.

190. Waste Management - the key waste streams generated during construction are likely to include residual sediment (although this will be reused), additional rock material and draining from the coastal work and drainage works. Contractors involved in construction and operational activities should be familiar with methods minimizing the impacts of clearing vegetation and on coastal sediments to minimize the footprint to that essential for the works and rehabilitate disturbed areas. A waste management monitoring program has been established for the project.

191. Chemical and Fuel Management - some types of chemicals and fuels such as diesel, unleaded petrol, grease and few chemicals are likely to be stored on-site and/or on vessels during construction. If not handled, stored or used appropriately, contamination of land and the surface water and groundwater systems could occur. Further, accidental discharge of

hazardous materials during construction and operation activities is a potential risk to the local environment. Accordingly, all oil, grease, diesel, petrol and chemicals should be stored off site within a bounded area. A Material Safety Data Sheet Register will be developed for all chemicals and fuels retained on site and the handling and storage of hazardous material will be in accordance with the relevant legislation and best management practices.

192. Social Management Measures - the project has been designed with the assistance of stakeholders and aims to provide benefits to the broader community. The core work under Output 1 is focussed directly on developing a national dialogue, the success of which depends on effective and broad-based consultation processes. The project does not require involuntary resettlement or acquisition of land although there may be temporary impacts on land during construction activities. If dissatisfaction or conflicts arise over construction activities, early recognition and appropriate actions will be taken to avoid or minimize conflicts. Community consultation is crucial, and the Stakeholder Engagement Strategy (see Annex) ensures close and systematic consultations with all stakeholders for increased participation and representation during project design and implementation alike.

193. Archaeological, Indigenous and Cultural Heritage - no cultural heritage places, buildings or monuments are known to exist in areas where the project will be undertaken. The Archaeological, Indigenous and Cultural Heritage monitoring program that has been developed for the projects will ensure that cultural heritage awareness training is provided to all construction site personnel (including contractors), identify and collect any cultural heritage items worthy of protection, consult with the relevant Museums about any important Archaeological, Indigenous and/or Cultural Heritage material discovered during construction and will cease work in the area where any human remains are discovered.

194. Emergency Management Measures - in the event of actions occurring, which may result in serious health, safety, and environmental (catastrophic) damage, emergency response or contingency actions will be implemented as soon as possible to limit the extent of environmental damage.

G.2. Gender assessment and action plan (max. 500 words, approximately 1 page)

Provide a summary of the gender assessment and project/programme-level gender action plan that is aligned with the objectives of GCF's Gender Policy. Confirm a gender assessment and action plan exists describing the process used to develop both documents. Provide information on the key findings (who is vulnerable and why) and key recommendations (how to address the vulnerability identified) of the gender assessment. Indicate if stakeholder consultations have taken place and describe the key inputs integrated into the action plan, including: how addressing the vulnerability will ensure equal participation and benefits from funds investment; key gender-related results to be expected from the project/programme with targets; implementation arrangements that the AE has put in place to ensure activities are implemented and expected outcomes will be achieved, monitored and evaluated.
Provide the full gender assessment and project-level gender action plan as annex 8.

195. The gender assessment and analysis undertaken during the design of this project (and updated during 2023), led to including gender considerations across project outputs and activities and will serve as the basis for gender-responsive implementation of the project. The assessment and analysis will require continuous updates and revision throughout the lifetime of the project to ensure that project does not negatively affect local social dynamics but contributes to enhancing coastal resilience of local communities, including the marginalized and socially vulnerable groups. During the project design phase, consultations were held with a diverse range of stakeholders and partners – including MEIDECC, Ministry of Internal Affairs (Women's Affairs and Gender Equality Division), Ministry of Lands and Natural Resources (MLNR), Prime Minister's office (Department of Local Governments), Ministry of Finance, etc., In addition, there were discussions with development partners such as ADB, EU, SPREP and NGOs such as the Civil Society for Tonga (CSFT), the Mainstreaming of Rural Development Innovation (MORDI), and the Vava'u Environmental Protection Association (VEPA). The analysis also drew on data from different assessment and survey conducted by the Government of Tonga – such as 2019 Multiple Indicator Cluster Survey, 2018 National Disability Survey etc., as well as policy directives provided by the WEGET Policy, the Tonga Strategic Development Framework II (TSDF), Joint National Action Plan on Climate Change and Disaster Risk (2018-2028), and National Climate Change Policy. This gender assessment and stakeholder consultations have identified the following areas as particularly important dimensions to be reflected in the implementation strategy of the project:

196. Gender Analysis:
- The different needs in responding to extreme weather events faced by women and men, as well as other vulnerable groups;
- Analysis of the gendered-based division of labour which can be reflected in the engagement of women, men and youth in project activities;
- Reflecting women's access to, and control over, environmental resources and the goods and services that they provide into the design of ecosystem restoration and monitoring activities;
- The need to collect sex and age disaggregated data during the project implementation so that gender- and age-responsive decisions can be supported;

197. Voice and participation:
- Ensure equitable participation of women, men, youth and elderly people at community-level adaptation actions;

- Create opportunities for women to voice their views on transformative agenda for adaptation, and in planning, monitoring and implementation of climate and risk informed Community Development Plans (CDPs) (output 1).
- Ensure the participation of women's groups, youth groups and church groups in community mobilization;
- Create opportunities for participation of vulnerable and people with disabilities in all project consultations and activities.
- Adjusting advocacy and awareness raising approach to different population groups; for sensitive topics, women-only, men-only, or youth-only meetings may be needed and organizing consultations at specific times of the day may be advised for ensuring women's participation;
- Include all stakeholders involved in the project to develop awareness raising / training aimed at drawing attention to the implication of climate resilience adaptation and gender equality;
- Undertake community discussions and dialogue in relation to gender and social inclusion in climate and disaster resilience.
- Organize design workshops (targeting women and people with disabilities) to develop transformative adaptation solutions.

The full gender assessment and project-level gender action plan is provided at Annex 8.

G.3. Financial management and procurement (max. 500 words, approximately 1 page)

Describe the project/programme's financial management including the financial monitoring systems, financial accounting, auditing, and disbursement structure and methods. Refer to section B.4 on implementation arrangements as necessary.

Articulate any procurement issues that may require attention, e.g. procurement implementation arrangements and the role of the AE under the respective proposal, articulation of procurement risk assessment undertaken and how that will be managed by the AE or the implementing agency. Provide a detailed procurement plan as annex 10.

198. The financial management and procurement of this project will follow UNDP financial rules and regulations available here: https://info.undp.org/global/documents/frm/Financial-Rules-and-Regulations_E.pdf. Further guidance is outlined in the financial resources management section of the UNDP Programme and Operations Policies and Procedures.

199. UNDP has comprehensive procurement policies in place as outlined in the 'Contracts and Procurement' section of UNDP's Programme and Operations Policies and Procedures (POPP). The policies outline formal procurement standards and guidelines across each phase of the procurement process, and they apply to all procurements in UNDP. See here: https://popp.undp.org/. Procurement of services and goods will be done in a cost effective and reliable way and by applying following principles: **Best Value for Money**, which consists of the selection of the offer that best meets the end-users' needs and that presents the best return on investment; **Fairness, Integrity and Transparency,** which ensures that competitive processes are fair, open, and rules-based. All potential vendors will be treated equally, and the process will feature clear evaluation criteria, unambiguous solicitation instructions, realistic requirements, and rules and procedures that are easy to understand; **Effective International Competition,** understood as giving all potential vendors timely and adequate information on UNDP requirements, as well as equal opportunity to participate in procurement actions; and **In the best interest of UNDP**, which means that any business transactions must conform to the mandates and principles of UNDP and the United Nations.

200. Under the DIM implementation arrangement, procurement of goods and services will be carried out by UNDP and the UNDP procurement procedures will be applied (POPP), as outlined in the Procurement Plan. Additional details on the UNDP procurement thresholds and methods are provided in the Annex 10. Procurement Plan.

201. The project will be implemented following the **Direct Implementation Modality (DIM)** guidelines available here.
202. All projects will be audited following the UNDP financial rules and regulations noted above and applicable audit guidelines and policies, informed by and together with any specific requirements agreed in the AMA. According to the current audit policies, UNDP will be appointing the auditors. In UNDP, scheduled audits are performed during the project cycle as per UNDP assurance/audit plans, based on the implementing partner's risk rating and UNDP's guidelines. A scheduled audit is used to determine whether the funds transferred to the implementing partner were used for the appropriate purpose and in accordance with the work plan. A scheduled audit can consist of a financial audit and/or an internal control audit.

203. UNDP's Construction Works Policy will be applied to all construction works under this project.

G.4. Disclosure of funding proposal

Note: The Information Disclosure Policy (IDP) provides that the GCF will apply a presumption in favour of disclosure for all information and documents relating to the GCF and its funding activities. Under the IDP, project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Information provided in confidence is one of the exceptions, but this exception should not be applied broadly to an entire document if the document contains specific, segregable portions that can be disclosed without prejudice or harm.

Indicate below whether or not the funding proposal includes confidential information.

☒ No confidential information: The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.

☐ With confidential information: The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided in confidence. Accordingly, the accredited entity is providing to the Secretariat the following two copies of the funding proposal, including all annexes:

- full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and
- redacted copy for disclosure on the GCF website.

The funding proposal can only be processed upon receipt of the two copies above, if containing confidential information.

ANNEXES

H.1. Mandatory annexes

☐ Annex 1 NDA no-objection letter(s)

☐ Annex 2 Feasibility study - and a market study, if applicable

☐ Annex 3 Economic and/or financial analyses in spreadsheet format

☐ Annex 4 Detailed budget plan

☐ Annex 5 Implementation timetable including key project/programme milestones

☐ Annex 6 E&S document corresponding to the E&S category (A, B or C; or I1, I2 or I3):
- ☐ Environmental and Social Impact Assessment (ESIA) or
- ☐ Environmental and Social Management Plan (ESMP) or
- ☐ Environmental and Social Management System (ESMS)
- ☐ Others (please specify – e.g. Resettlement Action Plan, Resettlement Policy Framework, Indigenous People's Plan, Land Acquisition Plan, etc.)

☐ Annex 7 Summary of consultations and stakeholder engagement plan

☐ Annex 8 Gender assessment and project/programme-level action plan

☐ Annex 9 Legal due diligence (regulation, taxation and insurance)

☐ Annex 10 Procurement plan

☐ Annex 11 Monitoring and evaluation plan

☐ Annex 12 AE fee request

☐ Annex 13 Co-financing commitment letter, if applicable

☐ Annex 14 Term sheet including a detailed disbursement schedule and, if applicable, repayment schedule

H.2. Other annexes as applicable

☐ Annex 15 Evidence of internal approval

☐ Annex 16 Map(s) indicating the location of proposed interventions

☐ Annex 17 Multi-country project/programme information

☐ Annex 18 Appraisal, due diligence or evaluation report for proposals based on up-scaling or replicating a pilot project

☐ Annex 19 Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity

☐ Annex 20 First level AML/CFT (KYC) assessment

☐ Annex 21 Operations manual (Operations and maintenance)

☐ Annex 22 Assessment of GHG emission reductions and their monitoring and reporting (for mitigation and cross cutting-projects)[125]

☐ Annex 23 Note on beneficiaries

** Please note that a funding proposal will be considered complete only upon receipt of all the applicable supporting documents.*

[125] Annex 22 is mandatory for mitigation and cross-cutting projects.

No-objection letter issued by the national designated authority(ies) or focal point(s)

MINISTRY OF METEOROLOGY, ENERGY,
INFORMATION, DISASTER MANAGEMENT,
ENVIRONMENT, COMMUNICATIONS,
CLIMATE CHANGE AND CERT (MEIDECC)
NUKU'ALOFA, TONGA

Ref: DCC/11/23
To: The Green Climate Fund ("GCF")
Date: 22 November, 2023.

Re: Funding proposal for the GCF by the United Nations Development Programme regarding the Coastal Resilience Project in Tonga.

Dear Sir/Madam,

We refer to the project titled *GCF Coastal Resilience Project* in Tonga as included in the funding proposal submitted by the United Nations Development Programme to us on 8 November, 2023.

The undersigned is the duly authorized representative of Ministry of Meteorology, Energy, Information, Disaster Management, Environment, Climate Change and Communications (MEIDECC), the National Designated Authority of Tonga.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the project as included in the funding proposal.

By communicating our no-objection, it is implied that:

(a) The Government of Tonga has no-objection to the project as included in the funding proposal;
(b) The project as included in the funding proposal is in conformity with the national priorities, strategies, and plans of Tonga;
(c) In accordance with the GCF's environmental and social safeguards, the project as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the project as included in the funding proposal has been duly followed.

We acknowledge that this letter will be made publicly available on the GCF website.

Yours Sincerely,

MINISTRY OF METEOROLOGY, ENERGY, INFORMATION, DISASTER MANAGEMENT, ENVIRONMENT, CLIMATE CHANGE & COMMUNICATIONS
MEIDECC TONGA

Mr. Sione Pulotu 'Akau'ola.
Tonga NDA to GCF, and
Chief Executive Officer for MEIDECC.
TONGA.

Environmental and social safeguards report form pursuant to para. 17 of the IDP

Basic project or programme information	
Project or programme title	Tonga Coastal Resilience
Existence of subproject(s) to be identified after GCF Board approval	No
Sector (public or private)	Public
Accredited entity	United Nations Development Programme (UNDP)
Environmental and social safeguards (ESS) category	Category B
Location – specific location(s) of project or target country or location(s) of programme	Kingdom of Tonga Coastal protection measures implemented in: Hahake Coast, Tongatapu
Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	Tuesday, June 11, 2024
Language(s) of disclosure	English and Tongan
Explanation on language	English and Tongan are both official languages of Tonga.
Link to disclosure	English: https://undpngddlsprod01.blob.core.windows.net/pdc/01001855-PPMFP-UNDP-Tonga-20240426-5942-Annex%206b-ESMF_Rev1a-Clean.docx Tongan: https://www.undp.org/sites/g/files/zskgke326/files/2024-07/gcf_tonga_coastal_adaptation_project_tonga_only_draft_1.pdf
Other link(s)	**Link to project website** https://www.undp.org/pacific/projects/tonga-coastal-resilience **Environmental and Social Management Framework Summary** English: https://undpngddlsprod01.blob.core.windows.net/pdc/01001855-PPMExecutive%20Summary%20for%20Translation%20(1).docx Tongan: https://undpngddlsprod01.blob.core.windows.net/pdc/01001855-PPMExecutive%20Summary%20for%20Translation%20_Tongan%20version%20only_Draft%203%20_incl%20footnote.pdf
Remarks	An outline of the ESIA consistent with the requirements for a Category B project to be conducted during

	implementation is contained in the "Environmental and Social Management Framework (ESMF)"
Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity's website	Tuesday, June 11, 2024
Language(s) of disclosure	English and Tongan
Explanation on language	English and Tongan are both the official languages of Tonga.
Link to disclosure	English: https://undpngddlsprod01.blob.core.windows.net/pdc/01001855-PPMFP-UNDP-Tonga-20240426-5942-Annex%206b-ESMF_Rev1a-Clean.docx Tongan: https://www.undp.org/sites/g/files/zskgke326/files/2024-07/gcf_tonga_coastal_adaptation_project_tonga_only_draft_1.pdf
Other link(s)	**Link to project website** https://www.undp.org/pacific/projects/tonga-coastal-resilience **Environmental and Social Management Framework Summary** English: https://undpngddlsprod01.blob.core.windows.net/pdc/01001855-PPMExecutive%20Summary%20for%20Translation%20(1).docx Tongan: https://undpngddlsprod01.blob.core.windows.net/pdc/01001855-PPMExecutive%20Summary%20for%20Translation%20_Tongan%20version%20only_Draft%203%20_incl%20footnote.pdf
Remarks	An ESMP consistent with the requirements for a Category B project is contained in the "Environmental and Social Management Framework" for completion during implementation.
Environmental and Social Management System (ESMS) (if applicable)	
Date of disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Any other relevant ESS reports, e.g. Resettlement Action Plan (RAP), Resettlement Policy Framework (RPF), Indigenous Peoples Plan (IPP), Indigenous Peoples Planning Framework (IPPF) (if applicable)	

Description of report/disclosure on accredited entity's website	Social and Environmental Safeguards Procedure (SESP) / Tuesday, June 11, 2024
Language(s) of disclosure	English and Tongan
Explanation on language	English and Tongan are both official languages of Tonga. The translated document includes the risks as captured in the SESP.
Link to disclosure	English: https://undpngddlsprod01.blob.core.windows.net/pdc/01001855-PPMDocuSign_Tonga%20GCF_SESP_Sep%202023_Nov%208.docx%20(1).pdf Tongan: https://www.undp.org/sites/g/files/zskgke326/files/2024-07/01001855-ppmdocusign_tonga_gcf_sesp_sep_2023_nov_8.docx_1_tonga_only_draft_1.pdf
Other link(s)	**Link to project website** https://www.undp.org/pacific/projects/tonga-coastal-resilience
Remarks	The SESP is a UNDP document that is used for identifying social and environmental risks and mitigation measures. It does not replace or conflict with GCF ESS requirements and policies.
Disclosure in locations convenient to affected peoples (stakeholders)	
Date	12 June 2024 (for English documents and translation of summary documents) 3 July 2024 (for translations of full documents)
Place	1. Department of Climate Change of the Kingdom of Tonga (the main Government Office for the project and where the PMU will be located. It houses physical copies of previous and on-going climate change related project documents within their library which is accessible by members of the public.) MEDIECC, Level 3, Snaft Building, Taufa'ahau Road. Nuku'alofa 21°08'03.4"S 175°12'02.7"W 2. Tongatapu 10 Constituent and Member of Parliament Office. Manuka. (The project site for the coastal protection measures is within this Constituency. The Constituency Office houses the Constituent elected Member of Parliament together with staff. The Office is located on the outskirts of Manuka, within close proximity to the project site. The documents displaced within the Office is accessible to members of the public.) 21°07'51.7"S 175°07'16.4"W 3. Mo'unga'olive College. Taufa'ahau Road. Kolonga. (The school is located in Kolonga where some of the coastal

	protection measures will be undertaken, on the main Taufa'ahau Road, and is easily accessible to the public. The documents will be available from the schools' main office.) 21°07'41.0"S 175°03'49.7"W
Date of Board meeting in which the FP is intended to be considered	
Date of accredited entity's Board meeting	N/A
Date of GCF's Board meeting	Monday, July 15, 2024

Note: This form was prepared by the accredited entity stated above.

* Subsequent to the disclosure of this form to the Board and active observers on 14 June 2024, the following update has been made: The Environment Social Management Framework (ESMF) has been disclosed in Tongan on 10 July 2024.

Secretariat's assessment of FP234

Proposal name:	Tonga Coastal Resilience
Accredited entity:	United Nations Development Programme
Country:	Tonga
Project size:	Small

I. Overall assessment of the Secretariat

1. The funding proposal is presented to the Board for consideration with the following remarks:

Strengths	Points of caution
The project addresses multiple aspects of Tonga's long-term adaptation needs, including extensive community engagement, developing the country's first spatial adaptation plans, undertaking dynamic surveys using LiDAR (light detection and ranging) technologies to collect data and providing extensive capacity-building to analyse and translate data to support Tonga's long-term adaptation planning while deploying physical construction in highly exposed and vulnerable sections of the coastline. This integrated approach acknowledges the linkages and complementarity of various factors and interventions contributing to climate resilience.	The project is being implemented under the Direct Implementation Modality (DIM) as per United Nations Development Programme (UNDP) operational modalities. Hence, the Secretariat strongly recommended that UNDP, as accredited entity (AE), ensure proactive and continuous engagement with national and local authorities to assure full ownership and involvement of all main stakeholders, which is critical for the success of the project.
Strong focus on capacity-building to address gaps on how data will be integrated into government processes and decision-making. A strong cohort of government officials will be trained and their capacity strengthened to interpret, analyse and translate climate data to support evidence-based and climate risk informed planning and decision-making.	The sustainability of the sea defence infrastructure component will require regular and routine maintenance and allocation of budget after construction, which may not be readily available. The Secretariat has engaged with the AE on measures to ensure that the necessary strategy and plans and committed resources are in place for the long-term sustainability of these interventions, which is essential to maximize its effectiveness and to ensure its sustainability.
Strong local community participation and capacity-building will be promoted to ensure long-term sustainability and replicability.	Specific to Activity 3.1, the lack of and/or limited experience of the AE in implementing sea defence construction and in-house

Local community involvement and national consultations ensure that the project is informed by local knowledge, needs and priorities, which helps to enhance its relevance and effectiveness in addressing local challenges.	engineering capacity. Hence, the Secretariat recommended that the AE pay close attention to the design and supervision of the project, which should be done by an experienced engineering consultancy firm, and likewise the contracting of an experienced construction firm to implement this component.
The country's strong ownership has been demonstrated through the government's strong commitment throughout the project development phase and its advocacy and commitment to the post-construction operation and maintenance responsibilities.	The interventions under Output 3 are limited to hard engineering approaches since neither of the two project sites identified entails nature-based solutions such as coastal ecosystem restoration and conservation (e.g. mangroves).

2. The Board may wish to consider approving this funding proposal in accordance with the term sheet agreed between the Secretariat and the accredited entity, and, if considered appropriate, subject to the conditions set out in Annex II of document GCF/B.39/02.

II. Summary of the Secretariat's assessment

2.1 Project background

3. Climate change poses significant threats to Tonga, particularly through increased rainfall, tropical cyclones, sea level rise and extreme sea level events, necessitating adaptation measures for people, infrastructure and coastal ecosystems. Tongatapu, Tonga's most populous island, is especially at risk of frequent flooding and sea level rise induced permanent losses without adaptive interventions. Recognizing these challenges, a project proposed by the United Nations Development Programme (UNDP) aims to mitigate coastal inundation on Tongatapu through immediate coastal protection and long-term adaptation strategies.

4. The overall objective for the project is to build the long-term resilience of vulnerable coastal communities to the direct impacts of climate change in Tonga and build transformative adaptation capacities through strategic infrastructure work, long-term climate risk informed adaptation planning and engagement on long-term adaptation solutions.

5. The project cost for the proposed project is USD 23,919,409 in grants, of which GCF investment is USD 22,656,409 in grants.

6. The project is expected to reach 100,179 people (100 per cent of the population in Tonga), with 3,608 as direct beneficiaries.

7. The environmental and social safeguards (ESS) category for this project is B (medium).

2.2 Component-by-component analysis

8. The project is structured in three components and activities as follows:

Component 1: Strengthened knowledge, capacity and engagement for long-term adaptation planning supported through a multisectoral, multi-stakeholder engagement and dialogue platform (total cost: USD 4.21 million; GCF cost: USD 3.48 million in grants)

9. Component 1 activities are (1.1.) establish a national multi-stakeholder engagement platform for dialogue on co-creating long-term climate change adaptation strategies and solutions, including voluntary retreat; (1.2) develop village- and district-level participatory climate risk informed plans; and (1.3) build the capacity of local government, village committees and non-governmental organizations to integrate climate risks and adaptation needs into community-level planning, and inform future community development plans.

Component 2: Strengthened national and local capacities for effective monitoring and assessment of climate risks (total cost: USD 7.72 million; GCF cost: USD 7.46 million in grants)

10. Component 2 activities are (2.1) strengthen the mechanisms for collecting and analysing data and information for better-informed climate risk monitoring and coastal adaptation planning; and (2.2) improve the knowledge base of multisectoral, multiple stakeholders on adaptation planning strategies for long-term resilient planning and transformative adaptation for Tonga based on climate risks and projections.

Component 3: Reduced vulnerabilities of coastal communities in Hahake to climate hazards through coastal protection measures (total cost: USD 10.98 million; GCF cost: USD 10.7 million in loans)

11. Component 3 activities are (3.1) coastal protection measures constructed along 4 km coastline in Hahake; and (3.2) sharing lessons learned and best practices in climate-resilient coastal protection measures for scale-up at the national and regional level. The evaluation of the design for coastal adaptation and protection measures in Sub-activity 3.2.1 will inform the guidelines and standards to be developed as part of Sub-activity 3.2.3.

Project management (total cost: USD 1.01 million; GCF cost: USD 1.01 million in grants)

12. This activity refers to the support provided to a dedicated project management unit, (PMU) which is responsible for overseeing day-to-day execution activities under all components, including procurement and financial management. The PMU also conducts regular monitoring and evaluation of all activities implemented by the project.

III. Assessment of performance against investment criteria

3.1 Impact potential *Scale: N/A*

13. The overall objective of the project is to support the transformative adaptation agenda setting through a participatory process that builds policy consensus and promotes planning coherence. Towards this end, the project will focus on the bottom-up development planning process and linking development planning with spatial planning and land-use management to promote transformative adaptation.

14. The project will support 3,608 direct beneficiaries and 96,571 indirect beneficiaries, which together make up the entire population in Tonga. Half of all beneficiaries are women.

15. The figure of 3,608 direct beneficiaries represents 300 community members being trained in climate risk and adaptation for developing the community development plans and island-level spatial plans, 50 government officials receiving training in incorporating climate risks in planning and budgeting, and technical training on climate risk monitoring and data, and 3,258 individuals living directly adjacent to the coastal resilience works under Output 3.

16. Furthermore, the adaptation planning and strategies of the community development plans and island-level spatial plans and the multi-stakeholder engagement platform, along with the strengthening of the climate data analysis and adaptation planning for coastal adaptation, will indirectly benefit 96,571 people, ensuring that the project impacts Tonga's entire population both directly and indirectly.

3.2 Paradigm shift potential *Scale: N/A*

17. This project will tackle a range of existing challenges in Tonga, including complex land ownership, insufficient data, limited capacity to interpret climate risks and a lack of mechanisms for decision-making on future land-use planning with a climate perspective to fundamentally shift how Tonga addresses climate risks. The project approach to the sea defence infrastructure to be constructed adopts a hold the line approach while promoting longer-term adaptation measures such as land-use planning.

18. The project has potential for scaling up and replication at the national and regional level through the development of spatial adaptation plans and policies that are widely accepted by the public and politicians, enhancing processes to collect and analyse critical data over time. The project will also empower communities to direct their futures through community development plans and build practical coastal protection measures using locally sourced materials along the Hahake coastline.

19. It also has high potential for knowledge-sharing and learning across multiple stakeholders under all components.

3.3 Sustainable development potential *Scale: N/A*

20. The project contributes to Sustainable Development Goals 13 (Climate action) and 11 (Sustainable cities and communities).

21. The major social co-benefit is enhancing the overall well-being of the Tongan people. The project interventions aim to minimize disruptions to daily life and livelihoods, including economic activities, education, public and medical services, and cultural events. Additionally, incidents of coastal inundation and waterlogging directly impact public health by increasing waterborne diseases and overburdening the public health facilities in Tonga. Implementation of the funded activities will decrease incidence of coastal inundation and water accumulation, leading to improved public health in Tonga.

22. The main environmental co-benefit is increased biodiversity and reduced salinity levels in groundwater. Wave overtopping events and subsequent waterlogging can be a cause of saline water intrusion into the fragile groundwater resources in Tongatapu, where groundwater is still used for drinking purposes, or aggravate bacterial contamination by accelerating seepage from septic tanks and livestock pens.

23. The project also delivers gender co-benefits through a focus on ensuring strong participation of women and youth in the national dialogue, and related consultation and management processes, including developing long-term adaptation solutions through design thinking workshops, enhancing the capacity of women to shape local development and spatial planning.

3.4 Needs of the recipient *Scale: N/A*

24. Settlements in Tonga are concentrated in coastal areas and communities earn their livelihoods within one km of the coast. As a result, the livelihoods and critical infrastructure, such as hospitals, government buildings, schools and workplaces, located in the coastal zone are at risk of coastal erosion caused by inundation during storms. However, current budgetary capability of the government limits the long-term adaptation planning and infrastructure development that requires high upfront investment.

25. In addition, the main capacity gap in construction works has been identified, namely poor attention to design, construction methodology and materials selection. This project,

therefore, intends to address the key gap via a learning-by-doing approach, by seeking to use local suppliers and managing their work under competent coastal engineering supervision and undertaking a best practice work strategy that closely adheres to international standards.

26. Thus, GCF concessionality is expected to address the financial and technical capacity barriers.

3.5 Country ownership *Scale: N/A*

27. This project is included in Tonga's country programme.

28. The project supports the government policy in coastal adaptation and climate informed development planning and is in line with the national adaptation plan and other sectoral plans.

29. The project is strongly aligned with the country's nationally determined contribution objectives and the Joint National Action Plan 2 on Climate Change and Disaster Risk Management, and co-financing demonstrates the strong commitment of the Government of Tonga to the project as a top priority for the country.

30. No-objection letters have been issued by the country, confirming this support.

3.6 Efficiency and effectiveness *Scale: N/A*

31. The total project cost is USD 23.92 million with a GCF contribution of USD 22.66 million, benefiting the entire population of the country.

32. The economic analysis was carried out by computing net present value with the assumed 6 per cent discount rate (guided by the Asian Development Bank, a key development partner in Tonga) and economic internal rate of return. Given the nature of the public goods, the project itself will not generate any cash flow and, therefore, financial internal rate of return is not calculated.

33. The economic analysis focuses on estimating the expected benefits to be delivered by Output 3.1 (engineered flood protection measures) for which benefits can be clearly identified and quantified, while accounting for the whole cost (the cost of all activities) of the project. The economic internal rate of return is estimated at 8.38 per cent, which is above the social discount rates of 6 per cent.

34. According to the International Monetary Fund 2023 Article IV Consultation, Tonga's high vulnerability to natural disasters complicates its efforts to create the fiscal space necessary to finance development spending. Further, under the fiscal year 2024 budget, the primary balance is expected to revert to a deficit. Tonga faces the dual challenge of addressing immediate reconstruction needs while lacking the fiscal space to invest in long-term adaptation and resilience measures.

35. To ensure effectiveness, a selection of tested, locally appropriate and available options will be used, based on site-specific assessments and public consultations. In addition, strong local community participation and capacity-building will be promoted to ensure long-term sustainability for coastal adaptation planning and monitoring.

IV. Assessment of consistency with GCF safeguards and policies

4.1 Environmental and social safeguards

36. **Project background.** The project aims to increase resilience of vulnerable coastal communities in the country through building resilient infrastructure and capacities on climate risk information and adaptation planning. The project will include the construction of 4km of physical protection of vulnerable coastlines to reduce the impact of wave energy and/or soil erosion in the Northeastern Tongatapu (Hahake) and will be equipped with rock revetments preventing erosion, wave overtopping and/or subsequent coastal flooding. It will also undertake activities that will strengthen national and local capacities for effective monitoring, maintenance and community adaptation actions with respect to monitoring and the status of ecosystem functions; mapping and modelling coastal scenarios; and monitoring of coastal/marine ecosystems. Among the environmental and social co-benefits include improved coastal protection and increased biodiversity. The infrastructure intervention is expected to contribute to improved health and safety and quality of potable water as it will reduce the salinity level of the groundwater which is used for drinking by the community. By protecting the coast from inundation and erosion and wave overtopping that could result to waterlogging, the community will experience decreased incidents of water borne diseases.

37. **Environmental and social risk category and safeguard instrument.** The project has been screened against the AE's social and environmental screening process and deemed it equivalent to GCF category B given that the project does not require any land acquisition and/or resettlement and the activities will have potential risks/impacts that can be readily mitigated. An Environmental and Social Management Framework (ESMF) has been prepared which sets out the principles, rules, guidelines, and procedures for screening, assessing, and managing the potential social and environmental impacts of the interventions under the project. It also specifies the most likely applicable social and environmental policies and requirements, as well as how those requirements will be met through the conduct of screening, assessment, approval, mitigation, monitoring, and reporting of the project's social and environmental performance.

38. Compliance with GCF's Environmental and Social Safeguards (ESS) Standards. The discussions below describe how the project complies with the interim GCF standards:

39. **ESS1. Assessment and Management of Environmental and Social Risks and Impacts.** The main intervention that is expected to have potential limited adverse environmental and social impacts is Sub-Activity 3.1.2 (Construction of coastal protection measures in Hahake). An ESMF was prepared which will guide the project implementors in the conduct of safeguards related activities during implementation. An Environmental and Social Impact Assessment will be undertaken, and corresponding management plans will be prepared for the project and will be publicly disclosed.

40. **ESS2. Labor and Working Conditions.** The construction of the coastal protection measures will entail the use of local labour with minimal expectation of labour influx from the outside. Workers could be exposed to occupational health and safety risk particularly those who may be working on quarries and construction-related activities. Workers will be trained on occupational health and safety issues and will be provided with personal protective equipment. For potential labour-related concerns, a worker-based grievance redress mechanism (GRM) including a sexual exploitation, abuse and harassment (SEAH) Action Plan will be established and implemented.

41. **ESS 3. Resource Efficiency and Pollution Prevention.** Construction-related activities may result to generation of noise and vibration, dusts and sediments, effluent discharges to the sea, and generation of solid and liquid wastes. Soil and drainage excavations will potentially lead to soil erosion and loss in soil productivity. However, whenever feasible, most of the excavations will be reused in the establishment of the coastal structures. Sediment and erosion control measures will also be employed to prevent sedimentation in the nearby marine habitats. A water quality monitoring program will also be developed for the project.

42. **ESS4. Community Health, Safety and Security.** Given the anticipated increase in construction-related works in the area, the community may also experience increased safety risks. For instance, there will be increased vehicular and machinery movements around and within the construction sites and the community may be exposed to traffic hazards. The project plans to mitigate this by fencing work areas to minimize public access, training drivers to observe proper driving protocols, and inform nearby communities of ongoing construction activities through its continuous stakeholder engagement. Construction activities will also be limited during the daytime hours. Community management committees (CMCs) will also be formed to set up O&M funds for the maintenance of the infrastructure to prevent it from collapse that could result to accidents.

43. **ESS5. Land Acquisition and Involuntary Resettlement.** The project is not expected to acquire any land and/or result to resettlement as the land where the project activities will be undertaken is owned by the government. Should there be economic displacement that may be encountered during implementation such as temporary disruption or restriction to access the shorelines that could affect their livelihoods, the project has provided guidance to develop a livelihood action plan to ensure that the living standards of economically displaced persons are improved or at least restored.

44. **ESS6. Biodiversity Conservation and Sustainable Management of Living Natural Resources.** The project is not expected to have significant adverse direct impacts on flora and fauna in the target areas. Nevertheless, the installation of coastal protection infrastructure could potentially result to sedimentation of the marine habitat causing entrainment, impingement and/or entrapment of benthic macro-fauna and other coastal/marine organisms. Specific actions that may be considered include ensuring minimal vegetation clearance, avoiding introduction of new invasive/weed species during construction activities. The project will also consult with relevant organizations such as the Fanga'uta Stewardship Plan Steering Committee and the CMCs to be to develop feasible management strategies for the area. The ESMF also provide for the development of a Biodiversity Action Plan that can undertake biodiversity-related actions under the project.

45. **GCF Indigenous Peoples Policy and ESS 7 Indigenous Peoples.** Although there is no specific Indigenous Peoples status in the Tongan national political and legal framework, Tongans are considered under the AE and GCF standards as Indigenous Peoples. A stakeholder engagement plan was developed during the project's design phase. It will guide all actions pertaining to social and environmental safeguards implementation. It will be completed by a free, prior and informed consent protocol, to be developed together with local communities and Indigenous Peoples to enable communities to get extensive information about the project and associated possible positive and negative consequences. They will be encouraged and given the time to explicitly reflect on this information to be able to give their free, prior and informed consent. The protocol will then be applied to each activity of the project, as communities will be allowed to provide their consent to part of them, ask for modifications or withdraw their consent. In line with its roles and functions, the GCF Indigenous Peoples Advisory Group is available to provide advice to the AE, which is also encouraged to share emerging good practices and success stories with the Group.

46. **ESS8. Cultural Heritage**. While there are no known cultural heritage places, buildings or monuments that are identified in the project areas, a chance finds procedure will be developed as part of the construction stage ESMP and a cultural heritage awareness training will be provided to all construction site personnel (including contractors).

47. The AE screened the project's potential risks and impacts regarding SEAH safeguarding against a set of high-level indicators (as per annex 2 to the environmental and social management framework) and identified moderate SEAH risks of the proposed project given its regular interactions with project actors at information and training sessions and construction work. Following the SEAH risk screening by the AE, general mitigation measures have been

formulated with actions, timeline, responsible party and monitoring activities in a SEAH action plan, as part of the project's ESMF. The AE will address SEAH complaints through the project's grievance redress mechanism using a survivor-centred approach. Noting that SEAH response mechanisms were set up following the launch of Tonga's National Service Delivery Protocol, the project will work with Tonga's Women and Children Crisis Centre to assist the project to support SEAH survivors. Where additional support is deemed necessary, the AE will ensure that such support is made available.

48. **Implementation arrangements.** At the project level, the project will have a Project Board (also called the Project Steering Committee) whose function includes risk management and providing guidance on project risks including on compliance with environmental and social requirements of co-financed activities or activities taking place in the project's area of influence that have implications for the project. A project management unit (PMU) will also be established to plan, oversee, execute, evaluate, and report on the progress and performance of the project (including on safeguards) to the PSC and other stakeholders. The PMU's function includes ensuring that the project develops, implement and monitor the environmental and social management and associated plans as well as that a functioning GRM is established. The PMU will also ensure that capacity building activities of all relevant stakeholders on safeguards related concerns are undertaken. Fulltime in-country safeguards officer for construction and gender and safeguards officer will be employed and will be supported by part-time international safeguards specialist.

49. **Stakeholder engagement and GRM.** The AE has engaged with various government agencies, NGOs, development partners and the communities through information sessions, consultations and validation workshops. A stakeholder engagement plan is also prepared which outlines the strategy to communicate with stakeholders of the project including methods and techniques for engagement and monitoring and reporting provisions. A dedicated Communications Officer will also be engaged in the PMU to verse stakeholder engagement activities. A GRM has been included in the ESMF to address any complaints and/or grievances related to environmental and social safeguards. A two-tier system is provided where in addition to the project-level and national grievance redress mechanisms, aggrieved parties may also access the AE's Accountability Mechanism, which have both compliance and grievance functions. Grievances and complaints may also be raised with the GCF independent redress mechanism. In line with the GCF Indigenous Peoples Policy, the GCF indigenous peoples focal point will be available for assistance at any stage, including before a claim has been made.

4.2 Gender policy

50. The AE provided a gender assessment and action plan with the funding proposal and therefore complies with the requirements of the GCF Gender Policy. The project is designed to support the Government of Tonga's efforts to enhance coastal resilience and provide medium-term vulnerability reduction and adaptation benefits in the Hahake region as well as strengthen the overall adaptation capabilities of the government and the people of Tonga. As such, 3,609 people (of whom 1,804 are women) will benefit directly from activities in vulnerability reduction and/or resilience-building activities in Tongatapu.

51. The project-specific gender assessment and gender action plan provide an overview of the context in Tonga and highlights the intersectionality between gender equality, climate change and the economy, which are relevant to this project context. It further identifies gender mainstreaming opportunities and activities across the three output areas. The assessment is based on a desk review and stakeholder consultations with the Women's Affairs and Gender Equality Division of the Ministry of Internal Affairs and civil society organizations. It also leveraged on lessons learned and recommendations from past assessments undertaken by United Nations agencies, development partners, civil society organizations, etc. A range of

gender-specific activities, outputs and targets have been presented and the impact statement for the gender action plan is “gender responsive and participatory adaptation solutions enhance immediate and medium-term resilience, and long-term transformative adaptation capacities of women and girls in Tonga”. The intended outcome of the gender action plan is improved capacities and participatory spaces for women and girls to promote gender-responsive adaptation solutions.

52. Implementation of the gender action plan will be led by the AE, UNDP, which will work in close collaboration with responsible parties, namely the government ministries and departments, including the Women’s Affairs and Gender Equality Division of the Ministry of Internal Affairs. The PMU will be based in Tonga and will work with an international gender specialist.

4.3 Risks

4.3.1. Project-specific execution risks

53. Macroeconomic risk: market factors (inflation and currency variation) may trigger inflation on costs of the goods, services and labour that are essential to delivering the project activities and intended impact. The AE has adequately conducted first-level due diligence on project costs and risks in the funding proposal. The Secretariat will monitor budget utilization through the annual performance reports. In addition, the term sheet includes a clause to allow room for reallocation of budget lines whereby cost savings in one item may be utilized to cover price increases in another, to be approved by the Secretariat if more than 10 per cent variation of the agreed budget, except for Output 3.

54. Natural hazards: Tonga is prone to extreme events, including cyclones, earthquakes and tsunamis, that could expose the construction works during the implementation period to damage, destruction and/or failure and result in delays and cost overruns. As part of the project design, risk mitigation measures are in place such as the contract between the AE and the contractor to be engaged, which entails the use of lump-sum contracts and requirements for relevant insurances and performance bonds, as well as the project budget, to include provision for contingencies.

55. Implementation risk due to limited amount of workforce/labour, equipment and construction materials (rocks for revetment) available in Tonga for the type of construction work required in Hahake coast: capacity limitation of the Government of Tonga to implement the specific activities of the project is mitigated by the AE using the Direct Implementation Modality (DIM), where UNDP is also the executing entity, and by the project governance structure whereby the PMU will support the execution of project activities and will carry out quality assurance and contract management oversight along with UNDP. As part of project design, the AE has undertaken preliminary assessments in terms of availability and quality of construction material (rocks) and will continue close monitoring of the availability during the project approval and implementation phases, and the construction timeline allows for staggering of construction between the two areas on the Hahake coast.

56. Technical/implementation risk specific to Activity 3.1 (around 40 per cent of GCF financing): the construction work may not meet the required quality standards, leading to additional costs, project completion delays, challenges in proper operation and maintenance and safety issues partly owing to the lack of and/or limited experience of the AE in implementing sea defence construction and in-house engineering capacity. As mitigants, there will be separate design and construction phases, which will be implemented by an experienced engineering consultancy firm and an experienced construction firm respectively. In addition, the term sheet of the AE will ensure that the detailed technical assessments and relevant studies are

carried out prior to the commencement of any construction works and are reflected in the term sheet as a condition for first disbursement.

4.3.2. Compliance risk (medium risk)

57. Following a comprehensive assessment of various risk factors outlined in the proposal, it is evident that several robust measures are in place to mitigate compliance risks effectively. The proposal outlines a meticulous approach to addressing potential risks related to money-laundering, financing of terrorism, prohibited practices and other integrity concerns.

58. One significant aspect of the risk assessment pertains to the implementation of the project under the DIM by UNDP, leveraging its policy framework and adherence to United Nations principles. UNDP systematic screening processes against the United Nations Security Council Consolidated List, coupled with due diligence assessments such as the Partner Capacity Assessment Tool and the Harmonized Approach to Cash Transfers framework, demonstrate a proactive stance in identifying and mitigating risks associated with project partners and financial transactions.

59. Furthermore, the integration of anti-money-laundering/countering the financing of terrorism policies within the UNDP Enterprise Risk Management framework underscores a commitment to combating financial illicit activities. The emphasis on transparency, accountability and zero tolerance towards fraud and corruption is reiterated throughout various policies and procedures, including the UNDP Anti-Fraud Policy and the United Nations Supplier Code of Conduct.

60. Additionally, the establishment of grievance redress mechanisms, including provisions for addressing concerns related to social and environmental sustainability, underscores a commitment to accountability and stakeholder engagement. These mechanisms not only provide avenues for addressing grievances but also promote transparency and inclusivity in project implementation.

61. In conclusion, the Secretariat's review acknowledges the comprehensive risk mitigation measures outlined in the proposal, which align with Board-approved policies. While recognizing the inherent risks associated with such projects, the Secretariat has determined a medium risk rating based on the available information and finds no material issues or deviations warranting objection to the proposal's advancement. This assessment reflects a balanced understanding of the compliance landscape and underscores confidence in the proposal's ability to navigate potential risks effectively.

4.3.3. GCF portfolio concentration risk (low risk)

62. In the case of approval, the impact of this proposal on the GCF portfolio concentration in terms of results area and single proposal is not material.

Summary risk assessment	
Accredited entity/executing entity capability	Medium
Project-specific execution	Medium
GCF portfolio concentration	Low
Compliance risk	Medium

4.4 Fiduciary

63. Following a comprehensive review of the funding proposal for the Tonga Coastal Resilience funding proposal, several key financial management aspects have been carefully examined and evaluated.

64. First, the funds flow structure and reflow mechanisms outlined in the proposal suggest an appropriate implementation process, particularly regarding procurement, transport and construction of coastal protection measures. The use of responsible parties and appropriate instruments such as letters of agreement and responsible party agreements signifies a structured approach to resource allocation and accountability. Moreover, the direct involvement of UNDP through the DIM arrangement, coupled with its oversight responsibilities, reinforces the project's integrity and adherence to established standards.

65. Second, the differentiation of responsibilities between AEs and executing entities is clearly delineated, with UNDP serving as the implementing partner under the DIM modality. The role of UNDP in project planning, coordination, financial management and oversight underscores its commitment to effective execution while ensuring alignment with GCF guidelines and regulations. Additionally, the inclusion of responsible parties, government entities and civil society stakeholders enhances project ownership and fosters inclusive decision-making processes.

66. Third, the financial instrument composition, including grants and co-financing, reflects a strategic approach to resource mobilization, with contributions from the Ministry of Finance facilitating broader financial support for project activities. The mechanisms for refunding taxes, as detailed in the project documentation, ensure compliance with relevant taxation regulations, thereby mitigating financial risks and optimizing resource utilization.

67. Furthermore, rigorous risk assessment and mitigation measures are embedded throughout the proposal, encompassing procurement ethics, fraud prevention and internal audit mechanisms. UNDP adherence to procurement policies and financial regulations, coupled with its Enterprise Risk Management framework, enhances transparency, integrity and accountability across project implementation phases.

68. Moreover, the economic and financial analyses provide valuable insights into Tonga's vulnerability to climate change and the critical role of climate finance in fostering resilience-building efforts. By addressing knowledge gaps and enhancing national capacities, the project aims to facilitate informed decision-making and optimize resource allocation for sustainable adaptation measures.

69. In conclusion, the comprehensive financial management assessment underscores the robustness of the proposal’s financial framework, emphasizing accountability, transparency and stakeholder engagement. Moving forward, continued adherence to established protocols, effective oversight mechanisms and adaptive management practices will be essential for realizing the project's objectives and maximizing its impact on building climate resilience in Tonga.

4.5 Results monitoring and reporting

70. The proposal has formulated a theory of change showcasing the major climate adaptation pathways to achieve the goal statement for the project. It has also listed some important assumptions and risks that are to be monitored to ensure that the project outcomes related to transformative adaptation are realized. The logical framework is aligned with the Integrated Results Management Framework encompassing paradigm shift potential and enabling environment. The logical framework has selected two core adaptation indicators to measure complementary adaptation benefits in terms of beneficiaries and assets made resilient. Regarding the resilient assets indicator, the AE will be conducting a further estimation study

during the project implementation and will propose a reliable target. Project-specific indicators are complementary and will be measuring the overall performance of the project.

71. The AE will be ensuring that the monitoring, reporting and evaluation arrangements will comply with GCF results policies. The PMU and Project Manager will conduct the project's regular monitoring, while the UNDP regional office will oversee the evaluations and ex post technical assessments. An appropriate budget is set aside to cover the costs of monitoring and evaluation, and the AE has submitted an initial monitoring and evaluation plan providing further explanation of the budget calculation.

4.6 Legal assessment

72. The Accreditation Master Agreement was signed with the Accredited Entity on 5 August 2016, and became effective on 23 November 2016, which was amended and restated pursuant to a first amendment and restatement agreement dated 16 March 2023, and which became effective on 3 May 2023 (the “AMA”).The AE has provided a certificate confirming that it has obtained all internal approvals and it has the capacity and authority to implement the project.

73. The Accredited Entity has provided a certificate confirming that it has obtained all internal approvals and it has the capacity and authority to implement the project.

74. The proposed project will be implemented in the Kingdom of Tonga. The GCF has signed a bilateral agreement on privileges and immunities with the Government of the Kingdom of Tonga. It was signed and became effective on 25 May 2017.

75. To address the matters raised in this section, it is recommended that any approval by the Board is made subject to the following conditions:

(a) Signature of the funded activity agreement in a form and substance satisfactory to the Secretariat within 180 days from the date of Board approval; and

(b) Completion of the legal due diligence to the satisfaction of the Secretariat.

Independent Technical Advisory Panel's assessment of FP234

Proposal name:	Tonga Coastal Resilience
Accredited entity:	United Nations Development Programme
Country:	Tonga
Project size:	Small

I. Assessment of the independent Technical Advisory Panel

1. The funding proposal for Tonga Coastal Resilience is for a small public sector adaptation project in environmental and social safeguards risk category B, to be implemented over a 7-year period and with an overall project lifespan of 40 years.[1]

2. The funding proposal is submitted by the Government of Tonga through the Ministry of Meteorology, Energy, Information, Disaster Management, Environment, Climate Change and Communications (MEIDECC) as the national designated authority. The United Nations Development Programme (UNDP) is the accredited entity (AE) and functions as the executing entity. MEIDECC is on board with the use of the Direct Implementation Modality (DIM) approach, under which the project will be directly managed by UNDP. However, project implementation will also be co-located with the Government of Tonga (see para. 49 below). The total cost of the funding proposal is USD 23.919 million, comprising a request for GCF grant funds of USD 22.65 million, with a co-finance contribution of USD 1.2 million from the Government of Tonga and USD 63,000 from UNDP.

3. Tackling climate change is a priority for the Government of Tonga. The aim of the Tonga Coastal Resilience project is to build the long-term resilience of vulnerable coastal communities to the direct impacts of climate change. It proposes to do this through building dialogue platforms to achieve policy consensus, building national capability to manage and monitor climate risks, supporting participatory planning processes and promoting planning coherence at the community, district and island level. This will be backed by engineering works to strengthen coastal protection in two strategic locations along the coastline.

4. The project will focus on the largest and most populous island of Tongatapu, which faces significant flooding risks and permanent losses from sea level rise. Coastal flooding poses a risk to communities that are located on comparatively low-lying land exposed to marine hazards. The risks of flooding are likely to increase with future sea level rise, storm surges or intense rainfall, and more intense cyclones are causing increased vulnerability for housing and infrastructure, and coastal land loss, with flow-on effects on livelihoods and food security.

1.1 Impact potential *Scale: High*

[1] The assessment of the independent Technical Advisory Panel is based on the first submission of the funding proposal and its annexes, as received from the national designated authority on 3 May 2024. The project has been under development since 2017; however, its submission to GCF was delayed by various external events, including the coronavirus disease 2019 pandemic and other disaster recovery efforts in the country.

5. Tonga, a small island developing State (SIDS) located in the western South Pacific, comprises 170 small islands, many of which remain uninhabited. Tonga ranks as the 6th most vulnerable country (out of 185) globally with respect to climate change impacts according to the Notre Dame Global Adaptation Initiative.[2] The funding proposal comprehensively highlights Tonga's extreme vulnerability to coastal inundation and erosion through climate change, extreme weather events and natural disasters.

6. The country has experienced a series of climate disasters in recent years, namely Tropical Cyclones Ian (2014), Gita (2018) and Harold (2020). Located within the seismically active Pacific Ring of Fire, the islands are also affected by intense earthquake and volcanic activities which trigger tsunamis as additional coastal hazards. Recent non-climate-related disasters include the volcanic eruption in 2022, which generated tsunami waves of up to 42 m.

7. The funding proposal draws on historical data and future projections to demonstrate that climate change is leading to an increase in annual average atmospheric and sea surface temperatures and sea level rise, with changes in rainfall patterns and tropical cyclones in the Pacific region. In its response to questions from the independent Technical Advisory Panel (iTAP),[3] the AE clarified the routine use of a long-term approach to sea level rise projections for SIDS using the high-emissions Representative Concentration Pathway scenario of RCP8.5, which estimates sea level rise of 40–87 cm and increases in temperature of around 2.6 °C by 2100.

8. Project design has drawn on key priorities highlighted in Tonga's vulnerability and adaptation assessment, which identifies Tonga's vulnerability to the adverse impacts of climate change and presents the country's exposure and vulnerability to coastal inundation affecting communities, infrastructure and livelihoods. Vulnerable sectors include agriculture, fisheries, coastal areas, water resources, lands, infrastructure, disaster risk management, biodiversity and human health. The vulnerability and adaptation assessment report highlights the vulnerability of the low-lying coastal areas in Tonga and the adaptation measures in place to counter the adverse impacts of sea level rise and storm surges such as coastal erosion and inundation.

9. The funding proposal also refers to studies suggesting that significant displacement of communities could occur in the future. Tonga has a unique system of land ownership and management whereby, under the Constitution of Tonga, all land belongs to the King of Tonga. Decision-making around land use and voluntary retreat as an option for long-term climate adaptation remain sensitive topics. The influx of new settlers from other islands is resulting in ever greater strain on land resources in Tonga's capital city of Nuku'alofa and is increasing settlement into marginal, low-lying and highly flood-exposed areas.

10. The funding proposal notes that project implementation will be carried out in a collaborative manner with several government departments, notably the Department of Climate Change under MEIDECC, the Ministry of Lands and Natural Resources (MoLNR) and the Ministry of Infrastructure. Additionally, staff from other government departments and ministries, including the Ministry of Internal Affairs and the Ministry of Finance, will also benefit from training.

11. The project will be delivered through three interrelated outputs as follows:

(a) Output 1: Strengthened knowledge, capacity and engagement for incorporating climate risks into long-term adaptation planning supported through a multisectoral, multi-stakeholder engagement and dialogue platform;

[2] Available at https://gain.nd.edu/our-work/country-index/rankings/.
[3] The iTAP received written responses to its questions from the AE on 23 May 2024. This was followed by a clarificatory call between the iTAP and AE on 23 May 2024.

(b) Output 2: Strengthened national and local capacities for effective monitoring and assessment of climate risks; and

(c) Output 3: Reduced vulnerabilities of coastal communities in Hahake to climate hazards through coastal protection measures.

12. **Adaptation results areas**: the project aims to deliver impacts against the following two GCF adaptation results areas:

(a) Most vulnerable people and communities (60 per cent of GCF contribution); and

(b) Infrastructure and built environment (40 per cent of GCF contribution).

13. The project is expected to benefit the entire population of Tonga of 100,179 people, comprising 3,608 direct beneficiaries (of which 50 per cent are women) and 96,571 indirect beneficiaries (of which 50 per cent are women). The project also proposes to build the capacities of 300 community members on climate risk and adaptation planning at the local and island level, and to enhance the knowledge and technical skills of 25 government staff in incorporating climate change risks in planning and budgeting and a further 25 government staff on technical aspects of climate risk monitoring and data.

14. The first two outputs could have a medium to high impact and will largely depend on factors such as the absorptive capacity of local stakeholders, political will and the ability of the project to navigate sensitive issues related to the country's land ownership. The selection of project sites for physical coastal protection work under Output 3 was based on the areas considered to be most at risk in the Hahake region of north-eastern Tongatapu. Planned project measures involve installing 2.4 km of rock revetment along a section of the low-lying southern Hahake coastline to prevent flooding; and selected adaptation measures in western Hahake, such as overtopping protection and raising the roadway along 2 km of coastline. This is expected to substantially reduce the risk of flooding in the main villages located in these areas.

15. Adaptation impact potential is considered as high.

1.2 Paradigm shift potential *Scale: Medium to high*

16. The proposal supports Tonga in addressing climate change risks by combining long-term adaptation planning with short- to medium-term measures for minimizing the impacts of sea level rise and cyclones. The logic of the paradigm shift is clearly outlined in the theory of change. The iTAP appreciates that the project's comprehensive and participatory approach to adaptation measures on coastal climate risks takes into consideration the particular needs of Tonga and could potentially be replicable across other islands in the Pacific region.

17. The project starts from a low baseline in Tonga and aims to tackle several barriers, including limited capacities and mechanisms to plan for and manage climate risks and land-use planning at the national and community level, insufficient availability of quality data and limited skill sets to interpret climate risks and to understand coastal adaptation infrastructure.

18. Outputs 1 and 2 lay the groundwork for a fundamental change on how coastal adaptation is managed in Tonga in the face of sea level rise and related climate hazards. The national dialogue platform provides an avenue for finding long-term solutions to tackle sensitive issues such as those related to land-use planning and voluntary retreat (people and infrastructure) due to sea level rise and coastal inundation.

19. While all communities and villages across Tonga have completed community development plans (CDPs), these do not currently include measures to address climate risks. The proposal envisages support to participatory bottom-up planning processes, guidelines and capacity-building, enabling communities to integrate climate adaptation into CDPs. However,

the success of this approach and its scaling up will largely depend on Tonga's success in securing additional funding for implementing these climate activities with CDPs.

20. The input of high-quality data for considering sea level rise or wave impacts is critical to designing and informing adaptation decisions at the national, island and community level. Strengthening national capacities to collect, manage and interpret high-quality data on climate and coastal dynamics is also fundamental to achieving a paradigm shift.

21. The proposal refers to Tonga's weak institutional capacity and highlights the challenges with cross-governmental coordination on a project of this nature and scale (barrier 4). However, the proposal lacks specificity on the extent of involvement of each of these key agencies in the actual delivery of the respective outputs and on how coordination across agencies will work in practice during implementation. This is an important aspect that could affect sustainability of results.

22. Access to technology – such as LiDAR (Light Detection and Ranging) equipment – would also become a key enabler for local authorities to build internal capacity and knowledge in mapping coastal areas, rather than relying on external donor and technical resources for data. Enhanced capacities for coastal management could potentially lead to transformational change in the medium term but will largely depend on the extent to which system capabilities are able to be built and sustained. As part of project sustainability, data would be housed within the regional Pacific Community (SPC) agency, which has the capacity and the mandate for data archiving and analysis services for the Pacific Islands.

23. The proposed approach to involve local suppliers, alongside international technical experts, in construction and materials procurement could help to build and sustain engineering capacity beyond the project. The project also envisages drawing on voluntary services of local communities to help in periodic cleaning of culverts/drainage in north-eastern Tongatapu for the long-term functioning and visual monitoring of the revetment structure, especially after cyclone events, without which the long-term costs of these structures would be higher.

24. The iTAP notes that the project design considered three broad strategic options for addressing coastal vulnerability, namely retreat, accommodate and protect. Drawing on international best practice, the project opted for coastal protection strategies that aim to work with natural processes. These models are site-specific, but emerging best practices could be replicated across other vulnerable coastal areas in Tonga and offer broader learning for the Pacific region.

25. In response to a question from the iTAP, the AE clarified that the project considers two potential scenarios of maladaptation. The first relates to overdesign of coastal protection solutions without taking into consideration that long-term flooding in Hahake and other low-lying areas in Tonga is unavoidable; and the second to unchecked coastal development based on a false sense of safety from coastal protection measures constructed by the project. These risks would be managed through activities under Outputs 1 and 2 to strengthen informed planning and decision-making.

26. The funding proposal proposes to use compatible design specifications based on a 40-year design life for the rock revetments. The environmental effects of building the revetment works are considered to be negligible, and the design proposes to enhance environmental benefits by incorporating hard engineering interventions with coral restoration, mangrove plantation and coastal vegetation. However, the proposal identifies two site areas[4] that may face some transient issues arising from the construction phase, mainly through mobilization of

[4] One site involves revetment works between the villages of Nukuleka and Talafo'ou, within the catchment area of the Fanga'uta lagoon, which is a marine reserve area. The other includes fish habitat reserves that may fall between the two project sites.

beach material (coral sand) from rock placement activities. The proposal outlines measures to manage these risks. Additionally, signage and local communication programmes are intended to support behaviour change and encourage communities to refrain from activities that could undermine project objectives, such as by avoiding the use of invasive plant species.

27. While the proposal offers significant opportunities to deliver a paradigm shift, this will also depend on the ability of the project to deliver systemic change at scale, strengthen coordinated action across government agencies while being implemented by UNDP through its DIM approach, and for community-level adaptation actions to be fully resourced through the updated CDPs. The iTAP assesses the paradigm shift potential as medium to high.

1.3 Sustainable development potential *Scale: High*

28. The proposal highlights the project's contribution to the Sustainable Development Goals (SDGs), most notably to SDG 13 (Climate action). It also notes the project's direct contribution to SDG 11 (Sustainable cities and communities).

29. The proposal has adequately highlighted the co-benefits and potential risks, with more details captured in the environmental and social management framework (annex 6) and the gender assessment and gender action plan (annex 8). It also sets out the institutional arrangements for how these will be managed by the project. A refinement of key documents is proposed in the first six months (inception phase) of project implementation.

30. **Environmental co-benefits**: the project has been subjected to an environmental and social screening assessment and is designated as category B (moderate risk). The project does not require any land acquisition and/or resettlement, and all project activities will take place on land owned by the Crown.

31. The funding proposal lists several wider benefits arising from improved coastal protection works (Output 3), including increased biodiversity, reduced salinity levels in groundwater and improved access to the sea for communities that are dependent on the sea for their livelihoods. Additionally, improved data access under Output 2 could influence the quality and use of information and data in policies, planning and actions at the national, subnational and community level.

32. Annex 6 also highlights some potential risks from the installation of coastal protection infrastructure, which could lead to sediment movement, loss of habitat, changes in hydrodynamic processes, potential increases in erosion, entrainment, impingement and/or entrapment of marine organisms, including in or around protected areas, and damage to the environment from exploitation of quarries from where stones will be sourced. Mitigation measures are also included.

33. **Social and gender co-benefits**: the proposal can bring benefits to the country through the multi-stakeholder dialogue platform that could inform national policies, and directly to communities through the coastal protection and the strengthening of local-level planning to integrate climate and disaster risks. The physical infrastructure improvements could slow down erosion and wave overtopping, and reduce flooding and waterlogging, thereby bringing health benefits to local communities. The protection of the road in north-eastern Tongatapu would benefit communities as it is also used as an evacuation route.

34. There is growing recognition in Tonga of the need for women's political participation and leadership. The National Women's Empowerment and Gender Equality Tonga Policy and Strategic Action Plan (2019–2025) includes "Increased women's leadership and equitable political representation" as one of the five priority outcomes. In terms of gender and coastal resilience, the proposal highlights three areas of benefits: inclusion and participation in

decision-making processes related to coastal resilience (including infrastructure development); livelihoods; and early warning and disaster risk reduction. The proposal indicates that gender will be mainstreamed across all activities, with measures to ensure representation and participation of women and youth in shaping decisions on building climate resilience. The gender assessment notes that the CDP process has been instrumental in expanding women's participation in decision-making, and this approach will continue to inform project activities.

35. While Tonga's national political and legal framework does not delineate a separate status for Indigenous Peoples, the proposal refers to UNDP standards to outline potential risks that could indirectly affect the rights, livelihoods and heritage of Indigenous Peoples, women and youth. The proposal is accompanied by a gender action plan and a stakeholder engagement plan that will guide engagement and consultation across stakeholder groups.

36. **Economic co-benefits**: the funding proposal envisages the creation of 50 short-term jobs (35 male/15 female), particularly for the delivery of construction works under Output 3. It also notes that women and youth will be given priority in these jobs to ensure their representation. However, contractors may have limited capacities or not pay adequate attention to risks for vulnerable populations. In the short to medium term, the improved climate resilience of community infrastructure and assets could help to enhance economic services and operations.

37. The sustainable development potential of the project is assessed as high.

1.4 Needs of the recipient *Scale: High*

38. Tonga has one of the world's smallest economies, as measured by gross domestic product.[5] Its economy is highly dependent on remittances and climate-sensitive sectors such as agriculture, fisheries and tourism. Tonga has a high level of public debt, with an increasing burden of debt repayments and servicing. The country is highly reliant on donor funds, which are mainly used for funding developmental projects. The country is also continuing to recover from the impacts of the coronavirus disease 2019 pandemic.

39. Tonga remains vulnerable to climate and non-climate shocks that put a stress on its already overstretched resources and require higher capital expenditure on climate-resilient infrastructure. The Global Climate Risk Index 2020 cited Tonga as one of the countries experiencing the highest levels of climate-related loss as a percentage of gross domestic product over 1999–2018.[6]

40. Although the incidence of absolute poverty is low in Tonga, many experience economic hardship. According to Tonga's own multidimensional poverty measure, approximately one fifth of the population is considered poor owing to low living standards, with those in rural areas disproportionately worse off. Women also tend to be disproportionately affected by climate change and disaster events, despite receiving early warning information almost on a par with men. A joint assessment by the Kingdom of Tonga and UNDP[7] found that infrastructure projects tended to be gender blind or insufficiently gender sensitive as compared with projects concerned with provision of social services. Land ownership dynamics are sensitive and constrain the development of consensus-driven, climate-informed strategic land-use planning.

41. The population of Tongatapu is projected to grow over the next decade, partly as a result of relocation of communities from the outer low-lying islands because of climate change. This is likely to put more pressure on land resources in the capital city of Nuku'alofa leading to

[5] World Bank. 2023. Available at https://data.worldbank.org/country/tonga?view=chart
[6] Eckstein D et al. *Global Climate Risk Index 2020*..
[7] UNDP. 2016. Tonga Climate Finance and Risk Governance Assessment.

changes in land use and increasing patterns of settlement into marginal, low-lying and highly flood-exposed areas. Hence any development will require a long-term perspective that takes into account high risk areas that are prone to sea level rise and climate change induced flood inundation.

42. Despite having policies in place, government ministries lack adequate technical capacities to collect, analyse and use climate risk data, which limits their ability to mainstream climate risks into national policy design and coastal planning. Consequently, adaptation decisions are taken on an ad hoc basis and often guided by budget availability rather than technical feasibility. There is limited coordination and sharing of climate risk information between key government ministries.

43. Since 2012, Tonga has been promoting a bottom-up development planning process to help to provide an integrated government that is responsive to community needs. To date, all rural communities have developed CDPs which then feed into district development plans and upward into island development plans. CDPs are required to integrate key priorities of climate change and disaster risk management, as well as gender and social inclusion; however, this is yet to be carried out. Local communities lack capacities to undertake climate vulnerability assessments, design bottom-up plans for implementing climate-responsive and climate risk informed solutions and access adequate resources.

44. Needs of the recipient is assessed as high.

1.5 Country ownership *Scale: Medium to high*

45. The proposal was prepared in consultation with key stakeholders and is accompanied by a no-objection letter from the national designated authority of Tonga (annex 1) and UNDP (as AE) co-financing commitment letter (annex 13).

46. The proposal is aligned with and contributes to Tonga's national policy priorities, including on climate change. It also refers to the project's alignment with Tonga's national strategic development framework (2015–2025). The proposal clearly outlines the relevant policies and plans, particularly noting the country's national Climate Change Policy (2016),[8] which sets out an integrated approach to adaptation, mitigation and disaster risk reduction. It also outlines the project's potential contribution to the Joint National Action Plan 2 on Climate Change and Disaster Risk Management (2018-2028),[9] which serves as Tonga's national adaptation plan, and sets out six policy objectives and targets and an implementation strategy for the country to realize its vision of a resilient Tonga by 2035. The proposal notes that its activities for monitoring and data collection could potentially contribute to the Joint National Action Plan 2. The proposal also responds to Tonga's international commitments under the United Nations Framework Convention on Climate Change, as outlined in its third national communication and its second nationally determined contribution (both submitted in 2020).

47. Project design drew on consultations with key government ministries, including MIEDECC, the Ministry of Internal Affairs (Women's Affairs and Gender Equality Division), MoLNR and the Prime Minister's Office (Department of Local Governments).

48. Consultations were also undertaken with communities, key civil society organizations and international development partners. The proposal notes that representation of women and youth was ensured during the stakeholder consultations and inputs from the residents on the coastal protection measures were collected (see annex 7a). The funding proposal includes a stakeholder engagement plan (see annex 7b) as a basis for identifying, engaging and

[8] Government of Tonga. 2016. *Tonga Climate Change Policy: A Resilient Tonga by 2023*.
[9] Available at: https://climatechange.gov.to/?p=757

communicating with stakeholders that hold an interest or potential interest in the project work and objectives. The proposal intends to use existing mechanisms for community engagement, including *fono* (community meetings) and traditional community structures such as *talanga* (interactive listening and speaking), to raise public awareness of climate change and disaster risks faced by the country.

49. The project is being executed by the UNDP Pacific Multi-Country Office, a United Nations agency, with its base in another country (Fiji) some 800 km away; the project management unit and all its posts will be based in Nuku'alofa. UNDP has clarified to the iTAP the separation of functions in its role as both AE and executing entity and that under the DIM arrangement, fund flows, procurement and financial management will lie with UNDP. The AE has also explained to the iTAP that a capacity assessment undertaken for MEIDECC, which is mandated as the national responsible agency for climate change adaptation under the Tonga Climate Change Policy, indicated limited capacity in implementing the project. This result, and the decision to implement the project under the UNDP DIM arrangement, has been accepted by MEIDECC and the project management unit will be housed within the Ministry of Infrastructure.

50. The project governance structure makes provision for representation by different stakeholder groups, including MEIDECC, other government entities, civil society groups and industry associations. A proposed Project Board will bring together relevant government and non-government bodies, including MEIDECC, the Ministry of Infrastructure, MoLNR, Joint National Action Plan advisory members and the Civil Society Forum of Tonga, to guarantee full project ownership, participation in project strategic decision-making and alignment with national priorities. The proposal makes provision for a Project Board to meet at least once a year. Details on its establishment are proposed to be developed during project inception.

51. Paragraphs 85 and 92 of the funding proposal indicate that MEIDECC will play the role of Development Partner in the Project Board, and paragraph 93c further clarifies that this involves "representing the interests of the parties concerned that provide funding, strategic guidance and/or technical expertise to the project". Since the project is being executed by UNDP and not the Government of Tonga, it would be worth considering the possibility of MEIDECC chairing the Project Board, to help realize a joined-up strategic vision for the project and maximize enhancement of government capacity through implementation.

52. Overall, country ownership of the project is assessed as medium to high.

1.6 Efficiency and effectiveness *Scale: Medium to high*

53. The total cost of the funding proposal is USD 23,919,409, comprising a request for GCF grant funds of USD 22,656,409 with co-finance of USD 1,200,000 from the Government of Tonga. Additionally, the government has agreed to co-financing of USD 225,000 for operations and maintenance costs of coastal defence measures during project implementation with a further commitment for post-project maintenance over the remaining lifetime of 40 years (see para. 158 of the funding proposal). The funding proposal will also establish community volunteers for monitoring and reporting on the coastal infrastructure.

54. The economic analysis (annexes 3a and 3b) notes very high confidence that the project is economically efficient. The analysis suggests that costs should be maintained at the levels presented in the existing budget.

55. The iTAP recognizes that lessons learned from the implementation of this project are likely to be highly relevant to other SIDS in the Pacific region. The iTAP appreciates that, in year 7 of the project, UNDP will contribute USD 63,000 towards an international/regional lesson learning event on climate-resilient coastal protection and management.

56. The presentation of the budget could be made more consistent to aid its readability across the funding proposal and the relevant annexes. The detailed budget (annex 4) allocated the total project cost of USD 23,919,409 as follows: Output 1: USD 4,210,594; Output 2: USD 7,717,384; Output 3: USD 10,977, 385; and project management: USD 1,014,046. Monitoring and evaluation costs of USD 482,832 and USD 66,000 respectively are embedded within the outputs and are also separately presented in annex 11. The AE fee request of USD 1,585,949 (as per annex 12) also includes annual reporting and terminal evaluation costs.

57. The monitoring and evaluation (M&E) plan is not detailed enough. While the funding proposal touches on M&E, it mainly outlines processes to be undertaken and lead responsibilities but does not detail the overall approach to track results and evidence of a paradigm shift. Information on M&E (annex 11) are limited to listing data collection tools with indicative costs listed. There is limited discussion on the significance of selecting these indicators, how this links with the overall theory of change and the logical framework indicators, or how monitoring information will feed back into decision-making.

58. No baseline or target values are currently provided for core indicator 3, "Value of physical assets made more resilient to the effects of climate change", in section E.3 of the logical framework. The AE has clarified that owing to insufficient data on the economic values of the different types of assets, such as crops and public assets such as communications towers, groundwater pumping stations, schools, etc., the establishment of the baseline and target value will be done as part of the project implementation and will be completed and reported before the mid-term point of the project.

59. The iTAP appreciates that the project proposes to build on lessons and synergies with other donor-funded projects. The financial analysis (annex 3) clearly explains the value addition of this GCF grant and sets out the case for GCF concessional finance for generating benefits of a public good nature. The hard infrastructure under Output 3 fills a critical gap in seaward defences along the shoreline and is site-adjacent to the revetment previously constructed by Asian Development Bank resources. Infrastructure funded by GCF will replicate and build on this design as the coastal infrastructure funded by the Asian Development Bank demonstrated protection and prevented damage from the 2022 volcanic eruption and resulting tsunami referred to in paragraph 6 above.

60. The efficiency and effectiveness potential are assessed as medium to high.

II. Overall remarks from the independent Technical Advisory Panel

61. The iTAP considers that the proposal is sound, and recommends that the AE undertake the following:

(a) By project mid-term, clarify what additional avenues for funding could become available to finance the climate and disaster risk activities that will be integrated within CDPs to ensure that these activities are supported and taken up (see para. 19 above);

(b) During project inception, clarify the roles, entry points and involvement of the different government ministries across implementation of all outputs (see para. 21 above);

(c) Consider whether the Project Board could potentially be chaired by a senior official from MEIDECC to ensure that a joined-up strategic vision is realized through project implementation (see para. 51 above); and

(d) Strengthen and review the overall M&E plan during the project inception period, including the establishment of the baseline and target value for core indicator 3 by mid-term (see paras. 57 and 58 above).

62. The iTAP recommends that the Board approve this funding proposal.

Response from the accredited entity to the independent Technical Advisory Panel's assessment (FP234)

Proposal name:	Tonga Coastal Resilience
Accredited entity:	United Nations Development Programme
Country:	Tonga
Project size:	Small

Impact potential
We appreciate the detailed analysis and considerations of the iTAP with regards to the project's impact potential and its High rating. On the use of RCP8.5 and an expectation of reaching temperatures of 2.6 °C by 2100, clarifying that it is widely seen by science (at this time) that it is not so likely to achieve RCP8.5 across all indicators, such as global mean temperature increase. However, with regards to coastal infrastructure, increasingly coastal adaptation measures tend to be designed to withstand the worst-case scenarios using RCP8.5 for Sea Level Rise (SLR) projection figures to encompass the significant uncertainty for SLR as a result of ice cap melting and Antarctic ice sheet instability. The best advice available to guide coastal adaptation is therefore to use the pessimistic SLR projections of RCP8.5, as considered in this proposal, as a sound "no-regrets" approach.
Paradigm shift potential
The Medium to High rating is noted. We appreciate the iTAP's acknowledgment of how the paradigm shift has been outlined in the Theory of Change as this is a key pillar of strength of the project design. Furthermore, we appreciate the recognition of the important considerations of the broad strategic options to address coastal vulnerability that has been built into the design of the project, namely, retreat, accommodate and protect. We note the importance of detailing the involvement of key Government agencies during project implementation to address challenges associated with weak institutional capacity and cross-governmental coordination. As such, the project will further detail out the methods and processes of engaging key project stakeholders which will include Government agencies. This process will be undertaken in the initial project inception phase through dedicated support from the Project Management Unit personnel. The agreement on the involvement of Government agencies will thereby be responsive to capacities, roles and responsibilities, and have the opportunity for the appropriate level of agreement via the stakeholder engagement process. We recognise the important role the CDPs play in incorporating climate risks and adaptation measures, and the project will promote this by developing guidelines for mainstreaming climate change concerns into the CDPs. The importance of the long-term financing options is also noted for making the CDPs effective and meaningful, and in this regard, the project will clarify by the mid-term the potential avenues available to finance the climate and disaster risk activities that will be integrated within CDPs.

We are confirming that the project is designed to enhance environmental benefits by incorporating hard engineering interventions. with appropriate soft nature-based solutions to further the natural protective measures existing in the area of coastal protection measures, including the role of coastal vegetation. Clarifying, however, that there is no coral restoration work designed into the project.
Sustainable development potential
The High rating is noted.
Needs of the recipient
The High rating is noted.
Country ownership
The Medium to High rating is noted. We appreciate the recognition of the consultations that took place with Government ministries as well as with communities, key civil society organisations and development partners. With regards to the consultations with the key government ministries, we recognise the importance of noting the consultations, engagement and inputs from key ministries such as MEIDECC, Ministry of Internal Affairs, MoLNR, Prime Minister's Office as well as Ministry of Infrastructure and Ministry of Finance. We welcome the comments on the governance structure of the project as detailed out in the Funding Proposal, which includes composition of the Project Board, the exact representations to be detailed out during project inception as noted in the Assessment. Furthermore, the recommendation on the possibility of MEIDECC to co-chairing the Project Board is also appreciated and is a possibility that will be considered during project inception. We appreciate the recognition of all the arrangements that have been put in place to ensure maximum country ownership (as noted in the FP), including locating the full Project Management Unit in Tongatapu (partly in MEIDECC and part of the construction-related PMU within Ministry of Infrastructure) and that the Direct Implementation Modality (DIM) arrangement of the project is mainly related to mechanisms for fund flow, procurement, recruitment and financial management based on the results of the capacity assessment accepted by the Government. This arrangement does not take away from the Government ownership of the project.
Efficiency and effectiveness
The Medium to High rating is noted. We appreciate the recommendation to further strengthen and review the overall M&E plan during the project inception period and confirm that reviewing and updating the M&E Plan is part of the inception phase with the participation and inputs from all the key stakeholders. We confirm that data gaps related to the Core Indicator 3 "Value of physical assets made more resilient to the effects of climate change" have been analysed during project design, and that establishment of the baseline and target values by mid-term will be a priority for the project

management team during implementation through close engagement with the Government stakeholders.

Overall remarks from the independent Technical Advisory Panel:

ITAP's recommendation that the Board approves this Funding Proposal is greatly appreciated as well as its comprehensive analysis of the project via the lens of GCF's Investment Criteria. We believe this project will be a great support to the Kingdom of Tonga to build the long-term resilience of vulnerable coastal communities to the direct impacts of climate change.

We appreciate the following recommendations from iTAP to further strengthen the project's ability to meet its goals:

iTAP Recommendation: *By project mid-term, clarify what additional avenues for funding could become available to finance the climate and disaster risk activities that will be integrated within CDPs to ensure that these activities are supported and taken up.*

We appreciate this valuable recommendation, and this will be incorporated in the implementation of the related project activities as a way to enhance the project's support to the local government's capacities to integrate climate risks and adaptation needs into local planning tools.

iTAP Recommendation: *During project inception, clarify the roles, entry points and involvement of the different government ministries across implementation of all outputs.*

Confirming that this will be carried out as part of project inception through inputs from all the key stakeholders whereby the roles, entry points and involvement of Government Ministries and other stakeholders will be further detailed out and agreed.

iTAP Recommendation: *Consider whether the Project Board could potentially be chaired by a senior official from MEIDECC to ensure that a joined-up strategic vision is realized through project implementation.*

We appreciate this recommendation and recognize this is recommended as a means to further enhance the Government's ownership of the project. Confirming MEIDECC is part of the project board with a senior official representing (CEO) representing the Ministry. Furthermore, the potential for the senior official to co-chair the Board with UNDP will be considered during project inception.

iTAP Recommendation: *Strengthen and review the overall M&E plan during the project inception period, including the establishment of the baseline and target value for core indicator 3 by mid-term.*

Confirming that in addition to further review and update of the M&E Plan during implementation with the support of dedicated project team personnel, the establishment of the baseline and target value for core indicator 3 by mid-term will be prioritized.

Gender documentation for FP234

ANNEX 8

Gender Assessment and Gender action plan

Coastal Resilience in the Kingdom of Tonga

March 2024

I. Introduction

The proposed GCF project is designed to support the Government of Tonga's efforts to enhance coastal resilience and provide medium-term vulnerability reduction and adaptation benefits in the Hahake region as well as strengthen the overall adaptation capabilities of the government and the people of Tonga. The direct beneficiaries will be 39,427 people (of which 19,782 are women) who will benefit from activities in vulnerability reduction and/or resilience building activities in Tongatapu. Additionally, 100,179 people (i.e. the entire population of the country) will be the indirect beneficiaries of the project and are expected to benefit from strengthened adaptation capabilities of the government.

This Gender Assessment and Gender Action is prepared as part of the Coastal Resilience Project proposal package. The gender assessment aims to provide an overview of the gender context in the Kingdom of Tonga and highlights the intersectionality between gender equality, climate change, and economy, which may be relevant to the project context. The assessment also identifies gender mainstreaming opportunities within the project context. The Gender Action Plan includes specific gender-mainstreaming activities across the three output areas, as well as the gender equality results that will be achieved through the proposed project.

This gender assessment is based on:

- A desk review of relevant national policy documents, including the Tonga Strategic Development Framework 2015-2025 (TSDF),
- the National Women's Empowerment and Gender Equality Tonga (WEGET) Policy and Strategic Plan of Action 2019-2025
- the Joint National Action Plan 2 on Climate Change Adaptation and Disaster Risk Management (JNAP-2) 2018-2028
- Corporate plans (2020-2023) of all the ministries of the Government of Tonga.
- Community Development Plans of Hahake region, developed by the community in collaboration with the Prime Minister's Office and Mainstreaming of Rural Development Innovation Tonga Trust (MORDI).
- Lessons learned and recommendations from past assessments and studies on gender equality undertaken by the Government of Tonga, UN agencies, development partners, civil society organizations and academic organizations, and information available from programs and projects currently being implemented; and
- Stakeholder consultations (including with Women's Affairs and Gender Equality Division – WAGED, Ministry of Internal Affairs, and Civil Society Organizations)

II. Gender equality and social inclusion in Tonga

The Tongan society revolves around four core cultural values *faka'apa'apa* (respect), *feveitokai'aki* (reciprocity), *'ofa* (love) and *loto fakatokilalo* (humility). Traditionally, women in Tonga have held a high social status due to the *fahu* system. The '*fahu'* or the elder sister (or another chosen sister) holds a place of privilege and plays an important role in family decision-making.[1] At the same time, the traditional role of women revolved around home, family, and extended family, and focused on caring and nurturing.[2] In modern times, though Tongan women take part in all aspects of the society and exert greater influence in shaping Tongan politics and society, they continue to be limited by antiquated legal frameworks and practices.

[1] Government of the Kingdom of Tonga (2019), 'Gender Equality: Where do we stand?', prepared by the Women's Affairs Division, Ministry of Internal Affairs, for Pacific Community (SPC), Pg 4.
[2] ibid

At present, Tonga is facing a demographic shift due to net migration along with increasing vulnerability to climate change and disaster risks. The annual population growth rate is negative 0.1 percent (-0.1). Nearly quarter of the households (22.1 percent) are headed by women.[3] This demographic change makes it imperative for women to be able to engage economically and politically for sustained growth and development of the country.

Added to that, Tonga's vulnerability of the country to climate change and disaster risk is increasing. The overall score of Tonga on the ND-GAIN Index is 41.1 and it ranks 140 out of 185 countries. The lower the score the more vulnerable the country is to climate change and other global challenges, and less readiness to improve resilience. In terms of vulnerability alone, Tonga ranks as the 6th most vulnerable country out of 185 countries measured.[4] On the 2021 Global Climate Risk Index, Tonga ranks 77th out of 176 countries most affected by extreme weather events for the period 2000-2019.[5]

The disproportionate impact of climate change on women and other vulnerable groups is widely recognized. In Tonga, there is a need to understand and unpack the intersections of gender inequality with a range of other dimensions of vulnerability and resilience that can be amplified by climate change.

The government recognizes the importance of equal political and economic participation of men and women for resilient development and has prioritized achievement of **gender equality by 2025**. Gender and Social Inclusion (GESI) for resilient development is also included as a target in the Joint National Action Plan-2 on Climate Change and Disaster Risk Management (2018-2028). However, gaps remain in translating policy ambition to practical implementation measures to promote overall gender equality and women's well-being in the country.

Changing demography

The population of Tonga is largely homogenous. 98 percent of the population is ethnic Tongans, and the remaining 2 percent is comprised of immigrants from various countries, including China and other Pacific Island Countries.

According to the 2021 census, the number of women in Tonga is higher than the number of men. Women constitute 51.33 percent of the population, and men make up the remaining 48.66 percent of the population. The number of males for every 100 females fell from 99 in 2016 to 95 males in 2021[6], largely due to emigration. Emigration may also be increasing the dependency ration of Tonga which is estimated at 74 (meaning for every 100 working-aged persons, there are 74 dependents).[7] Since the median age of the population is 22 years, and 35 percent of the population is aged less than 15 years old, the child dependency ratio is high at 0.61, while the elderly dependency ration is only 0.13.[8] Despite a young population, the annual population growth rate remains negative due to emigration of working age population.

[3] Government of Tonga (2021), Census 2021: factsheet https://tongastats.gov.to/census-2/population-census-3/census-report-and-factsheet/
[4] University of Notre Dame, 'Notre Dame Global Adaptation Initiative', country rankings https://gain.nd.edu/our-work/country-index/rankings/
[5] GermanWatch (2021), Global Climate Risk Index 2021: Who suffers most from extreme weather events? Weather-related loss events in 2019 and 2000 to 2019' https://www.germanwatch.org/en/19777
[6] ibid
[7] Pacific Community (2023), 'Tonga 2021 Household Income and Expenditure Survey Report', Tonga Statistics Department, Government of Tonga
[8] Government of Tonga (2021), 'Tonga Poverty Assessment Report: Assessing progress towards the reduction of multi-dimensional, extreme and monetary poverty in the Kingdom of Tonga' https://tongastats.gov.to/statistics/social-statistics/poverty-in-tonga/#:~:text=The percent20income percent20poverty percent20line percent20is,in percent20a percent20low percent2Dincome percent20household.

The Tonga Strategic Development Framework II (2015-2025) indicates that emigration has been a "key characteristic of Tongan development for the last four decades".[9] Despite lack of reliable data, the Tongan diaspora is estimated to be as large, if not bigger, than the population of Tongans residing in the kingdom.[10]

Table 1: Sex Ratio: Males for every 100 females

	2016	2021
Tonga	99	95
Urban	97	93
Rural	100	95
Greater Nuku'alofa	98	93

Table 2: Annual Population Growth

	2016	2021
Tonga	-0.5	-0.1
Urban	-0.9	-1.8
Rural	-0.4	0.4
Greater Nuku'alofa	-0.5	-1.1

Source: Government of Tonga: Census Factsheet 2021

Migration may be shifting gender norms towards less traditional, patriarchal and more gender equal views.[11] But at the same time, traditional gender norms are deep rooted across the Pacific. In Tonga, despite seeming shift in gender norms, gender stereotypes remain pervasive and paternalistic social hierarchy continues to be reinforced by both custom and religion.[12]

Tonga and Human Development and Gender Inequality

There are several international indices that have been developed to quantify the concept of gender inequality. The United Nations Development Programme (UNDP) developed the Human Development Index (HDI) as a way to measure achievement in the basic dimensions of human development across countries. In addition, UNDP also uses the Gender Inequality Index (GII), Gender Development Index (GDI) as well as Inequality-adjusted Human Development Index (IHDI) to measure inequality.[13] The GII is a composite measure that reflects gender-based disadvantages in three dimensions: health (specifically reproductive health), empowerment (specifically access to education and political participation) and labour market. GII shows loss in potential human development due to inequality between female and male achievements in these dimensions.

The GDI measures difference in male and female achievements in three basic dimensions of human development: health, education, and command over economic resources (in other words estimated earned income). GDI is the ration of female Human Development Index (HDI) to male HDI.[14] Inequality HDI indicates the loss in human development due to inequality. The IHDI value equals the HDI value when there is no inequality across people but falls below the HDI value as inequality arises.[15]

Tonga has a GII of 0.631 in 2021 and ranks 160th place out of 170 – which means that Tonga has high level of inequality between men and women. But at the same time, the GDI value for Tonga as of 2021 is 0.965, which shows high equality in Human Development Index achievements between women and men.[16] Overall, Tonga scored 0.745 on the 2022 HDI and ranks 91 out of 191. HDI value for female is 0.728 and for male is 0.754. In other words, Tonga has high human development with little difference

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[9] Government of Tonga (2015), 'Tonga Strategic Development Framework 2015-2025' – A more progressive Tonga: Enhancing our inheritance, https://policy.asiapacificenergy.org/sites/default/files/TSDF percent20II.pdf Pg 37.
[10] ibid
[11] Ryan Edwards (2023), 'Migration seems to be shifting gender norms in Tonga', Development Policy Blog, 13th Jan 2023, https://devpolicy.org/migration-seems-to-be-shifting-gender-norms-in-tonga-20230113/
[12] UN Women (2022), 'Gender Equality Brief for Tonga' https://asiapacific.unwomen.org/sites/default/files/2022-12/UN_WOMEN_TONGA.pdf
[13] United Nations Development Programme. Human Development Report.https://hdr.undp.org/data-center/thematic-composite-indices/gender-inequality-index#/indicies/GII
[14] UNDP, 'Gender Development Index' https://www.undp.org/sites/g/files/zskgke326/files/migration/tr/UNDP-TR-EN-HDR-2019-FAQS-GDI.pdf
[15] UNDP, 'Inequality-adjusted Human Development Index' https://hdr.undp.org/inequality-adjusted-human-development-index#/indicies/IHDI
[16] http://hdr.undp.org/sites/default/files/hdr14-report-en-1.pdf; http://hdr.undp.org/en/composite/GDI

between male and female achievements, with women faring better in life expectancy (73.7 years compared to 68.4 years for men), and in education attainment (mean years of schooling is 11.5 years for women, and 11.2 years for men). However, when issues empowerment and economic participation are considered, gender inequality in Tonga becomes apparent. Gross National Income Per Capita is $4,842 for women and $8,845 for men based on 2017 Purchasing Power Parity.[17] Tonga ranks 62 out of 156 on the 2021 IHDI and scores 0.666 on IHDI which is below its 0.754 score on HDI. The human development loss of Tonga is 10.6 percent from HDI to IHDI.

At the same time, women tend to have higher perception of better life compared to men (91 and 88 percentage, respectively).[18]

Poverty[19]

Tonga has succeeded in virtually eradicating extreme poverty (SDG1.1.1), which is defined as people living on less than $1.90 per day. Poverty rates in Tonga have been steadily declining. In terms of monetary poverty (cost of basic needs), 20.6 percent of the population or one in five live below the National Poverty line (which is around $ 8.69 per day). Food poverty is negligible in the country. Tonga score on Gini Index is 0.271 which shows low levels of consumption inequality.

Poverty in Tonga is also highly spatially autocorrelated – meaning that the prevalence of poverty is geographically clustered and there is a major geographic difference in the extent of poverty across Tonga. Though the prevalence of poverty is higher in other islands of Tonga, 53 percent of the Tonga's poor people live in Tongatapu rural areas. Tongatapu is home to 74 percent of Tonga's population. Within Tongatapu, there is a large difference between rural and urban areas: poverty rate is 13.3 percent in urban areas against 21.1 percent in rural areas.

26 percent of the households are female headed households. At national level, poverty rate is lower for people living in female-headed households (18.3 percent against 21.4 percent for those living in male-headed households). In Tongatapu, female headed households' poverty rate is 14.5 percent, while it is five percent points higher for male-headed households. However, in the outer islands (except Ongo Niua) female headed households have higher rate of poverty. For instance, in rural Vava'u poverty rate of female headed households is 36.4 percent whereas for male headed households it is 24.4 percent.

Young people are more frequently affected by poverty. The highest poverty rates are found among the youngest age groups (0 to 20 years old) with a rate of around 23 percent, while it is less than 20 percent for older people. Therefore, the largest number of poor is among children under 20 years who account for 47 percent of the population and 53 percent of the poor.

Multi-dimensional poverty refers to various deprivations experienced by poor people in their daily life – such as poor health, lack of access to education, etc. Multi-dimensional poverty in Tonga is calculated through a survey developed using the Consensual Approach which identified 29 items (range of things, activities, and services) that are considered or regarded as necessities for life by the population. Of the 29 items 11 were for the adult population, 13 for children, and 5 household-items. The survey allows for classifying the population facing multi-dimensional poverty into four groups: the poor, not poor, the vulnerable due to income (i.e., low income but relatively high living standards), and the vulnerable due to low living standards (i.e., low living standards but high income).

[17] Ibid
[18] Govt of Tonga (2020), 'Tonga MICS 2019: Snapshot of Key Findings' https://tongastats.gov.to/survey/mics-survey/
[19] This section includes data from the 2021 Tonga Poverty Assessment Report by the Government of Tonga, except where otherwise indicated.

The prevalence of multi-dimensional poverty has decreased from 27 percent in 2016 to 24 percent in 2022. 41 percent of the population can be classified as multi-dimensionally non-poor. It is also important to note here that 90 percent of households receive remittances.

Table 1: Prevalence of multi-dimensional poverty: Total, Adults and Children 2022

Category	Total Population (%)	Adults (%)	Children (%)
Poor	24	21	28
Vulnerable deprivation	15	15	15
Vulnerable income	20	18	22
Not poor	41	46	35

Overall, women are just slightly more likely than men to be deprived in almost all items. Saving a small amount of money for themselves is the item with the biggest gap between men and women. At the same time, income poverty rates are slightly higher among males than females.

The child deprivation rates show that boys are slightly more likely than girls to be deprived. For example, deprivations of new clothes, books, and enough beds seem to be more prevalent among boys than girls. As for the rest of the items, the differences are small and not significant after considering the survey sampling error.

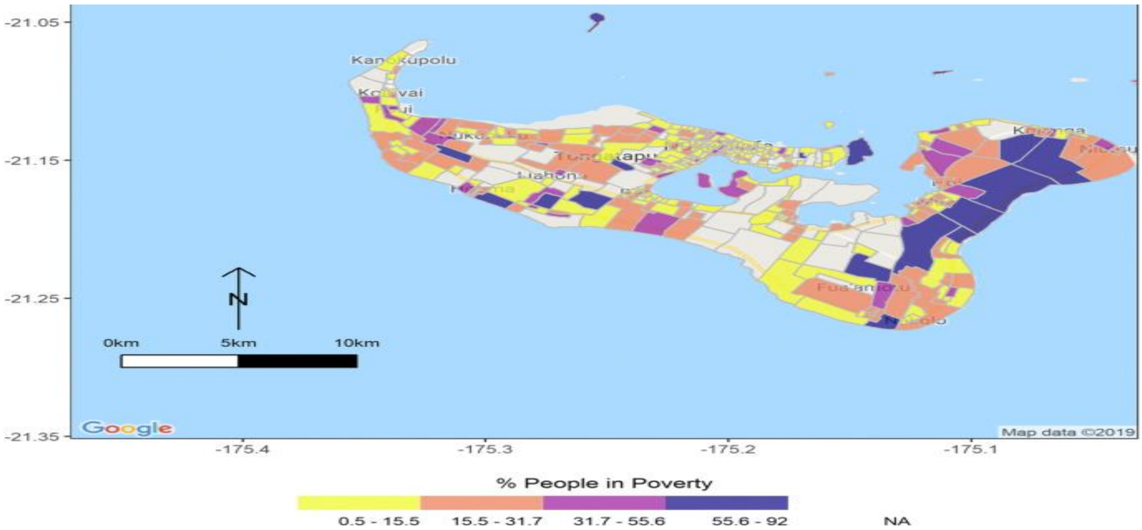

Figure 1: Poverty rate in Tongatapu, consensual method. Block-level estimates, Tongatapu 2016[20]

Political Participation

Women were granted voting rights following the amendment of the constitution during the reign of Queen Salote Tupou III in 1951. However, women representation in the national legislature is well below is the average for Pacific Island Countries. The Tonga National Parliament consists of 28 members of parliament, which includes the speaker, 9 nobles (elected from and only by the 33 hereditary nobles and the noble titles are inherited only by men), 17 elected members, and 1 appointed member.

At present, there are two female MPs (one elected in 2022 by-elections and one appointed) in the Parliament. Women's representation in the Tongan parliament is about 7 percent which is well below the global average of 26 percent.

[20] Nájera Catalán, H.E., Fifita, V.K. & Faingaanuku, W.(2019), 'Small-Area Multidimensional Poverty Estimates for Tonga 2016: Drawn from a Hierarchical Bayesian Estimator. *Appl. Spatial Analysis* **13**, 305–328 (2020). https://doi.org/10.1007/s12061-019-09304-8

Table 2: Women's political participation

	National Parliament (2022)	Local government (2023)
Women's representation number	2	7
Women's representation percentage	7.6 percent	3.9 percent

There has been little indication of national commitment towards adopting Temporary Special Measures (TSM) or providing incentives to improve women's representation in Parliament. The Tongan Women's Coalition, a nongovernmental organization, has been advocating for a TSM Bill to reserve seats for women in Parliament and made a submission to cabinet in 2017.

At the sub-national level, women's representation though remains low, is showing signs of improvement. Local government in Tonga consists of 23 district and 156 town officers, who are elected every three years. In 2023, 1 women district officer, and 6 women town officers were elected, which translates to 3.9 percent of seats in local government. This is considerable improvement from the 2016 local elections, where only one-woman district officer and one women town officer were elected (or about 1.1. percent of seats in local government).

However, it is important to note that none of 7 women local government officials elected were from the main Tongatapu island, where 74 percent of the country's population live. More research is needed to unpack the 2023 local election results. But a 2021 survey points to continued negative attitudes towards women's political participation and leadership. 92 percent of those surveyed considered men were more likely to have the 'right' skills and experience for parliament. (see box below on Voter's perception of women as leaders in Tonga).

Decision-making process continue to be firmly male driven. Access to certain spaces (such as kava bara) is restricted for women. These spaces are often where actual decisions are made and restricted access of women to these spaces limits their ability to fully engage in decision-making.

Voter's Perceptions of Women as Leaders in Tonga

- 61 percent of respondents considered that the Tonga family unit, "fāmili" is hierarchical, with men at the top of the hierarchy and therefore at the head of the Tongan family.
- 69 percent of respondents considered that mothers/women should stay at home with children while fathers/man should attend and participate in village (fono) meetings.
- 58 percent considered it appropriate that the father/man should go to work while the working mother/woman should stay home to look after a sick child.
- 80 percent of respondents felt that a woman, staying in her husband's village, could participate in village meetings if she has been involved in village activities.
- 65 percent of respondents considered that it was inappropriate for a mother to advise her husband to allow his daughters to inherit his land.
- 52 percent of respondents considered that it was appropriate for a Tongan mother to be a wage earner while the father remains at home to conduct domestic chores.
- 66 percent believed that fathers and mothers should have equal access to financial income.
- 80 percent recognized the privileged role of "mehekitanga" (father's sister) in the Tongan family.
- 57 percent believed that men should lead in the village while 57 percent believed that both males and females could lead in the workplace.
- 53 percent considered that both men and women could lead in parliament, but both male and female respondents were more likely to consider men as the 'best' leader in this area.
- 52 percent stated that they would vote for a male candidate over a female candidate with exactly the same qualifications.
- 92 percent considered men were more likely to have the right skills and experience for parliament.

Source: Ungatea Fonua Kata, Vanessa Lolohea (2021), 'Voters' Perceptions of Women as Leaders in Tonga', Balance of Power Program (DFAT), accessed via https://www.toksavepacificgender.net/research-paper/voters-perceptions-of-women-as-leaders-in-tonga/

As part of local governance, in addition to district and town officers, some villages have established councils to discuss priority issues and assist the district and town officers. The gender composition of these councils is not known.

The Community Development Plan (CDP) process has been instrumental in expanding women's participation in decision making. The CDP identified village women's practical priorities but women's strategic needs, such as the inclusion of women's concerns in village development committees, were not explored.[21] Community Development Committees in Tonga are responsible for community level decision making related to local development. There are community development committees with female chairs in some villages. In other villages, women can only influence the women's committee and subcommittees such as women's agriculture committees. More community awareness is required to enhance knowledge and skills of women. The exact number and structure of these community development committees is not available.

There have been several programme – supported by development partners, regional networks, and also local civil society groups – to support women's political participation. One such group is the FI-E-FI-A'a Fafine Tonga (FFFT) network to support and empower women and girls, which organized training of female candidates,[22] and a post-election panel in December 2021 to discuss female candidates' experiences and identify barriers for women's participation. Local civil society groups also signed a Memorandum of Understanding affirming solidarity in supporting government commitments to gender equality, and the participation and empowerment of women and girls in Tonga.[23]

In terms of women in leadership role, women make up 24 percent of CEOs and commissioners and 49 percent of the total employees of Public Service Commission-governed ministries.[24] In the private sector, 2021 analysis of women's representation on the boards of 19 organizations in Tonga, women were found to hold 19 percent of Director, 17 percent of Deputy Chair, and 10 percent of Board Chair positions. Women were more highly represented as Directors on the boards of private sector organizations (23 percent) than state-owned enterprises (18 percent).[25]

Tonga recognizes the need for enhancing women's political participation and leadership. The National Women's Empowerment and Gender Equality Tonga (WEGET) Policy and Strategic Action Plan (2019-2025) includes "Increased women's leadership and equitable political representation" as one of the five priority outcomes. The WEGET policy acknowledges the need for amending legislation and leadership training to encourage women's political participation at all levels. Under this outcome, the action plan includes various activities to increase women's representation in parliament and in elected local government offices, and increased participation of women in decision-making in all spheres. Some of the activities include encouraging CSOs, faith-based organizations and youth groups to make women's leadership role visible; providing capacity building support to women candidates and elected representatives, as well as building media's capacity to analyze gender issues among others.[26]

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[21] Government of the Kingdom of Tonga (2019), 'National Women's Empowerment and Gender Equality Tonga Policy and Strategic Action Plan', prepared by the Women's Affairs Division, Ministry of Internal Affairs, for Pacific Community (SPC), https://hrsd.spc.int/sites/default/files/2021-07/WEDGET_STRATEGIC_PLAN_OF_ACTION_2019_2025_Final.pdf

[22] Lepolo Taunisila and Sonia Palmieri (2022), 'Women's Candidate Training in the COVID Era: The 2021 Tonga Election' Australia National University, Department of Women's Affairs, In brief 2022/5, DOI: 10.25911/1QPG-DX55

[23] Pacific Data Hub (2022), 'Launch of the new FI-E-FI-A'a Fafine Tonga (FFFT) network to support and empower women'

[24] The Pacific Private Sector Development initiative- PSDI (2021). 'Leadership Matters: Benchmarking women in business leadership in the Pacific', https://www.pacificpsdi.org/assets/Uploads/PSDI-LeadershipMatters-Web3.pdf

[25] UN Women (2022), 'Gender Equality Brief for Tonga' https://asiapacific.unwomen.org/sites/default/files/2022-12/UN_WOMEN_TONGA.pdf

[26] Government of the Kingdom of Tonga (2019), 'National Women's Empowerment and Gender Equality Tonga Policy and Strategic Action Plan', https://hrsd.spc.int/sites/default/files/2021-07/WEDGET_STRATEGIC_PLAN_OF_ACTION_2019_2025_Final.pdf

Education

Though Tonga has high literacy rates at 99.4 percent, the overall performance of the country on education indicators need to improve.

The Government of Tonga recognizes right to education of every child between the ages of 4 and 18 years. The Education Act of 1974, and of 2013 (revised in 2020) enshrines the right of every child to receive quality education, and access to educational and vocational information and guidance.[27] The act also promotes equitable access and participation in technical and vocation education and training (TVET) for girls and marginalized groups (including students with disabilities).

Percentage of government expenditure on education stands at 11.9 percent (2020), and roughly about 6.6 percent of the GDP.[28] Primary school attendance is close to universal. The Gross Enrollment Ration (GER) was 112 percent for girls and 117 percent for boys in 2020[29] (GER can be over 100 percent as it includes students who may be older or younger than the official age group). However, GER and completion rates start to fall as students move to lower secondary and upper secondary.

The GER for secondary education was 95.6 percent for girls and 81.3 percent for boys in 2020. The lower secondary completion rate was 93.2 percent for girls and 90.7 percent for boys. 64.8 percent of girls and 61.3 percent of boys aged 7 to 14 years could demonstrate foundational reading skills, while 53.7 percent of girls and 51.3 percent of boys could demonstrate foundational numeracy skills.[30]

In upper secondary, the 2019 Multiple Indicators Cluster Survey recorded a Net Attendance Rates of just 64 percent (74 percent for girls and 55 percent for boys). Approximately a third of children were out of school at upper secondary level, with significantly higher rates among boys (41.1 percent) than girls (23.6 percent). The upper secondary completion rate was 44 percent (49 percent for girls and 38 percent for boys).[31]

The table below shows Tonga's education sector performance on different parity indices. A parity index value less than 1 indicates disparity in favour of the advantaged group (i.e. boys, richest, urban) and a value higher than 1 indicates disparity in favour of the disadvantaged group (i.e. girls, poorest, rural).

It is important to note here that higher education for young women does not translate into better employment outcomes due to gender barriers in labour markets, pervasive stereotypes regarding suitable occupations for men and women, and the expectation placed on women to engage in unpaid domestic and care work.[32]

Table 3: Education sector and parity indices (2019 MICS)

SDG indicator	MICS indicator	Definition & notes	Value		
			Primary	Lower Secondary	Upper Secondary
4.5.1	LN 11a	Gender Parity Indices (girls/boys)	1.00	1.09	1.34
4.5.1	LN 11b	Wealth Parity Indices (poorest/richest)	0.98	0.88	0.66
4.5.1	LN 11c	Area Parity Indices (rural/urban)	0.98	0.95	0.93

[27] Government of Tonga (2020), Education Act – 2020 Revised Edition https://ago.gov.to/cms/images/LEGISLATION/PRINCIPAL/2013/2013-0023/EducationAct_3.pdf?zoom_highlight=early+childhood+education#search=%22early%20childhood%20education%22
[28] UNESCO (2023), 'SDG 4 country profile: Tonga' https://uis.unesco.org/sites/default/files/country-profile/Tonga.pdf
[29] UN Women (2022), 'Gender Equality Brief for Tonga' https://asiapacific.unwomen.org/sites/default/files/2022-12/UN_WOMEN_TONGA.pdf
[30] ibid
[31] Ibid, see also, Govt of Tonga (2020), 'Tonga MICS 2019: Snapshot of Key Findings' https://tongastats.gov.to/survey/mics-survey/
[32] UN Women (2022), 'Gender Equality Brief for Tonga' https://asiapacific.unwomen.org/sites/default/files/2022-12/UN_WOMEN_TONGA.pdf

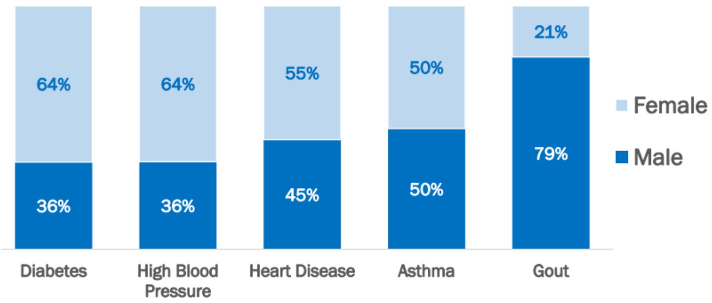
The state of education in the country was further affected by the pandemic and a series of natural disasters in recent years. Over 32,000 Tongan school children lost over approximately 510 hours each of in-person learning.[33] The marginalized and vulnerable groups were more affected by learning losses.

The learning loss negatively affects the social readiness of the country to mitigate and adapt to climate change. Tonga has one of the lowest scores on 2021 ND-GAIN index for social readiness (0.297), with education score at a mere 0.121. due to low rates of enrollment in tertiary education.[34] The country recognizes the need to improve education sector performance, and in 2023 organized the first national dialogue on 'transforming education'.

Health, including sexual and reproductive health.

Compared to other countries in the Western Pacific region, Tonga has relatively advanced progress on key health indicators. The life expectancy at birth in Tonga was 75 years for women and 69 years for men in 2019.[35] However, gaps exist in non-communicable diseases (NCDs) prevention and control There is a high prevalence of NCDs, especially diabetes and cardiovascular disease among women. Figure 2 shows sex disaggregated distribution of the top five NCDs. However, when it comes to mortality, 20.2 percent of female mortality and 29.5 percent of male mortality was attributed to cardiovascular disease, cancer, diabetes, and chronic respiratory disease in 2019.[36] 74 percent of the Tongan adult population was considered obese, and women (aged 25-59) and adults from higher quintiles were more likely to be obese.[37]

Figure 2: Distribution of top 5 NCDs by Sex, Census (2021)



On sexual and reproductive health, Tonga performs well in reproductive, maternal, newborn and child health (RMNCH), except in family planning and immunization.[38] The total fertility rate (TFR) in Tonga is 3.2, and the TFR among poorest quintile is higher (3.7) compared to richest quintile (2.0). Tonga has met SDG indicator 3.2.1 is on under 5 mortality rates. The target is less than 25 deaths for every 1000 live births by 2030, and Tonga under 5 mortality rate is 11.4 per 1000 live birth (2019). However, the country's faces significant challenges to meet maternal mortality rate target (less than 70 per 100,000 live births). There was a considerable uptick in the maternal mortality rates between 2018 and 2020.[39] The reasons for this are unclear, as skilled attendance at birth is 100 percent in urban areas, and 98 percent in rural areas. Institutional delivery is as high as 98 percent.[40] A more detailed analysis is required to understand the uptick in maternal mortality rates.

Adolescent Birth rate SDG 3.7.2 indicator is under target. 3.7: By 2030, ensure universal access to sexual and reproductive health-care services, including for family planning, information and education, and the integration of reproductive health into national strategies and programmes. As of 2019,

[33] UNICEF (2022), "let's re-imagine education: Tonga boosts efforts to ensure all its children are able to access quality education", https://www.unicef.org/pacificislands/press-releases/lets-re-imagine-education-tonga-boosts-efforts-ensure-all-its-children-are-able
[34] University of Notre Dame, 'Notre Dame Global Adaptation Initiative' Tonga country Profile https://gain.nd.edu/our-work/country-index/rankings/
[35] WHO, Tonga: Country Profile, https://www.who.int/countries/ton
[36] UN Women (2022), 'Gender Equality Brief for Tonga' https://asiapacific.unwomen.org/sites/default/files/2022-12/UN_WOMEN_TONGA.pdf
[37] Tonga statistics department (2022), 'Tonga 2021 Household Income and Expenditure Survey Report'
[38] WHO (2018), 'UHC and SDG Country Profile 2018: Tonga' https://iris.who.int/bitstream/handle/10665/272327/WPR-2018-DHS-024-ton-eng.pdf?isAllowed=y&sequence=1
[39] SDG Index, 'Tonga' https://dashboards.sdgindex.org/profiles/tonga/indicators
[40] Govt of Tonga (2020), 'Tonga MICS 2019: Snapshot of Key Findings' https://tongastats.gov.to/survey/mics-survey/

adolescent birth rate is 30 per 1000 women aged 15-19. The proportion of women aged 20-24 years giving birth before age 18 in Eua division is 17 percent compared to national three percent. [41]

Abortion is illegal in Tonga. More importance needs to be given to reducing adolescent fertility and addressing the multiple factors underlying sexual and reproductive health and the social and economic well-being of adolescents. Preventing births very early in a woman's life is an important measure to improve maternal health and reduce infant mortality.[42] 22.5 percent of women aged 15-49 had unmet needs for family planning.

Among 15–49-year-olds, 10.8 percent of women reported having a sexually transmitted infection (STI) or symptoms of an STI in the previous 12 months. Many sources on sexual and reproductive health and adolescent pregnancy across the Pacific note that the local culture regards sexuality as taboo. There is much cultural stigma and shame surrounding sex. This stigma prevents the communication of all topics connected to sex. This prevents adolescents and vulnerable groups such as women and disabled people from acquiring knowledge about sex and sexuality, as well as limiting access to services and knowledge of sexual and reproductive health.[43]

Climate change induced changes to rainfall patterns, and growing intensity of disasters in the country is not only affecting people's lives and livelihoods but also impacting people's health. Tonga is experiencing increased incidences of vector borne diseases such as dengue fever. Access to safe drinking water is also compromised due to various disasters, sea-level rise and saltwater intrusion, particularly in low-lying areas with shallow and thin groundwater levels. Vulnerability of health care facilities is also rising due to increasing disaster risks. Without adequate measures to protect health care facilities, provision of health services may be affected in the face of sudden onset extreme weather events. Women and vulnerable groups may be at risk of not receiving adequate health care in the future.

Economic participation

The economy of Tonga is highly dependent on climate sensitive sectors such agriculture, fisheries and tourism and a limited resource base that is sensitive to external shocks. Pandemic and series of disasters led to contraction of economy by 3 percent in 2021, real GDP is estimated to have expanded by 2.6 percent in 2023 and is expected to grow further in 2024. Remittances as a share of GDP in Tonga is 37.7 percent. Tonga's public debt-to-GDP ratio is projected to rise and remain above the 70 percent debt-distress benchmark starting in FY2033. This mainly reflects significant development spending needs over the long term to achieve its climate resilience and Sustainable Development Goals. [44]

The economy faces significant labour shortages due to labour participation in seasonal workers programs in Australia and New Zealand.

The agriculture sector's share of GDP is 16.32 percent in 2021. The 2021 Household Income and Expenditure Survey shows that around 80 percent of households were participating in primary activities: 63 percent participated in agricultural activities, 10 percent in fisheries, 64 percent in livestock. Another 37 participated in handicraft/home-processed food production.[45] There are significant gender differences in the types of primary activities engaged by individuals. 98 percent of females engaged in primary activities reported doing handicraft work, and whereas cropping is exclusively undertaken by males.[46]

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[41] Govt of Tonga (2020), 'Tonga MICS 2019: Snapshot of Key Findings' https://tongastats.gov.to/survey/mics-survey/
[42] ibid
[43] UN Women (2022), 'Gender Equality Brief for Tonga' https://asiapacific.unwomen.org/sites/default/files/2022-12/UN_WOMEN_TONGA.pdf
[44] IMF (2023), 'Tonga: Staff concluding statement of the 2023 Article IV mission' https://www.imf.org/en/News/Articles/2023/08/04/tonga-staff-concluding-statement-of-the-2023-article-iv-mission
[45] Tonga statistics department (2022), 'Tonga 2021 Household Income and Expenditure Survey Report'
[46] ibid

Table 4: Economic participation in Tonga[47]

	Women	Men
Labour force participation rate (15+)	38.4 %	56.2 %
Youth labour force participation rate (15-24)	20.9 %	28.1%
Unemployment rate Age 15+	3.6%	2.6%
Youth unemployment rate (15-24)	13%	5.7%
Youth not in education, employment or training (15-24)	31.5%	29%
Informal employment rate (15+)	75.2%	71.7%

The agriculture sector's share of GDP is 16.32 percent in 2021. The 2021 Household Income and Expenditure Survey shows that around 80 percent of households were participating in primary activities: 63 percent participated in agricultural activities, 10 percent in fisheries, 64 percent in livestock. Another 37 participated in handicraft/home-processed food production.[48] There are significant gender differences in the types of primary activities engaged by individuals. 98 percent of females engaged in primary activities reported doing handicraft work, and whereas cropping is exclusively undertaken by males.[49]

Labour force participation rate (see table 6) for women is lower than men. Overall, women are less likely to be active labour force participants. The rate of unemployment (people not working but who are looking for jobs) is very low (3 percent) and similar among men and women. According to the 2018 Labour Force Survey (latest to be conducted in 2023), of the women who participate in formal economy, most are employed in the manufacturing sector (40.9 percent of female and 4.4 percent of male employment), the administrative and support services sector (11.1 percent of female employment and 7.3 percent of male employment) and the education sector (10.9 percent of female employment and 4.5 percent of male employment). Men were most commonly employed in the agriculture, forestry and fishing sector (33.8 percent of male employment and 1.9 percent of female employment) and the construction sector (15 percent of male employment and 0.3 percent of female employment).[50]

Access to resources

90 percent of the households in Tonga receive remittance, which accounts for 37.7 percent of the GDP. In Tonga, around 60 percent of household income come from work, meaning cash from employer or business (33.8 percent), sale from rural activities such as agriculture, fishing, livestock, handicraft (22.1 percent) and subsistence from the latter activities (4.8 percent). Interpersonal solidarities (gifts and remittances) represent important source of income (30 percent) as the shares of cash gifts received or remittances and gifts in kind are respectively 15.3 percent and 14.5 percent.

Income sources vary considerably across localities. Income from employment comprises a much higher share of income in Tongatapu (42.9 percent and 34.4 percent respectively in urban and rural areas) while it varies from 24 percent to 28 percent in outer islands. For the latter, income comes mainly from rural activities (agriculture, fishing, livestock and handicraft) which provide with cash money as well as means of subsistence. Cumulatively, cash and subsistence from these activities account for 38 percent to 45 percent of income in outer islands.

The above context demonstrates the importance of access to land to engage income generating activities (other than formal employment and handicrafts production). However, by tradition and by law, land rights are guaranteed only to men. Tongan men are guaranteed land rights to an *'api kolo*

[47] UN Women (2022), 'Gender Equality Brief for Tonga' https://asiapacific.unwomen.org/sites/default/files/2022-12/UN_WOMEN_TONGA.pdf
[48] Tonga statistics department (2022), 'Tonga 2021 Household Income and Expenditure Survey Report'
[49] ibid
[50] UN Women (2022), 'Gender Equality Brief for Tonga' https://asiapacific.unwomen.org/sites/default/files/2022-12/UN_WOMEN_TONGA.pdf

(town allotment) and an *'api 'uta* (tax or country allotment). In Tonga, inheritance passes through the male successors, and women are excluded from land ownership, with the exception of when there are no males in the ancestry. The only access to land that women have is temporary and is limited by social constraints. In Tonga, women are part of the traditional *fahu system* where the brother is obligated to take care of his sister and her children.

The difficulties that women face in the contemporary world, in terms of land ownership, could be considered as obstacles if they wish to pursue business opportunities. The second Royal Land Commission of Inquiry (2008-2012) recommended reforms that would allow women rights to town allotments for housing.[51] The commission did not propose any changes to rural allotments. Lack of access to land impedes women's economic empowerment and also impacts their access to credit and financial services. The lack of land rights for women in Tonga may prevent women from accessing financial services, as they do not have land to use as collateral for bank loans. The Tonga Development Bank has introduced loan products with low interest rates and financial support services to support women's microenterprises. Previous entrepreneurial training programmes delivered by Tonga Skills and the Ministry of Labour and Commerce have included business and financial literacy training for women.[52]

Gender and connectivity

Digital connectivity is critical aspect of modern life – as it opens up access to socio-economic opportunities. In Tonga, 53 percent of women and 47 percent of men own a mobile phone, and 53.8 percent of women, and 46.2 percent of men use mobile data.[53] However, these figures are well below Tonga's target of universal internet access by 2025. Tonga also lacks sex or gender disaggregated data and gender analysis, regarding many of the ways in which these new technologies are intersecting with gender relations, and cultural transformations, are underway. The country also wants to translate mobile use to actual qualifications in ICT technology and engineering, innovation, research and development, including energy, and climate change. [54]

Women with Disabilities

Percentage of the population with disability is 6.4 percent (male 5.7 percent; female 7 percent).[55] Gender disaggregated data for disability is sparse. The 2018 Tonga Disability Survey shows that disability prevalence is 2.2 percent among children aged 2-4, 2.0 percent among children aged 5-17 and 11.4 percent among population aged 18+. Out of the total population with disabilities, Tongatapu rural recorded the highest prevalence rate of 47.1 percent; Tongatapu urban at 21.3 percent; Vava'u at 13.4 percent; Ha'apai at 10.9 percent; 'Eua at 6.4 percent and Ongo Niua at 0.7 percent. A substantial proportion of those with a disability occurring early in life (about 12 percent) were the result of preventable diseases and medical conditions, which can be read as a gap in post-natal, pre-natal and early childhood health services.[56]

People with disability are one of the most marginalized and vulnerable groups in Tongan society, often facing stigma, exclusion and discrimination, and lack voice and influence over community and national resources.[57]

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[51] Government of the Kingdom of Tonga (2019), 'Gender Equality: Where do we stand?',
[52] UN Women (2022), 'Gender Equality Brief for Tonga' https://asiapacific.unwomen.org/sites/default/files/2022-12/UN_WOMEN_TONGA.pdf
[53] Government of Tonga (2021), Census 2021: factsheet https://tongastats.gov.to/census-2/population-census-3/census-report-and-factsheet/
[54] Government of Tonga (2023), 'National Statement by Lord Vaea, Minister for Internal Affairs of the Kingdom of Tonga' CSW 67, https://www.forumsec.org/2023/03/10/35252/
[55] Tonga Statistics Department (2022) 'Tonga 2021 Census of Population and Housing' Volume 1: Basic Tables, https://tongastats.gov.to/download/272/census-report-and-factsheet/7646/census-2021-preliminary-results_dec-2021-2.pdf
[56] Tonga Statistics Department (2018), 2018 Tonga Disability Report'
[57] RNZ (2023), **Disabled face discrimination in Tonga according to survey.** Article by Finau Fonua, 6 Jan 2023

Households with persons with disability had higher average and median expenditure than households without persons with disability. In terms of access to education, 94 percent of persons with disabilities have ever attended school in comparison to 100 percent of persons without disabilities. Connectivity wise, 91 percent of people without disability used a mobile phone against 75 percent for people with a disability. Related to livelihood, 59 percent of people with disability undertook handicrafts as their main activities. 60 percent of people with disabilities reported doing household/family duties as their most prevalent activity, compared to 42 percent of females, 32 percent of all the population aged 15+ years.[58]

It is likely that persons with disabilities are more engaged in household/family duties due to lack of access to other spaces. A high proportion of people with disability, notably women, find it very difficult to go out in public places, including participating in the election, and shopping, implying that much needs to be done to make public places, facilities for compulsory events like voting and commercial precincts accessible to people with disability.[59]

The 2018 survey also notes that a higher proportion of persons without disabilities able to participate in Government decision-making with 94.2 percent compared to 68.4 for those with disabilities, indicating that Government decision making could be more inclusive. There are no significant differences between men and women with functional disability and participation rates, and the disparities between those with and without functional difficulties are stark. However, it is interesting to note women with disability find it difficult to participate in household decision making than women with no functional disability.[60]

Given the above context, the project recognizes the need for ensuring participation of people (especially women) with disabilities in setting the transformative adaptation agenda and in coastal resilience activities.

Gender Based Violence[61]

Gender Based Violence (GBV) is a major issue across the pacific. The 2019 Multiple Indicator Cluster Survey (MICS) provides latest data on some forms of GBV in Tonga, especially domestic violence.

Data shows that more women than men justify domestic violence. About 2-in-5 women and 1-in-5 men aged 15-49 years justified husbands beating their wives for specific reasons, such as, going out without telling husband; neglecting children; arguing with husband; refuses sex with husband or burns the food. More women in rural areas (43 percent) justified wife beating than women in urban areas (20 percent).

Close to 3-in-10 ever married woman aged 15-49 years has in their lifetime experienced emotional, physical or sexual violence at the hands of their current or most recent husband/partner, and 2-in-10 in the last 12 months preceding the 2019 MICS survey. One-in-20 women aged 15-49 years have ever experienced sexual violence and one-in-50 during the last 12 months (preceding the 2019 MICS survey). Of those women aged 15-49 years who have experienced any physical or sexual violence, only 18 percentage sought help to stop violence, and 60 percentage never sought help or told anyone about the violence faced.

1 percent of the women aged 15-49 indicated that experienced non-partner sexual violence in the 12 months preceding the 2019 survey, and 3 percent indicated having experienced non-partner sexual violence in their lifetime (SDG 5.2.2.). The 2009 National Study on Domestic Violence against Women

[58] Tonga statistics department (2022), 'Tonga 2021 Household Income and Expenditure Survey Report'
[59] Tonga Statistics Department (2018), 2018 Tonga Disability Report'
[60] ibid
[61] This section includes data from Govt of Tonga (2020), 'Tonga MICS 2019: Snapshot of Key Findings' https://tongastats.gov.to/survey/mics-survey/, except where otherwise indicated.

in Tonga found that women in Tonga experience violence by non-partners (especially fathers and teachers) three times more often than by partners.[62]

The non-partner and intimate partner violence at home could be an explanation for 22 percent of women in urban areas and 13 percent of women in rural areas not feeling safe at home.

Feeling of safety (SDG 16.1.4) 97 percent of boys and 81 percent of girls ages 15 to 19 feel safe walking alone in their neighborhood after dark. When ages 15 to 49 are considered 78 percent of women in urban areas and 87 percent of women in rural areas feel safe, compared to 97 percent and 99 percent of men in urban and rural areas respectively. At home too, 78 percent of women in urban areas, 79 percent of women in rural areas feel safe, as opposed to 99 percent of men and 98 percent in urban and rural areas respectively.

13 percent of adolescent girls, and 4 percent of boys ages 15 to 19 who ever felt discriminated or harassed based on their gender. The figure for adults 15 to 49, is 5 percent of women in urban, 13 percent in rural area, compared to 2 percent and 3 percent for men in urban and rural areas respectively.

More mothers/female caretakers believe that a child needs physical punishment in order to educate them properly compared to male caretakers (37 and 28 percentage, respectively).

A 2017 government survey highlighted that cyber bullying and crime are growing and should be seen as new form of gender-based violence.[63] Tonga plans to develop a law to curb cybercrime.

The impact of violence on health, socio-economic development of women is very high as it prevents women from having an equal opportunity to participate in their society. The government of Tonga, development partners, CSO/NGOs as well as faith-based organizations are responding to address GBV and also provide needed services to GBV survivors. The Tonga Women and Children's Crisis Centre (WCCC), founded in 2009, provides counselling services, refuge and advocacy for women and children survivors/victims. The Family Protection Act 2013 (FPA) provides the legal framework for women to seek protection, security and justice from domestic violence, with financial resources for implementation of the FPA and the No Drop policy by the police.[64]

Gender, Climate Change and Disaster Risk

Tonga's Climate Change Policy recognizes the different impacts of climate change on men and women. JNAP-2 (2018-2028) recognizes, in line with the WEGET policy, that considerations of gender, disability and vulnerability must be placed at the center of all planning and climate change, disaster preparedness and response activities. JNAP-2 target 17 is on achieving gender equality and social inclusion (GESI) for resilient development.

Globally women and children are 14 times more likely to die from natural disasters than men. In the 2009 Tonga Tsunami, 70 percent of those who died were women. In 2022 Tsunami triggered by the Hunga-Tonga-Hunga-Ha'apai volcano eruption, of the four fatalities, 3 were women (in other words 75 percent of the fatalities were women). 80 percent of the population was affected by tropical cyclone Gita (2018), which destroyed 800 houses and damaged an additional 4000 houses. However, in accessing relief measures, significantly more women (7124) than men (1410) accessed shelter relief and non-food items.[65] This data can be interpreted in different ways – a) that more women are

[62] Government of the Kingdom of Tonga (2019), 'Gender Equality: Where do we stand?'
[63] Government of Tonga (2023), 'National Statement by Lord Vaea, Minister for Internal Affairs of the Kingdom of Tonga' CSW 67, https://www.forumsec.org/2023/03/10/35252/
[64] Government of the Kingdom of Tonga (2019), 'Gender Equality: Where do we stand?'
[65] ibid

vulnerable than men b) traditional and cultural attitudes allow for women to readily access relief instead of men.

Economically, women are more vulnerable as nearly 75 percent are engaged in informal sector, including handicraft work. Climate change and disaster risks are likely to adversely impact subsistence farming and access to natural resources (such as weaving materials) used for handicrafts. Women also face increased risk of GBV post-disaster. Women participate in much smaller numbers in formal and informal decision-making spaces on disaster risk management and climate change adaption, and, as a consequence, are often less likely to receive critical information on emergency preparedness or to participate in decision-making bodies that finalize relocation plans and locations of evacuation centers. Disaster preparedness committees are largely made up of men resulting in decisions and discussions that fail to take into account the specific needs of women and other marginalized groups.[66]

Concerted efforts are required to improve women's capacity and access to decision making spaces. Response and preparedness measures will have to respond to specific needs of women.

In addition, the elderly, youth and people with disabilities are also vulnerable to climate change and disasters. Elderly are physically more susceptible to risks of injury and death from natural disasters. As the proportion of people over the age of 60 in Tonga is expected to steadily increase to 13 percent by 2050, special measures are required to reduce their vulnerability. As mentioned in the section above on disabilities, climate change and disaster risk reduction measures will have to be designed to ensure participation as well as respond to the specific needs of people with disabilities.

Gender and coastal resilience (with a focus on Hahake)

As elaborated in the funding proposal, Tonga faces significant climate change risks and is particularly vulnerable to sea-level rise (SLR), increasing temperatures, and increased frequency of storms and tropical cyclones. The largest island Tongatapu, for example, where 74% of the population reside, faces significant frequent flooding risks, as well as permanent losses from SLR – under a 0.5m SLR scenario are expected to be 6 percent, and under a 1.0m SLR scenario to be 25 percent [67] - in the absence of future adaptation measures.

The Hahake peninsula area, where the coastal protection works will be constructed through the Tonga Coastal Resilience Project, is particularly vulnerable to SLR and flooding. As indicated in the Funding Proposal, the area has extremely low laying ground that is less than 1m above sea level and is exposed to wave overtopping and marine flooding. In the recent 2022 tsumani, 1.2m tsunami struck Nuku'alofa and Nukuleka and Manuka (Kolongo) on Tongatapu [68]. The proposed coastal protection works will be in two sections: Nukuleka in the west (2300m) and Kolonga in the northeast (2025m). Other parts of the shoreline in the Hahake peninsula area have had revetment work in 2018-19. The Kolonga section is particularly vulnerable as it is a narrow strip of land that prevents marine water ingress into the extensive freshwater swamp landward, which is used for cultivating swamp taro and tree crops, and is a main source of drinking water. The road along the shoreline (especially Kolonga section) is used by the western low-lying villages as an escape route for tsunami and cyclone alerts.

Related to gender and coastal resilience, it can be viewed in three broad areas – inclusion and participation in decision making processes related to coastal resilience (including infrastructure development), livelihood, and early warning and disaster risk reduction.

- Engagement in decision-making processes: Coastal resiliency involves reducing ecological and social economic risks of climate change and hazards. The two types of decision-making

[66] ibid
[67] ADB, *Multi-Hazard Disaster Risk Assessment, Tongatapu*, Risk Assessment Summary Report (Asian Development Bank (ADB), 2021).
[68] Funding proposal Tonga Coastal Resilience Project. Submission to GCF March 2024. Quoting BoM, 'National Tsunami Bulletin', *Bureau of Meteorology (Australia)*, 2022 <http://www.bom.gov.au/tsunami/national.shtml#nationalBulletin0> [accessed 18 September 2023].

processes in Tonga that promote community-based resilience are the CDPs and district plans, and the Special Management Areas (SMAs). While the CDP process has provided an avenue for women's participation, overall women's engagement remains limited (see section on political participation). Added to that, CDPs (and by extension the district and island plans) are also limited in scope in incorporating climate risks and promoting long-term adaptation planning. Local communities, particularly women, understanding of long-term risks is limited which prevents an informed engagement on long-term risks and potential solutions. Enhancing women's knowledge and awareness of climate change risks and creating spaces for women to develop local climate solutions is necessary for enhancing women's climate change resilience.

Kolonga and Nukuleka also are declared Special Management Areas (SMAs). SMAs are part of a community-based fishery programme in Tonga, where designated communities are granted legal rights to manage their coastal fishery resources.[69] Within each SMA is at least one fish habitat reserve (FHR) or no-take area is placed for the conservation and replenishment of marine species, no-one including the community are allowed to fish or harvest from the FHR.

The SMA programme is gender-responsive and promotes inclusion of women in the SMA management, use and conservation. Community councils that govern SMAs are made up of both men (60 percent) and women (40 percent).[70] However, gaps remain in full engagement of women in SMA. In a response to survey question on the inclusion, recognition, and support for women' knowledge and engagement in fisheries management – more men than women found the inclusion and recognition of women's role as being supported by SMA management, decision-making and leadership. Women believed that less change in the area of equal engagement of women in SMA has occurred compared to men. Women accounted for only 35 percent of those trained on fisheries management and alternative fishing activities. The insights from the findings from the survey on the SMA programme on women's engagement in decision making could be useful for strengthening women's participation. However, more analysis is required to identify the different roles of men and women in the SMA management and how meaningful engagement of women can be promoted to enhance coastal resilience.

Beyond community level engagement, spaces for broader national-level engagement on long-term adaptation planning that takes a critical look at the various risks and solutions are not present. Even the National Infrastructure Investment Plan (2021-2030) does not elaborate on enhancing women's participation in infrastructure development (including climate resilience infrastructure development), beyond initial community consultation. A core priority project identified in the infrastructure investment plan was to build a women's center for handicraft processing.

- Livelihood: As indicated in the sections on economic participation, and access to resources, women's economic participation is lower than men. In coastal areas as well, significantly more men than women are engaged in reef fishing and other types of fishing and gleaning for shells and invertebrates for household consumption. Women are mostly engaged in harvesting sea grass and other materials for making handicrafts.[71] Unlike in other islands, women in the main Tongatapu island have more opportunities to participate in the formal economy (see section on economic participation). Nevertheless, coastal areas provide vital raw materials for household consumption and for manufacturing of traditional goods.

[69] ibid
[70] ibid
[71] Pacific Community -SPC (2022), 'Household Survey of Special Management Area Communities in Tonga: Assessment for the monitoring and evaluation of the SMA programme", Tonga Ministry of Fisheries, and Vava'u Environmental Protection Association (VEPA)

Additionally, as mentioned above, the freshwater swamp on the landward side on the Kolonga section is a key agricultural area and a drinking water source. Protecting the swamp through coastal protection works will ensure continued food security of the population in that area, at least short-term till long-term adaptation measures can be identified and supported.

- Disaster risk reduction: Women are disproportionately affected by disasters. In Tonga, though men and women receive early-warning information almost on par, women continue to be affected more which reduces their resilience to disaster risks. Gender-based violence is prevalent, particularly following a disaster.[72] More than 9 in 10 women experienced mental health issues because of disasters.[73] Infrastructure and road damage, especially in rural areas, impacts women more by limiting their mobility and increasing their vulnerability.
  The coastal protection wo, especially in the Kolonga section will protect an important evacuation route and will help in reducing women's vulnerability during disaster.

III. Mechanisms to promote gender equality in Tonga – Legal and administrative frameworks.

Tonga is not one of the signatory countries to the Convention on the Elimination of all Forms of discrimination against Women (CEDAW), as that would require amending the land ownership system, which restricts women's rights to own land. However, commitments have been made to a number of international and regional gender equality conventions. These include the following:

- Beijing Platform for Action of Women (Sep 1995);
- Millennium Development Goals (Sep 2005);
- Commonwealth Plan of Action for Gender Equality (1995);
- United Nations General Assembly Special Sessions (UNGASS) on the Implementation of the Outcomes of the World Summit for Social Development (May 2000); and on the Implementation of the Outcomes of the International Conference on Population and Development (ICPD+5) (Hague, February 1999);
- Declaration and Program of Action of the UN World Summit for Social Development, Copenhagen, Denmark, (March 1995);
- UN International Conference on Population and Develo
- pment, Cairo, Egypt, (1994);
- The Pacific Platform for Action on Women and Development (1994);
- The Pacific Plan and various Pacific Islands Forum Communiqués, and
- Pacific Islands Forum Leaders' Declaration on Gender Equality (2012)
- SPC Pacific Platform for Action on Gender Equality and Women's Human Rights (2018-2030)

Despite existence of some laws discriminate against women, notably those related to land ownership and the distribution of property and wealth during divorce, Tonga is committed to promoting gender equality. Tonga pledged its commitment to the UN Sustainable Development Goals (SDGs), including achievement of SDG 5 on gender equality and women's empowerment.

Nationally, the Tonga Strategic Development Framework II (2015-2025) aims to promote a more progressive Tonga supporting a higher quality of life for all through inclusive and sustainable growth and development. TSDFII (2015-2025) includes seven national outcomes, and 29 organizational outcomes grouped into five pillars. National Outcome 3 focuses on 'inclusive, sustainable and empowering human development with gender equality'.

[72] UNDRR(2022), 'Disaster Risk Reduction in the Kingdom of Tonga: Status Report' https://www.undrr.org/media/83687/download?startDownload=true
[73] UN Women (2022), 'Gender and environment Survey Report 2022: Kingdom of Tonga https://data.unwomen.org/sites/default/files/documents/Publications/2023/Tonga-gender-env-survey.pdf

The revised Gender National Women's Empowerment and Gender Equality Tonga (WEGET) Policy and Strategic Plan of Action 2019-2025 overall objective is to support achievement of the national outcome 3. The policy has five priority outcomes:

1. Enabling environment for mainstreaming gender across government policies, programmes and services, corporate budgeting and monitoring and evaluation
2. Families and communities prosper from gender equality.
3. Equitable access to economic assets and employment
4. Increased women's leadership and equitable political representation
5. Create equal conditions to respond to natural disasters, environmental challenges and climate change.

People with disabilities and vulnerable groups are integral to the five (5) priority outcomes.

The Women's Affairs and Gender Equality Division (WAGED) within the Ministry of Internal Affairs has the overall mandate for implementing WEGET policy and supporting other line ministries to mainstream gender. In addition to the policy, the department also produced the Gender Mainstreaming Handbook (2019), and a status report on gender equality in Tonga titled, 'Gender Equality: Where do we stand?'(2019). The handbook provides sector guidance to line ministries and identifies 6 entry points for mainstream gender. Additional capacity development support may be required for different ministries and departments to fully utilize the handbook and mainstream gender. In the 2019 performance audit by the Office of the Auditor General on Tonga's Preparedness for Implementation of Sustainable Development Goals, recognized gender as a cross-cutting issues and the role of Ministry of Internal Affairs (where WAGED is located) role in mainstreaming gender, along with MEIDECC role in mainstreaming climate change. The audit report called for performance audit of gender and climate change, and addressing any gaps identified.

However, with COVID19, implementation of the above initiatives had stalled. In addition, the WEGET policy also identifies a weak enabling environment for gender mainstreaming in the country. WEGET policy refers to UNDP's Tonga Climate Finance and Risk Governance Assessment conducted in 2016, which found that only 6 percent of the projects assessed had targeted action to advance gender equality, and 17 percent of projects assessed mainstreamed gender and have a component on gender analysis to inform at least one gender responsive activity. However, the majority of the projects (44 percent) were gender blind, and 22 percent projects had limited gender elements.[74]

The assessment found that departments and agencies involved in infrastructure or technical activities were significantly more likely to have projects that were gender blind or insufficiently gender sensitive, than those by ministries involved in the provision of human services. Infrastructure projects are not usually intended to target specific social vulnerabilities but do have the capacity to impact upon local community and social dynamics inadvertently or indirectly, sometimes significantly – a key finding for this project implementation. This observation demonstrates the need to build gender and social inclusion mainstreaming into whole-of-government processes and to ensure that agencies have appropriate and accessible technical expertise to call upon.[75] Inability to mainstream gender could affect Tonga's position to meet gender and social inclusion requirements of the main climate funds, not to ensure that vulnerable groups are adequately supported to prepare for, survive and recover from the impacts of disasters and climate change.[76]

[74] UNDP (2016), Tonga Climate Finance and Risk Governance Assessment https://www.undp.org/pacific/publications/tonga-climate-financing-and-risk-governance-assessment
[75] ibid
[76] ibid

Other factors that have impacted the weak enabling environment for gender mainstreaming included constant changes to the accountable ministers and delays in recruitment of CEOs. Additionally, lack of gender budgets in corporate plans; understaffing and under resourcing of WEGAD; lack of institutionalized data collection especially disaggregated data by relevant ministries; lack of public sector awareness and capacity to mainstream gender impacts the overall ability of the government to empower women and promote gender equality in the country.[77]

Despite challenges, the government is committed to promoting gender equality and there is buy-in from key ministries to mainstream gender. Specifically, the Joint National Action Plan on Climate Change and Disaster Risk (2018-2028) recognizes the importance of integrating gender and vulnerability considerations in all planning, and climate change, disaster preparedness and response activities. JNAP-2 target 17 is on achieving gender equality and social inclusion (GESI) for resilient development.

IV. Gender Analysis and Recommendations

The gender assessment and analysis undertaken during the design of this project led to including gender considerations across project outputs and activities and will serve as the basis for gender-responsive implementation of the project. The assessment and analysis will require continuous updates and revision throughout the lifetime of the project to ensure that project does not negatively affect local social dynamics but contributes to enhancing coastal resilience of local communities, including the marginalized and socially vulnerable groups.

During the project design phase, consultations were held with a diverse range of stakeholders and partners – including MEIDECC, Ministry of Internal Affairs (Women's Affairs and Gender Equality Division), Ministry of Lands and Natural Resources (MLNR), Prime Minister's office (Department of Local Governments), Ministry of Finance, etc., In addition, there were discussions with development partners such as ADB, EU, SPREP and NGOs such as the Civil Society for Tonga (CSFT), the Mainstreaming of Rural Development Innovation (MORDI), and the Vava'u Environmental Protection Association (VEPA). The analysis also drew on data from different assessment and survey conducted by the Government of Tonga – such as 2019 Multiple Indicator Cluster Survey, 2018 National Disability Survey etc., as well as policy directives provided by the WEGET Policy, the Tonga Strategic Development Framework II (TSDF), Joint National Action Plan on Climate Change and Disaster Risk (2018-2028), and National Climate Change Policy.

The gender analysis, through stakeholder engagement and consultations, enabled the following:

- Acknowledgement of the need for expanding public spaces for equitable engagement of men, women, youth etc., in setting the transformative agenda for adaptation.
- Identification of project activities that reinforce women's, men's and youth's empowerment in social, institutional, and economic landscape particularly in the context of coastal vulnerability reduction, ecosystem restoration and resilience building,
- Initial awareness raising among women's groups, youth groups, church groups and other vulnerable groups about the project,
- Eliciting women's and other vulnerable groups' concerns about coastal erosion and flood related risks; and
- Identifying specific roles and responsibilities that can be played by women and other vulnerable groups during the project implementation and verifying their commitment for their support.
- Ensuring that gender analysis (differentiated impact on men and women) is integrated into all capacity development activities.

[77] Government of Tonga (2019),' National Women's Empowerment and Gender Equality Tonga Policy and Strategic Plan of Action 2019–2025' https://hrsd.spc.int/sites/default/files/2021-07/WEDGET_STRATEGIC_PLAN_OF_ACTION_2019_2025_Final.pdf

- Ensuring the grievances mechanism is accessible to all groups of society.

This gender assessment and stakeholder consultations have identified the following areas as particularly important dimensions to be reflected in the implementation strategy of the project:

Gender Analysis

- The different needs in responding to extreme weather events faced by women and men, as well as other vulnerable groups;
- Analysis of the gendered-based division of labour which can be reflected in the engagement of women, men and youth in project activities;
- Reflecting women's access to, and control over, environmental resources and the goods and services that they provide into the design of ecosystem restoration and monitoring activities;
- The need to collect sex and age disaggregated data during the project implementation so that gender- and age-responsive decisions can be supported;

Voice and participation

- Ensure equitable participation of women, men, youth and elderly people at community-level adaptation actions;
- Create opportunities for women to voice their views on transformative agenda for adaptation, and in planning, monitoring and implementation of climate and risk informed Community Development Plans (CDPs) (component 1).
- Ensure the participation of women's groups, youth groups and church groups in community mobilization;
- Create opportunities for participation of vulnerable and people with disabilities in all project consultations and activities.
- Adjusting advocacy and awareness raising approach to different population groups; for sensitive topics, women-only, men-only, or youth-only meetings may be needed and organizing consultations at specific times of the day may be advised for ensuring women's participation;
- Include all stakeholders involved in the project to develop awareness raising / training aimed at drawing attention to the implication of climate resilience adaptation and gender equality;
- Undertake community discussions and dialogue in relation to gender and social inclusion in climate and disaster resilience.
- Organize design workshops (targeting women and people with disabilities) to develop transformative adaptation solutions.

V. Implementation arrangements for the Gender Action Plan

UNDP – as the Accredited Entity – will be responsible for the overall implementation of the Gender Action Plan (GAP), in close collaboration with the responsible parties – the Department of Climate Change under the Ministry of Meteorology, Energy, Information, Disaster Management, Environment, Climate Change and Communications (MEIDECC), the Ministry of Lands and Natural Resources (MLNR) and Ministry of Infrastructure (MoI).

Ministry of Internal Affairs (MIA) which is also part of the Joint National Action Plan for Climate Change- 2 or JNAP -2 (2018-2028) Task Force is another key partner in the implementation of the project, and the GAP. Especially, the Women's Affairs Division, and the Department of Local Government, both within the MIA will provide coordination support for mainstreaming gender across all three output areas.

The Project Management Unit (PMU) for the project will be based in Tonga and an International Gender Specialist will be part of the PMU. Gender technical expertise and assistance (including from civil society organizations) will be sought to supplement the available skill set and experience of the PMU in implementing the GAP. Areas where expertise of CSOs will be sought (indicative) are highlighted in the GAP below.

The GAP also acknowledges existing gender capacity gaps in the country. Though gender inclusivity is one of the guiding principles of JNAP-2, and Gender equality and social inclusion (GESI) for resilient development is one of the 22 targets of JNAP-2, MEIDECC which serves as the secretariat for the JNAP-2 and the main responsible partner of the project, does not have dedicated gender experts. The Women's Affairs Division does not have the technical expertise on gender and climate change and enhancing resilience. Therefore, as part of the implementation of the GAP, the PMU (especially the Gender Specialist) will sensitize responsible parties and partners on the GAP and organize capacity development training on gender responsive climate resilience.

Gender Action Plan

Impact	**Gender responsive and participatory adaptation solutions enhances immediate and medium-term resilience, and long-term transformative adaptation capacities of women and girls in Tonga .**							
Outcome	**Improved capacities and participatory spaces for women and girls promote gender-responsive transformative adaption solutions.**							
Project outputs	**Gender Targets and Activities**	**Baseline**	**Timeline**	**Budget**	**Responsibility**	**Means of Verifying the achievement of targets**	**Potential risk and barriers in the delivery of targets**	**Risk mitigating measures**
Output 1: Knowledge, capacity and engagement for incorporating climate risks into long-term adaptation planning strengthened and supported through multi-sectoral, multi-stakeholder engagement and dialogue platform	1.1.1. The engagement plan and guidelines for bottom-up national engagement process will include specific guidance on engaging women, girls and other vulnerable groups. Target: one gender responsive stakeholder engagement plan	0	Year 2-6	10,000 (for consultant to assist the drafting of the stakeholder engagement plan)	UNDP, MEIDECC	Gender responsive stakeholder engagement plan (final document)	Low risk level. however, capacities to translate the policy ambition of gender and social inclusion (JNAP target 17) into plans may be limited among government ministries.	The project will build on JNAP's commitment to promote gender and social inclusion and hire an expert to support development of a gender responsive stakeholder engagement plan
	1.1.2. National awareness campaign will include specific campaigns and workshops targeting women and girls to co-create solutions.	0	Year 2- 6	500,000 (100,000x 5 years for campaigns and workshops)	UNDP, MEIDECC, NGOs/ CSOs	Sex disaggregated data of national awareness campaign and workshop attendees in semi-annual	Community stakeholder may not see women's participation as a priority and discourage	The project will work with women leaders, women officials, and women's organizations to raise awareness, and organize consultations.

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Outcome	**Improved capacities and participatory spaces for women and girls promote gender-responsive transformative adaption solutions.**							
Project outputs	**Gender Targets and Activities**	**Baseline**	**Timeline**	**Budget**	**Responsibility**	**Means of Verifying the achievement of targets**	**Potential risk and barriers in the delivery of targets**	**Risk mitigating measures**
	Target: at least 3 campaign activities (women-led talk shows, or women's forum or media advertisements targeting women and their role in shaping national adaption strategies) 3 design thinking workshops held for women/women leaders/ women organizations targeting at least 75 women	0				GAP progress reports	women's participation	The project will also raise awareness of men and male leaders on the importance and effectiveness of gender equality and women's participation in decision making through targeted messaging.
	1.1.3. Terms of reference for national adaptation planning engagement platform will be gender-responsive. Target: 1 gender-responsive ToR developed	0	Year 2	-	UNDP, MEIDECC, Women's Affairs Division (Ministry of Internal Affairs)	Gender responsive ToR (final document)		No issue anticipated. The project will support Women's Affairs Division role in the National Adaptation Planning engagement platform

Impact	Gender responsive and participatory adaptation solutions enhances immediate and medium-term resilience, and long-term transformative adaptation capacities of women and girls in Tonga .							
Outcome	**Improved capacities and participatory spaces for women and girls promote gender-responsive transformative adaption solutions.**							
Project outputs	**Gender Targets and Activities**	**Baseline**	**Timeline**	**Budget**	**Responsibility**	**Means of Verifying the achievement of targets**	**Potential risk and barriers in the delivery of targets**	**Risk mitigating measures**
	1.1.4. participation of women's affair division supported as per the ToR for the national adaption planning engagement platform Target: 1 meeting (as per the ToR) attended	0	Year 2-6	-	UNDP, Women's Affairs Division (Ministry of Internal Affairs), Women's organizations	Meeting attendee lists, GAP semi annual progress reports		No issue anticipated but any discrepancies will be reported
	1.1.5. gender-responsive policy options to incorporate climate resilience and land use principles into national frameworks Target: 1 gender-responsive guidance note, and 1 gender sensitive land use policy developed	0	Year 3-6	20,000 (for consultants)	UNDP, MEIDECC, Women's Affairs Division (Ministry of Internal Affairs), Women's organizations	Guidance note and policy document (final document)	Land use is a politically and culturally sensitive issue. Gender differentiated access to land needs to be considered in developing a gender sensitive land use policy. National actors, and community stakeholders may be reluctant to	Commitment for promoting gender equality and social inclusion is expressed in the JNAP. MEIDECC and Department of Climate change is keen on supporting gender-sensitive land use policy. The project team will build on the commitment of government, and raise

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Project outputs	**Gender Targets and Activities**	**Baseline**	**Timeline**	**Budget**	**Responsibility**	**Means of Verifying the achievement of targets**	**Potential risk and barriers in the delivery of targets**	**Risk mitigating measures**
							develop a gender-sensitive land-use policy	awareness on the importance of gender-sensitive land-use for long-term adaptation through the national awareness campaign (1.1.2). capacity development trainings (output 2) will also highlight the importance of gender mainstreaming and developing gender-responsive policies.
	1.2.1. Gender-responsive guidelines developed for participatory village and district spatial plans, and	0	Year 3-5	10,000 (for consultant) 10,000 (for 1 village level, and 1	UNDP, MEIDECC, Ministry of Lands, NGOs/CSOs,	1 guideline (final document), and sex disaggregated data of participants	Community stakeholder may discourage women's participation as gender is	The project team will sensitize and raise awareness on the importance of women's

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Outcome	**Improved capacities and participatory spaces for women and girls promote gender-responsive transformative adaption solutions.**							
Project outputs	**Gender Targets and Activities**	**Baseline**	**Timeline**	**Budget**	**Responsibility**	**Means of Verifying the achievement of targets**	**Potential risk and barriers in the delivery of targets**	**Risk mitigating measures**
	target 50 percent of female participation Target: 1 gender-responsive spatial planning guidelines developed 50 percent of men and women engaged in 1 village level and at least 30 percent of men and women of the district engaged in 1 district level participatory spatial planning			district level participatory process)	Women's Affairs Division	included in the semi-annual GAP progress report	not considered a priority in spatial planning due to traditional gender norms related to land access	participation in defining the spatial plans for long-term adaptation. The project team may involve key stakeholders such as NGOs/CSOs and Women's Affair Division to lead community awareness campaigns
	1.3.1 Updated gender-and climate-responsive CDP guidelines developed. Target: 1 gender-responsive CDP guideline	0	Year 3-5	10,000 (for consultant)	UNDP, MEIDECC, Women's Affairs Division, Department of Local Governments	1 gender-responsive CDP guideline (final document)		No issue anticipated as previous CDPs were developed through participatory process that involved women in local communities,

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Outcome	Improved capacities and participatory spaces for women and girls promote gender-responsive transformative adaption solutions.							
Project outputs	**Gender Targets and Activities**	**Baseline**	**Timeline**	**Budget**	**Responsibility**	**Means of Verifying the achievement of targets**	**Potential risk and barriers in the delivery of targets**	**Risk mitigating measures**
								but any discrepancy will be reported
	1.3.2 Develop gender-responsive capacity building package on the new gender and climate-responsive CDP guideline. Target: 1 gender-responsive capacity development package	0	Year 2-6	10,000 (for gender consultant or women's organization to work with the consultant developing the overall capacity development package)	UNDP, Department of Local Government, Women's Affairs Division	Final capacity development package (training materials, guidelines and other documents)		No issue anticipated but any discrepancy will be reported
	1.3.3. support equitable participation of decision makers and officials (men and women) in the capacity development trainings* Target: Equitable participation of men and women decision makers* *see note below	0	Year 2-6	-	UNDP	Sex disaggregated data of training attendees included in the semi-annual GAP report		No issue anticipated but any discrepancy will be reported

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Project outputs	**Gender Targets and Activities**	**Baseline**	**Timeline**	**Budget**	**Responsibility**	**Means of Verifying the achievement of targets**	**Potential risk and barriers in the delivery of targets**	**Risk mitigating measures**
	1.3.4 Voices of and women's specific issues highlighted in the national event on climate responsive CDPs Target: 1 session on women and climate and gender responsive CDPs held at the national event	0	Year 6-7	15,000	UNDP	Meeting agenda, Presentations and other media materials from the event highlighting women's voice and issues		No issue anticipated but any discrepancy will be reported
Output 2 Mechanisms for collecting and analyzing data and information for better-informed climate risk monitoring and coastal adaptation planning	2.1.1 Develop and implement gender-responsive capacity development strategy and plan for coastal data management and use. ** Target: 1 gender responsive capacity development strategy and plan	0	Year 1-3	10,000	UNDP	Final strategy and plan documents		No issue anticipated but any discrepancy will be reported

Impact	**Gender responsive and participatory adaptation solutions enhances immediate and medium-term resilience, and long-term transformative adaptation capacities of women and girls in Tonga .**							
Outcome	**Improved capacities and participatory spaces for women and girls promote gender-responsive transformative adaption solutions.**							
Project outputs	**Gender Targets and Activities**	**Baseline**	**Timeline**	**Budget**	**Responsibility**	**Means of Verifying the achievement of targets**	**Potential risk and barriers in the delivery of targets**	**Risk mitigating measures**
	2.1.2 develop knowledge materials on climate risks and impact on women living in coastal areas. Target: 1 knowledge product	0	Year 3-6	10,000 (for consultant)	UNDP	Final Knowledge product		No issue anticipated but any discrepancy will be reported
	2.1.3. Support inclusion of gender and disability considerations as part of the multi-stakeholder coordination and collaboration on coastal adaptation. Target: 1 guidance on integrating gender considerations developed for the coordination body/committee	0	year 4	10,000 (for consultant)	UNDP	Final guidance note		No issue anticipated but any discrepancy will be reported
	2.2.1. Integrate gender considerations in the assessment on trade offs between	0	Year 3-6	10,000 (for consultant)	UNDP	Final assessment		No issue anticipated but any discrepancy will be reported

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Outcome	**Improved capacities and participatory spaces for women and girls promote gender-responsive transformative adaption solutions.**							
Project outputs	**Gender Targets and Activities**	**Baseline**	**Timeline**	**Budget**	**Responsibility**	**Means of Verifying the achievement of targets**	**Potential risk and barriers in the delivery of targets**	**Risk mitigating measures**
	adaptation solution and current land-use considering food-security, water security and livelihoods of women. Target: gender-responsive assessment developed							
5Output 3: Vulnerabilities of coastal communities in Hahake to climate hazards reduced through coastal protection measures	3.1.1 50 percent of participation of women in the consultations and/or awareness workshops held on the design of the coastal protection measure. Target: 50 percent of women in Hakake engaged in the design of the coastal protection measure.	0	Year 2-6	10,000 (for consultations)	UNDP, MEIDECC, Ministry of Lands	Sex disaggregated attendees list included in the semi-annual report of the GAP		No issue anticipated on women's participation related to coastal protection measures, but any discrepancy will be reported
	3.1.2 (a) Requirements for contractors to hire at least 20 percent	0	Year 2-6	Contractor's budget	Contractor	Sex disaggregated pay sheet		No issue anticipated but any discrepancy will be reported

Impact	Gender responsive and participatory adaptation solutions enhances immediate and medium-term resilience, and long-term transformative adaptation capacities of women and girls in Tonga .							
Outcome	**Improved capacities and participatory spaces for women and girls promote gender-responsive transformative adaption solutions.**							
Project outputs	**Gender Targets and Activities**	**Baseline**	**Timeline**	**Budget**	**Responsibility**	**Means of Verifying the achievement of targets**	**Potential risk and barriers in the delivery of targets**	**Risk mitigating measures**
	women in both technical and non-technical work and ensure equal pay for equal work. Target: 20 percent of work force are women							
	3.1.2 (b) Provide safety gear, and women-friendly toilet and sanitation facilities. Target: all workers (men and women) have safety and protection gear and access to proper toilet and sanitation facilities	0	Year 2-6	Contractor's budget	Contractor	Site visits and inspection reports		No issue anticipated but any discrepancy will be reported
	3.1.2 (c) Establish grievance redressal mechanisms accessible to all (women, men, youth, elderly, people with disability) to voice complaints during the	0	Year 2-6	Contractor's budget (at construction site) 10,000 UNDP (for grievance	Contract UNDP	GAP progress report highlighting complaints received	Potential reluctance to report grievances, especially if Sexual exploitation,	UNDP will stipulate contractors, as part of the contract, to follow UNDP policies on SES, prevention of

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Outcome	**Improved capacities and participatory spaces for women and girls promote gender-responsive transformative adaption solutions.**							
Project outputs	**Gender Targets and Activities**	**Baseline**	**Timeline**	**Budget**	**Responsibility**	**Means of Verifying the achievement of targets**	**Potential risk and barriers in the delivery of targets**	**Risk mitigating measures**
	design and construction phase as part of the ESMF implementation strategy. (also linked to 1.1.2, and 1.2.1) Target: Two grievance redressal mechanisms established (one at construction site, and one for UNDP activities as indicted in the ESMF)			redressal related to 1.1.2, and 1.2.1)			abuse, and/or harassment	SEAH and the project ESMF. The project will share information on its zero-tolerance policy against SEAH, and on how to report grievance, including dedicated phone numbers, on UNDP website, on partner websites (MEIDECC) and social media as well as at the project locations.
	3.2.2 Highlight any gender-specific lessons learned in the final report and share lessons at	0	Year 7	25,000 For consultant	UNDP	Final lessons learned report, and gender specific lessons also highlighted in the semi-		No issue anticipated.

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Outcome	**Improved capacities and participatory spaces for women and girls promote gender-responsive transformative adaption solutions.**							
Project outputs	**Gender Targets and Activities**	**Baseline**	**Timeline**	**Budget**	**Responsibility**	**Means of Verifying the achievement of targets**	**Potential risk and barriers in the delivery of targets**	**Risk mitigating measures**
	international/regional conference. Target: gender-specific lessons learned identified and included in the report					annual, and final GAP reports		
				670,000				
	*Note on activity 1.3.3. Given the high number of women officials in the government of Tonga, rather than indicating 30 or 50 percent of participants should be women, the focus should on the participation of relevant decision makers in capacity development trainings to strengthen their capacity to mainstream gender and gender-responsive decision making.							
	**Note on activity 2.1.1. Use of LiDAR and satellite imagery is technical and there is no scope for mainstreaming gender. However, in the analysis of potential impact of climate change risks, it is important to mainstream gender to understand gender specific impacts.							